

Fig. 1 Free-body diagram of cable portion CB.

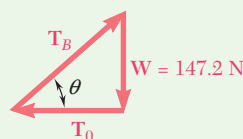


Fig. 2 Force triangle for cable portion CB.

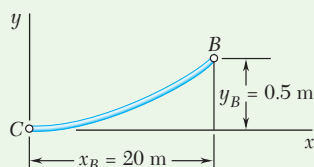


Fig. 3 Dimensions used to determine length of cable.

Sample Problem 7.9

A light cable is attached to a support at A, passes over a small frictionless pulley at B, and supports a load **P**. The sag of the cable is 0.5 m and the mass per unit length of the cable is 0.75 kg/m. Determine (a) the magnitude of the load **P**, (b) the slope of the cable at B, (c) the total length of the cable from A to B. Since the ratio of the sag to the span is small, assume the cable is parabolic. Also, neglect the weight of the portion of cable from B to D.

STRATEGY: Because the pulley is frictionless, the load **P** is equal in magnitude to the tension in the cable at B. You can determine the tension using the methods of this section and then use that value to determine the slope and length of the cable.

MODELING and ANALYSIS:

a. Load P. Denote the lowest point of the cable by C and draw the free-body diagram of the portion CB of cable (Fig. 1). Assuming the load is uniformly distributed along the horizontal, you have

$$w = (0.75 \text{ kg/m})(9.81 \text{ m/s}^2) = 7.36 \text{ N/m}$$

The total load for the portion CB of cable is

$$W = wx_B = (7.36 \text{ N/m})(20 \text{ m}) = 147.2 \text{ N}$$

This load acts halfway between C and B. Summing moments about B gives you

$$+\circlearrowleft \sum M_B = 0: \quad (147.2 \text{ N})(10 \text{ m}) - T_0(0.5 \text{ m}) = 0 \quad T_0 = 2944 \text{ N}$$

From the force triangle (Fig. 2), you obtain

$$\begin{aligned} T_B &= \sqrt{T_0^2 + W^2} \\ &= \sqrt{(2944 \text{ N})^2 + (147.2 \text{ N})^2} = 2948 \text{ N} \end{aligned}$$

Since the tension on each side of the pulley is the same, you end up with $P = T_B = 2948 \text{ N}$ ◀

b. Slope of Cable at B. The force triangle also tells us that

$$\tan \theta = \frac{W}{T_0} = \frac{147.2 \text{ N}}{2944 \text{ N}} = 0.05 \quad \theta = 2.9^\circ \quad \blacktriangleleft$$

c. Length of Cable. Applying Eq. (7.10) between C and B (Fig. 3) gives you

$$\begin{aligned} s_B &= x_B \left[1 + \frac{2}{3} \left(\frac{y_B}{x_B} \right)^2 + \cdots \right] \\ &= (20 \text{ m}) \left[1 + \frac{2}{3} \left(\frac{0.5 \text{ m}}{20 \text{ m}} \right)^2 + \cdots \right] = 20.00833 \text{ m} \end{aligned}$$

The total length of the cable between A and B is twice this value. Thus,

$$\text{Length} = 2s_B = 40.0167 \text{ m} \quad \blacktriangleleft$$

REFLECT and THINK: Notice that the length of the cable is only very slightly more than the length of the span between A and B. This means that the cable must be very taut, which is consistent with the relatively large value of load **P** (compared to the weight of the cable).

SOLVING PROBLEMS ON YOUR OWN

In the problems of this section, you will apply the equations of equilibrium to *cables that lie in a vertical plane*. We assume that a cable cannot resist bending, so the force of tension in the cable is always directed along the cable.

A. In the first part of this lesson, we considered **cables subjected to concentrated loads**. Since we assume the weight of the cable is negligible, the cable is straight between loads.

Your solution will consist of the following steps.

1. Draw a free-body diagram of the entire cable showing the loads and the horizontal and vertical components of the reaction at each support. Use this free-body diagram to write the corresponding equilibrium equations.

2. You will have four unknown components and only three equations of equilibrium (see Fig. 7.13). You must therefore find an additional piece of information, such as the *position* of a point on the cable or the *slope* of the cable at a given point.

3. After you have identified the point of the cable where the additional information exists, cut the cable at that point, and draw a free-body diagram of one of the two resulting portions of the cable.

a. If you know the position of the point where you have cut the cable, set $\Sigma M = 0$ about that point for the new free body. This will yield the additional equation required to solve for the four unknown components of the reactions [Sample Prob. 7.8].

b. If you know the slope of the portion of the cable you have cut, set $\Sigma F_x = 0$ and $\Sigma F_y = 0$ for the new free body. This will yield two equilibrium equations that, together with the original three, you can solve for the four reaction components and for the tension in the cable where it has been cut.

4. To find the elevation of a given point of the cable and the slope and tension at that point once you have found the reactions at the supports, you should cut the cable at that point and draw a free-body diagram of one of the two resulting portions of the cable. Setting $\Sigma M = 0$ about the given point yields its elevation. Writing $\Sigma F_x = 0$ and $\Sigma F_y = 0$ yields the components of the tension force from which you can find its magnitude and direction.

5. For a cable supporting vertical loads only, the horizontal component of the tension force is the same at any point. It follows that, for such a cable, the maximum tension occurs in the steepest portion of the cable.

(continued)

B. In the second portion of this section, we considered **cables carrying a load that is uniformly distributed along the horizontal**. The shape of the cable is then parabolic.

Your solution will use one or more of the following concepts.

1. Place the origin of coordinates at the lowest point of the cable and direct the x and y axes to the right and upward, respectively. Then *the equation of the parabola* is

$$y = \frac{wx^2}{2T_0} \quad (7.8)$$

The minimum cable tension occurs at the origin, where the cable is horizontal. The maximum tension is at the support where the slope is maximum.

2. If the supports of the cable have the same elevation, the sag h of the cable is the vertical distance from the lowest point of the cable to the horizontal line joining the supports. To solve a problem involving such a parabolic cable, use Eq. (7.8) for one of the supports; this equation can be solved for one unknown.

3. If the supports of the cable have different elevations, you will have to write Eq. (7.8) for each of the supports (see Fig. 7.17).

4. To find the length of the cable from the lowest point to one of the supports, you can use Eq. (7.10). In most cases, you will need to compute only the first two terms of the series.

Problems

- 7.93** Three loads are suspended as shown from the cable $ABCDE$. Knowing that $d_C = 4$ m, determine (a) the components of the reaction at E , (b) the maximum tension in the cable.

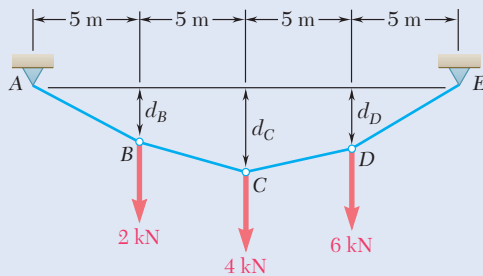


Fig. P7.93 and P7.94

- 7.94** Knowing that the maximum tension in cable $ABCDE$ is 25 kN, determine the distance d_C .
- 7.95** If $d_C = 8$ ft, determine (a) the reaction at A , (b) the reaction at E .
- 7.96** If $d_C = 4.5$ ft, determine (a) the reaction at A , (b) the reaction at E .
- 7.97** Knowing that $d_C = 3$ m, determine (a) the distances d_B and d_D , (b) the reaction at E .
- 7.98** Determine (a) distance d_C for which portion DE of the cable is horizontal, (b) the corresponding reactions at A and E .
- 7.99** Knowing that $d_C = 15$ ft, determine (a) the distances d_B and d_D , (b) the maximum tension in the cable.

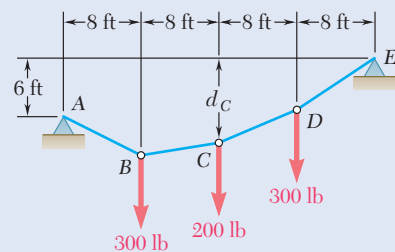


Fig. P7.95 and P7.96

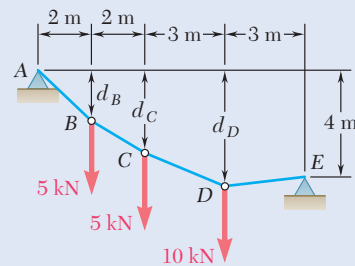


Fig. P7.97 and P7.98

- 7.100** Determine (a) the distance d_C for which portion BC of the cable is horizontal, (b) the corresponding components of the reaction at E .

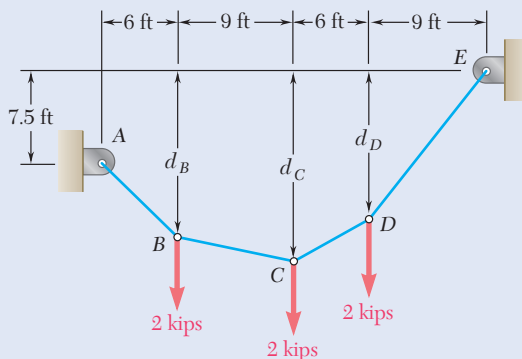


Fig. P7.99 and P7.100

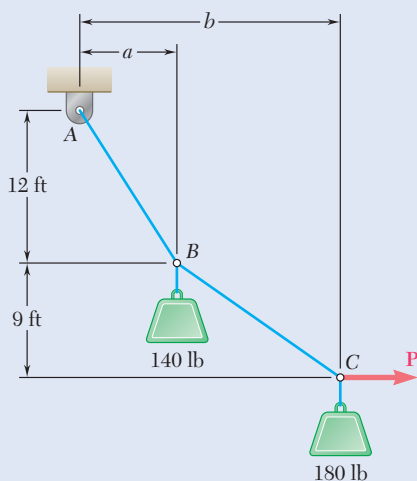


Fig. P7.103 and P7.104

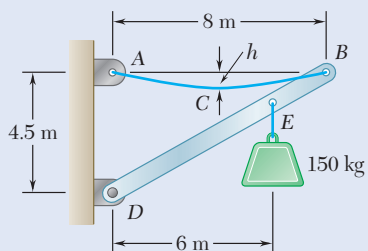


Fig. P7.108

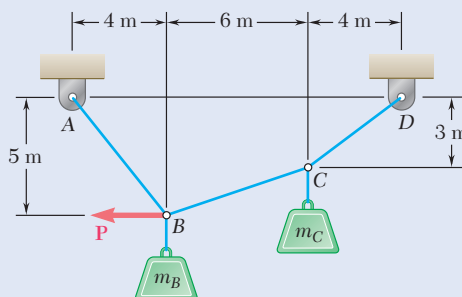


Fig. P7.101 and P7.102

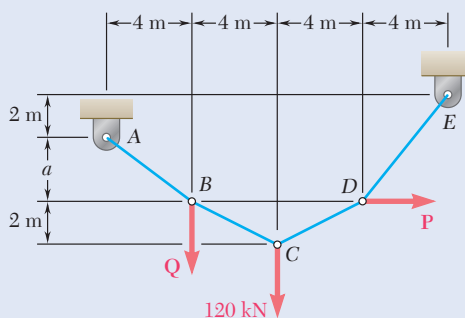


Fig. P7.105 and P7.106

- 7.101** Knowing that $m_B = 70$ kg and $m_C = 25$ kg, determine the magnitude of the force $\mathbf{P}$ required to maintain equilibrium.
- 7.102** Knowing that $m_B = 18$ kg and $m_C = 10$ kg, determine the magnitude of the force $\mathbf{P}$ required to maintain equilibrium.
- 7.103** Cable ABC supports two loads as shown. Knowing that $b = 21$ ft, determine (a) the required magnitude of the horizontal force $\mathbf{P}$, (b) the corresponding distance a .
- 7.104** Cable ABC supports two loads as shown. Determine the distances a and b when a horizontal force $\mathbf{P}$ of magnitude 200 lb is applied at C .
- 7.105** If $a = 3$ m, determine the magnitudes of $\mathbf{P}$ and $\mathbf{Q}$ required to maintain the cable in the shape shown.
- 7.106** If $a = 4$ m, determine the magnitudes of $\mathbf{P}$ and $\mathbf{Q}$ required to maintain the cable in the shape shown.
- 7.107** An electric wire having a mass per unit length of 0.6 kg/m is strung between two insulators at the same elevation that are 60 m apart. Knowing that the sag of the wire is 1.5 m, determine (a) the maximum tension in the wire, (b) the length of the wire.
- 7.108** The total mass of cable ACB is 20 kg. Assuming that the mass of the cable is distributed uniformly along the horizontal, determine (a) the sag h , (b) the slope of the cable at A .

- 7.109** The center span of the George Washington Bridge, as originally constructed, consisted of a uniform roadway suspended from four cables. The uniform load supported by each cable was $w = 9.75$ kips/ft along the horizontal. Knowing that the span L is 3500 ft and that the sag h is 316 ft, determine for the original configuration (a) the maximum tension in each cable, (b) the length of each cable.
- 7.110** The center span of the Verrazano-Narrows Bridge consists of two uniform roadways suspended from four cables. The design of the bridge allows for the effect of extreme temperature changes that cause the sag of the center span to vary from $h_w = 386$ ft in winter to $h_s = 394$ ft in summer. Knowing that the span is $L = 4260$ ft, determine the change in length of the cables due to extreme temperature changes.
- 7.111** Each cable of the Golden Gate Bridge supports a load $w = 11.1$ kips/ft along the horizontal. Knowing that the span L is 4150 ft and that the sag h is 464 ft, determine (a) the maximum tension in each cable, (b) the length of each cable.
- 7.112** Two cables of the same gauge are attached to a transmission tower at B . Since the tower is slender, the horizontal component of the resultant of the forces exerted by the cables at B is to be zero. Knowing that the mass per unit length of the cables is 0.4 kg/m, determine (a) the required sag h , (b) the maximum tension in each cable.
- 7.113** A 76-m length of wire having a mass per unit length of 2.2 kg/m is used to span a horizontal distance of 75 m. Determine (a) the approximate sag of the wire, (b) the maximum tension in the wire. [Hint: Use only the first two terms of Eq. (7.10).]
- 7.114** A cable of length $L + \Delta$ is suspended between two points that are at the same elevation and a distance L apart. (a) Assuming that Δ is small compared to L and that the cable is parabolic, determine the approximate sag in terms of L and Δ . (b) If $L = 100$ ft and $\Delta = 4$ ft, determine the approximate sag. [Hint: Use only the first two terms of Eq. (7.10).]
- 7.115** The total mass of cable AC is 25 kg. Assuming that the mass of the cable is distributed uniformly along the horizontal, determine the sag h and the slope of the cable at A and C .

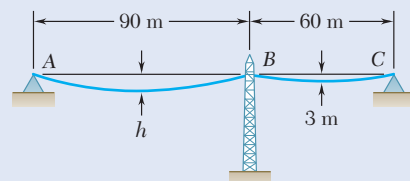


Fig. P7.112

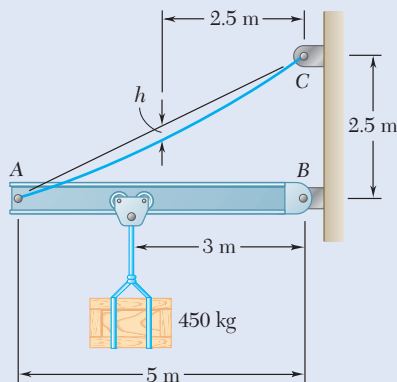


Fig. P7.115

- 7.116** Cable ACB supports a load uniformly distributed along the horizontal as shown. The lowest point C is located 9 m to the right of A . Determine (a) the vertical distance a , (b) the length of the cable, (c) the components of the reaction at A .

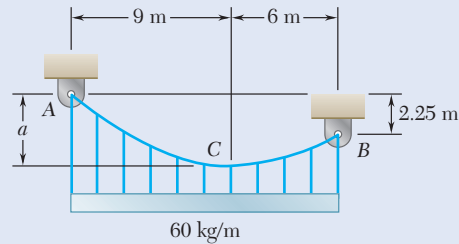


Fig. P7.116

- 7.117** Each cable of the side spans of the Golden Gate Bridge supports a load $w = 10.2$ kips/ft along the horizontal. Knowing that for the side spans the maximum vertical distance h from each cable to the chord AB is 30 ft and occurs at midspan, determine (a) the maximum tension in each cable, (b) the slope at B .

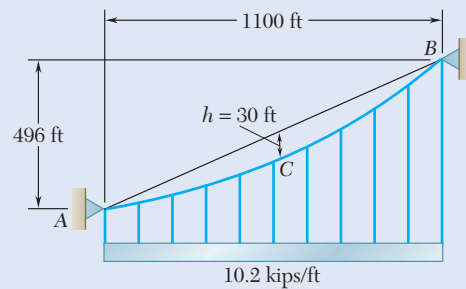


Fig. P7.117

- 7.118** A steam pipe weighing 45 lb/ft that passes between two buildings 40 ft apart is supported by a system of cables as shown. Assuming that the weight of the cable system is equivalent to a uniformly distributed loading of 5 lb/ft, determine (a) the location of the lowest point C of the cable, (b) the maximum tension in the cable.

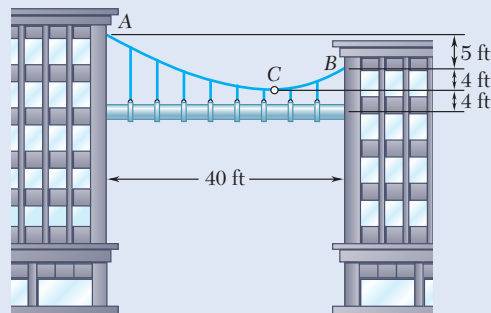


Fig. P7.118

- *7.119** A cable AB of span L and a simple beam $A'B'$ of the same span are subjected to identical vertical loadings as shown. Show that the magnitude of the bending moment at a point C' in the beam is equal to the product $T_0 h$, where T_0 is the magnitude of the horizontal component of the tension force in the cable and h is the vertical distance between point C and the chord joining the points of support A and B .

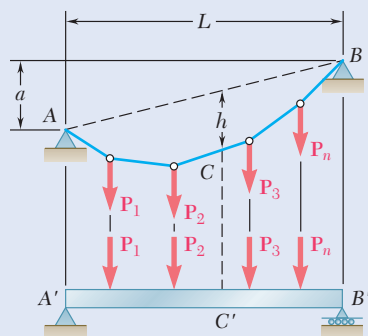


Fig. P7.119

- 7.120 through 7.123** Making use of the property established in Prob. 7.119, solve the problem indicated by first solving the corresponding beam problem.

7.120 Prob. 7.94.

7.121 Prob. 7.97a.

7.122 Prob. 7.99a.

7.123 Prob. 7.100a.

- *7.124** Show that the curve assumed by a cable that carries a distributed load $w(x)$ is defined by the differential equation $d^2y/dx^2 = w(x)/T_0$, where T_0 is the tension at the lowest point.
- *7.125** Using the property indicated in Prob. 7.124, determine the curve assumed by a cable of span L and sag h carrying a distributed load $w = w_0 \cos(\pi x/L)$, where x is measured from midspan. Also determine the maximum and minimum values of the tension in the cable.
- *7.126** If the weight per unit length of the cable AB is $w_0/\cos^2 \theta$, prove that the curve formed by the cable is a circular arc. (*Hint*: Use the property indicated in Prob. 7.124.)

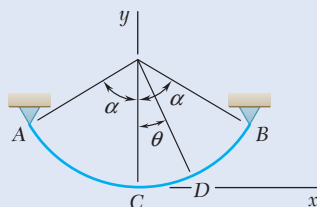


Fig. P7.126

*7.5 CATENARY CABLES

Let us now consider a cable AB carrying a load that is **uniformly distributed along the cable itself** (Fig. 7.18a). Cables hanging under their own weight are loaded in this way. We denote the load per unit length by w (measured along the cable) and express it in N/m or lb/ft. The magnitude W of the total load carried by a portion of cable with a length of s , extending

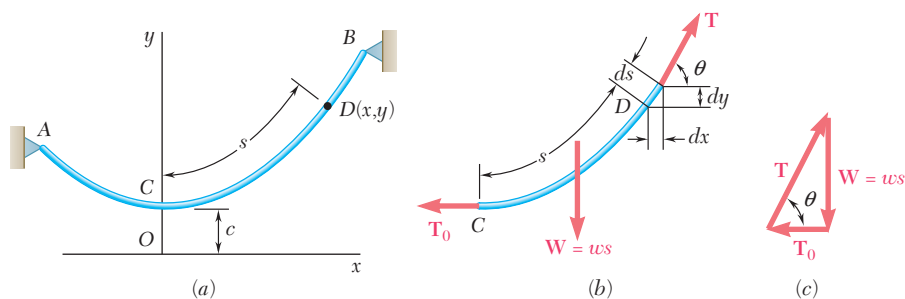


Fig. 7.18 (a) A cable carrying a load uniformly distributed along the cable; (b) free-body diagram of a portion of the cable CD ; (c) force triangle for part (b).

from the lowest point C to some point D , is $W = ws$. Substituting this value for W in formula (7.6), we obtain the tension at D , as

$$T = \sqrt{T_0^2 + w^2 s^2}$$

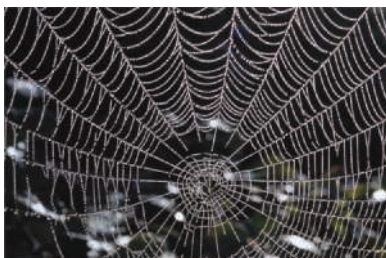
In order to simplify the subsequent computations, we introduce the constant $c = T_0/w$. This gives us

$$T_0 = wc \quad W = ws \quad T = w\sqrt{c^2 + s^2} \quad (7.11)$$

The free-body diagram of the portion of cable CD is shown in Fig. 7.18b. However, we cannot use this diagram directly to obtain the equation of the curve assumed by the cable, because we do not know the horizontal distance from D to the line of action of the resultant $\mathbf{W}$ of the load. To obtain this equation, we note that the horizontal projection



(a) High-voltage power lines



(b) A spider's web



(c) The Gateway Arch

Photo 7.5 Catenary cables occur in nature as well as in engineered structures. (a) High-voltage power lines, common all across the country and in much of the world, support only their own weight. (b) Catenary cables can be as delicate as the silk threads of a spider's web. (c) The Gateway to the West Arch in St. Louis is an inverted catenary arch cast in concrete (which is in compression instead of tension).

of a small element of cable of length ds is $dx = ds \cos \theta$. Observing from Fig. 7.18c that $\cos \theta = T_0/T$ and using Eq. (7.11), we have

$$dx = ds \cos \theta = \frac{T_0}{T} ds = \frac{wc ds}{w\sqrt{c^2 + s^2}} = \frac{ds}{\sqrt{1 + s^2/c^2}}$$

Selecting the origin O of the coordinates at a distance c directly below C (Fig. 7.18a) and integrating from $C(0, c)$ to $D(x, y)$, we obtain[†]

$$x = \int_0^s \frac{ds}{\sqrt{1 + s^2/c^2}} = c \left[\sinh^{-1} \frac{s}{c} \right]_0^s = c \sinh^{-1} \frac{s}{c}$$

This equation, which relates the length s of the portion of cable CD and the horizontal distance x , can be written in the form

Length of catenary cable

$$s = c \sinh \frac{x}{c} \quad (7.15)$$

We can now obtain the relation between the coordinates x and y by writing $dy = dx \tan \theta$. Observing from Fig. 7.18c that $\tan \theta = W/T_0$ and using (7.11) and (7.15), we have

$$dy = dx \tan \theta = \frac{W}{T_0} dx = \frac{s}{c} dx = \sinh \frac{x}{c} dx$$

Integrating from $C(0, c)$ to $D(x, y)$ and using Eqs. (7.12) and (7.13), we obtain

$$y - c = \int_0^x \sinh \frac{x}{c} dx = c \left[\cosh \frac{x}{c} \right]_0^x = c \left(\cosh \frac{x}{c} - 1 \right)$$

$$y - c = c \cosh \frac{x}{c} - c$$

[†]This integral appears in all standard integral tables. The function

$$z = \sinh^{-1} u$$

(read “arc hyperbolic sine u ”) is the inverse of the function $u = \sinh z$ (read “hyperbolic sine z ”). This function and the function $v = \cosh z$ (read “hyperbolic cosine z ”) are defined as

$$u = \sinh z = \frac{1}{2}(e^z - e^{-z}) \quad v = \cosh z = \frac{1}{2}(e^z + e^{-z})$$

Numerical values of the functions $\sinh z$ and $\cosh z$ are listed in tables of hyperbolic functions and also may be computed on most calculators, either directly or from the definitions. Refer to any calculus text for a complete description of the properties of these functions. In this section, we use only the following properties, which are easy to derive from the definitions:

$$\frac{d \sinh z}{dz} = \cosh z \quad \frac{d \cosh z}{dz} = \sinh z \quad (7.12)$$

$$\sinh 0 = 0 \quad \cosh 0 = 1 \quad (7.13)$$

$$\cosh^2 z - \sinh^2 z = 1 \quad (7.14)$$

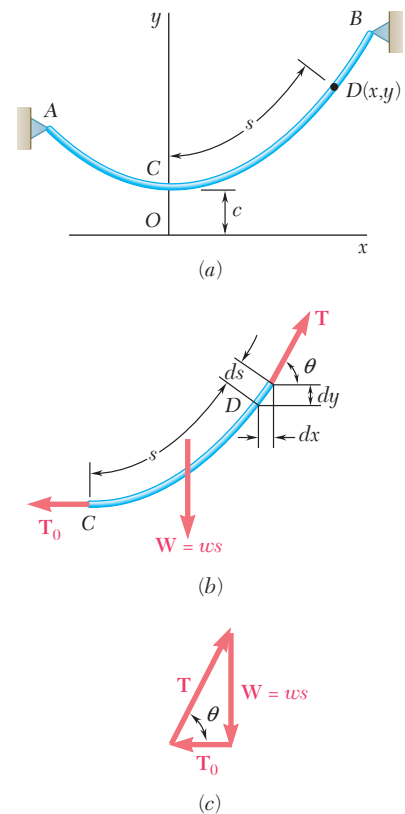


Fig. 7.18 (continued).

which reduces to

Equation of catenary cable

$$y = c \cosh \frac{x}{c} \quad (7.16)$$

This is the equation of a **catenary** with vertical axis. The ordinate c of the lowest point C is called the *parameter* of the catenary. By squaring both sides of Eqs. (7.15) and (7.16), subtracting, and taking Eq. (7.14) into account, we obtain the following relation between y and s :

$$y^2 - s^2 = c^2 \quad (7.17)$$

Solving Eq. (7.17) for s^2 and carrying into the last of the relations in Eqs. (7.11), we write these relations as

$$T_0 = wc \quad W = ws \quad T = wy \quad (7.18)$$

The last relation indicates that the tension at any point D of the cable is proportional to the vertical distance from D to the horizontal line representing the x axis.

When the supports A and B of the cable have the same elevation, the distance L between the supports is called the *span* of the cable and the vertical distance h from the supports to the lowest point C is called the *sag* of the cable. These definitions are the same as those given for parabolic cables; note that, because of our choice of coordinate axes, the sag h is now

$$h = y_A - c \quad (7.19)$$

Also note that some catenary problems involve transcendental equations, which must be solved by successive approximations (see Sample Prob. 7.10). When the cable is fairly taut, however, we can assume that the load is uniformly distributed *along the horizontal* and replace the catenary by a parabola. This greatly simplifies the solution of the problem, and the error introduced is small.

When the supports A and B have different elevations, the position of the lowest point of the cable is not known. We can then solve the problem in a manner similar to that indicated for parabolic cables by noting that the cable must pass through the supports and that $x_B - x_A = L$ and $y_B - y_A = d$, where L and d denote, respectively, the horizontal and vertical distances between the two supports.

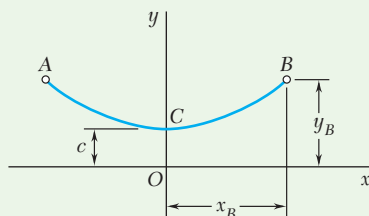
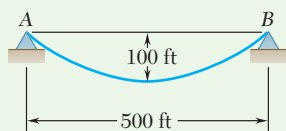


Fig. 1 Cable geometry.

Sample Problem 7.10

A uniform cable weighing 3 lb/ft is suspended between two points *A* and *B* as shown. Determine (a) the maximum and minimum values of the tension in the cable, (b) the length of the cable.

STRATEGY: This is a cable carrying only its own weight that is supported by its ends at the same elevation. You can use the analysis in this section to solve the problem.

MODELING and ANALYSIS:

Equation of Cable. Place the origin of coordinates at a distance *c* below the lowest point of the cable (Fig. 1). The equation of the cable is given by Eq. (7.16), as

$$y = c \cosh \frac{x}{c}$$

The coordinates of point *B* are

$$x_B = 250 \text{ ft} \quad y_B = 100 + c$$

Substituting these coordinates into the equation of the cable, you obtain

$$100 + c = c \cosh \frac{250}{c}$$

$$\frac{100}{c} + 1 = \cosh \frac{250}{c}$$

Determine the value of *c* by substituting successive trial values, as shown in the following table.

<i>c</i>	$\frac{250}{c}$	$\frac{100}{c}$	$\frac{100}{c} + 1$	$\cosh \frac{250}{c}$
300	0.833	0.333	1.333	1.367
350	0.714	0.286	1.286	1.266
330	0.758	0.303	1.303	1.301
328	0.762	0.305	1.305	1.305

Taking *c* = 328, you have

$$y_B = 100 + c = 428 \text{ ft}$$

a. Maximum and Minimum Values of the Tension. Using Eqs. (7.18), you obtain

$$T_{\min} = T_0 = wc = (3 \text{ lb/ft})(328 \text{ ft}) \quad T_{\min} = 984 \text{ lb} \quad \blacktriangleleft$$

$$T_{\max} = T_B = wy_B = (3 \text{ lb/ft})(428 \text{ ft}) \quad T_{\max} = 1284 \text{ lb} \quad \blacktriangleleft$$

b. Length of Cable. You can find one-half of the length of the cable by solving Eq. (7.17). Hence,

$$y_B^2 - s_{CB}^2 = c^2 \quad s_{CB}^2 = y_B^2 - c^2 = (428)^2 - (328)^2 \quad s_{CB} = 275 \text{ ft}$$

The total length of the cable is therefore

$$s_{AB} = 2s_{CB} = 2(275 \text{ ft}) \quad s_{AB} = 550 \text{ ft} \quad \blacktriangleleft$$

REFLECT and THINK: The sag in the cable is one-fifth of the cable's span, so it is not very taut. The weight of the cable is $ws = (3 \text{ lb/ft})(550 \text{ ft}) = 1650 \text{ lb}$, while its maximum tension is only 1284 lb. This demonstrates that the total weight of a cable can exceed its maximum tension.

SOLVING PROBLEMS ON YOUR OWN

In the last section of this chapter, we described how to solve problems involving a *cable carrying a load uniformly distributed along the cable*. The shape assumed by the cable is a catenary and is defined by

$$y = c \cosh \frac{x}{c} \quad (7.16)$$

1. Keep in mind that the origin of coordinates for a catenary is located at a distance c directly below its lowest point. The length of the cable from the origin to any point is expressed as

$$s = c \sinh \frac{x}{c} \quad (7.15)$$

2. You should first identify all of the known and unknown quantities. Then consider each of the equations listed in the text (Eqs. 7.15 through 7.19) and solve an equation that contains only one unknown. Substitute the value found into another equation, and solve that equation for another unknown.

3. If the sag h is given, use Eq. (7.19) to replace y by $h + c$ in Eq. (7.16) if x is known [Sample Prob. 7.10] or in Eq. (7.17) if s is known, and solve the resulting equation for the constant c .

4. Many of the problems you will encounter will involve the solution by trial and error of an equation involving a hyperbolic sine or cosine. You can make your work easier by keeping track of your calculations in a table, as in Sample Prob. 7.10, or by applying a numerical methods approach using a computer or calculator.

Problems

- 7.127** A 25-ft chain with a weight of 30 lb is suspended between two points at the same elevation. Knowing that the sag is 10 ft, determine (a) the distance between the supports, (b) the maximum tension in the chain.
- 7.128** A 500-ft-long aerial tramway cable having a weight per unit length of 2.8 lb/ft is suspended between two points at the same elevation. Knowing that the sag is 125 ft, find (a) the horizontal distance between the supports, (b) the maximum tension in the cable.
- 7.129** A 40-m cable is strung as shown between two buildings. The maximum tension is found to be 350 N, and the lowest point of the cable is observed to be 6 m above the ground. Determine (a) the horizontal distance between the buildings, (b) the total mass of the cable.

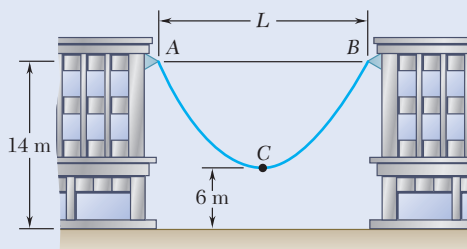


Fig. P7.129

- 7.130** A 50-m steel surveying tape has a mass of 1.6 kg. If the tape is stretched between two points at the same elevation and pulled until the tension at each end is 60 N, determine the horizontal distance between the ends of the tape. Neglect the elongation of the tape due to the tension.
- 7.131** A 20-m length of wire having a mass per unit length of 0.2 kg/m is attached to a fixed support at A and to a collar at B. Neglecting the effect of friction, determine (a) the force $\mathbf{P}$ for which $h = 8$ m, (b) the corresponding span L .
- 7.132** A 20-m length of wire having a mass per unit length of 0.2 kg/m is attached to a fixed support at A and to a collar at B. Knowing that the magnitude of the horizontal force applied to the collar is $P = 20$ N, determine (a) the sag h , (b) the span L .
- 7.133** A 20-m length of wire having a mass per unit length of 0.2 kg/m is attached to a fixed support at A and to a collar at B. Neglecting the effect of friction, determine (a) the sag h for which $L = 15$ m, (b) the corresponding force $\mathbf{P}$.

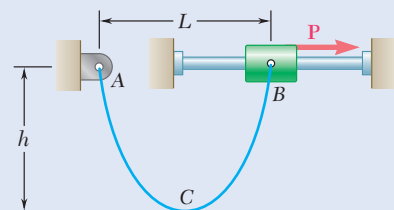


Fig. P7.131, P7.132, and P7.133

- 7.134** Determine the sag of a 30-ft chain that is attached to two points at the same elevation that are 20 ft apart.

- 7.135** A counterweight D is attached to a cable that passes over a small pulley at A and is attached to a support at B . Knowing that $L = 45$ ft and $h = 15$ ft, determine (a) the length of the cable from A to B , (b) the weight per unit length of the cable. Neglect the weight of the cable from A to D .

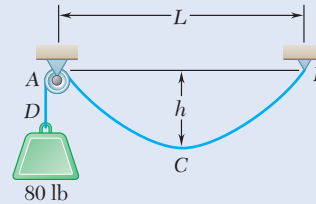


Fig. P7.135

- 7.136** A 90-m wire is suspended between two points at the same elevation that are 60 m apart. Knowing that the maximum tension is 300 N, determine (a) the sag of the wire, (b) the total mass of the wire.

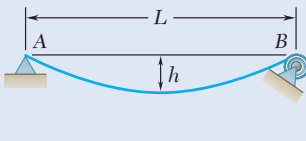


Fig. P7.138

- 7.137** A cable weighing 2 lb/ft is suspended between two points at the same elevation that are 160 ft apart. Determine the smallest allowable sag of the cable if the maximum tension is not to exceed 400 lb.

- 7.138** A uniform cord 50 in. long passes over a pulley at B and is attached to a pin support at A . Knowing that $L = 20$ in. and neglecting the effect of friction, determine the smaller of the two values of h for which the cord is in equilibrium.

- 7.139** A motor M is used to slowly reel in the cable shown. Knowing that the mass per unit length of the cable is 0.4 kg/m, determine the maximum tension in the cable when $h = 5$ m.

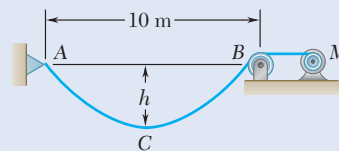


Fig. P7.139 and P7.140

- 7.140** A motor M is used to slowly reel in the cable shown. Knowing that the mass per unit length of the cable is 0.4 kg/m, determine the maximum tension in the cable when $h = 3$ m.

- 7.141** The cable ACB has a mass per unit length of 0.45 kg/m. Knowing that the lowest point of the cable is located at a distance $a = 0.6$ m below the support A , determine (a) the location of the lowest point C , (b) the maximum tension in the cable.

- 7.142** The cable ACB has a mass per unit length of 0.45 kg/m. Knowing that the lowest point of the cable is located at a distance $a = 2$ m below the support A , determine (a) the location of the lowest point C , (b) the maximum tension in the cable.

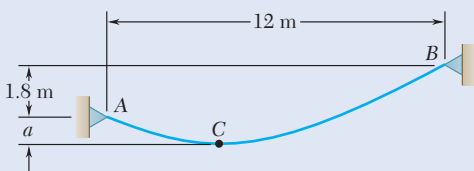


Fig. P7.141 and P7.142

7.143 A uniform cable weighing 3 lb/ft is held in the position shown by a horizontal force $\mathbf{P}$ applied at B . Knowing that $P = 180$ lb and $\theta_A = 60^\circ$, determine (a) the location of point B , (b) the length of the cable.

7.144 A uniform cable weighing 3 lb/ft is held in the position shown by a horizontal force $\mathbf{P}$ applied at B . Knowing that $P = 150$ lb and $\theta_A = 60^\circ$, determine (a) the location of point B , (b) the length of the cable.

7.145 To the left of point B , the long cable $ABDE$ rests on the rough horizontal surface shown. Knowing that the mass per unit length of the cable is 2 kg/m, determine the force $\mathbf{F}$ when $a = 3.6$ m.

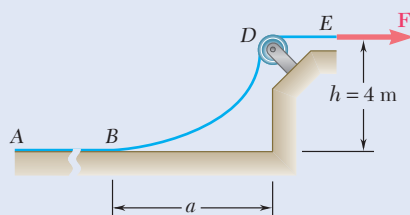


Fig. P7.145 and P7.146

7.146 To the left of point B , the long cable $ABDE$ rests on the rough horizontal surface shown. Knowing that the mass per unit length of the cable is 2 kg/m, determine the force $\mathbf{F}$ when $a = 6$ m.

***7.147** The 10-ft cable AB is attached to two collars as shown. The collar at A can slide freely along the rod; a stop attached to the rod prevents the collar at B from moving on the rod. Neglecting the effect of friction and the weight of the collars, determine the distance a .

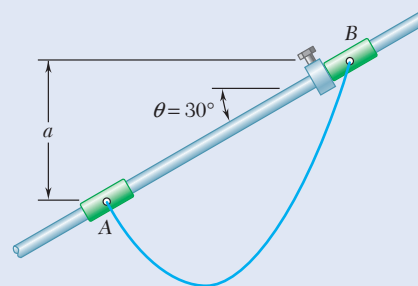


Fig. P7.147

***7.148** Solve Prob. 7.147 assuming that the angle θ formed by the rod and the horizontal is 45° .

7.149 Denoting the angle formed by a uniform cable and the horizontal by θ , show that at any point (a) $s = c \tan \theta$, (b) $y = c \sec \theta$.

***7.150** (a) Determine the maximum allowable horizontal span for a uniform cable with a weight per unit length of w if the tension in the cable is not to exceed a given value T_m . (b) Using the result of part a, determine the maximum span of a steel wire for which $w = 0.25$ lb/ft and $T_m = 8000$ lb.

***7.151** A cable has a mass per unit length of 3 kg/m and is supported as shown. Knowing that the span L is 6 m, determine the two values of the sag h for which the maximum tension is 350 N.

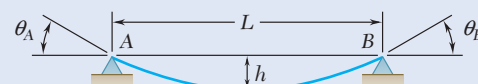


Fig. P7.151, P7.152 and P7.153

***7.152** Determine the sag-to-span ratio for which the maximum tension in the cable is equal to the total weight of the entire cable AB .

***7.153** A cable with a weight per unit length of w is suspended between two points at the same elevation that are a distance L apart. Determine (a) the sag-to-span ratio for which the maximum tension is as small as possible, (b) the corresponding values of θ_B and T_m .

Review and Summary

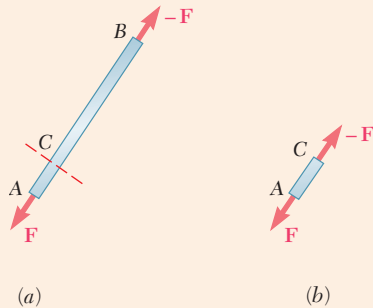


Fig. 7.19

In this chapter, you learned to determine the internal forces that hold together the various parts of a given member in a structure.

Forces in Straight Two-Force Members

Considering first a **straight two-force member** AB [Sec. 7.1], recall that such a member is subjected at A and B to equal and opposite forces F and $-F$ directed along AB (Fig. 7.19a). Cutting member AB at C and drawing the free-body diagram of portion AC , we concluded that the internal forces existing at C in member AB are equivalent to an **axial force** $-F$ equal and opposite to F (Fig. 7.19b). Note that, in the case of a two-force member that is not straight, the internal forces reduce to a force-couple system and not to a single force.

Forces in Multi-Force Members

Consider next a **multi-force member** AD (Fig. 7.20a). Cutting it at J and drawing the free-body diagram of portion JD , we concluded that the internal forces at J are equivalent to a force-couple system consisting of the **axial force** F , the **shearing force** V , and a couple M (Fig. 7.20b). The magnitude of the shearing force measures the **shear** at point J , and the moment of the couple is referred to as the **bending moment** at J . Since an equal and opposite force-couple system is obtained by considering the free-body diagram of portion AJ , it is necessary to specify which portion of member AD is used when recording the answers [Sample Prob. 7.1].

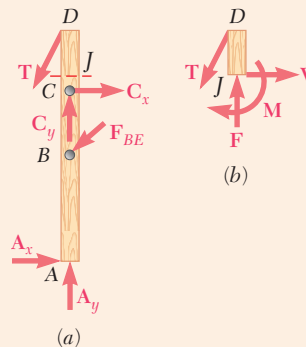


Fig. 7.20

Forces in Beams

Most of the chapter was devoted to the analysis of the internal forces in two important types of engineering structures: beams and cables. **Beams** are usually long, straight prismatic members designed to support loads applied at various points along the member. In general, the loads are perpendicular to the axis of the beam and produce only shear and bending in the beam. The loads may be either **concentrated** at specific points or **distributed** along the entire length or a portion of the beam. The beam itself may be supported in various ways; since only statically determinate beams are considered in this

text, we limited our analysis to that of simply supported beams, overhanging beams, and cantilever beams [Sec. 7.2].

Shear and Bending Moment in a Beam

To obtain the shear V and bending moment M at a given point C of a beam, we first determine the reactions at the supports by considering the entire beam as a free body. We then cut the beam at C and use the free-body diagram of one of the two resulting portions to determine V and M . In order to avoid any confusion regarding the sense of the shearing force V and couple M (which act in opposite directions on the two portions of the beam), we adopted the sign convention illustrated in Fig. 7.21 [Sec. 7.2B]. Once we have determined the values of the shear and bending moment at a few selected points of the beam, it is usually possible to draw a **shear diagram** and a **bending-moment diagram** representing, respectively, the shear and bending moment at any point of the beam [Sec. 7.2C]. When a beam is subjected to concentrated loads only, the shear is of constant value between loads, and the bending moment varies linearly between loads [Sample Prob. 7.2]. When a beam is subjected to distributed loads, the shear and bending moment vary quite differently [Sample Prob. 7.3].

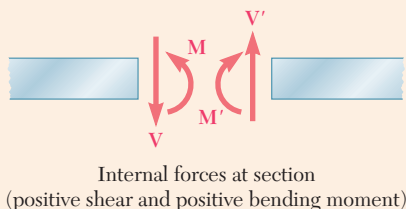


Fig. 7.21

Relations among Load, Shear, and Bending Moment

Construction of the shear and bending-moment diagrams is simplified by taking into account the following relations. Denoting the distributed load per unit length by w (assumed positive if directed downward), we have [Sec. 7.3]:

$$\frac{dV}{dx} = -w \quad (7.1)$$

$$\frac{dM}{dx} = V \quad (7.3)$$

In integrated form, these equations become

$$V_D - V_C = -(\text{area under load curve between } C \text{ and } D) \quad (7.2')$$

$$M_D - M_C = \text{area under shear curve between } C \text{ and } D \quad (7.4')$$

Equation (7.2') makes it possible to draw the shear diagram of a beam from the curve representing the distributed load on that beam and the value of V at one end of the beam. Similarly, Eq. (7.4') makes it possible to draw the bending-moment diagram from the shear diagram and the value of M at one end of the beam. However, discontinuities are introduced in the shear diagram by concentrated loads and in the bending-moment diagram by concentrated couples, none of which are accounted for in these equations [Sample Probs. 7.4 and 7.7]. Finally, we note from Eq. (7.3) that the points of the beam where the bending moment is maximum or minimum are also the points where the shear is zero [Sample Prob. 7.5].

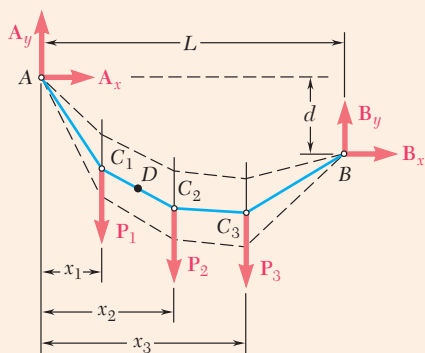


Fig. 7.22

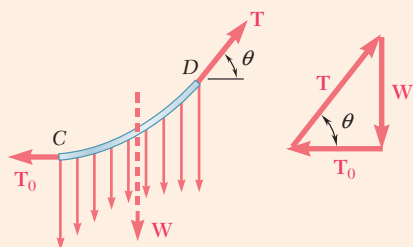


Fig. 7.23

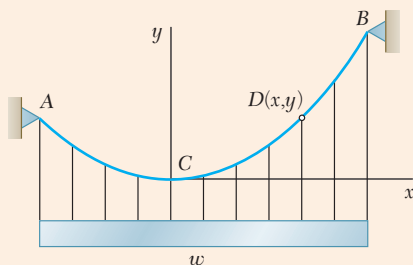


Fig. 7.24

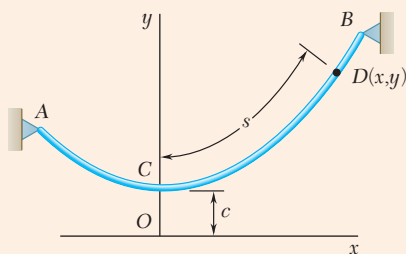


Fig. 7.25

Cables with Concentrated Loads

The second half of the chapter was devoted to the analysis of **flexible cables**. We first considered a cable of negligible weight supporting **concentrated loads** [Sec. 7.4A]. Using the entire cable AB as a free body (Fig. 7.22), we noted that the three available equilibrium equations were not sufficient to determine the four unknowns representing the reactions at supports A and B. However, if the coordinates of a point D of the cable are known, we can obtain an additional equation by considering the free-body diagram of portion AD or DB of the cable. Once we have determined the reactions at the supports, we can find the elevation of any point of the cable and the tension in any portion of the cable from the appropriate free-body diagram [Sample Prob. 7.8]. We noted that the horizontal component of the force **T** representing the tension is the same at any point of the cable.

Cables with Distributed Loads

We next considered cables carrying **distributed loads** [Sec. 7.4B]. Using as a free body a portion of cable CD extending from the lowest point C to an arbitrary point D of the cable (Fig. 7.23), we observed that the horizontal component of the tension force **T** at D is constant and equal to the tension T_0 at C, whereas its vertical component is equal to the weight W of the portion of cable CD. The magnitude and direction of **T** were obtained from the force triangle:

$$T = \sqrt{T_0^2 + W^2} \quad \tan \theta = \frac{W}{T_0} \quad (7.6)$$

Parabolic Cable

In the case of a load uniformly distributed along the horizontal—as in a suspension bridge (Fig. 7.24)—the load supported by portion CD is $W = wx$, where w is the constant load per unit horizontal length [Sec. 7.4C]. We also found that the curve formed by the cable is a **parabola** with equation

$$y = \frac{wx^2}{2T_0} \quad (7.8)$$

and that the length of the cable can be found by using the expansion in series given in Eq. (7.10) [Sample Prob. 7.9].

Catenary

In the case of a load uniformly distributed along the cable itself—e.g., a cable hanging under its own weight (Fig. 7.25)—the load supported by portion CD is $W = ws$, where s is the length measured along the cable and w is the constant load per unit length [Sec. 7.5]. Choosing the origin O of the coordinate axes at a distance $c = T_0/w$ below C, we derived the relations

$$s = c \sinh \frac{x}{c} \quad (7.15)$$

$$y = c \cosh \frac{x}{c} \quad (7.16)$$

$$y^2 - s^2 = c^2 \quad (7.17)$$

$$T_0 = wc \quad W = ws \quad T = wy \quad (7.18)$$

These equations can be used to solve problems involving cables hanging under their own weight [Sample Prob. 7.10]. Equation (7.16), which defines the shape of the cable, is the equation of a **catenary**.

Review Problems

7.154 and 7.155 Knowing that the turnbuckle has been tightened until the tension in wire AD is 850 N, determine the internal forces at the point indicated:

7.154 Point J

7.155 Point K

7.156 Two members, each consisting of a straight and a quarter-circular portion of rod, are connected as shown and support a 75-lb load at A . Determine the internal forces at point J .

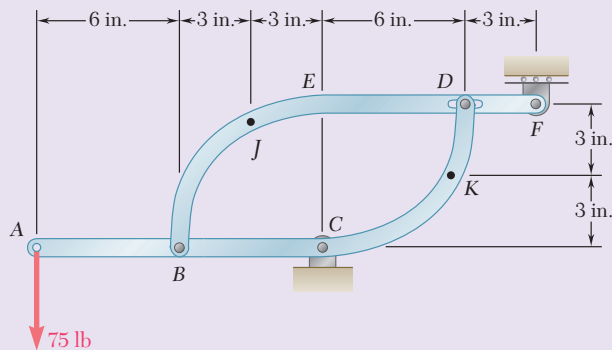


Fig. P7.156

7.157 Knowing that the radius of each pulley is 150 mm, that $\alpha = 20^\circ$, and neglecting friction, determine the internal forces at (a) point J , (b) point K .

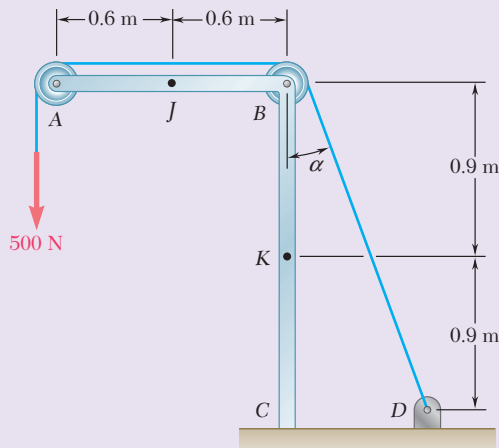


Fig. P7.157

7.158 For the beam shown, determine (a) the magnitude P of the two upward forces for which the maximum absolute value of the bending moment in the beam is as small as possible, (b) the corresponding value of $|M|_{\max}$.

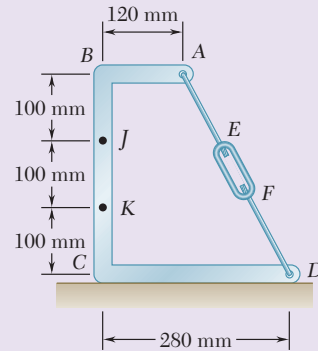


Fig. P7.154 and P7.155

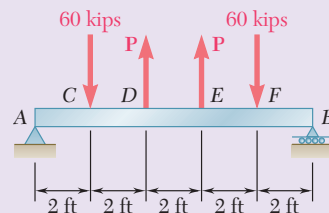


Fig. P7.158

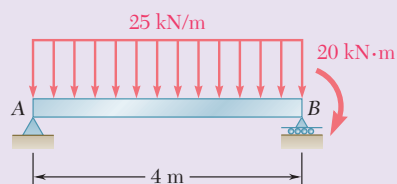


Fig. P7.159

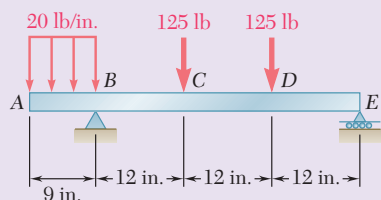


Fig. P7.160

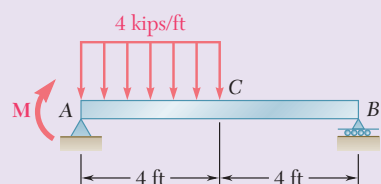


Fig. P7.161

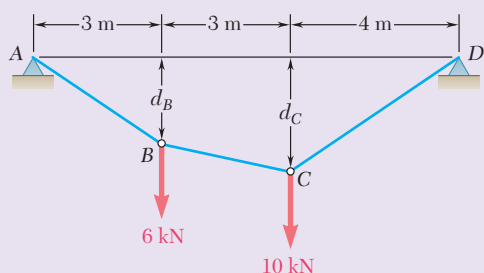


Fig. P7.163

7.159 For the beam and loading shown, (a) draw the shear and bending-moment diagrams, (b) determine the magnitude and location of the maximum absolute value of the bending moment.

7.160 For the beam and loading shown, (a) draw the shear and bending-moment diagrams, (b) determine the maximum absolute values of the shear and bending moment.

7.161 For the beam shown, draw the shear and bending-moment diagrams, and determine the magnitude and location of the maximum absolute value of the bending moment, knowing that (a) $M = 0$, (b) $M = 24$ kip-ft.

7.162 The beam AB , which lies on the ground, supports the parabolic load shown. Assuming the upward reaction of the ground to be uniformly distributed, (a) write the equations of the shear and bending-moment curves, (b) determine the maximum bending moment.

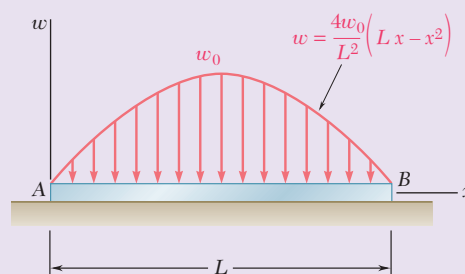


Fig. P7.162

7.163 Two loads are suspended as shown from the cable $ABCD$. Knowing that $d_B = 1.8$ m, determine (a) the distance d_C , (b) the components of the reaction at D , (c) the maximum tension in the cable.

7.164 A wire having a mass per unit length of 0.65 kg/m is suspended from two supports at the same elevation that are 120 m apart. If the sag is 30 m, determine (a) the total length of the wire, (b) the maximum tension in the wire.

7.165 A 10 -ft rope is attached to two supports A and B as shown. Determine (a) the span of the rope for which the span is equal to the sag, (b) the corresponding angle θ_B .

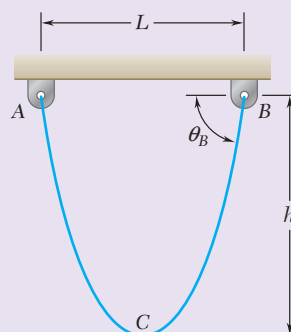


Fig. P7.165



8

Friction

The tractive force that a railroad locomotive can develop depends upon the frictional resistance between the drive wheels and the rails. When the potential exists for wheel slip to occur, such as when a train travels upgrade over wet rails, sand is deposited on top of the railhead to increase this friction.

Introduction

8.1 THE LAWS OF DRY FRICTION

8.1A Coefficients of Friction

8.1B Angles of Friction

8.1C Problems Involving Dry Friction

8.2 WEDGES AND SCREWS

8.2A Wedges

8.2B Square-Threaded Screws

*8.3 FRICTION ON AXLES, DISKS, AND WHEELS

8.3A Journal Bearings and Axle Friction

8.3B Thrust Bearings and Disk Friction

8.3C Wheel Friction and Rolling Resistance

8.4 BELT FRICTION

Objectives

- **Examine** the laws of dry friction and the associated coefficients and angles of friction.
- **Consider** the equilibrium of rigid bodies where dry friction at contact surfaces is modeled.
- **Apply** the laws of friction to analyze problems involving wedges and square-threaded screws.
- **Study** engineering applications of the laws of friction, such as in modeling axle, disk, wheel, and belt friction.

Introduction

In the previous chapters, we assumed that surfaces in contact are either *frictionless* or *rough*. If they are frictionless, the force each surface exerts on the other is normal to the surfaces, and the two surfaces can move freely with respect to each other. If they are rough, tangential forces can develop that prevent the motion of one surface with respect to the other.

This view is a simplified one. Actually, no perfectly frictionless surface exists. When two surfaces are in contact, tangential forces, called **friction forces**, always develop if you attempt to move one surface with respect to the other. However, these friction forces are limited in magnitude and do not prevent motion if you apply sufficiently large forces. Thus, the distinction between frictionless and rough surfaces is a matter of degree. You will see this more clearly in this chapter, which is devoted to the study of friction and its applications to common engineering situations.

There are two types of friction: **dry friction**, sometimes called *Coulomb friction*, and **fluid friction** or *viscosity*. Fluid friction develops

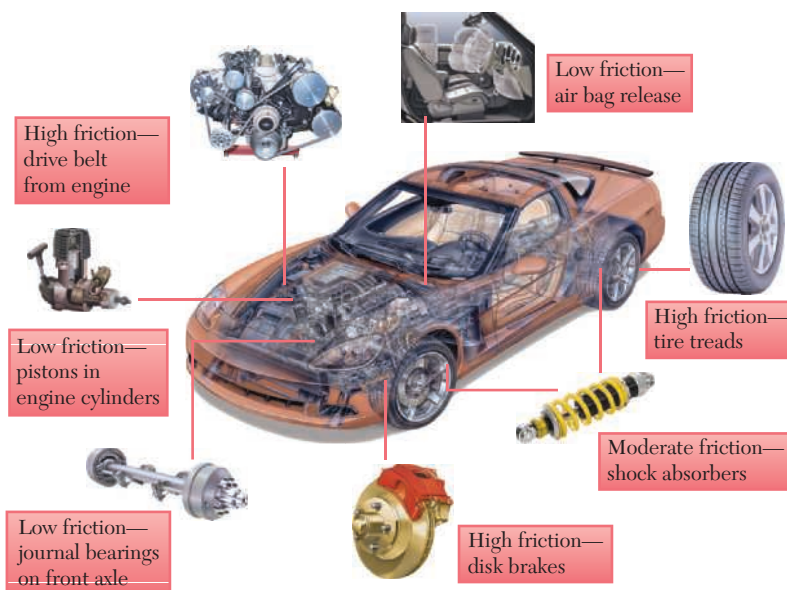


Photo 8.1 Examples of friction in an automobile. Depending upon the application, the degree of friction is controlled by design engineers.

between layers of fluid moving at different velocities. This is of great importance in analyzing problems involving the flow of fluids through pipes and orifices or dealing with bodies immersed in moving fluids. It is also basic for the analysis of the motion of *lubricated mechanisms*. Such problems are considered in texts on fluid mechanics. The present study is limited to dry friction, i.e., to situations involving rigid bodies that are in contact along *unlubricated* surfaces.

In the first section of this chapter, we examine the equilibrium of various rigid bodies and structures, assuming dry friction at the surfaces of contact. Afterward, we consider several specific engineering applications where dry friction plays an important role: wedges, square-threaded screws, journal bearings, thrust bearings, rolling resistance, and belt friction.

8.1 THE LAWS OF DRY FRICTION

We can illustrate the laws of dry friction by the following experiment. Place a block of weight $\mathbf{W}$ on a horizontal plane surface (Fig. 8.1a). The forces acting on the block are its weight $\mathbf{W}$ and the reaction of the surface. Since the weight has no horizontal component, the reaction of the surface also has no horizontal component; the reaction is therefore *normal* to the surface and is represented by $\mathbf{N}$ in Fig. 8.1a. Now suppose that you apply a horizontal force $\mathbf{P}$ to the block (Fig. 8.1b). If $\mathbf{P}$ is small, the block does not move; some other horizontal force must therefore exist, which balances $\mathbf{P}$. This other force is the **static-friction force** $\mathbf{F}$, which is actually the resultant of a great number of forces acting over the entire surface of contact between the block and the plane. The nature of these forces is not known exactly, but we generally assume that these forces are due to the irregularities of the surfaces in contact and, to a certain extent, to molecular attraction.

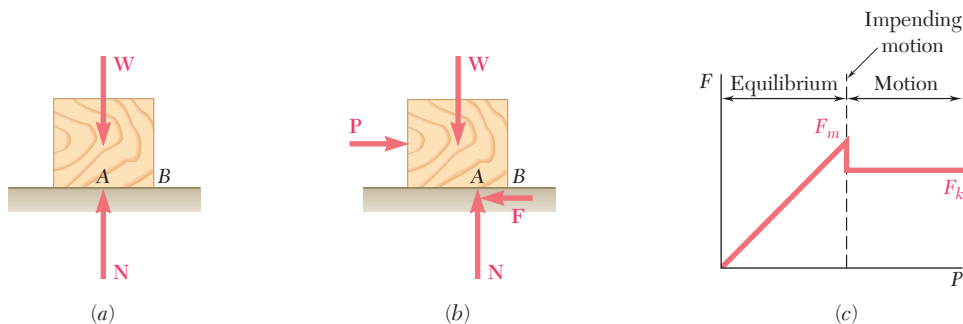


Fig. 8.1 (a) Block on a horizontal plane, friction force is zero; (b) a horizontally applied force $\mathbf{P}$ produces an opposing friction force $\mathbf{F}$; (c) graph of $\mathbf{F}$ with increasing $\mathbf{P}$.

If you increase the force $\mathbf{P}$, the friction force $\mathbf{F}$ also increases, continuing to oppose $\mathbf{P}$, until its magnitude reaches a certain *maximum value* F_m (Fig. 8.1c). If $\mathbf{P}$ is further increased, the friction force cannot balance it anymore, and the block starts sliding. As soon as the block has started in motion, the magnitude of $\mathbf{F}$ drops from F_m to a lower value F_k . This happens because less interpenetration occurs between the irregularities of the surfaces in contact when these surfaces move with respect to each other. From then on, the block keeps sliding with increasing velocity while the friction force, denoted by $\mathbf{F}_k$ and called the **kinetic-friction force**, remains approximately constant.

Note that, as the magnitude F of the friction force increases from 0 to F_m , the point of application A of the resultant $\mathbf{N}$ of the normal forces of contact moves to the right. In this way, the couples formed by $\mathbf{P}$ and $\mathbf{F}$ and by $\mathbf{W}$ and $\mathbf{N}$, respectively, remain balanced. If $\mathbf{N}$ reaches B before F reaches its maximum value F_m , the block starts to tip about B before it can start sliding (see Sample Prob. 8.4).

8.1A Coefficients of Friction

Experimental evidence shows that the maximum value F_m of the static-friction force is proportional to the normal component N of the reaction of the surface. We have

Static friction

$$F_m = \mu_s N \quad (8.1)$$

where μ_s is a constant called the **coefficient of static friction**. Similarly, we can express the magnitude F_k of the kinetic-friction force in the form

Kinetic friction

$$F_k = \mu_k N \quad (8.2)$$

where μ_k is a constant called the **coefficient of kinetic friction**. The coefficients of friction μ_s and μ_k do not depend upon the area of the surfaces in contact. Both coefficients, however, depend strongly on the *nature* of the surfaces in contact. Since they also depend upon the exact condition of the surfaces, their value is seldom known with an accuracy greater than 5%. Approximate values of coefficients of static friction for various combinations of dry surfaces are given in Table 8.1. The corresponding values of the coefficient of kinetic friction are about 25% smaller. Since coefficients of friction are dimensionless quantities, the values given in Table 8.1 can be used with both SI units and U.S. customary units.

Table 8.1 Approximate Values of Coefficient of Static Friction for Dry Surfaces

Metal on metal	0.15–0.60
Metal on wood	0.20–0.60
Metal on stone	0.30–0.70
Metal on leather	0.30–0.60
Wood on wood	0.25–0.50
Wood on leather	0.25–0.50
Stone on stone	0.40–0.70
Earth on earth	0.20–1.00
Rubber on concrete	0.60–0.90

From this discussion, it appears that four different situations can occur when a rigid body is in contact with a horizontal surface:

1. The forces applied to the body do not tend to move it along the surface of contact; there is no friction force (Fig. 8.2a).
2. The applied forces tend to move the body along the surface of contact but are not large enough to set it in motion. We can find the static-friction force $\mathbf{F}$ that has developed by solving the equations of equilibrium for the body. Since there is no evidence that $\mathbf{F}$ has reached its maximum

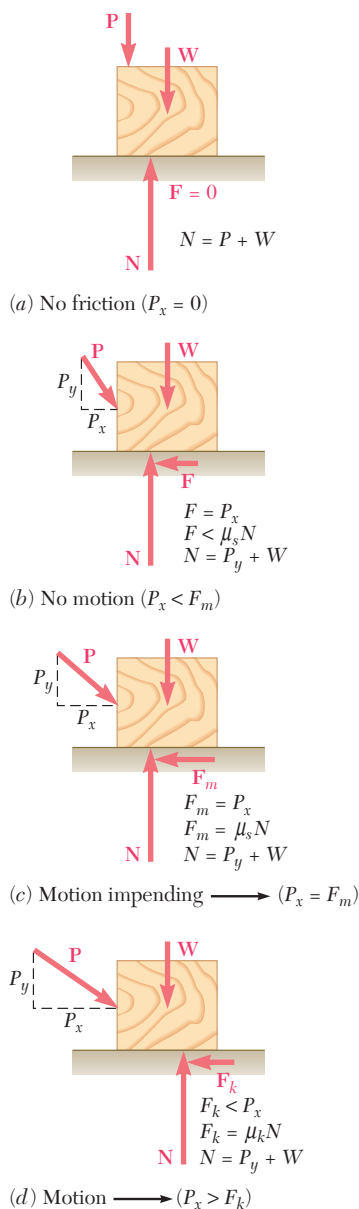


Fig. 8.2 (a) Applied force is vertical, friction force is zero; (b) horizontal component of applied force is less than F_m , no motion occurs; (c) horizontal component of applied force equals F_m , motion is impending; (d) horizontal component of applied force is greater than F_k , forces are unbalanced and motion continues.

value, the equation $F_m = \mu_s N$ *cannot be used* to determine the friction force (Fig. 8.2b).

3. The applied forces are such that the body is just about to slide. We say that *motion is impending*. The friction force $\mathbf{F}$ has reached its maximum value F_m and, together with the normal force $\mathbf{N}$, balances the applied forces. Both the equations of equilibrium and the equation $F_m = \mu_s N$ *can be used*. Note that the friction force has a sense opposite to the sense of impending motion (Fig. 8.2c).
4. The body is sliding under the action of the applied forces, and the equations of equilibrium no longer apply. However, $\mathbf{F}$ is now equal to $\mathbf{F}_k$, and we can use the equation $F_k = \mu_k N$. The sense of $\mathbf{F}_k$ is opposite to the sense of motion (Fig. 8.2d).

8.1B Angles of Friction

It is sometimes convenient to replace the normal force $\mathbf{N}$ and the friction force $\mathbf{F}$ by their resultant $\mathbf{R}$. Let's see what happens when we do that.

Consider again a block of weight $\mathbf{W}$ resting on a horizontal plane surface. If no horizontal force is applied to the block, the resultant $\mathbf{R}$ reduces to the normal force $\mathbf{N}$ (Fig. 8.3a). However, if the applied force $\mathbf{P}$ has a horizontal component $\mathbf{P}_x$ that tends to move the block, force $\mathbf{R}$ has a horizontal component $\mathbf{F}$ and, thus, forms an angle ϕ with the normal to the surface (Fig. 8.3b). If you increase $\mathbf{P}_x$ until motion becomes impending, the angle between $\mathbf{R}$ and the vertical grows and reaches a maximum value (Fig. 8.3c). This value is called the **angle of static friction** and is denoted by ϕ_s . From the geometry of Fig. 8.3c, we note that

Angle of static friction

$$\tan \phi_s = \frac{F_m}{N} = \frac{\mu_s N}{N}$$

$\tan \phi_s = \mu_s$

(8.3)

If motion actually takes place, the magnitude of the friction force drops to F_k ; similarly, the angle between $\mathbf{R}$ and $\mathbf{N}$ drops to a lower value ϕ_k , which is called the **angle of kinetic friction** (Fig. 8.3d). From the geometry of Fig. 8.3d, we have

Angle of kinetic friction

$$\tan \phi_k = \frac{F_k}{N} = \frac{\mu_k N}{N}$$

$\tan \phi_k = \mu_k$

(8.4)

Another example shows how the angle of friction can be used to advantage in the analysis of certain types of problems. Consider a block resting on a board and subjected to no other force than its weight $\mathbf{W}$ and the reaction $\mathbf{R}$ of the board. The board can be given any desired inclination. If the board is horizontal, the force $\mathbf{R}$ exerted by the board on the block is perpendicular to the board and balances the weight $\mathbf{W}$ (Fig. 8.4a). If the board is given a small angle of inclination θ , force $\mathbf{R}$ deviates from the perpendicular to the board by angle θ and continues to balance $\mathbf{W}$ (Fig. 8.4b). The reaction $\mathbf{R}$ now has a normal component $\mathbf{N}$ with a magnitude of $N = W \cos \theta$ and a tangential component $\mathbf{F}$ with a magnitude of $F = W \sin \theta$.

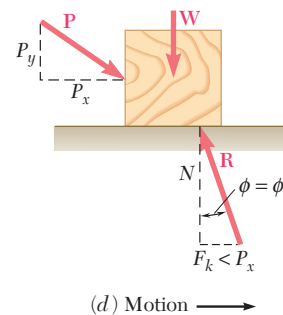
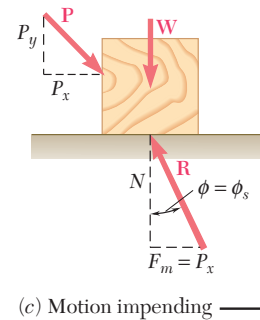
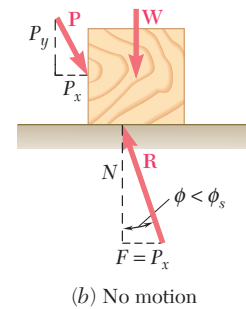
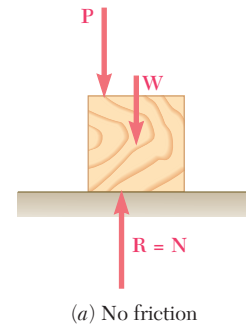


Fig. 8.3 (a) Applied force is vertical, friction force is zero; (b) applied force is at an angle, its horizontal component balanced by the horizontal component of the surface resultant; (c) impending motion, the horizontal component of the applied force equals the maximum horizontal component of the resultant; (d) motion, the horizontal component of the resultant is less than the horizontal component of the applied force.

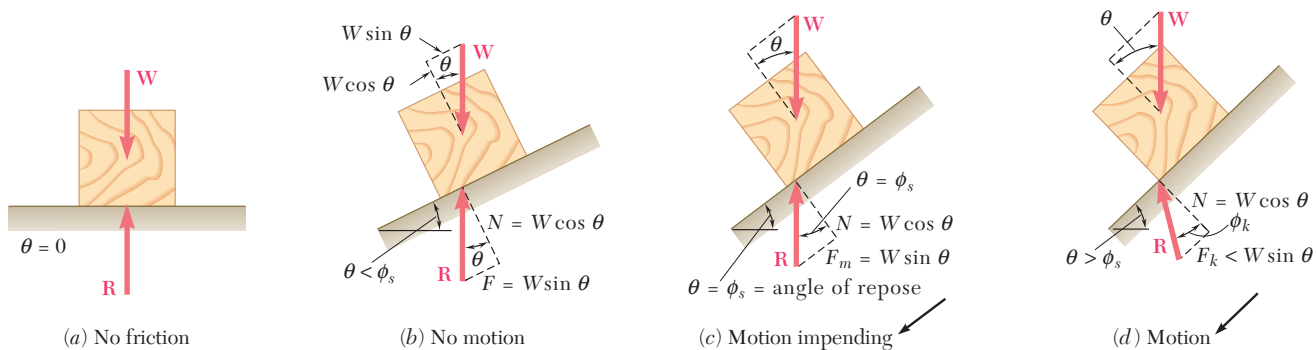


Fig. 8.4 (a) Block on horizontal board, friction force is zero; (b) board's angle of inclination is less than angle of static friction, no motion; (c) board's angle of inclination equals angle of friction, motion is impending; (d) angle of inclination is greater than angle of friction, forces are unbalanced and motion occurs.

If we keep increasing the angle of inclination, motion soon becomes impending. At that time, the angle between $\mathbf{R}$ and the normal reaches its maximum value $\theta = \phi_s$ (Fig. 8.4c). The value of the angle of inclination corresponding to impending motion is called the **angle of repose**. Clearly, the angle of repose is equal to the angle of static friction ϕ_s . If we further increase the angle of inclination θ , motion starts and the angle between $\mathbf{R}$ and the normal drops to the lower value ϕ_k (Fig. 8.4d). The reaction $\mathbf{R}$ is not vertical anymore, and the forces acting on the block are unbalanced.



Photo 8.2 The coefficient of static friction between a crate and the inclined conveyor belt must be sufficiently large to enable the crate to be transported without slipping.

8.1C Problems Involving Dry Friction

Many engineering applications involve dry friction. Some are simple situations, such as variations on the block sliding on a plane just described. Others involve more complicated situations, as in Sample Prob. 8.3. Many problems deal with the stability of rigid bodies in accelerated motion and will be studied in dynamics. Also, several common machines and mechanisms can be analyzed by applying the laws of dry friction, including wedges, screws, journal and thrust bearings, and belt transmissions. We will study these applications in the following sections.

The methods used to solve problems involving dry friction are the same that we used in the preceding chapters. If a problem involves only a motion of translation with no possible rotation, we can usually treat the body under consideration as a particle and use the methods of Chap. 2. If the problem involves a possible rotation, we must treat the body as a rigid body and use the methods of Chap. 4. If the structure considered is made of several parts, we must apply the principle of action and reaction, as we did in Chap. 6.

If the body being considered is acted upon by more than three forces (including the reactions at the surfaces of contact), the reaction at each surface is represented by its components $\mathbf{N}$ and $\mathbf{F}$, and we solve the problem using the equations of equilibrium. If only three forces act on the body under consideration, it may be more convenient to represent each reaction by the single force $\mathbf{R}$ and solve the problem by using a force triangle.

Most problems involving friction fall into one of the following three groups.

1. All applied forces are given, and we know the coefficients of friction; we are to determine whether the body being considered remains at rest or slides. The friction force $\mathbf{F}$ required to maintain equilibrium is

unknown (its magnitude is *not* equal to $\mu_s N$) and needs to be determined, together with the normal force N , by drawing a free-body diagram and solving the equations of equilibrium (Fig. 8.5a). We then compare the value found for the magnitude F of the friction force with the maximum value $F_m = \mu_s N$. If F is smaller than or equal to F_m , the body remains at rest. If the value found for F is larger than F_m , equilibrium cannot be maintained and motion takes place; the actual magnitude of the friction force is then $F_k = \mu_k N$.

2. All applied forces are given, and we know the motion is impending; we are to determine the value of the coefficient of static friction. Here again, we determine the friction force and the normal force by drawing a free-body diagram and solving the equations of equilibrium (Fig. 8.5b). Since we know that the value found for F is the maximum value F_m , we determine the coefficient of friction by solving the equation $F_m = \mu_s N$.
3. The coefficient of static friction is given, and we know that the motion is impending in a given direction; we are to determine the magnitude or the direction of one of the applied forces. The friction force should be shown in the free-body diagram with a *sense opposite to that of the impending motion* and with a magnitude $F_m = \mu_s N$ (Fig. 8.5c). We can then write the equations of equilibrium and determine the desired force.

As noted previously, when only three forces are involved, it may be more convenient to represent the reaction of the surface by a single force $\mathbf{R}$ and to solve the problem by drawing a force triangle. Such a solution is used in Sample Prob. 8.2.

When two bodies A and B are in contact (Fig. 8.6a), the forces of friction exerted, respectively, by A on B and by B on A are equal and opposite (Newton's third law). In drawing the free-body diagram of one of these bodies, it is important to include the appropriate friction force with its correct sense. Observe the following rule: *The sense of the friction force acting on A is opposite to that of the motion (or impending motion) of A as observed from B* (Fig. 8.6b). (It is therefore the same as the motion of B as observed from A .) The sense of the friction force acting on B is determined in a similar way (Fig. 8.6c). Note that the motion of A as observed from B is a *relative motion*. For example, if body A is fixed and body B moves, body A has a relative motion with respect to B . Also, if both B and A are moving down but B is moving faster than A , then body A is observed, from B , to be moving up.

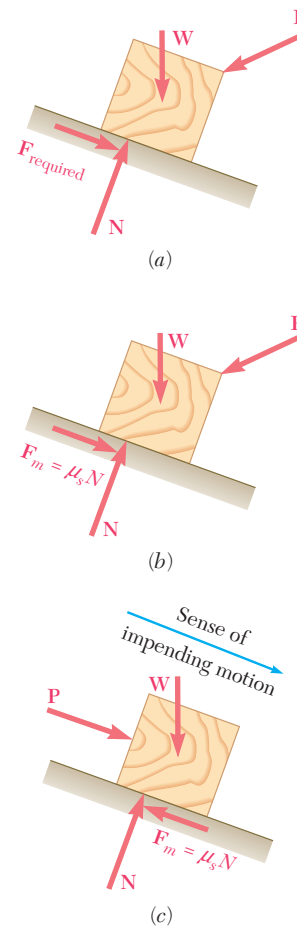


Fig. 8.5 Three types of friction problems: (a) given the forces and coefficient of friction, will the block slide or stay? (b) given the forces and that motion is impending, determine the coefficient of friction; (c) given the coefficient of friction and that motion is impending, determine the applied force.

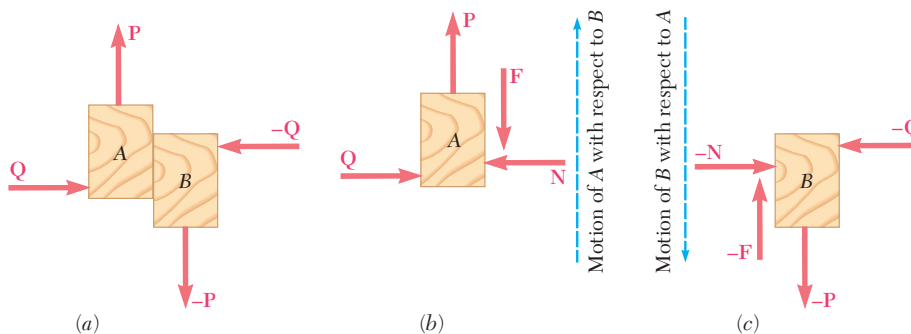


Fig. 8.6 (a) Two blocks held in contact by forces; (b) free-body diagram for block A , including direction of friction force; (c) free-body diagram for block B , including direction of friction force.

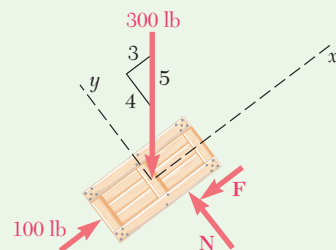
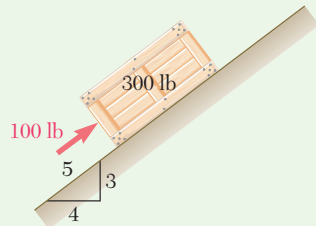


Fig. 1 Free-body diagram of crate showing assumed direction of friction force.

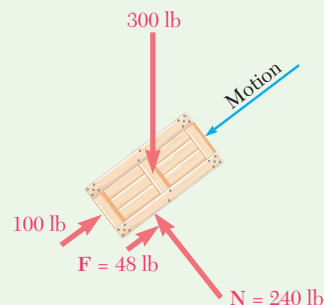


Fig. 2 Free-body diagram of crate showing actual friction force.

Sample Problem 8.1

A 100-lb force acts as shown on a 300-lb crate placed on an inclined plane. The coefficients of friction between the crate and the plane are $\mu_s = 0.25$ and $\mu_k = 0.20$. Determine whether the crate is in equilibrium, and find the value of the friction force.

STRATEGY: This is a friction problem of the first type: You know the forces and the friction coefficients and want to determine if the crate moves. You also want to find the friction force.

MODELING and ANALYSIS

Force Required for Equilibrium. First determine the value of the friction force *required to maintain equilibrium*. Assuming that $\mathbf{F}$ is directed down and to the left, draw the free-body diagram of the crate (Fig. 1) and solve the equilibrium equations:

$$+\nearrow \Sigma F_x = 0: \quad 100 \text{ lb} - \frac{3}{5}(300 \text{ lb}) - F = 0$$

$$F = -80 \text{ lb} \quad \mathbf{F} = 80 \text{ lb} \nearrow$$

$$+\nwarrow \Sigma F_y = 0: \quad N - \frac{4}{5}(300 \text{ lb}) = 0$$

$$N = +240 \text{ lb} \quad \mathbf{N} = 240 \text{ lb} \nwarrow$$

The force $\mathbf{F}$ required to maintain equilibrium is an 80-lb force directed up and to the right; the tendency of the crate is thus to move down the plane.

Maximum Friction Force. The magnitude of the maximum friction force that may be developed between the crate and the plane is

$$F_m = \mu_s N \quad F_m = 0.25(240 \text{ lb}) = 60 \text{ lb}$$

Since the value of the force required to maintain equilibrium (80 lb) is larger than the maximum value that may be obtained (60 lb), equilibrium is not maintained and *the crate will slide down the plane*.

Actual Value of Friction Force. The magnitude of the actual friction force is

$$F_{\text{actual}} = F_k = \mu_k N = 0.20(240 \text{ lb}) = 48 \text{ lb}$$

The sense of this force is opposite to the sense of motion; the force is thus directed up and to the right (Fig. 2):

$$\mathbf{F}_{\text{actual}} = 48 \text{ lb} \nearrow \blacktriangleleft$$

Note that the forces acting on the crate are not balanced. Their resultant is

$$\frac{3}{5}(300 \text{ lb}) - 100 \text{ lb} - 48 \text{ lb} = 32 \text{ lb} \swarrow$$

REFLECT and THINK: This is a typical friction problem of the first type. Note that you used the coefficient of static friction to determine if the crate moves, but once you found that it does move, you needed the coefficient of kinetic friction to determine the friction force.

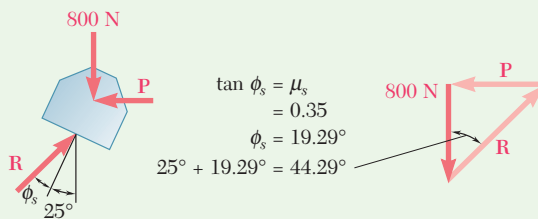
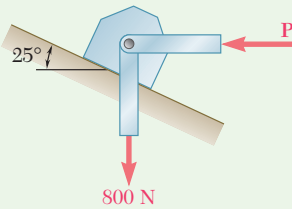


Fig. 1 Free-body diagram of block and its force triangle—motion impending up incline.

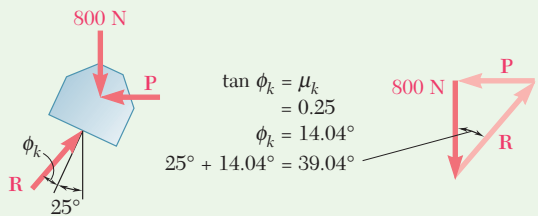


Fig. 2 Free-body diagram of block and its force triangle—motion continuing up incline.

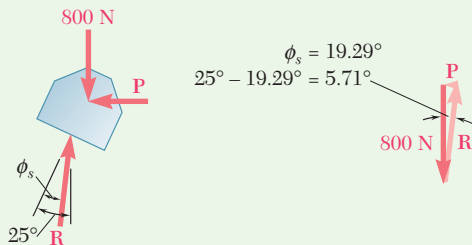


Fig. 3 Free-body diagram of block and its force triangle—motion prevented down the slope.

Sample Problem 8.2

A support block is acted upon by two forces as shown. Knowing that the coefficients of friction between the block and the incline are $\mu_s = 0.35$ and $\mu_k = 0.25$, determine the force **P** required to (a) start the block moving up the incline; (b) keep it moving up; (c) prevent it from sliding down.

STRATEGY: This problem involves practical variations of the third type of friction problem. You can approach the solutions through the concept of the angles of friction.

MODELING:

Free-Body Diagram. For each part of the problem, draw a free-body diagram of the block and a force triangle including the 800-N vertical force, the horizontal force **P**, and the force **R** exerted on the block by the incline. You must determine the direction of **R** in each separate case. Note that, since **P** is perpendicular to the 800-N force, the force triangle is a right triangle, which easily can be solved for **P**. In most other problems, however, the force triangle will be an oblique triangle and should be solved by applying the law of sines.

ANALYSIS:

a. Force *P* to Start Block Moving Up. In this case, motion is impending up the incline, so the resultant is directed at the angle of static friction (Fig. 1). Note that the resultant is oriented to the left of the normal such that its friction component (not shown) is directed opposite the direction of impending motion.

$$P = (800 \text{ N}) \tan 44.29^\circ \quad \mathbf{P = 780 \text{ N} \leftarrow}$$

b. Force *P* to Keep Block Moving Up. Motion is continuing, so the resultant is directed at the angle of kinetic friction (Fig. 2). Again, the resultant is oriented to the left of the normal such that its friction component is directed opposite the direction of motion.

$$P = (800 \text{ N}) \tan 39.04^\circ \quad \mathbf{P = 649 \text{ N} \leftarrow}$$

c. Force *P* to Prevent Block from Sliding Down. Here, motion is impending down the incline, so the resultant is directed at the angle of static friction (Fig. 3). Note that the resultant is oriented to the right of the normal such that its friction component is directed opposite the direction of impending motion.

$$P = (800 \text{ N}) \tan 5.71^\circ \quad \mathbf{P = 80.0 \text{ N} \leftarrow}$$

REFLECT and THINK: As expected, considerably more force is required to begin moving the block up the slope than is necessary to restrain it from sliding down the slope.

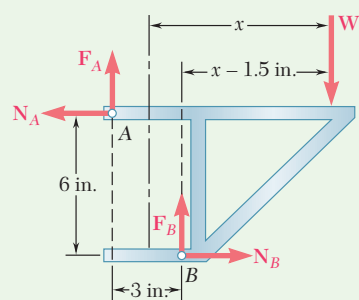
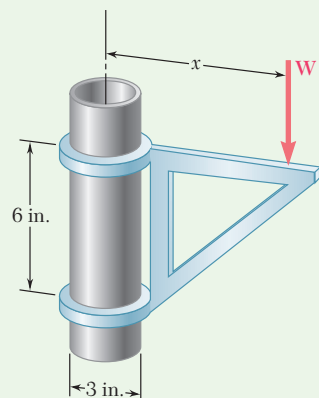


Fig. 1 Free-body diagram of bracket.

Sample Problem 8.3

The movable bracket shown may be placed at any height on the 3-in.-diameter pipe. If the coefficient of static friction between the pipe and bracket is 0.25, determine the minimum distance x at which the load W can be supported. Neglect the weight of the bracket.

STRATEGY: In this variation of the third type of friction problem, you know the coefficient of static friction and that motion is impending. Since the problem involves consideration of resistance to rotation, you should apply both moment equilibrium and force equilibrium.

MODELING:

Free-Body Diagram. Draw the free-body diagram of the bracket (Fig. 1). When W is placed at the minimum distance x from the axis of the pipe, the bracket is just about to slip, and the forces of friction at A and B have reached their maximum values:

$$F_A = \mu_s N_A = 0.25 N_A$$

$$F_B = \mu_s N_B = 0.25 N_B$$

ANALYSIS:

Equilibrium Equations.

$$\begin{aligned} \rightarrow \Sigma F_x = 0: \quad N_B - N_A &= 0 \\ N_B &= N_A \end{aligned}$$

$$\begin{aligned} + \uparrow \Sigma F_y = 0: \quad F_A + F_B - W &= 0 \\ 0.25N_A + 0.25N_B &= W \end{aligned}$$

Since N_B is equal to N_A ,

$$\begin{aligned} 0.50N_A &= W \\ N_A &= 2W \end{aligned}$$

$$\begin{aligned} + \curvearrowright \Sigma M_B = 0: \quad N_A(6 \text{ in.}) - F_A(3 \text{ in.}) - W(x - 1.5 \text{ in.}) &= 0 \\ 6N_A - 3(0.25N_A) - Wx + 1.5W &= 0 \\ 6(2W) - 0.75(2W) - Wx + 1.5W &= 0 \end{aligned}$$

Dividing through by W and solving for x , you have

$$x = 12 \text{ in.} \quad \blacktriangleleft$$

REFLECT and THINK: In a problem like this, you may not figure out how to approach the solution until you draw the free-body diagram and examine what information you are given and what you need to find. In this case, since you are asked to find a distance, the need to evaluate moment equilibrium should be clear.

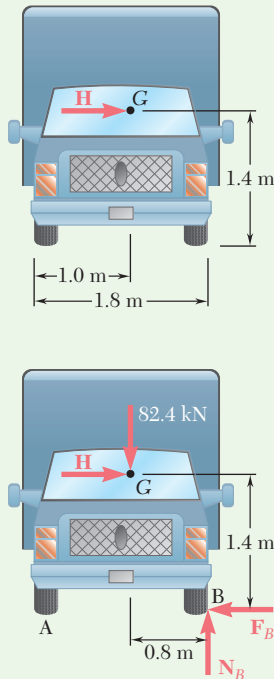


Fig. 1 Free-body diagram of truck.

Sample Problem 8.4

An 8400-kg truck is traveling on a level horizontal curve, resulting in an effective lateral force $\mathbf{H}$ (applied at the center of gravity G of the truck). Treating the truck as a rigid system with the center of gravity shown, and knowing that the distance between the outer edges of the tires is 1.8 m, determine (a) the maximum force $\mathbf{H}$ before tipping of the truck occurs, (b) the minimum coefficient of static friction between the tires and roadway such that slipping does not occur before tipping.

STRATEGY: For the direction of $\mathbf{H}$ shown, the truck would tip about the outer edge of the right tire. At the verge of tip, the normal force and friction force are zero at the left tire, and the normal force at the right tire is at the outer edge. You can apply equilibrium to determine the value of $\mathbf{H}$ necessary for tip and the required friction force such that slipping does not occur.

MODELING: Draw the free-body diagram of the truck (Fig. 1), which reflects impending tip about point B . Obtain the weight of the truck by multiplying its mass of 8400 kg by $g = 9.81 \text{ m/s}^2$; that is, $W = 82\,400 \text{ N}$ or 82.4 kN.

ANALYSIS:

Free Body: Truck (Fig. 1).

$$+\circlearrowleft \sum M_B = 0: \quad (82.4 \text{ kN})(0.8 \text{ m}) - H(1.4 \text{ m}) = 0$$

$$H = +47.1 \text{ kN}$$

$$H = 47.1 \text{ kN} \rightarrow \blacktriangleleft$$

$$+\rightarrow \sum F_x = 0: \quad 47.1 \text{ kN} - F_B = 0$$

$$F_B = +47.1 \text{ kN}$$

$$+\uparrow \sum F_y = 0: \quad N_B - 82.4 \text{ kN} = 0$$

$$N_B = +82.4 \text{ kN}$$

Minimum Coefficient of Static Friction. The magnitude of the maximum friction force that can be developed is

$$F_m = \mu_s N_B = \mu_s (82.4 \text{ kN})$$

Setting this equal to the friction force required, $F_B = 47.1 \text{ kN}$, gives

$$\mu_s (82.4 \text{ kN}) = 47.1 \text{ kN} \quad \mu_s = 0.572 \quad \blacktriangleleft$$

REFLECT and THINK: Recall from physics that $\mathbf{H}$ represents the force due to the centripetal acceleration of the truck (of mass m), and its magnitude is

$$H = m(v^2/\rho)$$

where

v = velocity of the truck

ρ = radius of curvature

In this problem, if the truck were traveling around a curve of 100-m radius (measured to G), the velocity at which it would begin to tip would be 23.7 m/s (or 85.2 km/h). You will learn more about this aspect in your study of dynamics.

SOLVING PROBLEMS ON YOUR OWN

In this section, you studied and applied the **laws of dry friction**. Previously, you had encountered only (a) frictionless surfaces that could move freely with respect to each other or (b) rough surfaces that allowed no motion relative to each other.

A. In solving problems involving dry friction, keep the following ideas in mind.

1. The reaction $\mathbf{R}$ exerted by a surface on a free body can be resolved into a normal component $\mathbf{N}$ and a tangential component $\mathbf{F}$. The tangential component is known as the **friction force**. When a body is in contact with a fixed surface, the direction of the friction force $\mathbf{F}$ is opposite to that of the actual or impending motion of the body.

a. No motion will occur as long as F does not exceed the maximum value $F_m = \mu_s N$, where μ_s is the **coefficient of static friction**.

b. Motion will occur if a value of F larger than F_m is required to maintain equilibrium. As motion takes place, the actual value of F drops to $F_k = \mu_k N$, where μ_k is the **coefficient of kinetic friction** [Sample Prob. 8.1].

c. Motion may also occur at a value of F smaller than F_m if tipping of the rigid body is a possibility [Sample Prob. 8.4]

2. When only three forces are involved, you might prefer an alternative approach to the analysis of friction [Sample Prob. 8.2]. The reaction $\mathbf{R}$ is defined by its magnitude R and the angle ϕ it forms with the normal to the surface. No motion occurs as long as ϕ does not exceed the maximum value ϕ_s , where $\tan \phi_s = \mu_s$. Motion does occur if a value of ϕ larger than ϕ_s is required to maintain equilibrium, and the actual value of ϕ drops to ϕ_k , where $\tan \phi_k = \mu_k$.

3. When two bodies are in contact, you must determine the sense of the actual or impending relative motion at the point of contact. On each of the two bodies, a friction force $\mathbf{F}$ is in a direction opposite to that of the actual or impending motion of the body as seen from the other body (see Fig. 8.6).

B. Methods of solution. The first step in your solution is to draw a free-body diagram of the body under consideration, resolving the force exerted on each surface where friction exists into a normal component $\mathbf{N}$ and a friction force $\mathbf{F}$. If several bodies are involved, draw a free-body diagram for each of them, labeling and directing the forces at each surface of contact, as described for analyzing frames in Chap. 6.

The problem you have to solve may fall in one of the following three categories.

1. You know all the applied forces and the coefficients of friction, and you must determine whether equilibrium is maintained. In this situation, the friction force is unknown and *cannot be assumed to be equal* to $\mu_s N$.

a. Write the equations of equilibrium to determine N and F .

b. Calculate the maximum allowable friction force, $F_m = \mu_s N$. If $F \leq F_m$, equilibrium is maintained. If $F \geq F_m$, motion occurs, and the magnitude of the friction force is $F_k = \mu_k N$ [Sample Prob. 8.1].

2. You know all the applied forces, and you must find the smallest allowable value of μ_s for which equilibrium is maintained. Assume that motion is impending, and determine the corresponding value of μ_s .

a. Write the equations of equilibrium to determine N and F .

b. Since motion is impending, $F = F_m$. Substitute the values found for N and F into the equation $F_m = \mu_s N$ and solve for μ_s [Sample Prob. 8.4].

3. The motion of the body is impending and μ_s is known; you must find some unknown quantity, such as a distance, an angle, the magnitude of a force, or the direction of a force.

a. Assume a possible motion of the body and, on the free-body diagram, draw the friction force in a direction opposite to that of the assumed motion.

b. Since motion is impending, $F = F_m = \mu_s N$. Substituting the known value for μ_s , you can express F in terms of N on the free-body diagram, thus eliminating one unknown.

c. Write and solve the equilibrium equations for the unknown you seek [Sample Prob. 8.3].

Problems

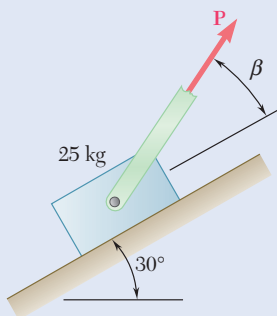


Fig. P8.F1

FREE-BODY PRACTICE PROBLEMS

8.F1 Knowing that the coefficient of friction between the 25-kg block and the incline is $\mu_s = 0.25$, draw the free-body diagram needed to determine both the smallest value of P required to start the block moving up the incline and the corresponding value of β .

8.F2 Two blocks A and B are connected by a cable as shown. Knowing that the coefficient of static friction at all surfaces of contact is 0.30 and neglecting the friction of the pulleys, draw the free-body diagrams needed to determine the smallest force P required to move the blocks.

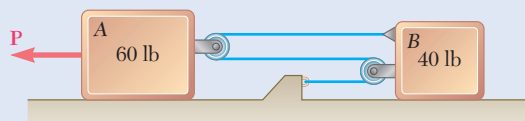


Fig. P8.F2

8.F3 A cord is attached to and partially wound around a cylinder with a weight of W and radius r that rests on an incline as shown. Knowing that $\theta = 30^\circ$, draw the free-body diagram needed to determine both the tension in the cord and the smallest allowable value of the coefficient of static friction between the cylinder and the incline for which equilibrium is maintained.

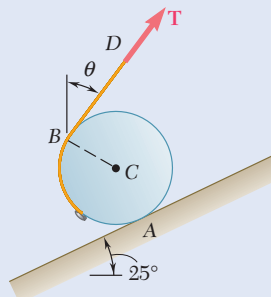


Fig. P8.F3

8.F4 A 40-kg packing crate must be moved to the left along the floor without tipping. Knowing that the coefficient of static friction between the crate and the floor is 0.35, draw the free-body diagram needed to determine both the largest allowable value of α and the corresponding magnitude of the force P .

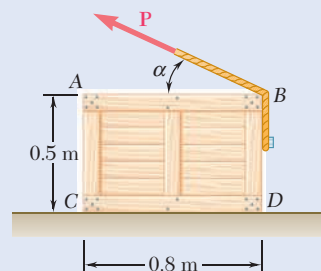


Fig. P8.F4

END-OF-SECTION PROBLEMS

- 8.1** Determine whether the block shown is in equilibrium, and find the magnitude and direction of the friction force when $P = 150 \text{ N}$.

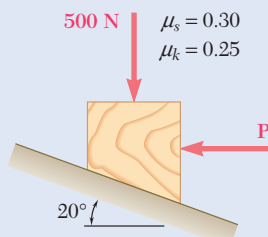


Fig. P8.1 and P8.2

- 8.2** Determine whether the block shown is in equilibrium, and find the magnitude and direction of the friction force when $P = 400 \text{ N}$.
- 8.3** Determine whether the block shown is in equilibrium, and find the magnitude and direction of the friction force when $P = 120 \text{ lb}$.
- 8.4** Determine whether the block shown is in equilibrium, and find the magnitude and direction of the friction force when $P = 80 \text{ lb}$.

- 8.5** Determine the smallest value of P required to (a) start the block up the incline, (b) keep it moving up.

- 8.6** The 20-lb block A hangs from a cable as shown. Pulley C is connected by a short link to block E , which rests on a horizontal rail. Knowing that the coefficient of static friction between block E and the rail is $\mu_s = 0.35$ and neglecting the weight of block E and the friction in the pulleys, determine the maximum allowable value of θ if the system is to remain in equilibrium.

- 8.7** The 10-kg block is attached to link AB and rests on a moving belt. Knowing that $\mu_s = 0.30$ and $\mu_k = 0.25$ and neglecting the weight of the link, determine the magnitude of the horizontal force P that should be applied to the belt to maintain its motion (a) to the left as shown, (b) to the right.

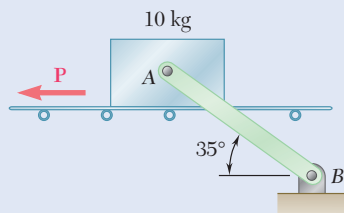


Fig. P8.7

- 8.8** Considering only values of θ less than 90° , determine the smallest value of θ required to start the block moving to the right when (a) $W = 75 \text{ lb}$, (b) $W = 100 \text{ lb}$.

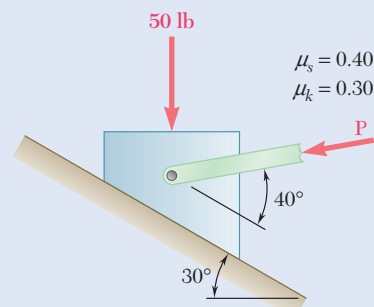


Fig. P8.3, P8.4, and P8.5

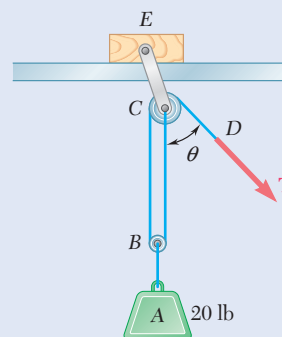


Fig. P8.6

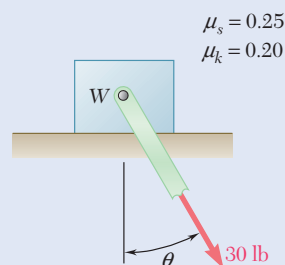


Fig. P8.8

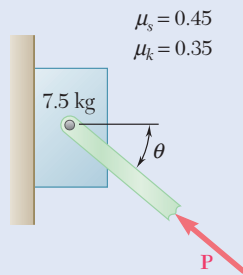


Fig. P8.9 and P8.10

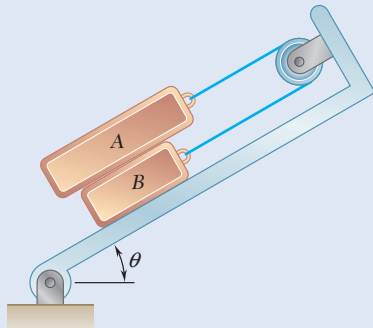


Fig. P8.11 and P8.12

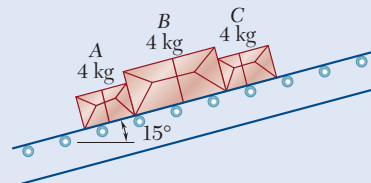


Fig. P8.13

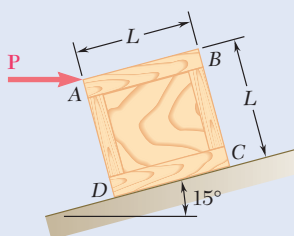


Fig. P8.15

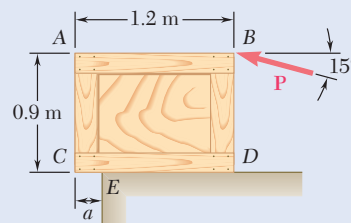


Fig. P8.16

- 8.9** Knowing that $\theta = 40^\circ$, determine the smallest force P for which equilibrium of the 7.5-kg block is maintained.
- 8.10** Knowing that $P = 100$ N, determine the range of values of θ for which equilibrium of the 7.5-kg block is maintained.
- 8.11** The 50-lb block A and the 25-lb block B are supported by an incline that is held in the position shown. Knowing that the coefficient of static friction is 0.15 between the two blocks and zero between block B and the incline, determine the value of θ for which motion is impending.
- 8.12** The 50-lb block A and the 25-lb block B are supported by an incline that is held in the position shown. Knowing that the coefficient of static friction is 0.15 between all surfaces of contact, determine the value of θ for which motion is impending.
- 8.13** Three 4-kg packages A , B , and C are placed on a conveyor belt that is at rest. Between the belt and both packages A and C , the coefficients of friction are $\mu_s = 0.30$ and $\mu_k = 0.20$; between package B and the belt, the coefficients are $\mu_s = 0.10$ and $\mu_k = 0.08$. The packages are placed on the belt so that they are in contact with each other and at rest. Determine which, if any, of the packages will move and the friction force acting on each package.
- 8.14** Solve Prob. 8.13 assuming that package B is placed to the right of both packages A and C .
- 8.15** A uniform crate with a mass of 30 kg must be moved up along the 15° incline without tipping. Knowing that force P is horizontal, determine (a) the largest allowable coefficient of static friction between the crate and the incline, (b) the corresponding magnitude of force P .
- 8.16** A worker slowly moves a 50-kg crate to the left along a loading dock by applying a force P at corner B as shown. Knowing that the crate starts to tip about edge E of the loading dock when $a = 200$ mm, determine (a) the coefficient of kinetic friction between the crate and the loading dock, (b) the corresponding magnitude P of the force.

- 8.17** A half-section of pipe weighing 200 lb is pulled by a cable as shown. The coefficient of static friction between the pipe and the floor is 0.40. If $\alpha = 30^\circ$, determine (a) the tension T required to move the pipe, (b) whether the pipe will slide or tip.

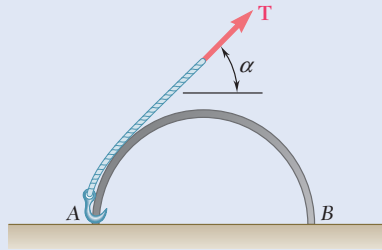


Fig. P8.17

- 8.18** A 120-lb cabinet is mounted on casters that can be locked to prevent their rotation. The coefficient of static friction between the floor and each caster is 0.30. Assuming that the casters at both A and B are locked, determine (a) the force $\mathbf{P}$ required to move the cabinet to the right, (b) the largest allowable value of h if the cabinet is not to tip over.
- 8.19** Wire is being drawn at a constant rate from a spool by applying a vertical force $\mathbf{P}$ to the wire as shown. The spool and the wire wrapped on the spool have a combined weight of 20 lb. Knowing that the coefficients of friction at both A and B are $\mu_s = 0.40$ and $\mu_k = 0.30$, determine the required magnitude of force $\mathbf{P}$.

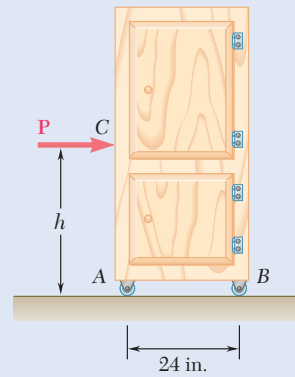


Fig. P8.18

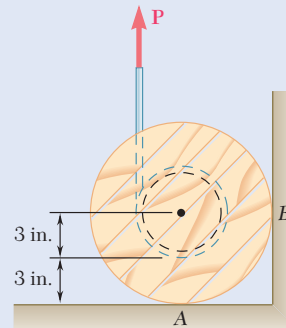


Fig. P8.19

- 8.20** Solve Prob. 8.19 assuming that the coefficients of friction at B are zero.
- 8.21** The cylinder shown has a weight W and radius r . Express in terms of W and r the magnitude of the largest couple $\mathbf{M}$ that can be applied to the cylinder if it is not to rotate, assuming the coefficient of static friction to be (a) zero at A and 0.30 at B , (b) 0.25 at A and 0.30 at B .

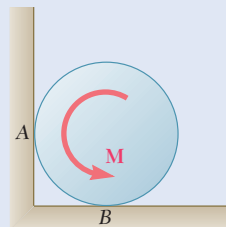


Fig. P8.21 and P.22

- 8.22** The cylinder shown has a weight W and radius r , and the coefficient of static friction μ_s is the same at A and B . Determine the magnitude of the largest couple $\mathbf{M}$ that can be applied to the cylinder if it is not to rotate.
- 8.23 and 8.24** End A of a slender, uniform rod with a length of L and weight W bears on a surface as shown, while end B is supported by a cord BC . Knowing that the coefficients of friction are $\mu_s = 0.40$ and $\mu_k = 0.30$, determine (a) the largest value of θ for which motion is impending, (b) the corresponding value of the tension in the cord.

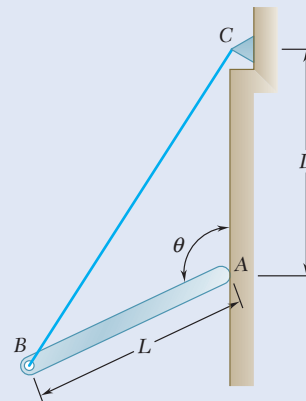


Fig. P8.23

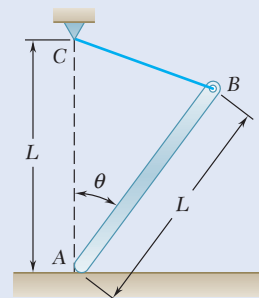


Fig. P8.24

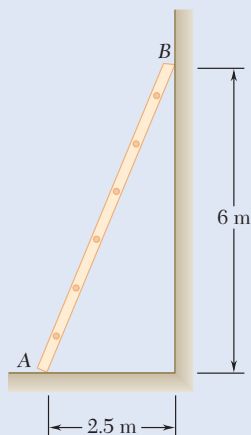


Fig. P8.25 and P8.26

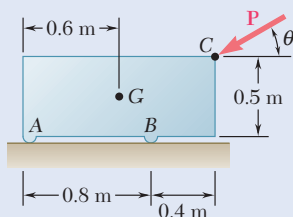


Fig. P8.28

8.25 A 6.5-m ladder AB leans against a wall as shown. Assuming that the coefficient of static friction μ_s is zero at B , determine the smallest value of μ_s at A for which equilibrium is maintained.

8.26 A 6.5-m ladder AB leans against a wall as shown. Assuming that the coefficient of static friction μ_s is the same at A and B , determine the smallest value of μ_s for which equilibrium is maintained.

8.27 The press shown is used to emboss a small seal at E . Knowing that the coefficient of static friction between the vertical guide and the embossing die D is 0.30, determine the force exerted by the die on the seal.

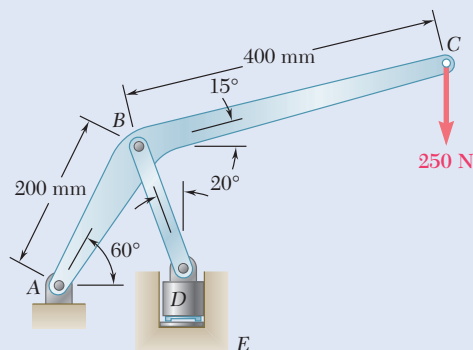


Fig. P8.27

8.28 The machine base shown has a mass of 75 kg and is fitted with skids at A and B . The coefficient of static friction between the skids and the floor is 0.30. If a force $\mathbf{P}$ with a magnitude of 500 N is applied at corner C , determine the range of values of θ for which the base will not move.

8.29 The 50-lb plate $ABCD$ is attached at A and D to collars that can slide on the vertical rod. Knowing that the coefficient of static friction is 0.40 between both collars and the rod, determine whether the plate is in equilibrium in the position shown when the magnitude of the vertical force applied at E is (a) $P = 0$, (b) $P = 20$ lb.

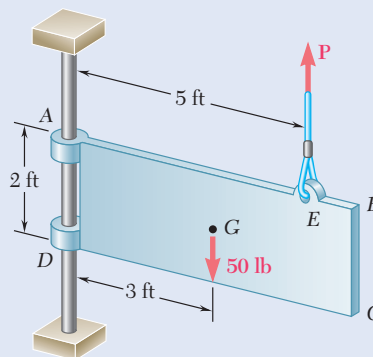


Fig. P8.29 and P8.30

8.30 In Prob. 8.29, determine the range of values of the magnitude P of the vertical force applied at E for which the plate will move downward.

- 8.31** A window sash weighing 10 lb is normally supported by two 5-lb sash weights. Knowing that the window remains open after one sash cord has broken, determine the smallest possible value of the coefficient of static friction. (Assume that the sash is slightly smaller than the frame and will bind only at points A and D.)

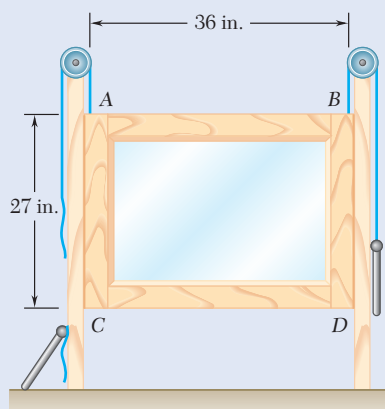


Fig. P8.31

- 8.32** A 500-N concrete block is to be lifted by the pair of tongs shown. Determine the smallest allowable value of the coefficient of static friction between the block and the tongs at F and G.

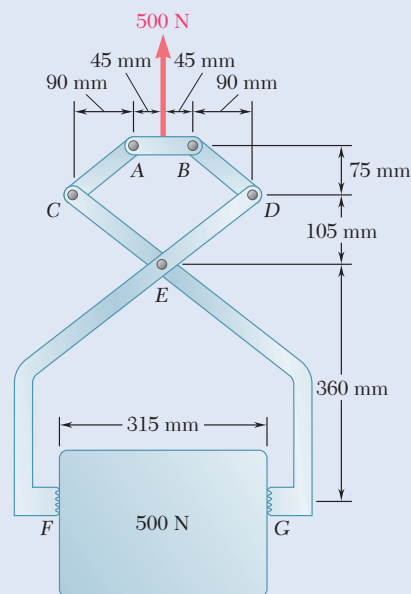


Fig. P8.32

- 8.33** A pipe with a diameter of 60 mm is gripped by the stillson wrench shown. Portions AB and DE of the wrench are rigidly attached to each other, and portion CF is connected by a pin at D. If the wrench is to grip the pipe and be self-locking, determine the required minimum coefficients of friction at A and C.

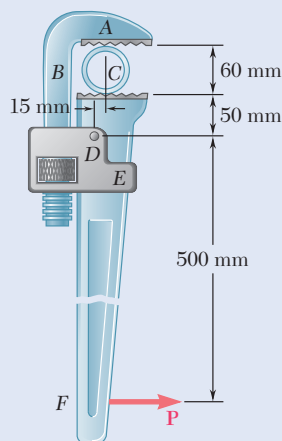


Fig. P8.33

- 8.34** A safety device used by workers climbing ladders fixed to high structures consists of a rail attached to the ladder and a sleeve that can slide on the flange of the rail. A chain connects the worker's belt to the end of an eccentric cam that can be rotated about an axle attached to the sleeve at C. Determine the smallest allowable common value of the coefficient of static friction between the flange of the rail, the pins at A and B, and the eccentric cam if the sleeve is not to slide down when the chain is pulled vertically downward.

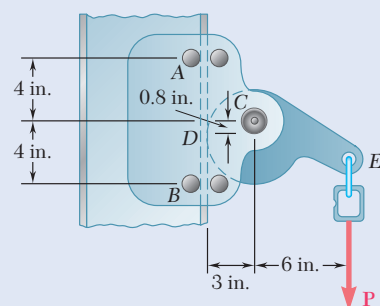


Fig. P8.34

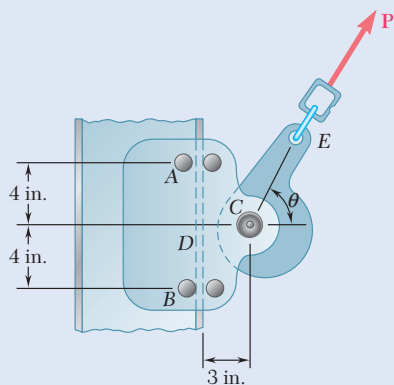


Fig. P8.35

- 8.35** To be of practical use, the safety sleeve described in Prob. 8.34 must be free to slide along the rail when pulled upward. Determine the largest allowable value of the coefficient of static friction between the flange of the rail and the pins at A and B if the sleeve is to be free to slide when pulled as shown in the figure. Assume (a) $\theta = 60^\circ$, (b) $\theta = 50^\circ$, (c) $\theta = 40^\circ$.

- 8.36** Two 10-lb blocks A and B are connected by a slender rod of negligible weight. The coefficient of static friction is 0.30 between all surfaces of contact, and the rod forms an angle $\theta = 30^\circ$ with the vertical. (a) Show that the system is in equilibrium when $P = 0$. (b) Determine the largest value of P for which equilibrium is maintained.

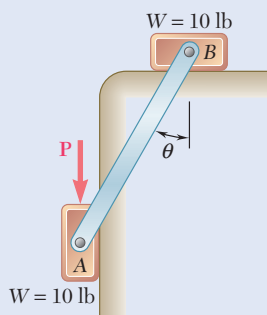


Fig. P8.36

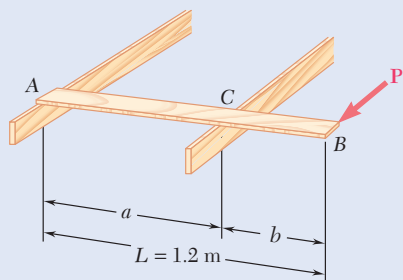


Fig. P8.37

- 8.37** A 1.2-m plank with a mass of 3 kg rests on two joists. Knowing that the coefficient of static friction between the plank and the joists is 0.30, determine the magnitude of the horizontal force required to move the plank when (a) $a = 750$ mm, (b) $a = 900$ mm.

- 8.38** Two identical uniform boards, each with a weight of 40 lb, are temporarily leaned against each other as shown. Knowing that the coefficient of static friction between all surfaces is 0.40, determine (a) the largest magnitude of the force P for which equilibrium will be maintained, (b) the surface at which motion will impend.

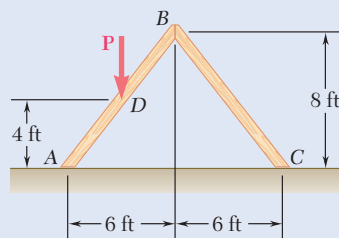


Fig. P8.38

- 8.39** Two rods are connected by a collar at B. A couple M_A with a magnitude of 15 N·m is applied to rod AB. Knowing that the coefficient of static friction between the collar and the rod is 0.30, determine the largest couple M_C for which equilibrium will be maintained.

- 8.40** In Prob. 8.39, determine the smallest couple M_C for which equilibrium will be maintained.

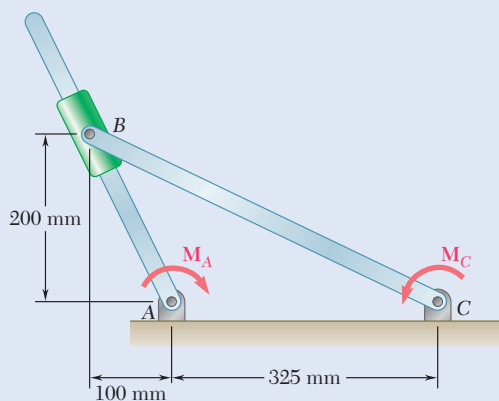


Fig. P8.39

- 8.41** A 10-ft beam, weighing 1200 lb, is to be moved to the left onto the platform as shown. A horizontal force $\mathbf{P}$ is applied to the dolly, which is mounted on frictionless wheels. The coefficients of friction between all surfaces are $\mu_s = 0.30$ and $\mu_k = 0.25$, and initially, $x = 2$ ft. Knowing that the top surface of the dolly is slightly higher than the platform, determine the force $\mathbf{P}$ required to start moving the beam. (*Hint:* The beam is supported at A and D .)

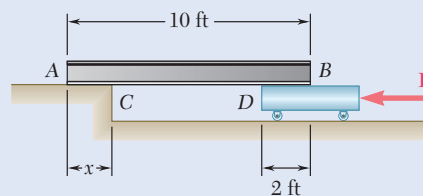


Fig. P8.41

- 8.42** (a) Show that the beam of Prob. 8.41 *cannot* be moved if the top surface of the dolly is slightly *lower* than the platform. (b) Show that the beam *can* be moved if two 175-lb workers stand on the beam at B , and determine how far to the left the beam can be moved.

- 8.43** Two 8-kg blocks A and B resting on shelves are connected by a rod of negligible mass. Knowing that the magnitude of a horizontal force $\mathbf{P}$ applied at C is slowly increased from zero, determine the value of P for which motion occurs and what that motion is when the coefficient of static friction between all surfaces is (a) $\mu_s = 0.40$, (b) $\mu_s = 0.50$.

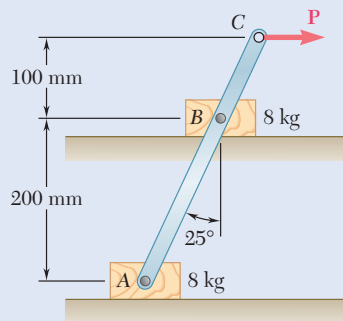


Fig. P8.43

- 8.44** A slender steel rod with a length of 225 mm is placed inside a pipe as shown. Knowing that the coefficient of static friction between the rod and the pipe is 0.20, determine the largest value of θ for which the rod will not fall into the pipe.

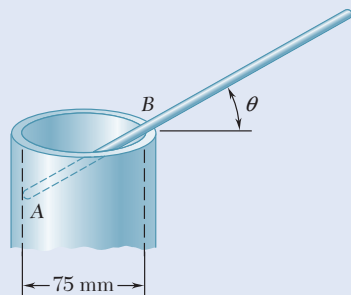


Fig. P8.44

- 8.45** In Prob. 8.44, determine the smallest value of θ for which the rod will not fall out of the pipe.

- 8.46** Two slender rods of negligible weight are pin-connected at C and attached to blocks A and B , each with a weight W . Knowing that $\theta = 80^\circ$ and that the coefficient of static friction between the blocks and the horizontal surface is 0.30, determine the largest value of P for which equilibrium is maintained.

- 8.47** Two slender rods of negligible weight are pin-connected at C and attached to blocks A and B , each with a weight W . Knowing that $P = 1.260W$ and that the coefficient of static friction between the blocks and the horizontal surface is 0.30, determine the range of values of θ between 0 and 180° for which equilibrium is maintained.

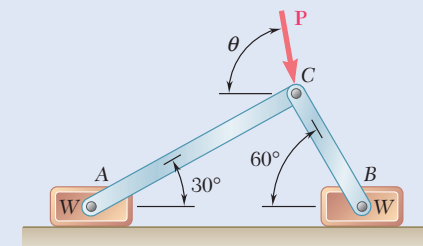
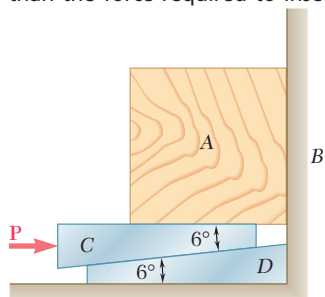


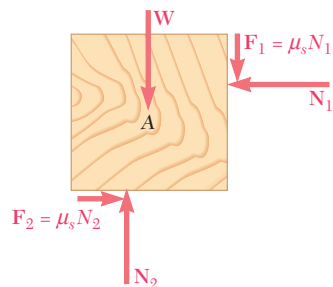
Fig. P8.46 and P8.47



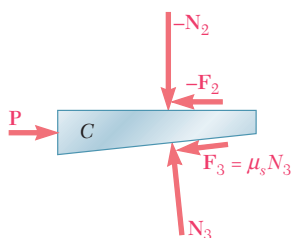
Photo 8.3 Wedges are used as shown to split tree trunks because the normal forces exerted by a wedge on the wood are much larger than the force required to insert the wedge.



(a)



(b)



(c)

Fig. 8.7 (a) A wedge C used to raise a block A ; (b) free-body diagram of block A ; (c) free-body diagram of wedge C . Note the directions of the friction forces.

8.2 WEDGES AND SCREWS

Friction is a key element in analyzing the function and operation of several types of simple machines. Here we examine the wedge and the screw, which are both extensions of the inclined plane we analyzed in Sect. 8.1.

8.2A Wedges

Wedges are simple machines used to raise large stone blocks and other heavy loads. These loads are raised by applying to the wedge a force usually considerably smaller than the weight of the load. In addition, because of the friction between the surfaces in contact, a properly shaped wedge remains in place after being forced under the load. In this way, you can use a wedge advantageously to make small adjustments in the position of heavy pieces of machinery.

Consider the block A shown in Fig. 8.7a. This block rests against a vertical wall B , and we want to raise it slightly by forcing a wedge C between block A and a second wedge D . We want to find the minimum value of the force $\mathbf{P}$ that we must apply to wedge C to move the block. We assume that we know the weight $\mathbf{W}$ of the block, which is either given in pounds or determined in newtons from the mass of the block expressed in kilograms.

We have drawn the free-body diagrams of block A and wedge C in Fig. 8.7b and c. The forces acting on the block include its weight and the normal and friction forces at the surfaces of contact with wall B and wedge C . The magnitudes of the friction forces $\mathbf{F}_1$ and $\mathbf{F}_2$ are equal, respectively, to $\mu_s N_1$ and $\mu_s N_2$, because the motion of the block must be started. It is important to show the friction forces with their correct sense. Since the block will move upward, the force $\mathbf{F}_1$ exerted by the wall on the block must be directed downward. On the other hand, since wedge C moves to the right, the relative motion of A with respect to C is to the left, and the force $\mathbf{F}_2$ exerted by C on A must be directed to the right.

Now consider the free body C in Fig. 8.7c. The forces acting on C include the applied force $\mathbf{P}$ and the normal and friction forces at the surfaces of contact with A and D . The weight of the wedge is small compared with the other forces involved and can be neglected. The forces exerted by A on C are equal and opposite to the forces $\mathbf{N}_2$ and $\mathbf{F}_2$ exerted by C on A , so we denote them, respectively, by $-\mathbf{N}_2$ and $-\mathbf{F}_2$; the friction force $-\mathbf{F}_2$ therefore must be directed to the left. We check that the force $\mathbf{F}_3$ exerted by D is also directed to the left.

We can reduce the total number of unknowns involved in the two free-body diagrams to four if we express the friction forces in terms of the normal forces. Then, since block A and wedge C are in equilibrium, we obtain four equations that we can solve to obtain the magnitude of $\mathbf{P}$. Note that, in the example considered here, it is more convenient to replace each pair of normal and friction forces by their resultant. Each free body is then subjected to only three forces, and we can solve the problem by drawing the corresponding force triangles (see Sample Prob. 8.5).

8.2B Square-Threaded Screws

Square-threaded screws are frequently part of jacks, presses, and other mechanisms. Their analysis is similar to the analysis of a block sliding along an inclined plane. (Screws are also commonly used as fasteners, but the threads on these screws are shaped differently.)

Consider the jack shown in Fig. 8.8. The screw carries a load W and is supported by the base of the jack. Contact between the screw and the base takes place along a portion of their threads. By applying a force P on the handle, the screw can be made to turn and to raise the load W .

In Fig. 8.9a, we have unwrapped the thread of the base and shown it as a straight line. We obtained the correct slope by horizontally drawing the product $2\pi r$, where r is the mean radius of the thread, and vertically drawing the **lead** L of the screw, i.e., the distance through which the screw advances in one turn. The angle θ this line forms with the horizontal is the **lead angle**. Since the force of friction between two surfaces in contact does not depend upon the area of contact, we can assume a much smaller than actual area of contact between the two threads, which allows us to represent the screw as the block shown in Fig. 8.9a. Note that, in this analysis of the jack, we neglect the small friction force between cap and screw.

The free-body diagram of the block includes the load W , the reaction R of the base thread, and a horizontal force Q , which has the same effect as the force P exerted on the handle. The force Q should have the same moment as P about the axis of the screw, so its magnitude should be $Q = Pa/r$. We can obtain the value of force Q , and thus that of force P required to raise load W , from the free-body diagram shown in Fig. 8.9a. The friction angle is taken to be equal to ϕ_s , since presumably the load is raised through a succession of short strokes. In mechanisms providing for the continuous rotation of a screw, it may be desirable to distinguish between the force required to start motion (using ϕ_s) and that required to maintain motion (using ϕ_k).

If the friction angle ϕ_s is larger than the lead angle θ , the screw is said to be *self-locking*; it will remain in place under the load. To lower the load, we must then apply the force shown in Fig. 8.9b. If ϕ_s is smaller than θ , the screw will unwind under the load; it is then necessary to apply the force shown in Fig. 8.9c to maintain equilibrium.

The lead of a screw should not be confused with its **pitch**. The **lead** is defined as the distance through which the screw advances in one turn; the **pitch** is the distance measured between two consecutive threads. Lead and pitch are equal in the case of *single-threaded* screws, but they are different in the case of *multiple-threaded* screws, i.e., screws having several independent threads. It is easily verified that for double-threaded screws the lead is twice as large as the pitch; for triple-threaded screws, it is three times as large as the pitch; etc.

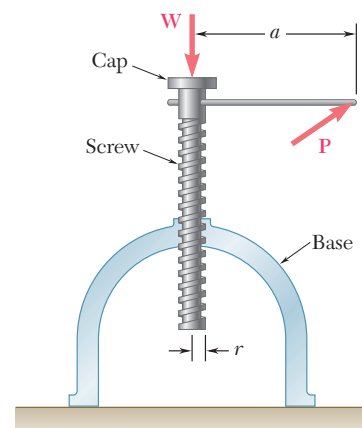


Fig. 8.8 A screw as part of a jack carrying a load W .

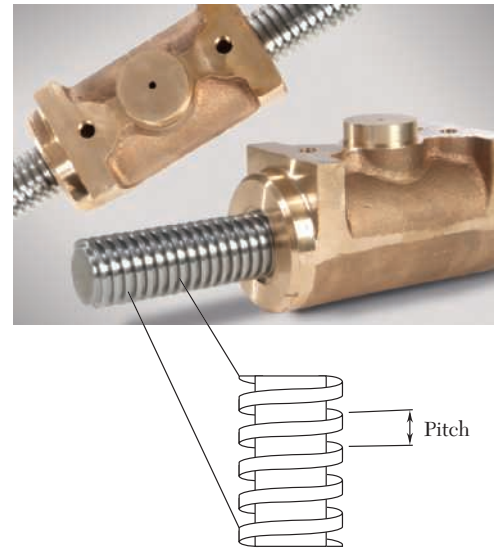


Photo 8.4 An example of a square-threaded screw, fitted to a sleeve, as might be used in an industrial application.

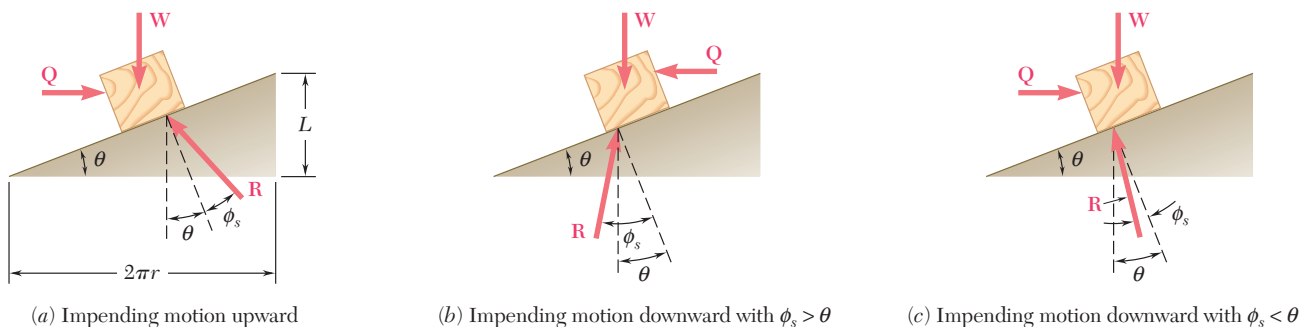


Fig. 8.9 Block-and-incline analysis of a screw. We can represent the screw as a block, because the force of friction does not depend on the area of contact between two surfaces.

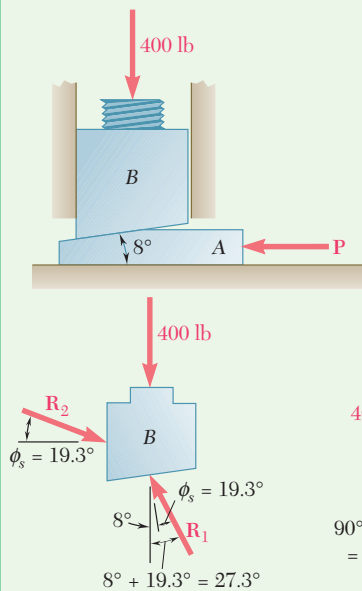


Fig. 1 Free-body diagram of block and its force triangle—block being raised.

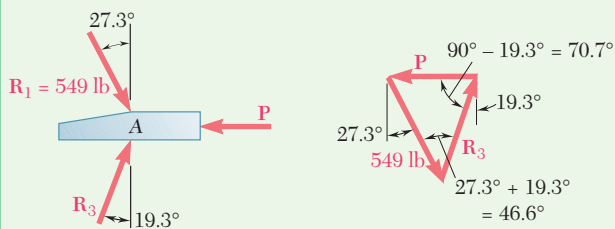


Fig. 2 Free-body diagram of wedge and its force triangle—block being raised.

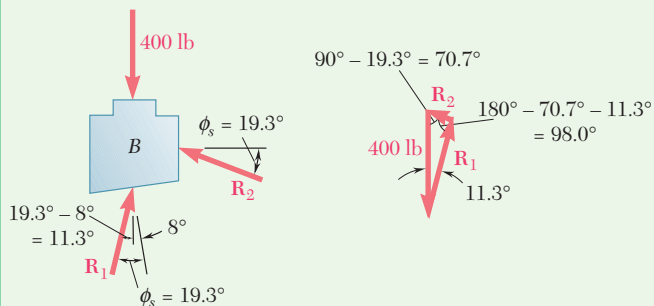


Fig. 3 Free-body diagram of block and its force triangle—block being lowered.

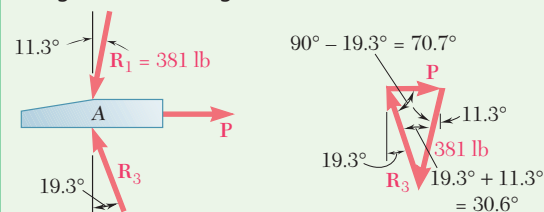


Fig. 4 Free-body diagram of wedge and its force triangle—block being lowered.

Sample Problem 8.5

The position of the machine block B is adjusted by moving the wedge A . Knowing that the coefficient of static friction is 0.35 between all surfaces of contact, determine the force P required to (a) raise block B , (b) lower block B .

STRATEGY: For both parts of the problem, normal forces and friction forces act between the wedge and the block. In part (a), you also have normal and friction forces at the left surface of the block; for part (b), they are on the right surface of the block. If you combine the normal and friction forces at each surface into resultants, you have a total of three forces acting on each body and can use force triangles to solve.

MODELING: For each part, draw the free-body diagrams of block B and wedge A together with the corresponding force triangles. Then use the law of sines to find the desired forces. Note that, since $\mu_s = 0.35$, the angle of friction is

$$\phi_s = \tan^{-1} 0.35 = 19.3^\circ$$

ANALYSIS: a. Force P to raise block

Free Body: Block B (Fig. 1). The friction force on block B due to wedge A is to the left, so the resultant R_1 is at an angle equal to the slope of the wedge plus the angle of friction.

$$\frac{R_1}{\sin 109.3^\circ} = \frac{400 \text{ lb}}{\sin 43.4^\circ} \quad R_1 = 549 \text{ lb}$$

Free Body: Wedge A (Fig. 2). The friction forces on wedge A are to the right.

$$\frac{P}{\sin 46.6^\circ} = \frac{549 \text{ lb}}{\sin 70.7^\circ} \quad P = 423 \text{ lb} \leftarrow$$

b. Force P to lower block

Free Body: Block B (Fig. 3). Now the friction force on block B due to wedge A is to the right, so the resultant R_1 is at an angle equal to the angle of friction minus the slope of the wedge.

$$\frac{R_1}{\sin 70.7^\circ} = \frac{400 \text{ lb}}{\sin 98.0^\circ} \quad R_1 = 381 \text{ lb}$$

Free Body: Wedge A (Fig. 4). The friction forces on wedge A are to the left.

$$\frac{P}{\sin 30.6^\circ} = \frac{381 \text{ lb}}{\sin 70.7^\circ} \quad P = 206 \text{ lb} \rightarrow$$

REFLECT and THINK: The force needed to lower the block is much less than the force needed to raise the block, which makes sense.

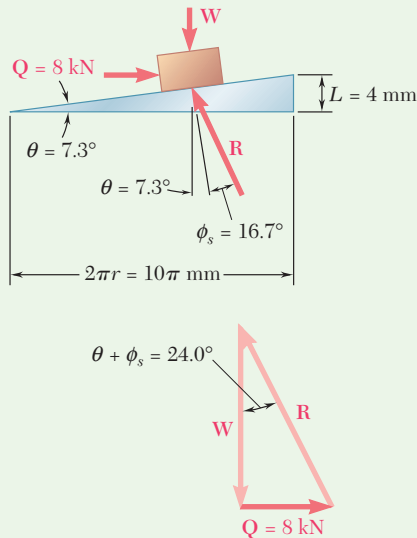
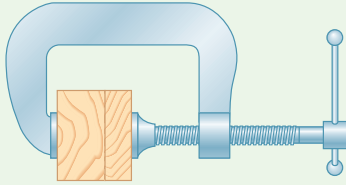


Fig. 1 Free-body diagram of block and its force triangle—clamp being tightened.

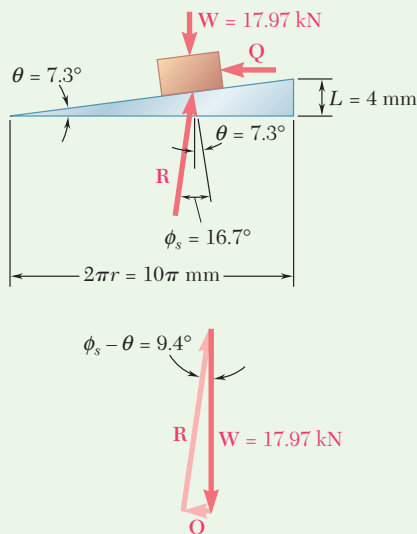


Fig. 2 Free-body diagram of block and its force triangle—clamp being loosened.

Sample Problem 8.6

A clamp is used to hold two pieces of wood together as shown. The clamp has a double square thread with a mean diameter of 10 mm and a pitch of 2 mm. The coefficient of friction between threads is $\mu_s = 0.30$. If a maximum couple of 40 N·m is applied in tightening the clamp, determine (a) the force exerted on the pieces of wood, (b) the couple required to loosen the clamp.

STRATEGY: If you represent the screw by a block, as in the analysis of this section, you can determine the incline of the screw from the geometry given in the problem, and you can find the force applied to the block by setting the moment of that force equal to the applied couple.

MODELING and ANALYSIS:

a. Force Exerted by Clamp. The mean radius of the screw is $r = 5$ mm. Since the screw is double-threaded, the lead L is equal to twice the pitch: $L = 2(2 \text{ mm}) = 4$ mm. Obtain the lead angle θ and the friction angle ϕ_s from

$$\tan \theta = \frac{L}{2\pi r} = \frac{4 \text{ mm}}{10\pi \text{ mm}} = 0.1273 \quad \theta = 7.3^\circ$$

$$\tan \phi_s = \mu_s = 0.30 \quad \phi_s = 16.7^\circ$$

You can find the force Q that should be applied to the block representing the screw by setting its moment Qr about the axis of the screw equal to the applied couple.

$$\begin{aligned} Q(5 \text{ mm}) &= 40 \text{ N}\cdot\text{m} \\ Q &= \frac{40 \text{ N}\cdot\text{m}}{5 \text{ mm}} = \frac{40 \text{ N}\cdot\text{m}}{5 \times 10^{-3} \text{ m}} = 8000 \text{ N} = 8 \text{ kN} \end{aligned}$$

Now you can draw the free-body diagram and the corresponding force triangle for the block (Fig. 1). Solve the triangle to find the magnitude of the force W exerted on the pieces of wood.

$$W = \frac{Q}{\tan(\theta + \phi_s)} = \frac{8 \text{ kN}}{\tan 24.0^\circ}$$

$$W = 17.97 \text{ kN} \quad \blacktriangleleft$$

b. Couple Required to Loosen Clamp. You can obtain the force Q required to loosen the clamp and the corresponding couple from the free-body diagram and force triangle shown in Fig. 2.

$$\begin{aligned} Q &= W \tan(\phi_s - \theta) = (17.97 \text{ kN}) \tan 9.4^\circ \\ &= 2.975 \text{ kN} \end{aligned}$$

$$\begin{aligned} \text{Couple} &= Qr = (2.975 \text{ kN})(5 \text{ mm}) \\ &= (2.975 \times 10^3 \text{ N})(5 \times 10^{-3} \text{ m}) = 14.87 \text{ N}\cdot\text{m} \end{aligned}$$

$$\text{Couple} = 14.87 \text{ N}\cdot\text{m} \quad \blacktriangleleft$$

REFLECT and THINK: In practice, you often have to determine the force effectively acting on a screw by setting the moment of that force about the axis of the screw equal to an applied couple. However, the rest of the analysis is mostly an application of dry friction. Also note that the couple required to loosen a screw is not the same as the couple required to tighten it.

SOLVING PROBLEMS ON YOUR OWN

In this section, you saw how to apply the laws of friction to the solution of problems involving **wedges** and **square-threaded screws**.

1. Wedges. Keep the following steps in mind when solving a problem involving a wedge.

a. First draw a free-body diagram of the wedge and of each of the other bodies involved. Carefully note the sense of the relative motion of all surfaces of contact and show each friction force acting in a *direction opposite* to the direction of that relative motion.

b. Show the maximum static friction force F_m at each surface if the wedge is to be inserted or removed, since motion will be impending in each of these cases.

c. The reaction R and the angle of friction, rather than the normal force and the friction force, are most useful in many applications. You can then draw one or more force triangles and determine the unknown quantities either graphically or by trigonometry [Sample Prob. 8.5].

2. Square-Threaded Screws. The analysis of a square-threaded screw is equivalent to the analysis of a block sliding on an incline. To draw the appropriate incline, you need to unwrap the thread of the screw and represent it as a straight line [Sample Prob. 8.6]. When solving a problem involving a square-threaded screw, keep the following steps in mind.

a. Do not confuse the pitch of a screw with the lead of a screw. The **pitch** of a screw is the distance between two consecutive threads, whereas the **lead** of a screw is the distance the screw advances in one full turn. The lead and the pitch are equal only in single-threaded screws. In a double-threaded screw, the lead is twice the pitch.

b. The couple required to tighten a screw is different from the couple required to loosen it. Also, screws used in jacks and clamps are usually *self-locking*; that is, the screw will remain stationary as long as no couple is applied to it, and a couple must be applied to the screw to loosen it [Sample Prob. 8.6].

Problems

8.48 The machine part ABC is supported by a frictionless hinge at B and a 10° wedge at C . Knowing that the coefficient of static friction is 0.20 at both surfaces of the wedge, determine (a) the force P required to move the wedge to the left, (b) the components of the corresponding reaction at B .

8.49 Solve Prob. 8.48 assuming that the wedge is moved to the right.

8.50 and 8.51 Two 8° wedges of negligible weight are used to move and position the 800-kg block. Knowing that the coefficient of static friction is 0.30 at all surfaces of contact, determine the smallest force P that should be applied as shown to one of the wedges.

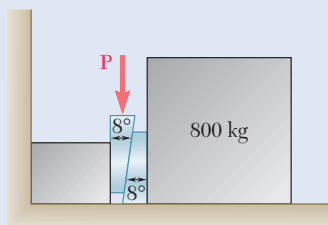


Fig. P8.50

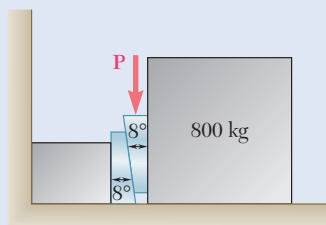


Fig. P8.51

8.52 The elevation of the end of the steel beam supported by a concrete floor is adjusted by means of the steel wedges E and F . The base plate CD has been welded to the lower flange of the beam, and the end reaction of the beam is known to be 18 kips. The coefficient of static friction is 0.30 between two steel surfaces and 0.60 between steel and concrete. If horizontal motion of the beam is prevented by the force Q , determine (a) the force P required to raise the beam, (b) the corresponding force Q .

8.53 Solve Prob. 8.52 assuming that the end of the beam is to be lowered.

8.54 Block A supports a pipe column and rests as shown on wedge B . Knowing that the coefficient of static friction at all surfaces of contact is 0.25 and that $\theta = 45^\circ$, determine the smallest force P required to raise block A .

8.55 Block A supports a pipe column and rests as shown on wedge B . Knowing that the coefficient of static friction at all surfaces of contact is 0.25 and that $\theta = 45^\circ$, determine the smallest force P for which equilibrium is maintained.

8.56 Block A supports a pipe column and rests as shown on wedge B . The coefficient of static friction at all surfaces of contact is 0.25. If $P = 0$, determine (a) the angle θ for which sliding is impending, (b) the corresponding force exerted on the block by the vertical wall.

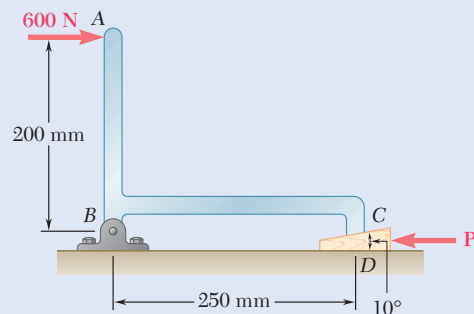


Fig. P8.48 and P8.49

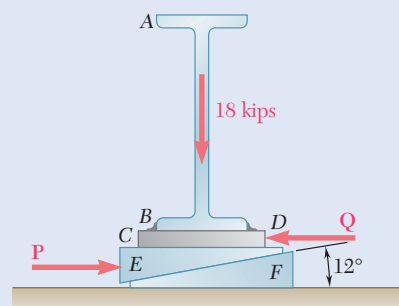


Fig. P8.52

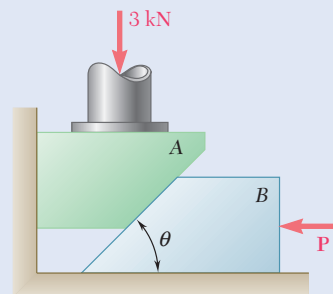


Fig. P8.54, P8.55, and P8.56

- 8.57** A wedge A of negligible weight is to be driven between two 100-lb blocks B and C resting on a horizontal surface. Knowing that the coefficient of static friction between all surfaces of contact is 0.35, determine the smallest force $\mathbf{P}$ required to start moving the wedge (a) if the blocks are equally free to move, (b) if block C is securely bolted to the horizontal surface.

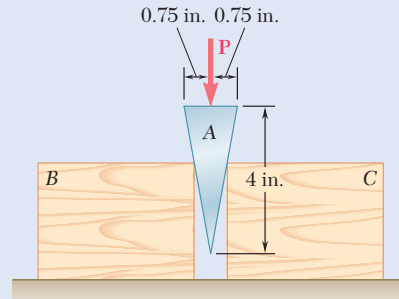


Fig. P8.57

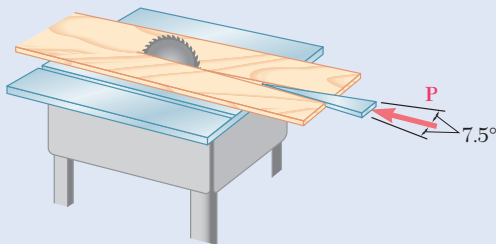


Fig. P8.58

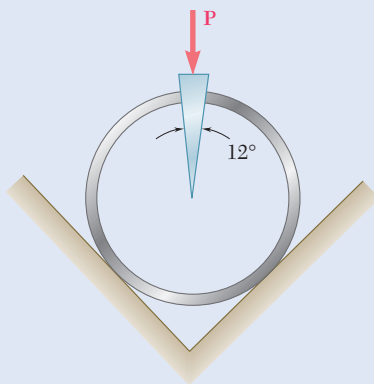


Fig. P8.59

- 8.58** A 15° wedge is forced into a saw cut to prevent binding of the circular saw. The coefficient of static friction between the wedge and the wood is 0.25. Knowing that a horizontal force $\mathbf{P}$ with a magnitude of 30 lb was required to insert the wedge, determine the magnitude of the forces exerted on the board by the wedge after insertion.

- 8.59** A 12° wedge is used to spread a split ring. The coefficient of static friction between the wedge and the ring is 0.30. Knowing that a force $\mathbf{P}$ with a magnitude of 120 N was required to insert the wedge, determine the magnitude of the forces exerted on the ring by the wedge after insertion.

- 8.60** The spring of the door latch has a constant of 1.8 lb/in. and in the position shown exerts a 0.6-lb force on the bolt. The coefficient of static friction between the bolt and the strike plate is 0.40; all other surfaces are well lubricated and may be assumed frictionless. Determine the magnitude of the force $\mathbf{P}$ required to start closing the door.

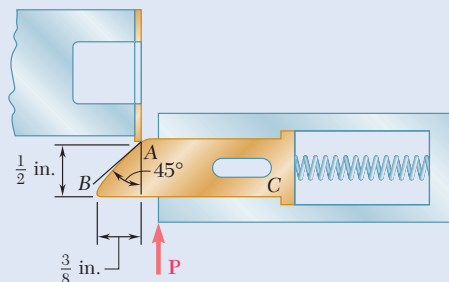


Fig. P8.60

- 8.61** In Prob. 8.60, determine the angle that the face of the bolt near B should form with line BC if the force $\mathbf{P}$ required to close the door is to be the same for both the position shown and the position when B is almost at the strike plate.

8.62 A 5° wedge is to be forced under a 1400-lb machine base at A . Knowing that the coefficient of static friction at all surfaces is 0.20, (a) determine the force $\mathbf{P}$ required to move the wedge, (b) indicate whether the machine base will move.

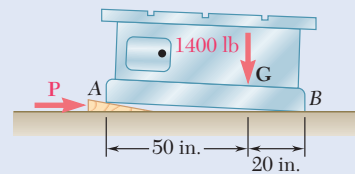


Fig. P8.62

8.63 Solve Prob. 8.62 assuming that the wedge is to be forced under the machine base at B instead of A .

8.64 A 15° wedge is forced under a 50-kg pipe as shown. The coefficient of static friction at all surfaces is 0.20. (a) Show that slipping will occur between the pipe and the vertical wall. (b) Determine the force $\mathbf{P}$ required to move the wedge.

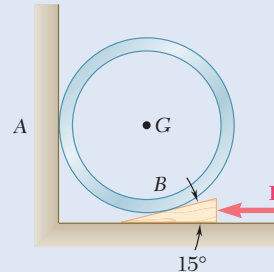


Fig. P8.64 and P8.65

***8.66** A 200-N block rests as shown on a wedge of negligible weight. The coefficient of static friction μ_s is the same at both surfaces of the wedge, and friction between the block and the vertical wall may be neglected. For $P = 100$ N, determine the value of μ_s for which motion is impending. (Hint: Solve the equation obtained by trial and error.)

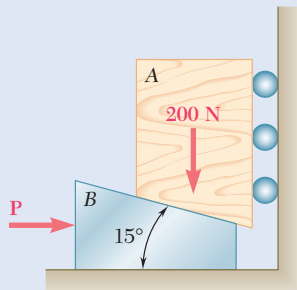


Fig. P8.66

***8.67** Solve Prob. 8.66 assuming that the rollers are removed and that μ_s is the coefficient of friction at all surfaces of contact.

8.68 Derive the following formulas relating the load $\mathbf{W}$ and the force $\mathbf{P}$ exerted on the handle of the jack discussed in Sec. 8.2B. (a) $P = (Wr/a) \tan(\theta + \phi_s)$ to raise the load; (b) $P = (Wr/a) \tan(\phi_s - \theta)$ to lower the load if the screw is self-locking; (c) $P = (Wr/a) \tan(\theta - \phi_s)$ to hold the load if the screw is not self-locking.

8.69 The square-threaded worm gear shown has a mean radius of 2 in. and a lead of 0.5 in. The large gear is subjected to a constant clockwise couple of 9.6 kip-in. Knowing that the coefficient of static friction between the two gears is 0.12, determine the couple that must be applied to shaft AB in order to rotate the large gear counterclockwise. Neglect friction in the bearings at A , B , and C .

8.70 In Prob. 8.69, determine the couple that must be applied to shaft AB in order to rotate the large gear clockwise.

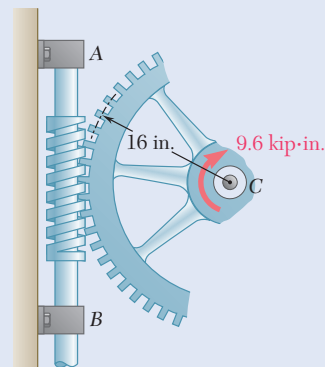


Fig. P8.69

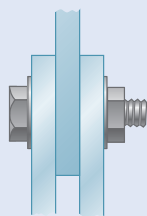


Fig. P8.71

8.71 High-strength bolts are used in the construction of many steel structures. For a 1-in.-nominal-diameter bolt, the required minimum bolt tension is 51 kips. Assuming the coefficient of friction to be 0.30, determine the required couple that should be applied to the bolt and nut. The mean diameter of the thread is 0.94 in., and the lead is 0.125 in. Neglect friction between the nut and washer, and assume the bolt to be square-threaded.

8.72 The position of the automobile jack shown is controlled by a screw ABC that is single-threaded at each end (right-handed thread at A , left-handed thread at C). Each thread has a pitch of 2.5 mm and a mean diameter of 9 mm. If the coefficient of static friction is 0.15, determine the magnitude of the couple M that must be applied to raise the automobile.

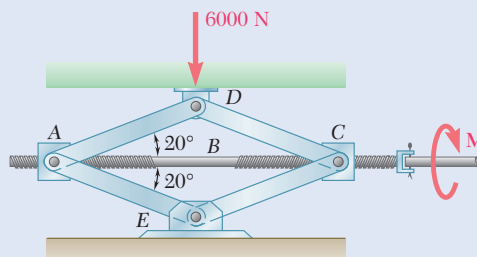


Fig. P8.72

8.73 For the jack of Prob. 8.72, determine the magnitude of the couple M that must be applied to lower the automobile.

8.74 The vise shown consists of two members connected by two double-threaded screws with a mean radius of 0.25 in. and pitch of 0.08 in. The lower member is threaded at A and B ($\mu_s = 0.35$), but the upper member is not threaded. It is desired to apply two equal and opposite forces of 120 lb on the blocks held between the jaws. (a) What screw should be adjusted first? (b) What is the maximum couple applied in tightening the second screw?

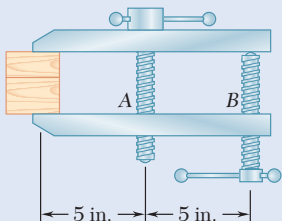


Fig. P8.74

8.75 The ends of two fixed rods A and B are each made in the form of a single-threaded screw with a mean radius of 6 mm and pitch of 2 mm. Rod A has a right-handed thread, and rod B has a left-handed thread. The coefficient of static friction between the rods and the threaded sleeve is 0.12. Determine the magnitude of the couple that must be applied to the sleeve in order to draw the rods closer together.



Fig. P8.75

8.76 Assuming that in Prob. 8.75 a right-handed thread is used on *both* rods A and B , determine the magnitude of the couple that must be applied to the sleeve in order to rotate it.

*8.3 FRICTION ON AXLES, DISKS, AND WHEELS

Journal bearings are used to provide lateral support to rotating shafts and axles. **Thrust bearings** are used to provide axial support to shafts and axles. If the journal bearing is fully lubricated, the frictional resistance depends upon the speed of rotation, the clearance between axle and bearing, and the viscosity of the lubricant. As indicated in Sec. 8.1, such problems are studied in fluid mechanics. However, we can apply the methods of this chapter to the study of axle friction when the bearing is not lubricated or only partially lubricated. In this case, we can assume that the axle and the bearing are in direct contact along a single straight line.

8.3A Journal Bearings and Axle Friction

Consider two wheels, each with a weight of W , rigidly mounted on an axle supported symmetrically by two journal bearings (Fig. 8.10a). If the wheels rotate, we find that, to keep them rotating at constant speed, it is

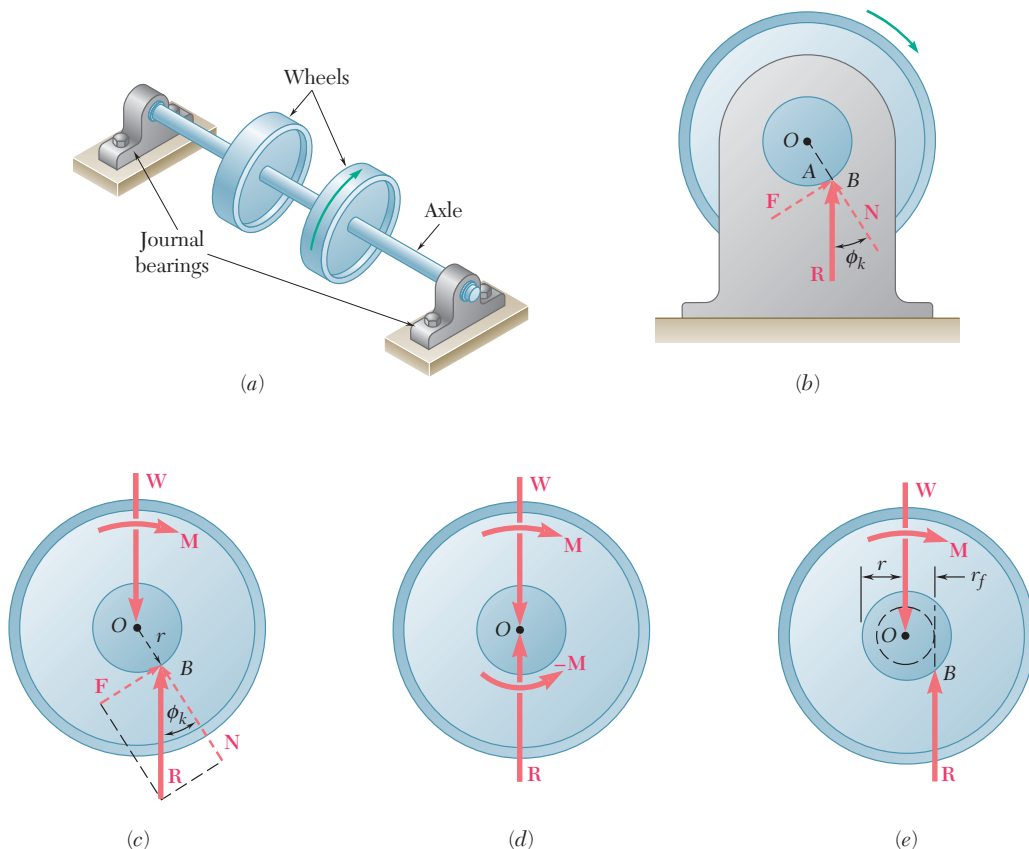


Fig. 8.10 (a) Two wheels supported by two journal bearings; (b) point of contact when the axle is rotating; (c) free-body diagram of one wheel and corresponding half axle; (d) frictional resistance produces a couple that opposes the couple maintaining the axle in motion; (e) graphical analysis with circle of friction.

necessary to apply a couple $\mathbf{M}$ to each of them. The free-body diagram in Fig. 8.10c represents one of the wheels and the corresponding half axle in projection on a plane perpendicular to the axle. The forces acting on the free body include the weight $\mathbf{W}$ of the wheel, the couple $\mathbf{M}$ required to maintain its motion, and a force $\mathbf{R}$ representing the reaction of the bearing. This force is vertical, equal, and opposite to $\mathbf{W}$, but it does not pass through the center O of the axle; $\mathbf{R}$ is located to the right of O at a distance such that its moment about O balances the moment $\mathbf{M}$ of the couple. Therefore, when the axle rotates, contact between the axle and bearing does not take place at the lowest point A . Instead, contact takes place at point B (Fig. 8.10b) or, rather, along a straight line intersecting the plane of the figure at B .

Physically, the location of contact is explained by the fact that, when the wheels are set in motion, the axle “climbs” in the bearings until slippage occurs. After sliding back slightly, the axle settles more or less in the position shown. This position is such that the angle between the reaction $\mathbf{R}$ and the normal to the surface of the bearing is equal to the angle of kinetic friction ϕ_k . The distance from O to the line of action of $\mathbf{R}$ is thus $r \sin \phi_k$, where r is the radius of the axle. Setting $\Sigma M_O = 0$ for the forces acting on the free body (the wheel), we obtain the magnitude of the couple $\mathbf{M}$ required to overcome the frictional resistance of one of the bearings:

$$M = Rr \sin \phi_k \quad (8.5)$$

For small values of the angle of friction, we can replace $\sin \phi_k$ by $\tan \phi_k$; that is, by μ_k . This gives us the approximate formula

$$M \approx Rr\mu_k \quad (8.6)$$

In the solution of certain problems, it may be more convenient to let the line of action of $\mathbf{R}$ pass through O , as it does when the axle does not rotate. In such a case, you need to add a couple $-\mathbf{M}$, with the same magnitude as the couple $\mathbf{M}$ but of opposite sense, to the reaction $\mathbf{R}$ (Fig. 8.10d). This couple represents the frictional resistance of the bearing.

If a graphical solution is preferred, you can readily draw the line of action of $\mathbf{R}$ (Fig. 8.10e) if you note that it must be tangent to a circle centered at O and with a radius

$$r_f = r \sin \phi_k \approx r\mu_k \quad (8.7)$$

This circle is called the **circle of friction** of the axle and bearing, and it is independent of the loading conditions of the axle.

8.3B Thrust Bearings and Disk Friction

Two types of thrust bearings are commonly used to provide axial support to rotating shafts and axles: (1) **end bearings** and (2) **collar bearings** (Fig. 8.11). In the case of collar bearings, friction forces develop between

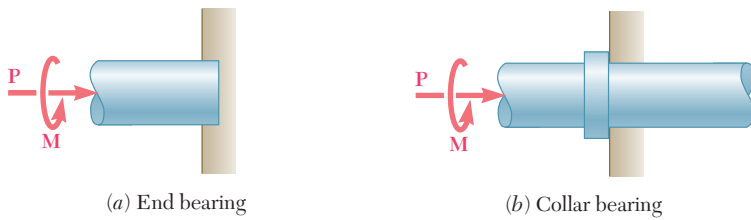


Fig. 8.11 In thrust bearings, an axial force keeps the rotating axle in contact with the support bearing.

the two ring-shaped areas in contact. In the case of end bearings, friction takes place over full circular areas or over ring-shaped areas when the end of the shaft is hollow. Friction between circular areas, called **disk friction**, also occurs in other mechanisms, such as disk clutches.

To obtain a formula for the most general case of disk friction, let us consider a rotating hollow shaft. A couple $\mathbf{M}$ keeps the shaft rotating at constant speed, while an axial force $\mathbf{P}$ maintains it in contact with a fixed bearing (Fig. 8.12). Contact between the shaft and the bearing takes place over a ring-shaped area with an inner radius of R_1 and an outer radius of R_2 . Assuming that the pressure between the two surfaces in contact is uniform, we find that the magnitude of the normal force $\Delta \mathbf{N}$ exerted on an element of area ΔA is $\Delta N = P \Delta A / A$, where $A = \pi (R_2^2 - R_1^2)$ and that the magnitude of the friction force $\Delta \mathbf{F}$ acting on ΔA is $\Delta F = \mu_k \Delta N$. Let's use r to denote the distance from the axis of the shaft to the element of area ΔA . Then the magnitude ΔM of the moment of $\Delta \mathbf{F}$ about the axis of the shaft is

$$\Delta M = r \Delta F = \frac{r \mu_k P \Delta A}{\pi (R_2^2 - R_1^2)}$$

Equilibrium of the shaft requires that the moment $\mathbf{M}$ of the couple applied to the shaft be equal in magnitude to the sum of the moments of the

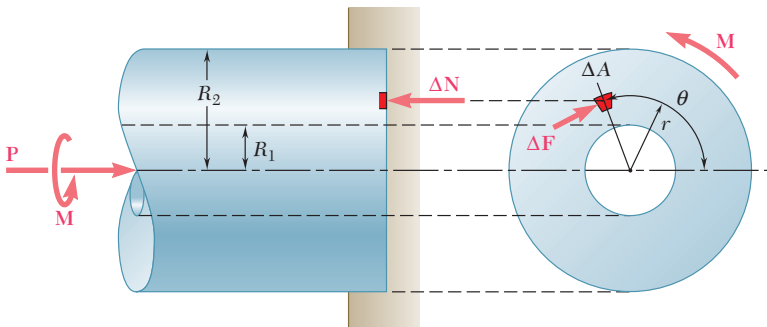


Fig. 8.12 Geometry of the frictional contact surface in a thrust bearing.

friction forces $\Delta \mathbf{F}$ opposing the motion of the shaft. Replacing ΔA by the infinitesimal element $dA = r d\theta dr$ used with polar coordinates and integrating over the area of contact, the expression for the magnitude of the couple $\mathbf{M}$ required to overcome the frictional resistance of the bearing is

$$\begin{aligned}
 M &= \frac{\mu_k P}{\pi(R_2^2 - R_1^2)} \int_0^{2\pi} \int_{R_1}^{R_2} r^2 dr d\theta \\
 &= \frac{\mu_k P}{\pi(R_2^2 - R_1^2)} \int_0^{2\pi} \frac{1}{3}(R_2^3 - R_1^3) d\theta \\
 M &= \frac{2}{3} \mu_k P \frac{R_2^3 - R_1^3}{R_2^2 - R_1^2} \quad (8.8)
 \end{aligned}$$

When contact takes place over a full circle with a radius of R , formula (8.8) reduces to

$$M = \frac{2}{3} \mu_k P R \quad (8.9)$$

This value of M is the same value we would obtain if contact between shaft and bearing took place at a single point located at a distance $2R/3$ from the axis of the shaft.

The largest couple that can be transmitted by a disk clutch without causing slippage is given by a formula similar to Eq. (8.9), where μ_k has been replaced by the coefficient of static friction μ_s .

8.3C Wheel Friction and Rolling Resistance

The wheel is one of the most important inventions of our civilization. Among many other uses, with a wheel we can move heavy loads with relatively little effort. Because the point where the wheel is in contact with the ground at any given instant has no relative motion with respect to the ground, use of the wheel avoids the large friction forces that would arise if the load were in direct contact with the ground. However, some resistance to the wheel's motion does occur. This resistance has two distinct causes. It is due to (1) a combined effect of axle friction and friction at the rim and (2) the fact that the wheel and the ground deform, causing contact between wheel and ground to take place over an area rather than at a single point.

To understand better the first cause of resistance to the motion of a wheel, consider a railroad car supported by eight wheels mounted on

axles and bearings. We assume the car is moving to the right at constant speed along a straight horizontal track. The free-body diagram of one of the wheels is shown in Fig. 8.13*a*. The forces acting on the free body include the load $\mathbf{W}$ supported by the wheel and the normal reaction $\mathbf{N}$ of the track. Since $\mathbf{W}$ passes through the center O of the axle, we represent the frictional resistance of the bearing by a counterclockwise couple $\mathbf{M}$ (see Sec. 8.3A). Then to keep the free body in equilibrium, we must add two equal and opposite forces $\mathbf{P}$ and $\mathbf{F}$, forming a clockwise couple of moment $-\mathbf{M}$. The force $\mathbf{F}$ is the friction force exerted by the track on the wheel, and $\mathbf{P}$ represents the force that should be applied to the wheel to keep it rolling at constant speed. Note that the forces $\mathbf{P}$ and $\mathbf{F}$ would not exist if there were no friction between the wheel and the track. The couple $\mathbf{M}$ representing the axle friction would then be zero; the wheel would slide on the track without turning in its bearing.

The couple $\mathbf{M}$ and the forces $\mathbf{P}$ and $\mathbf{F}$ also reduce to zero when there is no axle friction. For example, a wheel that is not held in bearings but rolls freely and at constant speed on horizontal ground (Fig. 8.13*b*) is subjected to only two forces: its own weight $\mathbf{W}$ and the normal reaction $\mathbf{N}$ of the ground. No friction force acts on the wheel regardless of the value of the coefficient of friction between wheel and ground. Thus, a wheel rolling freely on horizontal ground should keep rolling indefinitely.

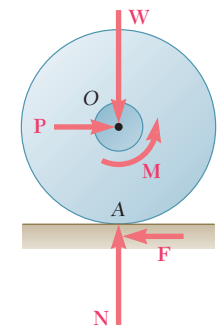
Experience, however, indicates that a free wheel does slow down and eventually come to rest. This is due to the second type of resistance mentioned at the beginning of this section, known as **rolling resistance**. Under the load $\mathbf{W}$, both the wheel and the ground deform slightly, causing the contact between wheel and ground to take place over a certain area. Experimental evidence shows that the resultant of the forces exerted by the ground on the wheel over this area is a force $\mathbf{R}$ applied at a point B , which is not located directly under the center O of the wheel but slightly in front of it (Fig. 8.13*c*). To balance the moment of $\mathbf{W}$ about B and to keep the wheel rolling at constant speed, it is necessary to apply a horizontal force $\mathbf{P}$ at the center of the wheel. Setting $\Sigma M_B = 0$, we obtain

$$Pr = Wb \quad (8.10)$$

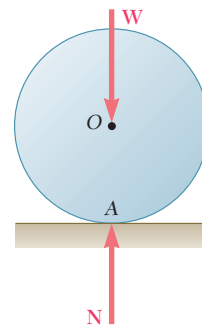
where r = radius of wheel

b = horizontal distance between O and B

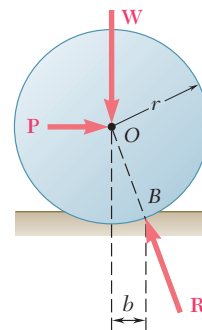
The distance b is commonly called the **coefficient of rolling resistance**. Note that b is not a dimensionless coefficient, since it represents a length; b is usually expressed in inches or in millimeters. The value of b depends upon several parameters in a manner that has not yet been clearly established. Values of the coefficient of rolling resistance vary from about 0.01 in. or 0.25 mm for a steel wheel on a steel rail to 5.0 in. or 125 mm for the same wheel on soft ground.



(a) Effect of axle friction



(b) Free wheel



(c) Rolling resistance

Fig. 8.13 (a) Free-body diagram of a rolling wheel, showing the effect of axle friction; (b) free-body diagram of a free wheel, not connected to an axle; (c) free-body diagram of a rolling wheel, showing the effect of rolling resistance.

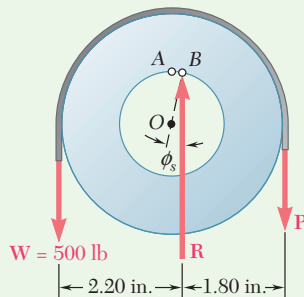


Fig. 1 Free-body diagram of pulley—smallest vertical force to raise the load.

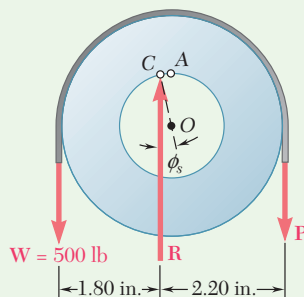


Fig. 2 Free-body diagram of pulley—smallest vertical force to hold load.

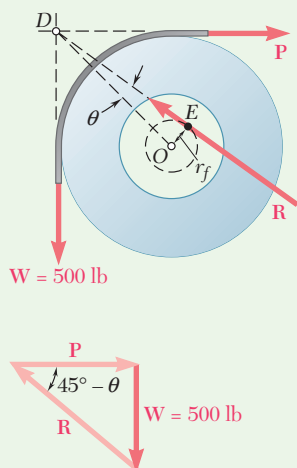


Fig. 3 Free-body diagram of pulley and force triangle—smallest horizontal force to raise load.

Sample Problem 8.7

A pulley with a diameter of 4 in. can rotate about a fixed shaft with a diameter of 2 in. The coefficient of static friction between the pulley and shaft is 0.20. Determine (a) the smallest vertical force **P** required to start raising a 500-lb load, (b) the smallest vertical force **P** required to hold the load, (c) the smallest horizontal force **P** required to start raising the same load.

STRATEGY: You can use the radius of the circle of friction to position the reaction of the pulley in each scenario and then apply the principles of equilibrium.

MODELING and ANALYSIS:

a. Vertical Force **P Required to Start Raising the Load.** When the forces in both parts of the rope are equal, contact between the pulley and shaft takes place at *A* (Fig. 1). When **P** is increased, the pulley rolls around the shaft slightly and contact takes place at *B*. Draw the free-body diagram of the pulley when motion is impending. The perpendicular distance from the center *O* of the pulley to the line of action of **R** is

$$r_f = r \sin \phi_s \approx r \mu_s \quad r_f \approx (1 \text{ in.})0.20 = 0.20 \text{ in.}$$

Summing moments about *B*, you obtain

$$+\circlearrowleft \Sigma M_B = 0: \quad (2.20 \text{ in.})(500 \text{ lb}) - (1.80 \text{ in.})P = 0$$

$$P = 611 \text{ lb} \quad \mathbf{P = 611 \text{ lb} \downarrow}$$

b. Vertical Force **P to Hold the Load.** As the force **P** is decreased, the pulley rolls around the shaft, and contact takes place at *C* (Fig. 2). Considering the pulley as a free body and summing moments about *C*, you find

$$+\circlearrowleft \Sigma M_C = 0: \quad (1.80 \text{ in.})(500 \text{ lb}) - (2.20 \text{ in.})P = 0$$

$$P = 409 \text{ lb} \quad \mathbf{P = 409 \text{ lb} \downarrow}$$

c. Horizontal Force **P to Start Raising the Load.** Since the three forces **W**, **P**, and **R** are not parallel, they must be concurrent (Fig. 3). The direction of **R** is thus determined from the fact that its line of action must pass through the point of intersection *D* of **W** and **P** and must be tangent to the circle of friction. Recall that the radius of the circle of friction is $r_f = 0.20 \text{ in.}$, so you can calculate the angle marked θ in Fig. 3 as

$$\sin \theta = \frac{OE}{OD} = \frac{0.20 \text{ in.}}{(2 \text{ in.})\sqrt{2}} = 0.0707 \quad \theta = 4.1^\circ$$

From the force triangle, you can determine

$$P = W \cot (45^\circ - \theta) = (500 \text{ lb}) \cot 40.9^\circ$$

$$= 577 \text{ lb} \quad \mathbf{P = 577 \text{ lb} \rightarrow}$$

REFLECT and THINK: Many elementary physics problems treat pulleys as frictionless, but when you do take friction into account, the results can be quite different, depending on the direction of motion, the directions of the forces involved, and especially the coefficient of friction.

SOLVING PROBLEMS ON YOUR OWN

In this section, we described several additional engineering applications of the laws of friction.

1. Journal bearings and axle friction. In journal bearings, the reaction does not pass through the center of the shaft or axle that is being supported. The distance from the center of the shaft or axle to the line of action of the reaction (Fig. 8.10) is defined by

$$r_f = r \sin \phi_k \approx r\mu_k$$

if motion is actually taking place.

It is defined by

$$r_f = r \sin \phi_s \approx r\mu_s$$

if motion is impending.

Once you have determined the line of action of the reaction, you can draw a free-body diagram and use the corresponding equations of equilibrium to complete the solution [Sample Prob. 8.7]. In some problems, it is useful to observe that the line of action of the reaction must be tangent to a circle with a radius of $r_f \approx r\mu_k$ or $r_f \approx r\mu_s$, which is known as the **circle of friction** [Sample Prob. 8.7, part c].

2. Thrust bearings and disk friction. In a thrust bearing, the magnitude of the couple required to overcome frictional resistance is equal to the sum of the moments of the *kinetic* friction forces exerted on the end of the shaft [Eqs. (8.8) and (8.9)].

An example of disk friction is the **disk clutch**. It is analyzed in the same way as a thrust bearing, except that to determine the largest couple that can be transmitted you must compute the sum of the moments of the maximum *static* friction forces exerted on the disk.

3. Wheel friction and rolling resistance. The rolling resistance of a wheel is caused by deformations of both the wheel and the ground. The line of action of the reaction **R** of the ground on the wheel intersects the ground at a horizontal distance b from the center of the wheel. The distance b is known as the **coefficient of rolling resistance** and is expressed in inches or millimeters.

4. In problems involving both rolling resistance and axle friction, the free-body diagram should show that the line of action of the reaction **R** of the ground on the wheel is tangent to the friction circle of the axle and intersects the ground at a horizontal distance from the center of the wheel equal to the coefficient of rolling resistance.

Problems

- 8.77** A lever of negligible weight is loosely fitted onto a 75-mm-diameter fixed shaft. It is observed that the lever will just start rotating if a 3-kg mass is added at *C*. Determine the coefficient of static friction between the shaft and the lever.

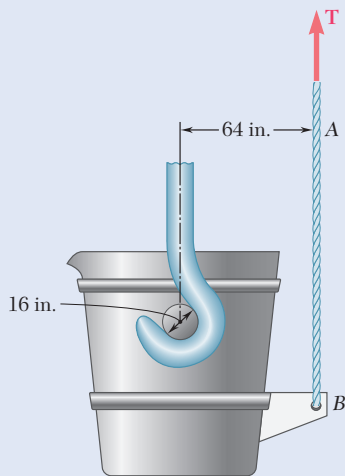


Fig. P8.78

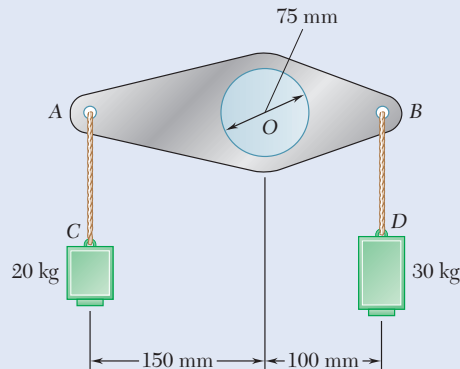


Fig. P8.77

- 8.78** A hot-metal ladle and its contents weigh 130 kips. Knowing that the coefficient of static friction between the hooks and the pinion is 0.30, determine the tension in cable *AB* required to start tipping the ladle.

- 8.79 and 8.80** The double pulley shown is attached to a 10-mm-radius shaft that fits loosely in a fixed bearing. Knowing that the coefficient of static friction between the shaft and the poorly lubricated bearing is 0.40, determine the magnitude of the force **P** required to start raising the load.

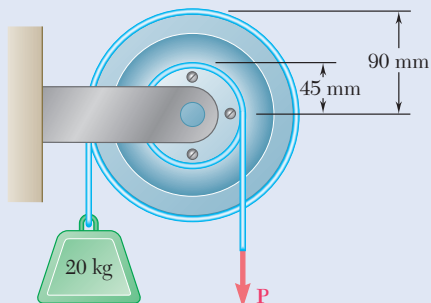


Fig. P8.79 and P8.81

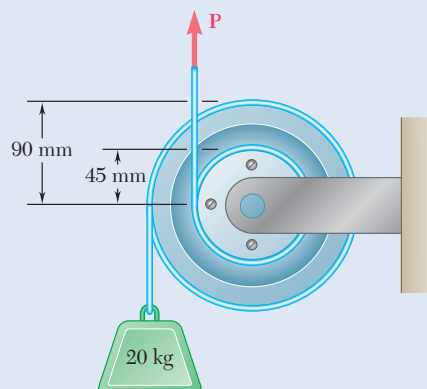


Fig. P8.80 and P8.82

- 8.81 and 8.82** The double pulley shown is attached to a 10-mm-radius shaft that fits loosely in a fixed bearing. Knowing that the coefficient of static friction between the shaft and the poorly lubricated bearing is 0.40, determine the magnitude of the smallest force **P** required to maintain equilibrium.

8.83 The block and tackle shown are used to raise a 150-lb load. Each of the 3-in.-diameter pulleys rotates on a 0.5-in.-diameter axle. Knowing that the coefficient of static friction is 0.20, determine the tension in each portion of the rope as the load is slowly raised.

8.84 The block and tackle shown are used to lower a 150-lb load. Each of the 3-in.-diameter pulleys rotates on a 0.5-in.-diameter axle. Knowing that the coefficient of static friction is 0.20, determine the tension in each portion of the rope as the load is slowly lowered.

8.85 A scooter is to be designed to roll down a 2 percent slope at a constant speed. Assuming that the coefficient of kinetic friction between the 25-mm-diameter axles and the bearings is 0.10, determine the required diameter of the wheels. Neglect the rolling resistance between the wheels and the ground.

8.86 The link arrangement shown is frequently used in highway bridge construction to allow for expansion due to changes in temperature. At each of the 60-mm-diameter pins *A* and *B*, the coefficient of static friction is 0.20. Knowing that the vertical component of the force exerted by *BC* on the link is 200 kN, determine (a) the horizontal force that should be exerted on beam *BC* to just move the link, (b) the angle that the resulting force exerted by beam *BC* on the link will form with the vertical.

8.87 and 8.88 A lever *AB* of negligible weight is loosely fitted onto a 2.5-in.-diameter fixed shaft. Knowing that the coefficient of static friction between the fixed shaft and the lever is 0.15, determine the force *P* required to start the lever rotating counterclockwise.

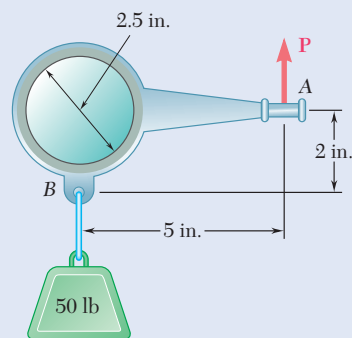


Fig. P8.87 and P8.89

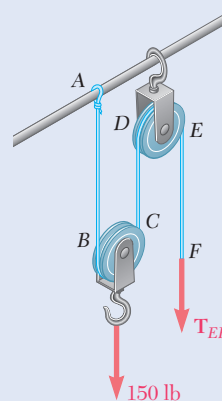


Fig. P8.83 and P8.84

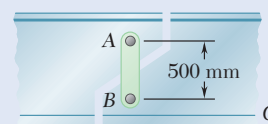


Fig. P8.86

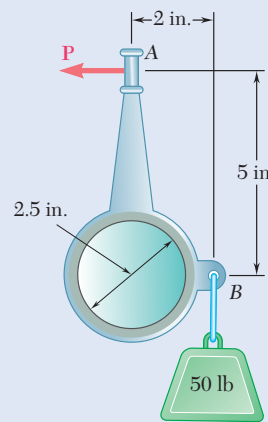


Fig. P8.88 and P8.90

8.89 and 8.90 A lever *AB* of negligible weight is loosely fitted onto a 2.5-in.-diameter fixed shaft. Knowing that the coefficient of static friction between the fixed shaft and the lever is 0.15, determine the force *P* required to start the lever rotating clockwise.

8.91 A loaded railroad car has a mass of 30 Mg and is supported by eight 800-mm-diameter wheels with 125-mm-diameter axles. Knowing that the coefficients of friction are $\mu_s = 0.020$ and $\mu_k = 0.015$, determine the horizontal force required (a) to start the car moving, (b) to keep the car moving at a constant speed. Neglect rolling resistance between the wheels and the rails.

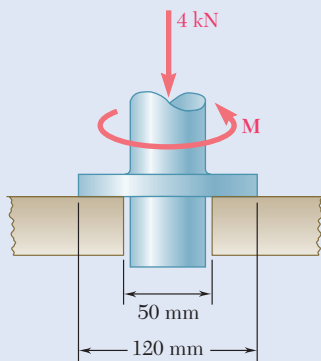


Fig. P8.92

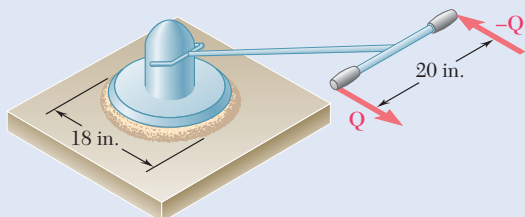


Fig. P8.93

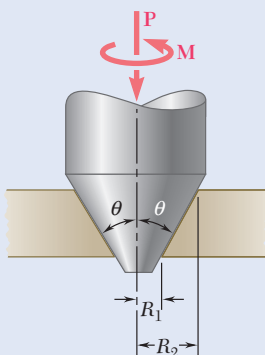


Fig. P8.96

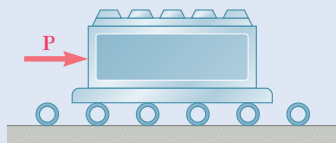


Fig. P8.100

8.92 Knowing that a couple of magnitude 30 N·m is required to start the vertical shaft rotating, determine the coefficient of static friction between the annular surfaces of contact.

8.93 A 50-lb electric floor polisher is operated on a surface for which the coefficient of kinetic friction is 0.25. Assuming that the normal force per unit area between the disk and the floor is uniformly distributed, determine the magnitude Q of the horizontal forces required to prevent motion of the machine.

***8.94** The frictional resistance of a thrust bearing decreases as the shaft and bearing surfaces wear out. It is generally assumed that the wear is directly proportional to the distance traveled by any given point of the shaft and thus to the distance r from the point to the axis of the shaft. Assuming then that the normal force per unit area is inversely proportional to r , show that the magnitude M of the couple required to overcome the frictional resistance of a worn-out end bearing (with contact over the full circular area) is equal to 75 percent of the value given by Eq. (8.9) for a new bearing.

***8.95** Assuming that bearings wear out as indicated in Prob. 8.94, show that the magnitude M of the couple required to overcome the frictional resistance of a worn-out collar bearing is

$$M = \frac{1}{2} \mu_k P (R_1 + R_2)$$

where P = magnitude of the total axial force
 R_1, R_2 = inner and outer radii of collar

***8.96** Assuming that the pressure between the surfaces of contact is uniform, show that the magnitude M of the couple required to overcome frictional resistance for the conical bearing shown is

$$M = \frac{2}{3} \frac{\mu_k P}{\sin \theta} \frac{R_2^3 - R_1^3}{R_2^2 - R_1^2}$$

8.97 Solve Prob. 8.93 assuming that the normal force per unit area between the disk and the floor varies linearly from a maximum at the center to zero at the circumference of the disk.

8.98 Determine the horizontal force required to move a 2500-lb automobile with 23-in.-diameter tires along a horizontal road at a constant speed. Neglect all forms of friction except rolling resistance, and assume the coefficient of rolling resistance to be 0.05 in.

8.99 Knowing that a 6-in.-diameter disk rolls at a constant velocity down a 2 percent incline, determine the coefficient of rolling resistance between the disk and the incline.

8.100 A 900-kg machine base is rolled along a concrete floor using a series of steel pipes with outside diameters of 100 mm. Knowing that the coefficient of rolling resistance is 0.5 mm between the pipes and the base and 1.25 mm between the pipes and the concrete floor, determine the magnitude of the force P required to slowly move the base along the floor.

8.101 Solve Prob. 8.85 including the effect of a coefficient of rolling resistance of 1.75 mm.

8.102 Solve Prob. 8.91 including the effect of a coefficient of rolling resistance of 0.5 mm.

8.4 BELT FRICTION

Another common application of dry friction concerns belts, which serve many different purposes in engineering, such as transmitting a torque from a lawn mower engine to its wheels. Some of the same analysis affects the design of band brakes and the operation of ropes and pulleys.

Consider a flat belt passing over a fixed cylindrical drum (Fig. 8.14a). We want to determine the relation between the values T_1 and T_2 of the tension in the two parts of the belt when the belt is just about to slide toward the right.

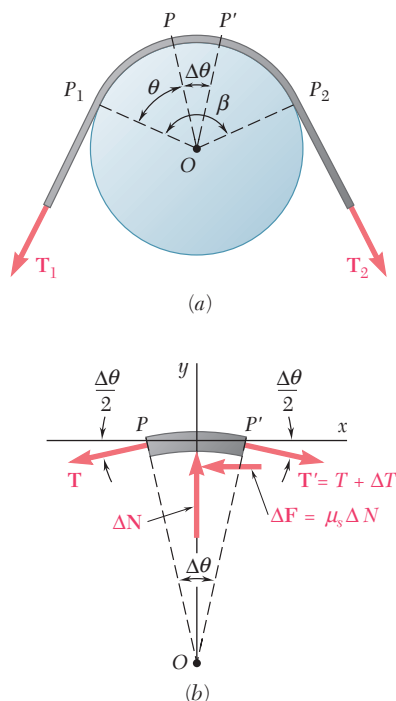


Fig. 8.14 (a) Tensions at the ends of a belt passing over a drum; (b) free-body diagram of an element of the belt, indicating the condition that the belt is about to slip to the right.

First we detach from the belt a small element PP' subtending an angle $\Delta\theta$. Denoting the tension at P by T and the tension at P' by $T + \Delta T$, we draw the free-body diagram of the element of the belt (Fig. 8.14b). Besides the two forces of tension, the forces acting on the free body are the normal component ΔN of the reaction of the drum and the friction force ΔF . Since we assume motion is impending, we have $\Delta F = \mu_s \Delta N$. Note that if $\Delta\theta$ approaches zero, the magnitudes ΔN and ΔF and the difference ΔT between the tension at P and the tension at P' also approach zero; the value T of the tension at P , however, remains unchanged. This observation helps in understanding our choice of notation.

Choosing the coordinate axes shown in Fig. 8.14b, we can write the equations of equilibrium for the element PP' as

$$\Sigma F_x = 0: \quad (T + \Delta T) \cos \frac{\Delta\theta}{2} - T \cos \frac{\Delta\theta}{2} - \mu_s \Delta N = 0 \quad (8.11)$$

$$\Sigma F_y = 0: \quad \Delta N - (T + \Delta T) \sin \frac{\Delta\theta}{2} - T \sin \frac{\Delta\theta}{2} = 0 \quad (8.12)$$

Solving Eq. (8.12) for ΔN and substituting into Eq. (8.11), we obtain after reductions

$$\Delta T \cos \frac{\Delta\theta}{2} - \mu_s(2T + \Delta T) \sin \frac{\Delta\theta}{2} = 0$$

Now we divide both terms by $\Delta\theta$. For the first term, we do this simply by dividing ΔT by $\Delta\theta$. We carry out the division of the second term by dividing the terms in parentheses by 2 and the sine by $\Delta\theta/2$. The result is

$$\frac{\Delta T}{\Delta\theta} \cos \frac{\Delta\theta}{2} - \mu_s \left(T + \frac{\Delta T}{2} \right) \frac{\sin(\Delta\theta/2)}{\Delta\theta/2} = 0$$

If we now let $\Delta\theta$ approach zero, the cosine approaches one and $\Delta T/2$ approaches zero, as noted above. The quotient of $\sin(\Delta\theta/2)$ over $\Delta\theta/2$ approaches one, according to a lemma derived in all calculus textbooks. Since the limit as $\Delta\theta$ approaches 0 of $\Delta T/\Delta\theta$ is equal to the derivative $dT/d\theta$ by definition, we get

$$\frac{dT}{d\theta} - \mu_s T = 0 \quad \frac{dT}{T} = \mu_s d\theta$$

Now we integrate both members of the last equation from P_1 to P_2 (see Fig. 8.14a). At P_1 , we have $\theta = 0$ and $T = T_1$; at P_2 , we have $\theta = \beta$ and $T = T_2$. Integrating between these limits, we have

$$\int_{T_1}^{T_2} \frac{dT}{T} = \int_0^\beta \mu_s d\theta$$

$$\ln T_2 - \ln T_1 = \mu_s \beta$$

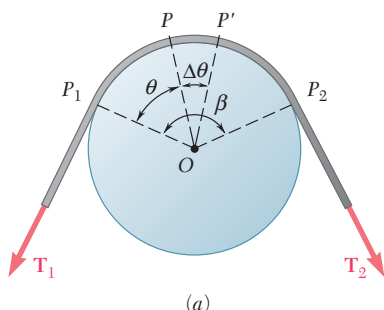


Fig. 8.14a (repeated)

Noting that the left-hand side is equal to the natural logarithm of the quotient of T_2 and T_1 , this reduces to

$$\ln \frac{T_2}{T_1} = \mu_s \beta \quad (8.13)$$

We can also write this relation in the form

**Belt friction,
impending slip**

$$\frac{T_2}{T_1} = e^{\mu_s \beta} \quad (8.14)$$



Photo 8.5 A sailor wraps a rope around the smooth post (called a *bollard*) in order to control the rope using much less force than the tension in the taut part of the rope.

The formulas we have derived apply equally well to problems involving flat belts passing over fixed cylindrical drums and to problems

involving ropes wrapped around a post or capstan. They also can be used to solve problems involving band brakes. (In this situation, the drum is about to rotate, but the band remains fixed.) The formulas also can be applied to problems involving belt drives. In these problems, both the pulley and the belt rotate; our concern is then to find whether the belt will slip; i.e., whether it will move with respect to the pulley.

Formulas (8.13) and (8.14) should be used only if the belt, rope, or brake is *about to slip*. Generally, it is easier to use Eq. (8.14) if you need to find T_1 or T_2 ; it is preferable to use Eq. (8.13) if you need to find either μ_s or the angle of contact β . Note that T_2 is always larger than T_1 . T_2 therefore represents the tension in that part of the belt or rope that *pulls*, whereas T_1 is the tension in the part that *resists*. Also observe that the angle of contact β must be expressed in *radians*. The angle of contact β may be larger than 2π ; for example, if a rope is wrapped n times around a post, β is equal to $2\pi n$.

If the belt, rope, or brake is actually slipping, you should use formulas similar to Eqs. (8.13) and (8.14) involving the coefficient of kinetic friction μ_k to find the difference in forces. If the belt, rope, or brake is not slipping and is not about to slip, none of these formulas can be used.

The belts used in belt drives are often V-shaped. In the V belt shown in Fig. 8.15a, contact between belt and pulley takes place along the sides of the groove. Again, we can obtain the relation between the values T_1 and T_2 of the tension in the two parts of the belt when the belt is just about to slip by drawing the free-body diagram of an element of the belt (Fig. 8.15b and c). Formulas similar to Eqs. (8.11) and (8.12) are derived, but the magnitude of the total friction force acting on the element is now $2\Delta F$, and the sum of the y components of the normal forces is $2\Delta N \sin(\alpha/2)$. Proceeding as previously, we obtain

$$\ln \frac{T_2}{T_1} = \frac{\mu_s \beta}{\sin(\alpha/2)} \quad (8.15)$$

or

$$\frac{T_2}{T_1} = e^{\mu_s \beta / \sin(\alpha/2)} \quad (8.16)$$

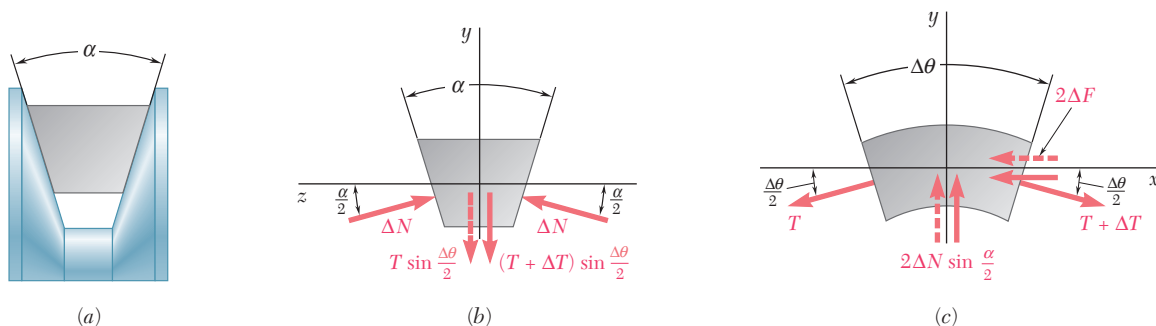


Fig. 8.15 (a) A V belt lying in the groove of a pulley; (b) free-body diagram of a cross-sectional element of the belt; (c) free-body diagram of a short length of belt.

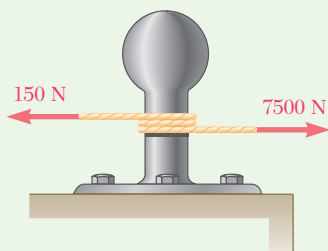


Photo 8.6 Dockworker mooring a ship using a hawser wrapped around a bollard.

Sample Problem 8.8

A hawser (a thick docking rope) thrown from a ship to a pier is wrapped two full turns around a bollard. The tension in the hawser is 7500 N; by exerting a force of 150 N on its free end, a dockworker can just keep the hawser from slipping. (a) Determine the coefficient of friction between the hawser and the bollard. (b) Determine the tension in the hawser that could be resisted by the 150-N force if the hawser were wrapped three full turns around the bollard.

STRATEGY: You are given the difference in forces and the angle of contact through which the friction acts. You can insert these data in the equations of belt friction to determine the coefficient of friction, and then you can use the result to determine the ratio of forces in the second situation.

MODELING and ANALYSIS:

a. Coefficient of Friction. Since slipping of the hawser is impending, we use Eq. (8.13):

$$\ln \frac{T_2}{T_1} = \mu_s \beta$$

Since the hawser is wrapped two full turns around the bollard, you have

$$\begin{aligned} \beta &= 2(2\pi \text{ rad}) = 12.57 \text{ rad} \\ T_1 &= 150 \text{ N} \quad T_2 = 7500 \text{ N} \end{aligned}$$

Therefore,

$$\begin{aligned} \mu_s \beta &= \ln \frac{T_2}{T_1} \\ \mu_s (12.57 \text{ rad}) &= \ln \frac{7500 \text{ N}}{150 \text{ N}} = \ln 50 = 3.91 \\ \mu_s &= 0.311 \end{aligned} \quad \mu_s = 0.311 \quad \blacktriangleleft$$

b. Hawser Wrapped Three Turns Around Bollard. Using the value of μ_s obtained in part a, you now have (Fig. 1)

$$\begin{aligned} \beta &= 3(2\pi \text{ rad}) = 18.85 \text{ rad} \\ T_1 &= 150 \text{ N} \quad \mu_s = 0.311 \end{aligned}$$

Substituting these values into Eq. (8.14), you obtain

$$\begin{aligned} \frac{T_2}{T_1} &= e^{\mu_s \beta} \\ \frac{T_2}{150 \text{ N}} &= e^{(0.311)(18.85)} = e^{5.862} = 351.5 \\ T_2 &= 52\,725 \text{ N} \end{aligned} \quad T_2 = 52.7 \text{ kN} \quad \blacktriangleleft$$

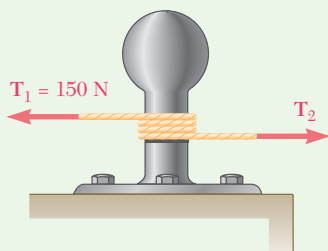


Fig. 1 Hawser wrapped three turns around bollard.

REFLECT and THINK: You can see how the use of a simple post or pulley can have an enormous effect on the magnitude of a force. This is why such systems are commonly used to control, load, and unload container ships in a harbor.

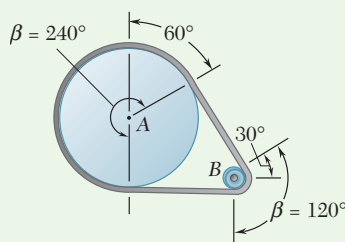
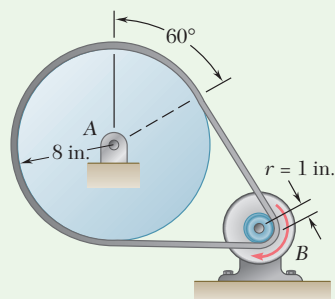


Fig. 1 Angles of contact for the pulleys.

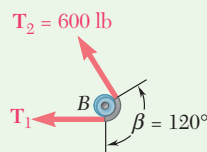


Fig. 2 Belt tensions at pulley B.

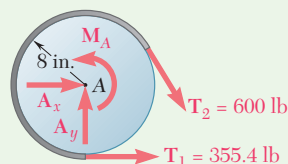


Fig. 3 Free-body diagram of pulley A.

Sample Problem 8.9

A flat belt connects pulley A, which drives a machine tool, to pulley B, which is attached to the shaft of an electric motor. The coefficients of friction are $\mu_s = 0.25$ and $\mu_k = 0.20$ between both pulleys and the belt. Knowing that the maximum allowable tension in the belt is 600 lb, determine the largest torque that the belt can exert on pulley A.

STRATEGY: The key to solving this problem is to identify the pulley where slippage would first occur, and then find the corresponding belt tensions when slippage is impending. The resistance to slippage depends upon the angle of contact β between pulley and belt, as well as upon the coefficient of static friction μ_s . Since μ_s is the same for both pulleys, slippage occurs first on pulley B, for which β is smaller (Fig. 1).

MODELING and ANALYSIS:

Pulley B. Using Eq. (8.14) with $T_2 = 600$ lb, $\mu_s = 0.25$, and $\beta = 120^\circ = 2\pi/3$ rad (Fig. 2), you obtain

$$\frac{T_2}{T_1} = e^{\mu_s \beta} \quad \frac{600 \text{ lb}}{T_1} = e^{0.25(2\pi/3)} = 1.688$$

$$T_1 = \frac{600 \text{ lb}}{1.688} = 355.4 \text{ lb}$$

Pulley A. Draw the free-body diagram of pulley A (Fig. 3). The couple M_A is applied to the pulley using the machine tool to which it is attached and is equal and opposite to the torque exerted by the belt. Setting the sum of the moments equal to zero gives

$$+\circlearrowleft \Sigma M_A = 0: \quad M_A - (600 \text{ lb})(8 \text{ in.}) + (355.4 \text{ lb})(8 \text{ in.}) = 0$$

$$M_A = 1957 \text{ lb}\cdot\text{in.} \quad M_A = 163.1 \text{ lb}\cdot\text{ft} \quad \blacktriangleleft$$

REFLECT and THINK: You may check that the belt does not slip on pulley A by computing the value of μ_s required to prevent slipping at A and verify that it is smaller than the actual value of μ_s . From Eq. (8.13), you have

$$\mu_s \beta = \ln \frac{T_2}{T_1} = \ln \frac{600 \text{ lb}}{355.4 \text{ lb}} = 0.524$$

Since $\beta = 240^\circ = 4\pi/3$ rad,

$$\frac{4\pi}{3} \mu_s = 0.524 \quad \mu_s = 0.125 < 0.25$$

SOLVING PROBLEMS ON YOUR OWN

In the preceding section, you studied **belt friction**. The problems you will solve include belts passing over fixed drums, band brakes in which the drum rotates when the band remains fixed, and belt drives.

1. Problems involving belt friction fall into one of the following two categories.

a. Problems in which slipping is impending. You can use one of the following formulas involving the *coefficient of static friction* μ_s ,

$$\ln \frac{T_2}{T_1} = \mu_s \beta \quad (8.13)$$

or

$$\frac{T_2}{T_1} = e^{\mu_s \beta} \quad (8.14)$$

b. Problems in which slipping is occurring. You can obtain the formulas to be used from Eqs. (8.13) and (8.14) by replacing μ_s with the *coefficient of kinetic friction* μ_k .

2. As you start solving a belt-friction problem, remember these conventions:

a. The angle β must be expressed in radians. In a belt-and-drum problem, this is the angle subtending the arc of the drum on which the belt is wrapped.

b. The larger tension is always denoted by T_2 and the smaller tension is denoted by T_1 .

c. The larger tension occurs at the end of the belt which is in the direction of the motion, or impending motion, of the belt relative to the drum.

3. In each of the problems you will be asked to solve, three of the four quantities T_1 , T_2 , β , and μ_s (or μ_k) will either be given or readily found, and you will then solve the appropriate equation for the fourth quantity. You will encounter two kinds of problems.

a. Find μ_s between belt and drum, knowing that slipping is impending. From the given data, determine T_1 , T_2 , and β ; substitute these values into Eq. (8.13) and solve for μ_s [Sample Prob. 8.8, part a]. Follow the same procedure to find the smallest value of μ_s for which slipping will not occur.

b. Find the magnitude of a force or couple applied to the belt or drum, knowing that slipping is impending. The given data should include μ_s and β . If it also includes T_1 or T_2 , use Eq. (8.14) to find the other tension. If neither T_1 nor T_2 is known but some other data is given, use the free-body diagram of the belt-drum system to write an equilibrium equation that you can solve simultaneously with Eq. (8.14) for T_1 and T_2 . You then will be able to find the magnitude of the specified force or couple from the free-body diagram of the system. Follow the same procedure to determine the largest value of a force or couple that can be applied to the belt or drum if no slipping is to occur [Sample Prob. 8.9].

Problems

8.103 A rope having a weight per unit length of 0.4 lb/ft is wound $2\frac{1}{2}$ times around a horizontal rod. Knowing that the coefficient of static friction between the rope and the rod is 0.30 , determine the minimum length x of rope that should be left hanging if a 100-lb load is to be supported.

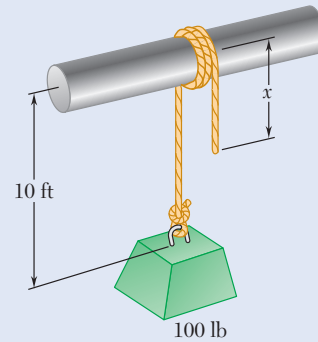


Fig. P8.103

8.104 A hawser is wrapped two full turns around a bollard. By exerting an 80-lb force on the free end of the hawser, a dockworker can resist a force of 5000 lb on the other end of the hawser. Determine (a) the coefficient of static friction between the hawser and the bollard, (b) the number of times the hawser should be wrapped around the bollard if a $20,000\text{-lb}$ force is to be resisted by the same 80-lb force.

8.105 Two cylinders are connected by a rope that passes over two fixed rods as shown. Knowing that the coefficient of static friction between the rope and the rods is 0.40 , determine the range of the mass m of cylinder D for which equilibrium is maintained.

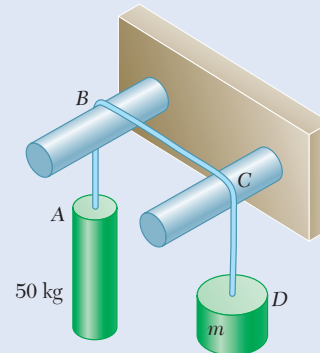


Fig. P8.105 and P8.106

8.106 Two cylinders are connected by a rope that passes over two fixed rods as shown. Knowing that for cylinder D upward motion impends when $m = 20 \text{ kg}$, determine (a) the coefficient of static friction between the rope and the rods, (b) the corresponding tension in portion BC of the rope.

8.107 Knowing that the coefficient of static friction is 0.25 between the rope and the horizontal pipe and 0.20 between the rope and the vertical pipe, determine the range of values of P for which equilibrium is maintained.

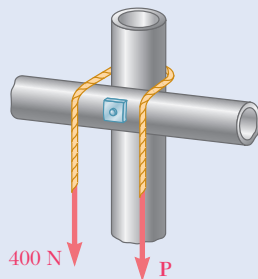


Fig. P8.107 and P8.108

8.108 Knowing that the coefficient of static friction is 0.30 between the rope and the horizontal pipe and that the smallest value of P for which equilibrium is maintained is 80 N , determine (a) the largest value of P for which equilibrium is maintained, (b) the coefficient of static friction between the rope and the vertical pipe.

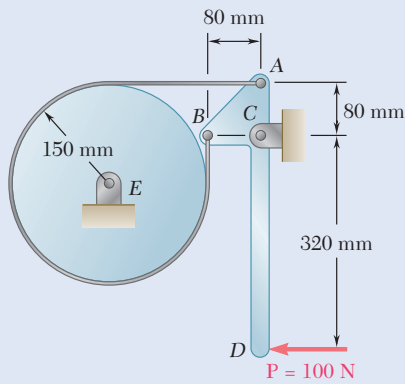


Fig. P8.109

8.109 A band belt is used to control the speed of a flywheel as shown. Determine the magnitude of the couple being applied to the flywheel, knowing that the coefficient of kinetic friction between the belt and the flywheel is 0.25 and that the flywheel is rotating clockwise at a constant speed. Show that the same result is obtained if the flywheel rotates counterclockwise.

8.110 The setup shown is used to measure the output of a small turbine. When the flywheel is at rest, the reading of each spring scale is 14 lb. If a 105-lb-in. couple must be applied to the flywheel to keep it rotating clockwise at a constant speed, determine (a) the reading of each scale at that time, (b) the coefficient of kinetic friction. Assume that the length of the belt does not change.

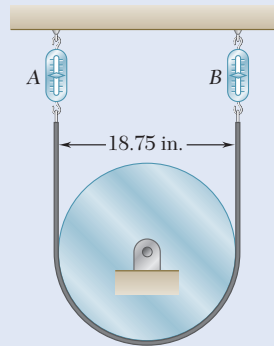


Fig. P8.110 and P8.111

8.111 The setup shown is used to measure the output of a small turbine. The coefficient of kinetic friction is 0.20, and the reading of each spring scale is 16 lb when the flywheel is at rest. Determine (a) the reading of each scale when the flywheel is rotating clockwise at a constant speed, (b) the couple that must be applied to the flywheel. Assume that the length of the belt does not change.

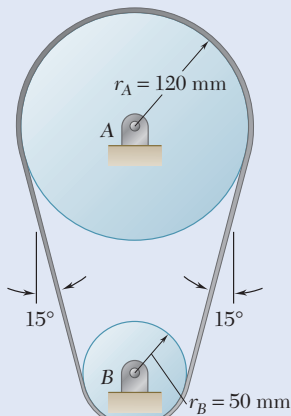


Fig. P8.112

8.112 A flat belt is used to transmit a couple from drum B to drum A. Knowing that the coefficient of static friction is 0.40 and that the allowable belt tension is 450 N, determine the largest couple that can be exerted on drum A.

8.113 A flat belt is used to transmit a couple from pulley A to pulley B. The radius of each pulley is 60 mm, and a force of magnitude $P = 900$ N is applied as shown to the axle of pulley A. Knowing that the coefficient of static friction is 0.35, determine (a) the largest couple that can be transmitted, (b) the corresponding maximum value of the tension in the belt.

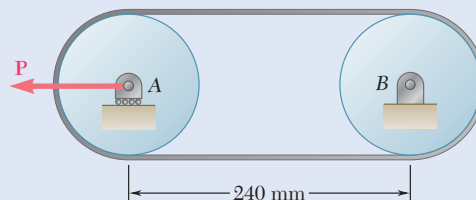


Fig. P8.113

8.114 Solve Prob. 8.113 assuming that the belt is looped around the pulleys in a figure eight.

8.115 The speed of the brake drum shown is controlled by a belt attached to the control bar AD . A force $\mathbf{P}$ with a magnitude of 25 lb is applied to the control bar at A . Determine the magnitude of the couple being applied to the drum knowing that the coefficient of kinetic friction between the belt and the drum is 0.25, that $a = 4$ in., and that the drum is rotating at a constant speed (*a*) counterclockwise, (*b*) clockwise.

8.116 The speed of the brake drum shown is controlled by a belt attached to the control bar AD . Knowing that $a = 4$ in., determine the maximum value of the coefficient of static friction for which the brake is not self-locking when the drum rotates counterclockwise.

8.117 The speed of the brake drum shown is controlled by a belt attached to the control bar AD . Knowing that the coefficient of static friction is 0.30 and that the brake drum is rotating counterclockwise, determine the minimum value of a for which the brake is not self-locking.

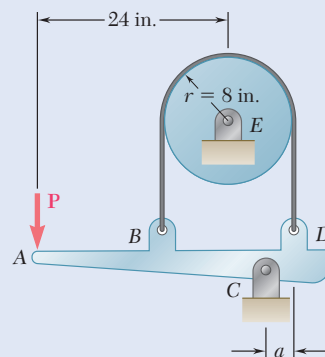


Fig. P8.115, P8.116, and P8.117

8.118 Bucket A and block C are connected by a cable that passes over drum B . Knowing that drum B rotates slowly counterclockwise and that the coefficients of friction at all surfaces are $\mu_s = 0.35$ and $\mu_k = 0.25$, determine the smallest combined mass m of the bucket and its contents for which block C will (*a*) remain at rest, (*b*) start moving up the incline, (*c*) continue moving up the incline at a constant speed.

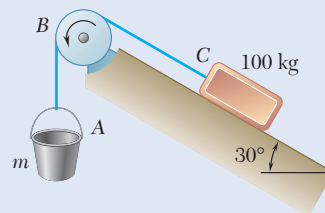


Fig. P8.118

8.119 Solve Prob. 8.118 assuming that drum B is frozen and cannot rotate.

8.120 and 8.122 A cable is placed around three parallel pipes. Knowing that the coefficients of friction are $\mu_s = 0.25$ and $\mu_k = 0.20$, determine (*a*) the smallest weight W for which equilibrium is maintained, (*b*) the largest weight W that can be raised if pipe B is slowly rotated counterclockwise while pipes A and C remain fixed.

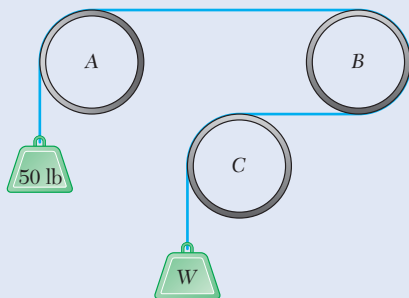


Fig. P8.120 and P8.121

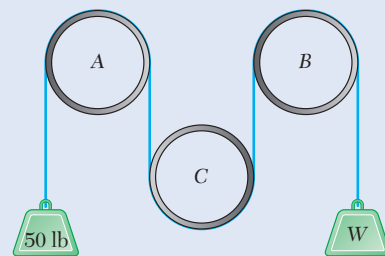


Fig. P8.122 and P8.123

8.121 and 8.123 A cable is placed around three parallel pipes. Two of the pipes are fixed and do not rotate; the third pipe is slowly rotated. Knowing that the coefficients of friction are $\mu_s = 0.25$ and $\mu_k = 0.20$, determine the largest weight W that can be raised (*a*) if only pipe A is rotated counterclockwise, (*b*) if only pipe C is rotated clockwise.

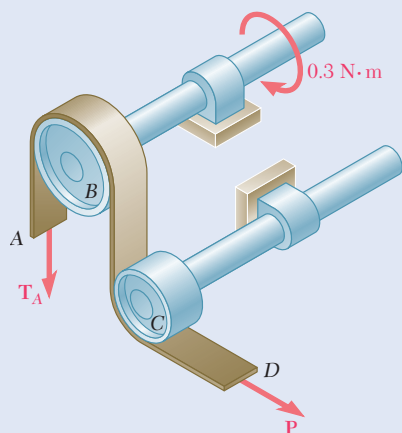


Fig. P8.124

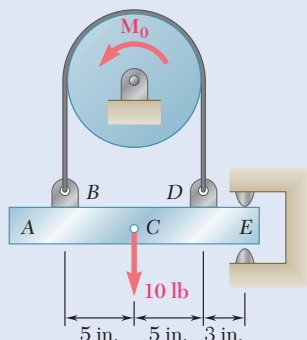


Fig. P8.128

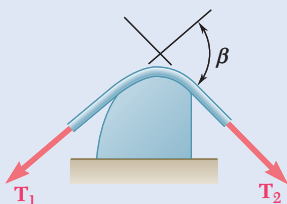


Fig. P8.130

8.124 A recording tape passes over the 20-mm-radius drive drum B and under the idler drum C . Knowing that the coefficients of friction between the tape and the drums are $\mu_s = 0.40$ and $\mu_k = 0.30$ and that drum C is free to rotate, determine the smallest allowable value of P if slipping of the tape on drum B is not to occur.

8.125 Solve Prob. 8.124 assuming that the idler drum C is frozen and cannot rotate.

8.126 The strap wrench shown is used to grip the pipe firmly without marring the external surface of the pipe. Knowing that the coefficient of static friction is the same for all surfaces of contact, determine the smallest value of μ_s for which the wrench will be self-locking when $a = 200$ mm, $r = 30$ mm, and $\theta = 65^\circ$.

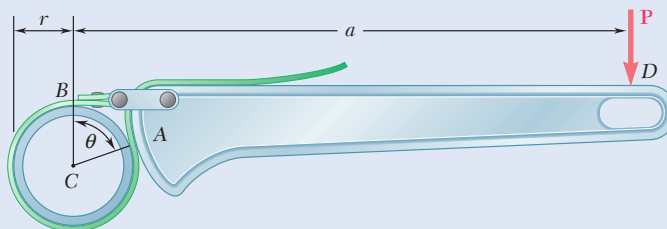


Fig. P8.126

8.127 Solve Prob. 8.126 assuming that $\theta = 75^\circ$.

8.128 The 10-lb bar AE is suspended by a cable that passes over a 5-in.-radius drum. Vertical motion of end E of the bar is prevented by the two stops shown. Knowing that $\mu_s = 0.30$ between the cable and the drum, determine (a) the largest counterclockwise couple $\mathbf{M}_0$ that can be applied to the drum if slipping is not to occur, (b) the corresponding force exerted on end E of the bar.

8.129 Solve Prob. 8.128 assuming that a clockwise couple $\mathbf{M}_0$ is applied to the drum.

8.130 Prove that Eqs. (8.13) and (8.14) are valid for any shape of surface provided that the coefficient of friction is the same at all points of contact.

8.131 Complete the derivation of Eq. (8.15), which relates the tension in both parts of a V belt.

8.132 Solve Prob. 8.112 assuming that the flat belt and drums are replaced by a V belt and V pulleys with $\alpha = 36^\circ$. (The angle α is as shown in Fig. 8.15a.)

8.133 Solve Prob. 8.113 assuming that the flat belt and pulleys are replaced by a V belt and V pulleys with $\alpha = 36^\circ$. (The angle α is as shown in Fig. 8.15a.)

Review and Summary

This chapter was devoted to the study of **dry friction**, i.e., to problems involving rigid bodies in contact along unlubricated surfaces.

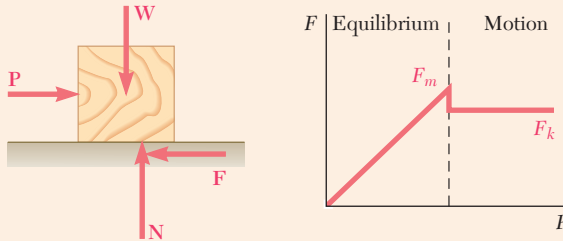


Fig. 8.16

Static and Kinetic Friction

If we apply a horizontal force $\mathbf{P}$ to a block resting on a horizontal surface [Sec. 8.1], we note that at first the block does not move. This shows that a **friction force $\mathbf{F}$** must have developed to balance $\mathbf{P}$ (Fig. 8.16). As the magnitude of $\mathbf{P}$ increases, the magnitude of $\mathbf{F}$ also increases until it reaches a maximum value F_m . If $\mathbf{P}$ is further increased, the block starts sliding, and the magnitude of $\mathbf{F}$ drops from F_m to a lower value F_k . Experimental evidence shows that F_m and F_k are proportional to the normal component N of the reaction of the surface. We have

$$F_m = \mu_s N \quad F_k = \mu_k N \quad (8.1, 8.2)$$

where μ_s and μ_k are called, respectively, the **coefficient of static friction** and the **coefficient of kinetic friction**. These coefficients depend on the nature and the condition of the surfaces in contact. Approximate values of the coefficients of static friction are given in Table 8.1.

Angles of Friction

It is sometimes convenient to replace the normal force $\mathbf{N}$ and the friction force $\mathbf{F}$ by their resultant $\mathbf{R}$ (Fig. 8.17). As the friction force increases and reaches its maximum value $F_m = \mu_s N$, the angle ϕ that $\mathbf{R}$ forms with the normal to the surface increases and reaches a maximum value ϕ_s , which is called the **angle of static friction**. If motion actually takes place, the magnitude of $\mathbf{F}$ drops to F_k ; similarly, the angle ϕ drops to a lower value ϕ_k , which is called the **angle of kinetic friction**. As shown in Sec. 8.1B, we have

$$\tan \phi_s = \mu_s \quad \tan \phi_k = \mu_k \quad (8.3, 8.4)$$

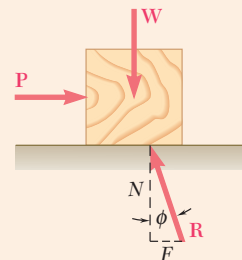


Fig. 8.17

Problems Involving Friction

When solving equilibrium problems involving friction, you should keep in mind that the magnitude F of the friction force is equal to $F_m = \mu_s N$ *only if the body is about to slide* [Sec. 8.1C]. *If motion is not impending*, you should

treat F and N as independent unknowns to be determined from the equilibrium equations (Fig. 8.18a). You should also check that the value of F required to maintain equilibrium is not larger than F_m ; if it were, the body would move, and the magnitude of the friction force would be $F_k = \mu_k N$ [Sample Prob. 8.1]. On the other hand, if motion is known to be impending, F has reached its maximum value $F_m = \mu_s N$ (Fig. 8.18b), and you should substitute this expression for F in the equilibrium equations [Sample Prob. 8.3]. When only three forces are involved in a free-body diagram, including the reaction $\mathbf{R}$ of the surface in contact with the body, it is usually more convenient to solve the problem by drawing a force triangle [Sample Prob. 8.2]. In some problems, impending motion can be due to tipping instead of slipping; the assessment of this condition requires a moment equilibrium analysis of the body [Sample Prob. 8.4].

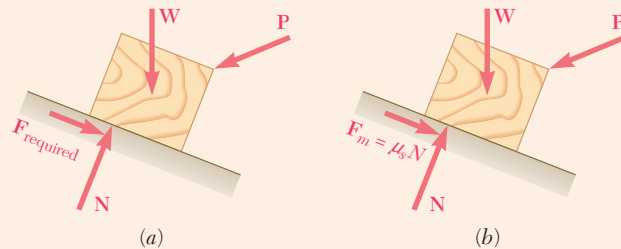


Fig. 8.18

When a problem involves the analysis of the forces exerted on each other by *two bodies A and B*, it is important to show the friction forces with their correct sense. The correct sense for the friction force exerted by B on A , for instance, is opposite to that of the *relative motion* (or impending motion) of A with respect to B [Fig. 8.6].

Wedges and Screws

In the later sections of this chapter, we considered several specific engineering applications where dry friction plays an important role. In the case of **wedges**, which are simple machines used to raise heavy loads [Sec. 8.2A], we must draw two or more free-body diagrams, taking care to show each friction force with its correct sense [Sample Prob. 8.5]. The analysis of **square-threaded screws**, which are frequently used in jacks, presses, and other mechanisms, is reduced to the analysis of a block sliding on an incline by unwrapping the thread of the screw and showing it as a straight line [Sec. 8.2B]. This is shown again in Fig. 8.19, where r denotes the *mean radius* of the thread, L is the *lead* of the screw (i.e., the distance through which the screw advances in one turn), W is the load, and Qr is equal to the couple exerted on the screw. We noted in the case of multiple-threaded screws that the lead L of the screw is *not* equal to its pitch, which is the distance measured between two consecutive threads.

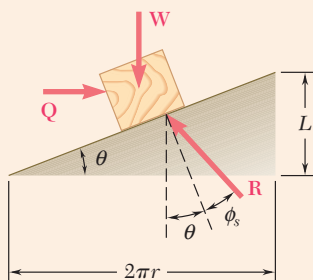


Fig. 8.19

Other engineering applications considered in this chapter were **journal bearings** and **axle friction** [Sec. 8.3A], **thrust bearings** and **disk friction** [Sec. 8.3B], **wheel friction** and **rolling resistance** [Sec. 8.3C], and **belt friction** [Sec. 8.4].

Belt Friction

In solving a problem involving a flat belt passing over a fixed cylinder, it is important to first determine the direction in which the belt slips or is about to slip. If the drum is rotating, the motion or impending motion of the belt should be determined *relative* to the rotating drum. For instance, if the belt shown in Fig. 8.20 is about to slip to the right relative to the drum, the friction

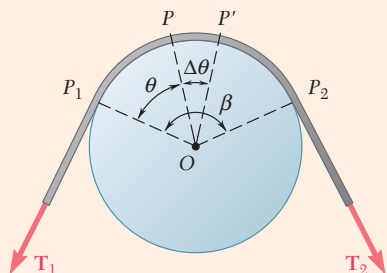


Fig. 8.20

forces exerted by the drum on the belt are directed to the left, and the tension is larger in the right-hand portion of the belt than in the left-hand portion. Denoting the larger tension by T_2 , the smaller tension by T_1 , the coefficient of static friction by μ_s , and the angle (in radians) subtended by the belt by β , we derived in Sec. 8.4 the formulas

$$\ln \frac{T_2}{T_1} = \mu_s \beta \quad (8.13)$$

$$\frac{T_2}{T_1} = e^{\mu_s \beta} \quad (8.14)$$

that we used in solving Sample Probs. 8.8 and 8.9. If the belt actually slips on the drum, the coefficient of static friction μ_s should be replaced by the coefficient of kinetic friction μ_k in both of these formulas.

Review Problems

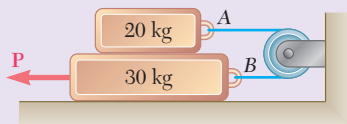


Fig. P8.134

8.134 and 8.135 The coefficients of friction are $\mu_s = 0.40$ and $\mu_k = 0.30$ between all surfaces of contact. Determine the smallest force $\mathbf{P}$ required to start the 30-kg block moving if cable AB (a) is attached as shown, (b) is removed.

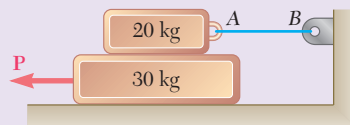


Fig. P8.135

8.136 A 120-lb cabinet is mounted on casters that can be locked to prevent their rotation. The coefficient of static friction between the floor and each caster is 0.30. If $h = 32$ in., determine the magnitude of the force $\mathbf{P}$ required to move the cabinet to the right (a) if all casters are locked, (b) if the casters at B are locked and the casters at A are free to rotate, (c) if the casters at A are locked and the casters at B are free to rotate.

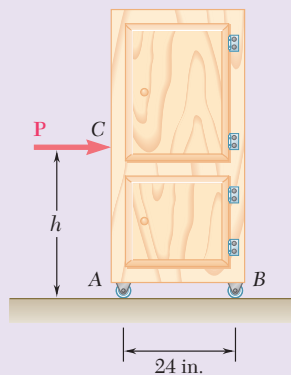


Fig. P8.136

8.137 A slender rod with a length of L is lodged between peg C and the vertical wall, and supports a load $\mathbf{P}$ at end A. Knowing that the coefficient of static friction between the peg and the rod is 0.15 and neglecting friction at the roller, determine the range of values of the ratio L/a for which equilibrium is maintained.

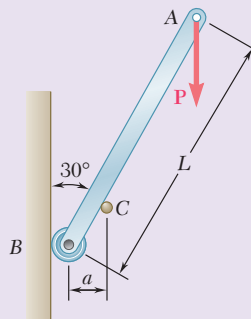


Fig. P8.137

- 8.138** The hydraulic cylinder shown exerts a force of 3 kN directed to the right on point B and to the left on point E . Determine the magnitude of the couple $\mathbf{M}$ required to rotate the drum clockwise at a constant speed.

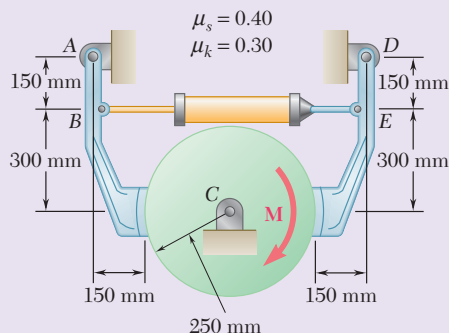


Fig. P8.138

- 8.139** A rod DE and a small cylinder are placed between two guides as shown. The rod is not to slip downward, however large the force $\mathbf{P}$ may be; i.e., the arrangement is said to be self-locking. Neglecting the weight of the cylinder, determine the minimum allowable coefficients of static friction at A , B , and C .

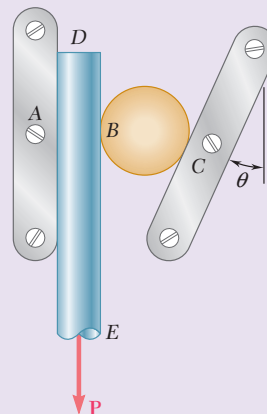


Fig. P8.139

- 8.140** Bar AB is attached to collars that can slide on the inclined rods shown. A force $\mathbf{P}$ is applied at point D located at a distance a from end A . Knowing that the coefficient of static friction μ_s between each collar and the rod upon which it slides is 0.30 and neglecting the weights of the bar and of the collars, determine the smallest value of the ratio a/L for which equilibrium is maintained.

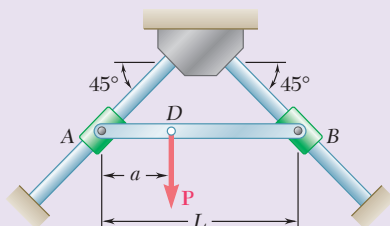


Fig. P8.140

- 8.141** Two 10° wedges of negligible weight are used to move and position the 400-lb block. Knowing that the coefficient of static friction is 0.25 at all surfaces of contact, determine the smallest force $\mathbf{P}$ that should be applied as shown to one of the wedges.

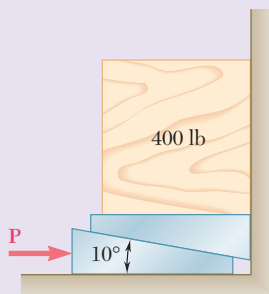


Fig. P8.141

- 8.142** A 10° wedge is used to split a section of a log. The coefficient of static friction between the wedge and the log is 0.35. Knowing that a force $\mathbf{P}$ with a magnitude of 600 lb was required to insert the wedge, determine the magnitude of the forces exerted on the wood by the wedge after insertion.

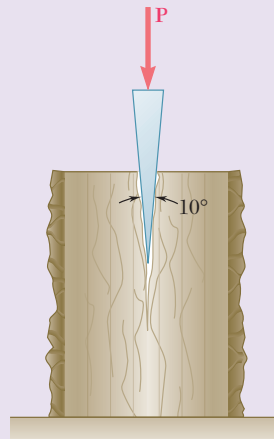


Fig. P8.142

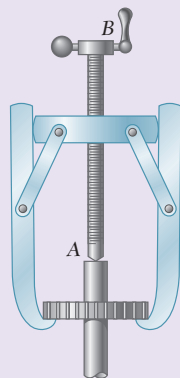


Fig. P8.143

- 8.143** In the gear-pulling assembly shown, the square-threaded screw AB has a mean radius of 15 mm and a lead of 4 mm. Knowing that the coefficient of static friction is 0.10, determine the couple that must be applied to the screw in order to produce a force of 3 kN on the gear. Neglect friction at end A of the screw.
- 8.144** A lever of negligible weight is loosely fitted onto a 30-mm-radius fixed shaft as shown. Knowing that a force $\mathbf{P}$ of magnitude 275 N will just start the lever rotating clockwise, determine (a) the coefficient of static friction between the shaft and the lever, (b) the smallest force $\mathbf{P}$ for which the lever does not start rotating counterclockwise.

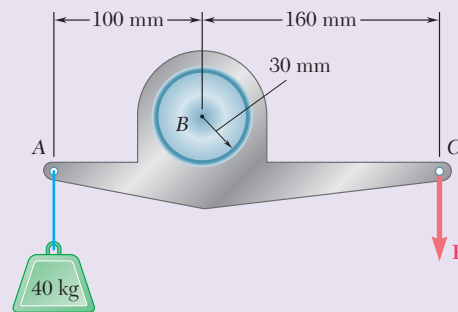


Fig. P8.144

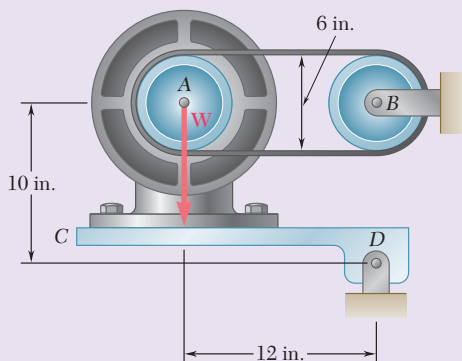


Fig. P8.145

- 8.145** In the pivoted motor mount shown, the weight $\mathbf{W}$ of the 175-lb motor is used to maintain tension in the drive belt. Knowing that the coefficient of static friction between the flat belt and drums A and B is 0.40 and neglecting the weight of platform CD , determine the largest couple that can be transmitted to drum B when the drive drum A is rotating clockwise.



9

Distributed Forces: Moments of Inertia

The strength of structural members used in the construction of buildings depends to a large extent on the properties of their cross sections. This includes the second moments of area, or moments of inertia, of these cross sections.

Introduction

9.1 MOMENTS OF INERTIA OF AREAS

9.1A Second Moment, or Moment of Inertia, of an Area

9.1B Determining the Moment of Inertia of an Area by Integration

9.1C Polar Moment of Inertia

9.1D Radius of Gyration of an Area

9.2 PARALLEL-AXIS THEOREM AND COMPOSITE AREAS

9.2A The Parallel-Axis Theorem

9.2B Moments of Inertia of Composite Areas

*9.3 TRANSFORMATION OF MOMENTS OF INERTIA

9.3A Product of Inertia

9.3B Principal Axes and Principal Moments of Inertia

*9.4 MOHR'S CIRCLE FOR MOMENTS OF INERTIA

9.5 MASS MOMENTS OF INERTIA

9.5A Moment of Inertia of a Simple Mass

9.5B Parallel-Axis Theorem for Mass Moments of Inertia

9.5C Moments of Inertia of Thin Plates

9.5D Determining the Moment of Inertia of a Three-Dimensional Body by Integration

9.5E Moments of Inertia of Composite Bodies

*9.6 ADDITIONAL CONCEPTS OF MASS MOMENTS OF INERTIA

9.6A Mass Products of Inertia

9.6B Principal Axes and Principal Moments of Inertia

9.6C Principal Axes and Moments of Inertia for a Body of Arbitrary Shape

Objectives

- **Describe** the second moment, or moment of inertia, of an area.
- **Determine** the rectangular and polar moments of inertia of areas and their corresponding radii of gyration by integration.
- **Develop** the parallel-axis theorem and apply it to determine the moments of inertia of composite areas.
- **Introduce** the product of inertia and apply it to analyze the transformation of moments of inertia when coordinate axes are rotated.
- **Describe** the moment of inertia of a mass with respect to an axis.
- **Apply** the parallel-axis theorem to facilitate mass moment of inertia computations.
- **Analyze** the transformation of mass moments of inertia when coordinate axes are rotated.

Introduction

In Chap. 5, we analyzed various systems of forces distributed over an area or volume. The three main types of forces considered were (1) weights of homogeneous plates of uniform thickness (Secs. 5.1 and 5.2); (2) distributed loads on beams and submerged surfaces (Sec. 5.3); and (3) weights of homogeneous three-dimensional bodies (Sec. 5.4). In all of these cases, the distributed forces were proportional to the elemental areas or volumes associated with them. Therefore, we could obtain the resultant of these forces by summing the corresponding areas or volumes, and we determined the moment of the resultant about any given axis by computing the first moments of the areas or volumes about that axis.

In the first part of this chapter, we consider distributed forces $\Delta \mathbf{F}$ where the magnitudes depend not only upon the elements of area ΔA on which these forces act but also upon the distance from ΔA to some given axis. More precisely, we assume the magnitude of the force per unit area $\Delta F/\Delta A$ varies linearly with the distance to the axis. Forces of this type arise in the study of the bending of beams and in problems involving submerged nonrectangular surfaces.

Starting with the assumption that the elemental forces involved are distributed over an area A and vary linearly with the distance y to the x axis, we will show that the magnitude of their resultant $\mathbf{R}$ depends upon the first moment Q_x of the area A . However, the location of the point where $\mathbf{R}$ is applied depends upon the *second moment*, or *moment of inertia*, I_x of the same area with respect to the x axis. You will see how to compute the moments of inertia of various areas with respect to given x and y axes. We also introduce the *polar moment of inertia* J_O of an area. To facilitate these computations, we establish a relation between the moment of inertia I_x

of an area A with respect to a given x axis and the moment of inertia I_x of the same area with respect to the parallel centroidal x' axis (a relation known as the parallel-axis theorem). You will also study the transformation of the moments of inertia of a given area when the coordinate axes are rotated.

In the second part of this chapter, we will explain how to determine the moments of inertia of various *masses* with respect to a given axis. Moments of inertia of masses are common in dynamics problems involving the rotation of a rigid body about an axis. To facilitate the computation of mass moments of inertia, we introduce another version of the parallel-axis theorem. Finally, we will analyze the transformation of moments of inertia of masses when the coordinate axes are rotated.

9.1 MOMENTS OF INERTIA OF AREAS

In the first part of this chapter, we consider distributed forces $\Delta \mathbf{F}$ whose magnitudes ΔF are proportional to the elements of area ΔA on which the forces act and, at the same time, vary linearly with the distance from ΔA to a given axis.

9.1A Second Moment, or Moment of Inertia, of an Area

Consider a beam with a uniform cross section that is subjected to two equal and opposite couples: one applied at each end of the beam. Such a beam is said to be in **pure bending**. The internal forces in any section of the beam are distributed forces whose magnitudes $\Delta F = ky \Delta A$ vary linearly with the distance y between the element of area ΔA and an axis passing through the centroid of the section. (This statement can be derived in a course on mechanics of materials.) This axis, represented by the x axis in Fig. 9.1, is known as the **neutral axis** of the section. The forces on one side of the neutral axis are forces of compression, whereas those on the other side are forces of tension. On the neutral axis itself, the forces are zero.

The magnitude of the resultant $\mathbf{R}$ of the elemental forces $\Delta \mathbf{F}$ that act over the entire section is

$$R = \int ky dA = k \int y dA$$

You might recognize this last integral as the **first moment** Q_x of the section about the x axis; it is equal to $\bar{y}A$ and is thus equal to zero, since the centroid of the section is located on the x axis. The system of forces $\Delta \mathbf{F}$ thus reduces to a couple. The magnitude M of this couple (bending moment) must be equal to the sum of the moments $\Delta M_x = y \Delta F = ky^2 \Delta A$ of the elemental forces. Integrating over the entire section, we obtain

$$M = \int ky^2 dA = k \int y^2 dA$$

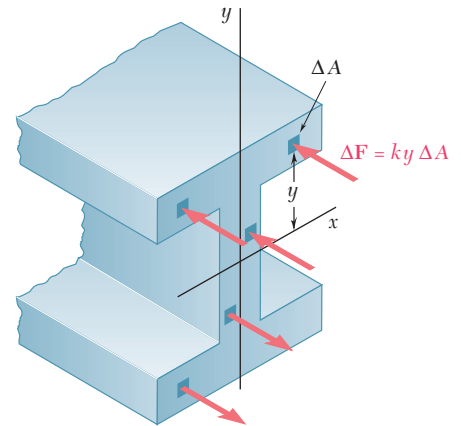


Fig. 9.1 Representative forces on a cross section of a beam subjected to equal and opposite couples at each end.

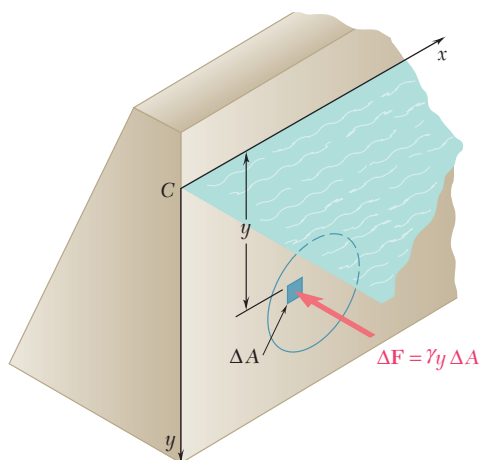


Fig. 9.2 Vertical circular gate, submerged under water, used to close the outlet of a reservoir.

This last integral is known as the **second moment**, or **moment of inertia**,[†] of the beam section with respect to the x axis and is denoted by I_x . We obtain it by multiplying each element of area dA by the *square of its distance* from the x axis and integrating over the beam section. Since each product $y^2 dA$ is positive, regardless of the sign of y , or zero (if y is zero), the integral I_x is always positive.

Another example of a second moment, or moment of inertia, of an area is provided by the following problem from hydrostatics. A vertical circular gate used to close the outlet of a large reservoir is submerged under water as shown in Fig. 9.2. What is the resultant of the forces exerted by the water on the gate, and what is the moment of the resultant about the line of intersection of the plane of the gate and the water surface (x axis)?

If the gate were rectangular, we could determine the resultant of the forces due to water pressure from the pressure curve, as we did in Sec. 5.3B. Since the gate is circular, however, we need to use a more general method. Denoting the depth of an element of area ΔA by y and the specific weight of water by γ , the pressure at an element is $p = \gamma y$, and the magnitude of the elemental force exerted on ΔA is $\Delta F = p \Delta A = \gamma y \Delta A$. The magnitude of the resultant of the elemental forces is thus

$$R = \int \gamma y dA = \gamma \int y dA$$

We can obtain this by computing the first moment $Q_x = \int y dA$ of the area of the gate with respect to the x axis. The moment M_x of the resultant must be equal to the sum of the moments $\Delta M_x = y \Delta F = \gamma y^2 \Delta A$ of the elemental forces. Integrating over the area of the gate, we have

$$M_x = \int \gamma y^2 dA = \gamma \int y^2 dA$$

Here again, the last integral represents the second moment, or moment of inertia, I_x of the area with respect to the x axis.

9.1B Determining the Moment of Inertia of an Area by Integration

We just defined the second moment, or moment of inertia, I_x of an area A with respect to the x axis. In a similar way, we can also define the moment of inertia I_y of the area A with respect to the y axis (Fig. 9.3a):

Moments of inertia of an area

$$I_x = \int y^2 dA \quad I_y = \int x^2 dA \quad (9.1)$$

[†]The term *second moment* is more proper than the term *moment of inertia*, which logically should be used only to denote integrals of mass (see Sec. 9.5). In engineering practice, however, moment of inertia is used in connection with areas as well as masses.

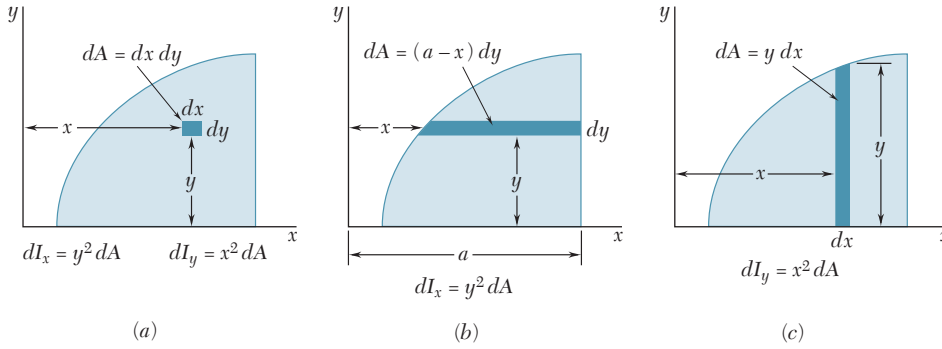


Fig. 9.3 (a) Rectangular moments of inertia dI_x and dI_y of an area dA ; (b) calculating I_x with a horizontal strip; (c) calculating I_y with a vertical strip.

We can evaluate these integrals, which are known as the **rectangular moments of inertia** of the area A , more easily if we choose dA to be a thin strip parallel to one of the coordinate axes. To compute I_x , we choose the strip parallel to the x axis, so that all points of the strip are at the same distance y from the x axis (Fig. 9.3b). We obtain the moment of inertia dI_x of the strip by multiplying the area dA of the strip by y^2 . To compute I_y , we choose the strip parallel to the y axis, so that all points of the strip are at the same distance x from the y axis (Fig. 9.3c). Then the moment of inertia dI_y of the strip is $x^2 dA$.

Moment of Inertia of a Rectangular Area. As an example, let us determine the moment of inertia of a rectangle with respect to its base (Fig. 9.4). Dividing the rectangle into strips parallel to the x axis, we have

$$\begin{aligned} dA &= b \, dy & dI_x &= y^2 b \, dy \\ I_x &= \int_0^h b y^2 \, dy = \frac{1}{3} b h^3 \end{aligned} \quad (9.2)$$

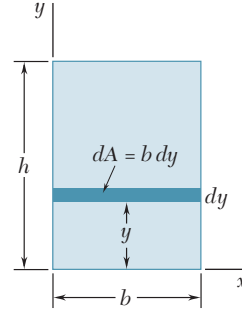


Fig. 9.4 Calculating the moment of inertia of a rectangular area with respect to its base.

Computing I_x and I_y Using the Same Elemental Strips. We can use Eq. (9.2) to determine the moment of inertia dI_x with respect to the x axis of a rectangular strip that is parallel to the y axis, such as the strip shown in Fig. 9.3c. Setting $b = dx$ and $h = y$ in formula (9.2), we obtain

$$dI_x = \frac{1}{3} y^3 \, dx$$

We also have

$$dI_y = x^2 dA = x^2 y \, dx$$

Thus, we can use the same element to compute the moments of inertia I_x and I_y of a given area (Fig. 9.5).

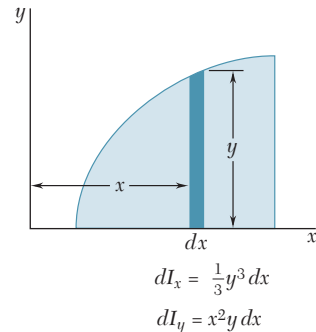


Fig. 9.5 Using the same strip element of a given area to calculate I_x and I_y .

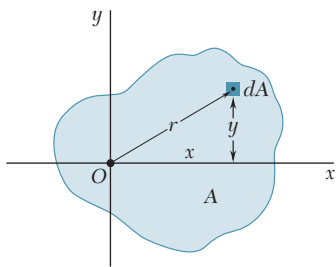
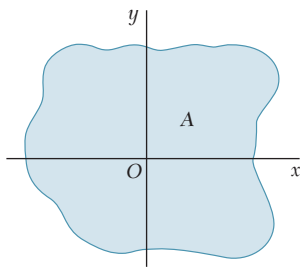
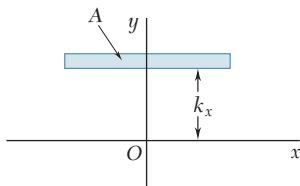


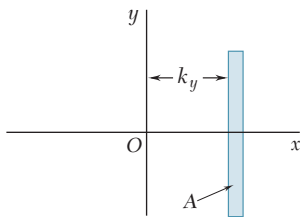
Fig. 9.6 Distance r used to evaluate the polar moment of inertia of area A .



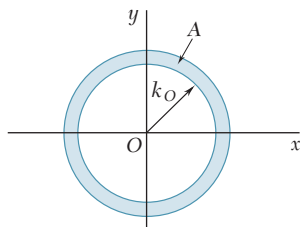
(a)



(b)



(c)



(d)

Fig. 9.7 (a) Area A with given moment of inertia I_x ; (b) compressing the area to a horizontal strip with radius of gyration k_x ; (c) compressing the area to a vertical strip with radius of gyration k_y ; (d) compressing the area to a circular ring with polar radius of gyration k_O .

9.1C Polar Moment of Inertia

An integral of great importance in problems concerning the torsion of cylindrical shafts and in problems dealing with the rotation of slabs is

Polar moment of inertia

$$J_O = \int r^2 dA \quad (9.3)$$

where r is the distance from O to the element of area dA (Fig. 9.6). This integral is called the **polar moment of inertia** of the area A with respect to the “pole” O .

We can compute the polar moment of inertia of a given area from the rectangular moments of inertia I_x and I_y of the area if these quantities are already known. Indeed, noting that $r^2 = x^2 + y^2$, we have

$$J_O = \int r^2 dA = \int (x^2 + y^2) dA = \int y^2 dA + \int x^2 dA$$

that is,

$$J_O = I_x + I_y \quad (9.4)$$

9.1D Radius of Gyration of an Area

Consider an area A that has a moment of inertia I_x with respect to the x axis (Fig. 9.7a). Imagine that we concentrate this area into a thin strip parallel to the x axis (Fig. 9.7b). If the concentrated area A is to have the same moment of inertia with respect to the x axis, the strip should be placed at a distance k_x from the x axis, where k_x is defined by the relation

$$I_x = k_x^2 A$$

Solving for k_x , we have

Radius of gyration

$$k_x = \sqrt{\frac{I_x}{A}} \quad (9.5)$$

The distance k_x is referred to as the **radius of gyration** of the area with respect to the x axis. In a similar way, we can define the radii of gyration k_y and k_O (Fig. 9.7c and d); we have

$$I_y = k_y^2 A \quad k_y = \sqrt{\frac{I_y}{A}} \quad (9.6)$$

$$J_O = k_O^2 A \quad k_O = \sqrt{\frac{J_O}{A}} \quad (9.7)$$

If we rewrite Eq. (9.4) in terms of the radii of gyration, we find that

$$k_O^2 = k_x^2 + k_y^2 \quad (9.8)$$

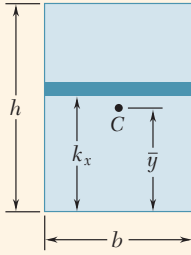


Fig. 9.8 Radius of gyration of a rectangle with respect to its base.

Concept Application 9.1

For the rectangle shown in Fig. 9.8, compute the radius of gyration k_x with respect to its base. Using formulas (9.5) and (9.2), you have

$$k_x^2 = \frac{I_x}{A} = \frac{\frac{1}{3}bh^3}{bh} = \frac{h^2}{3} \quad k_x = \frac{h}{\sqrt{3}}$$

The radius of gyration k_x of the rectangle is shown in Fig. 9.8. Do not confuse it with the ordinate $\bar{y} = h/2$ of the centroid of the area. The radius of gyration k_x depends upon the *second moment* of the area, whereas the ordinate $\bar{y}$ is related to the *first moment* of the area.

Sample Problem 9.1

Determine the moment of inertia of a triangle with respect to its base.

STRATEGY: To find the moment of inertia with respect to the base, it is expedient to use a differential strip of area parallel to the base. Use the geometry of the situation to carry out the integration.

MODELING: Draw a triangle with a base b and height h , choosing the x axis to coincide with the base (Fig. 1). Choose a differential strip parallel to the x axis to be dA . Since all portions of the strip are at the same distance from the x axis, you have

$$dI_x = y^2 dA \quad dA = l dy$$

ANALYSIS: Using similar triangles, you have

$$\frac{l}{b} = \frac{h-y}{h} \quad l = b \frac{h-y}{h} \quad dA = b \frac{h-y}{h} dy$$

Integrating dI_x from $y = 0$ to $y = h$, you obtain

$$\begin{aligned} I_x &= \int y^2 dA = \int_0^h y^2 b \frac{h-y}{h} dy = \frac{b}{h} \int_0^h (hy^2 - y^3) dy \\ &= \frac{b}{h} \left[h \frac{y^3}{3} - \frac{y^4}{4} \right]_0^h \end{aligned}$$

$$I_x = \frac{bh^3}{12} \quad \blacktriangleleft$$

REFLECT and THINK: This problem also could have been solved using a differential strip perpendicular to the base by applying Eq. (9.2) to express the moment of inertia of this strip. However, because of the geometry of this triangle, you would need two integrals to complete the solution.

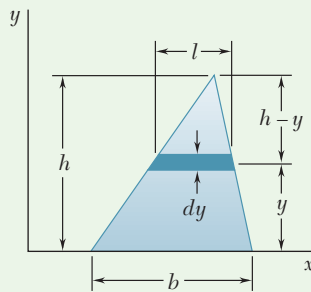


Fig. 1 Triangle with differential strip element parallel to its base.

Sample Problem 9.2

(a) Determine the centroidal polar moment of inertia of a circular area by direct integration. (b) Using the result of part (a), determine the moment of inertia of a circular area with respect to a diameter.

STRATEGY: Since the area is circular, you can evaluate part (a) by using an annular differential area. For part (b), you can use symmetry and Eq. (9.4) to solve for the moment of inertia with respect to a diameter.

MODELING and ANALYSIS:

a. Polar Moment of Inertia. Choose an annular differential element of area to be dA (Fig. 1). Since all portions of the differential area are at the same distance from the origin, you have

$$dJ_O = u^2 dA \quad dA = 2\pi u \, du$$

$$J_O = \int dJ_O = \int_0^r u^2 (2\pi u \, du) = 2\pi \int_0^r u^3 \, du \quad J_O = \frac{\pi}{2} r^4 \quad \blacktriangleleft$$

b. Moment of Inertia with Respect to a Diameter. Because of the symmetry of the circular area, $I_x = I_y$. Then from Eq. (9.4), you have

$$J_O = I_x + I_y = 2I_x \quad \frac{\pi}{2} r^4 = 2I_x \quad I_{\text{diameter}} = I_x = \frac{\pi}{4} r^4 \quad \blacktriangleleft$$

REFLECT and THINK: Always look for ways to simplify a problem by the use of symmetry. This is especially true for situations involving circles or spheres.

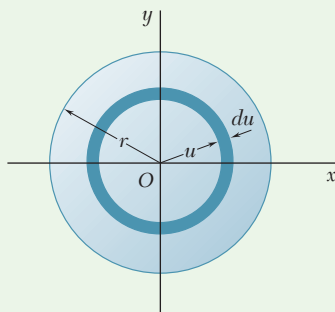
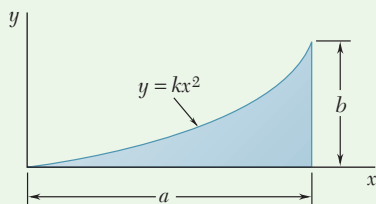


Fig. 1 Circular area with an annular differential element.

Sample Problem 9.3

(a) Determine the moment of inertia of the shaded region shown with respect to each of the coordinate axes. (Properties of this region were considered in Sample Prob. 5.4.) (b) Using the results of part (a), determine the radius of gyration of the shaded area with respect to each of the coordinate axes.

STRATEGY: You can determine the moments of inertia by using a single differential strip of area; a vertical strip will be more convenient. You can calculate the radii of gyration from the moments of inertia and the area of the region.



MODELING: Referring to Sample Prob. 5.4, you can find the equation of the curve and the total area using

$$y = \frac{b}{a^2}x^2 \quad A = \frac{1}{3}ab$$

ANALYSIS:

a. Moments of Inertia.

Moment of Inertia I_x . Choose a vertical differential element of area for dA (Fig. 1). Since all portions of this element are *not* at the same distance from the x axis, you must treat the element as a thin rectangle. The moment of inertia of the element with respect to the x axis is then

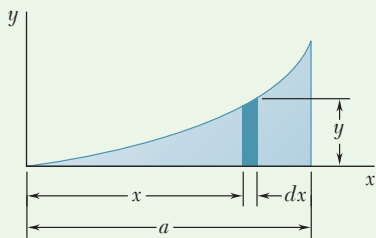


Fig. 1 Subject area with vertical differential strip element.

$$dI_x = \frac{1}{3}y^3 dx = \frac{1}{3}\left(\frac{b}{a^2}x^2\right)^3 dx = \frac{1}{3}\frac{b^3}{a^6}x^6 dx$$

$$I_x = \int dI_x = \int_0^a \frac{1}{3}\frac{b^3}{a^6}x^6 dx = \left[\frac{1}{3}\frac{b^3}{a^6}\frac{x^7}{7}\right]_0^a$$

$$I_x = \frac{ab^3}{21} \quad \blacktriangleleft$$

Moment of Inertia I_y . Use the same vertical differential element of area. Since all portions of the element are at the same distance from the y axis, you have

$$dI_y = x^2 dA = x^2(y dx) = x^2\left(\frac{b}{a^2}x^2\right) dx = \frac{b}{a^2}x^4 dx$$

$$I_y = \int dI_y = \int_0^a \frac{b}{a^2}x^4 dx = \left[\frac{b}{a^2}\frac{x^5}{5}\right]_0^a$$

$$I_y = \frac{a^3b}{5} \quad \blacktriangleleft$$

b. Radii of Gyration k_x and k_y . From the definition of radius of gyration, you have

$$k_x^2 = \frac{I_x}{A} = \frac{ab^3/21}{ab/3} = \frac{b^2}{7} \quad k_x = \sqrt{\frac{1}{7}}b \quad \blacktriangleleft$$

and

$$k_y^2 = \frac{I_y}{A} = \frac{a^3b/5}{ab/3} = \frac{3}{5}a^2 \quad k_y = \sqrt{\frac{3}{5}}a \quad \blacktriangleleft$$

REFLECT and THINK: This problem demonstrates how you can calculate I_x and I_y using the same strip element. However, the general mathematical approach in each case is distinctly different.

SOLVING PROBLEMS ON YOUR OWN

In this section, we introduced the **rectangular and polar moments of inertia of areas** and the corresponding **radii of gyration**. Although the problems you are about to solve may appear more appropriate for a calculus class than for one in mechanics, we hope that our introductory comments have convinced you of the relevance of moments of inertia to your study of a variety of engineering topics.

1. Calculating the rectangular moments of inertia I_x and I_y . We defined these quantities as

$$I_x = \int y^2 dA \quad I_y = \int x^2 dA \quad (9.1)$$

where dA is a differential element of area $dx dy$. The moments of inertia are the **second moments of the area**; it is for that reason that I_x , for example, depends on the perpendicular distance y to the area dA . As you study Sec. 9.1, you should recognize the importance of carefully defining the shape and the orientation of dA . Furthermore, you should note the following points.

a. You can obtain the moments of inertia of most areas by means of a single integration. You can use the expressions given in Figs. 9.3*b* and *c* and Fig. 9.5 to calculate I_x and I_y . Regardless of whether you use a single or a double integration, be sure to show the element dA that you have chosen on your sketch.

b. The moment of inertia of an area is always positive, regardless of the location of the area with respect to the coordinate axes. The reason is that the moment of inertia is obtained by integrating the product of dA and the *square* of distance. (Note how this differs from the first moment of the area.) Only when an area is *removed* (as in the case for a hole) does its moment of inertia enter in your computations with a minus sign.

c. As a partial check of your work, observe that the moments of inertia are equal to an area times the square of a length. Thus, every term in an expression for a moment of inertia must be a length to the fourth power.

2. Computing the polar moment of inertia J_O . We defined J_O as

$$J_O = \int r^2 dA \quad (9.3)$$

where $r^2 = x^2 + y^2$. If the given area has circular symmetry (as in Sample Prob. 9.2), it is possible to express dA as a function of r and to compute J_O with a single integration. When the area lacks circular symmetry, it is usually easier first to calculate I_x and I_y and then to determine J_O from

$$J_O = I_x + I_y \quad (9.4)$$

Lastly, if the equation of the curve that bounds the given area is expressed in polar coordinates, then $dA = r dr d\theta$, and you need to perform a double integration to compute the integral for J_O [see Prob. 9.27].

3. Determining the radii of gyration k_x and k_y and the polar radius of gyration k_O . These quantities are defined in Sec. 9.1D. You should realize that they can be determined only after you have computed the area and the appropriate moments of inertia. It is important to remember that k_x is measured in the y direction, whereas k_y is measured in the x direction; you should carefully study Sec. 9.1D until you understand this point.

Problems

9.1 through 9.4 Determine by direct integration the moment of inertia of the shaded area with respect to the y axis.

9.5 through 9.8 Determine by direct integration the moment of inertia of the shaded area with respect to the x axis.

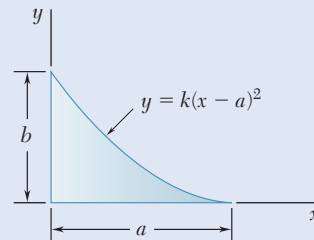


Fig. P9.1 and P9.5

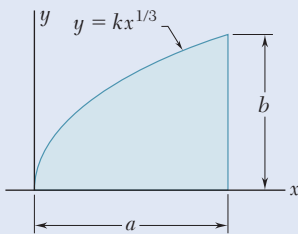


Fig. P9.2 and P9.6

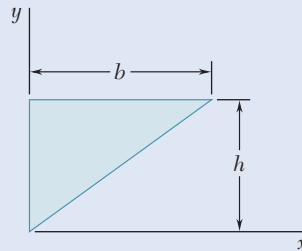


Fig. P9.3 and P9.7

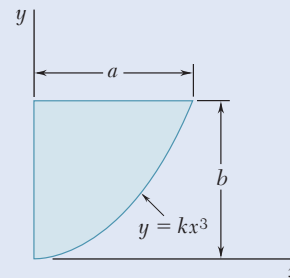


Fig. P9.4 and P9.8

9.9 through 9.11 Determine by direct integration the moment of inertia of the shaded area with respect to the x axis.

9.12 through 9.14 Determine by direct integration the moment of inertia of the shaded area with respect to the y axis.

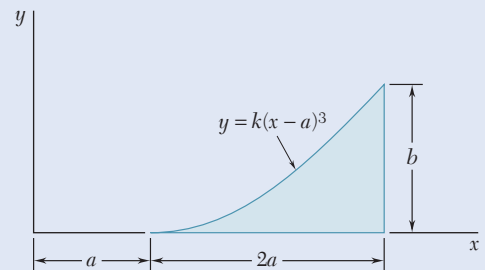


Fig. P9.9 and P9.12

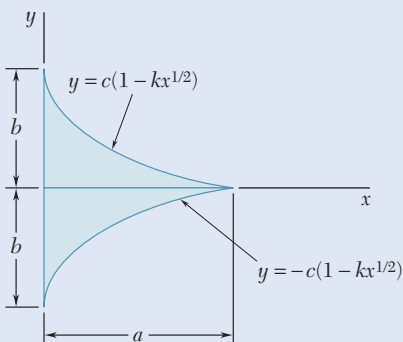


Fig. P9.10 and P9.13

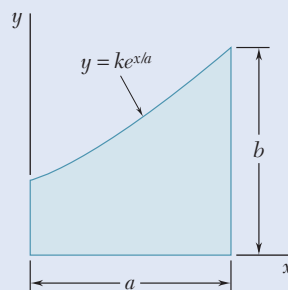


Fig. P9.11 and P9.14

9.15 and 9.16 Determine the moment of inertia and the radius of gyration of the shaded area shown with respect to the x axis.

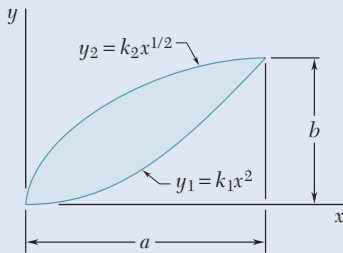


Fig. P9.15 and P9.17

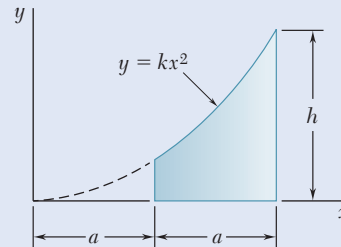


Fig. P9.16 and P9.18

9.17 and 9.18 Determine the moment of inertia and the radius of gyration of the shaded area shown with respect to the y axis.

9.19 Determine the moment of inertia and the radius of gyration of the shaded area shown with respect to the x axis.

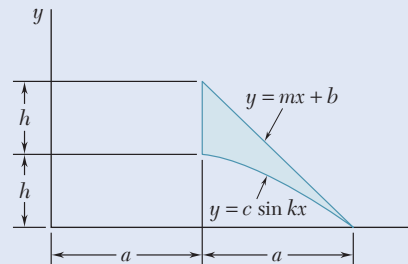


Fig. P9.19 and P9.20

9.20 Determine the moment of inertia and the radius of gyration of the shaded area shown with respect to the y axis.

9.21 and 9.22 Determine the polar moment of inertia and the polar radius of gyration of the shaded area shown with respect to point P .

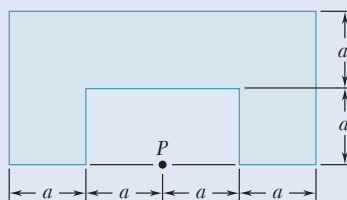


Fig. P9.21

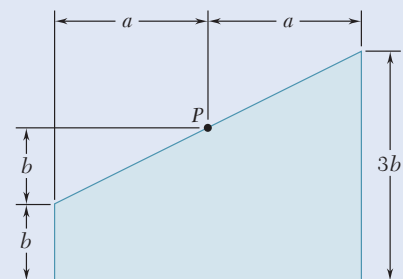


Fig. P9.22

9.23 and 9.24 Determine the polar moment of inertia and the polar radius of gyration of the shaded area shown with respect to point P .

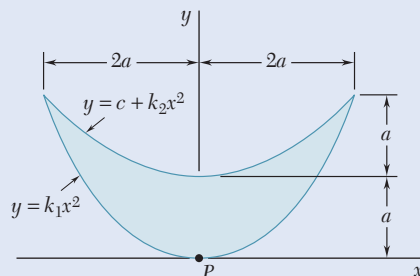


Fig. P9.23

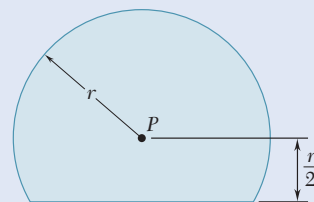


Fig. P9.24

9.25 (a) Determine by direct integration the polar moment of inertia of the annular area shown with respect to point O . (b) Using the result of part a, determine the moment of inertia of the given area with respect to the x axis.

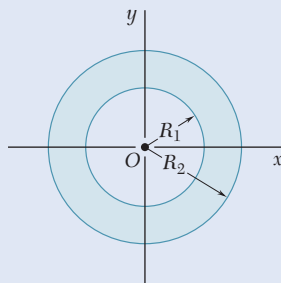


Fig. P9.25 and P9.26

9.26 (a) Show that the polar radius of gyration k_O of the annular area shown is approximately equal to the mean radius $R_m = (R_1 + R_2)/2$ for small values of the thickness $t = R_2 - R_1$. (b) Determine the percentage error introduced by using R_m in place of k_O for the following values of t/R_m : 1 , $\frac{1}{2}$, and $\frac{1}{10}$.

9.27 Determine the polar moment of inertia and the polar radius of gyration of the shaded area shown with respect to point O .

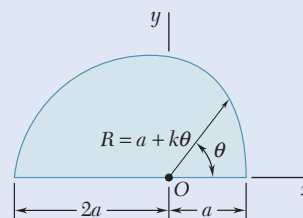


Fig. P9.27

9.28 Determine the polar moment of inertia and the polar radius of gyration of the isosceles triangle shown with respect to point O .

***9.29** Using the polar moment of inertia of the isosceles triangle of Prob. 9.28, show that the centroidal polar moment of inertia of a circular area of radius r is $\pi r^4/2$. (Hint: As a circular area is divided into an increasing number of equal circular sectors, what is the approximate shape of each circular sector?)

***9.30** Prove that the centroidal polar moment of inertia of a given area A cannot be smaller than $A^2/2\pi$. (Hint: Compare the moment of inertia of the given area with the moment of inertia of a circle that has the same area and the same centroid.)

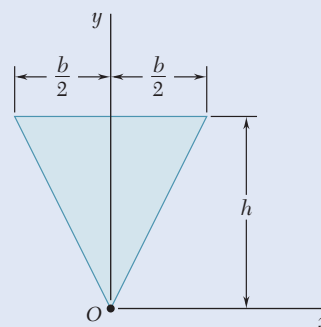


Fig. P9.28

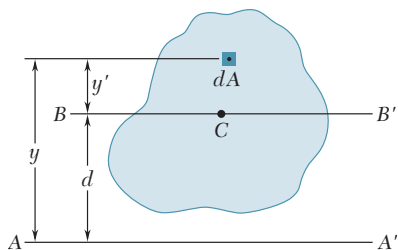


Fig. 9.9 The moment of inertia of an area A with respect to an axis AA' can be determined from its moment of inertia with respect to the centroidal axis BB' by a calculation involving the distance d between the axes.

9.2 PARALLEL-AXIS THEOREM AND COMPOSITE AREAS

In practice, we often need to determine the moment of inertia of a complicated area that can be broken down into a sum of simple areas. However, in doing these calculations, we have to determine the moment of inertia of each simple area with respect to the same axis. In this section, we first derive a formula for computing the moment of inertia of an area with respect to a centroidal axis parallel to a given axis. Then we show how you can use this formula for finding the moment of inertia of a composite area.

9.2A The Parallel-Axis Theorem

Consider the moment of inertia I of an area A with respect to an axis AA' (Fig. 9.9). We denote the distance from an element of area dA to AA' by y . This gives us

$$I = \int y^2 dA$$

Let us now draw through the centroid C of the area an axis BB' parallel to AA' ; this axis is called a *centroidal axis*. Denoting the distance from the element dA to BB' by y' , we have $y = y' + d$, where d is the distance between the axes AA' and BB' . Substituting for y in the previous integral, we obtain

$$\begin{aligned} I &= \int y^2 dA = \int (y' + d)^2 dA \\ &= \int y'^2 dA + 2d \int y' dA + d^2 \int dA \end{aligned}$$

The first integral represents the moment of inertia $\bar{I}$ of the area with respect to the centroidal axis BB' . The second integral represents the first moment of the area with respect to BB' , but since the centroid C of the area is located on this axis, the second integral must be zero. The last integral is equal to the total area A . Therefore, we have

Parallel-axis theorem

$$I = \bar{I} + Ad^2 \quad (9.9)$$

This formula states that the moment of inertia I of an area with respect to any given axis AA' is equal to the moment of inertia $\bar{I}$ of the area with respect to a centroidal axis BB' parallel to AA' *plus* the product of the area A and the square of the distance d between the two axes. This theorem is known as the **parallel-axis theorem**. Substituting $k^2 A$ for I and $\bar{k}^2 A$ for $\bar{I}$, we can also express this theorem as

$$k^2 = \bar{k}^2 + d^2 \quad (9.10)$$

A similar theorem relates the polar moment of inertia J_O of an area about a point O to the polar moment of inertia $\bar{J}_C$ of the same area about its centroid C . Denoting the distance between O and C by d , we have

$$J_O = \bar{J}_C + Ad^2 \quad \text{or} \quad k_O^2 = \bar{k}_C^2 + d^2 \quad (9.11)$$

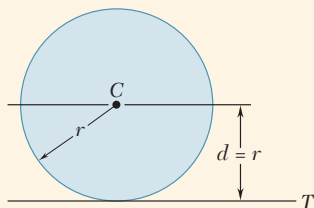


Fig. 9.10 Finding the moment of inertia of a circle with respect to a line tangent to it.

Concept Application 9.2

As an application of the parallel-axis theorem, let us determine the moment of inertia I_T of a circular area with respect to a line tangent to the circle (Fig. 9.10). We found in Sample Prob. 9.2 that the moment of inertia of a circular area about a centroidal axis is $\bar{I} = \frac{1}{4}\pi r^4$. Therefore, we have

$$I_T = \bar{I} + Ad^2 = \frac{1}{4}\pi r^4 + (\pi r^2)r^2 = \frac{5}{4}\pi r^4$$

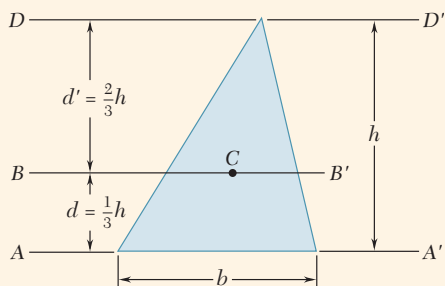


Fig. 9.11 Finding the centroidal moment of inertia of a triangle from the moment of inertia about a parallel axis.

Concept Application 9.3

We can also use the parallel-axis theorem to determine the centroidal moment of inertia of an area when we know the moment of inertia of the area with respect to a parallel axis. Consider, for instance, a triangular area (Fig. 9.11). We found in Sample Prob. 9.1 that the moment of inertia of a triangle with respect to its base AA' is equal to $\frac{1}{12}bh^3$. Using the parallel-axis theorem, we have

$$\begin{aligned} I_{AA'} &= \bar{I}_{BB'} + Ad^2 \\ \bar{I}_{BB'} &= I_{AA'} - Ad^2 = \frac{1}{12}bh^3 - \frac{1}{2}bh\left(\frac{1}{3}h\right)^2 = \frac{1}{36}bh^3 \end{aligned}$$

Note that we *subtracted* the product Ad^2 from the given moment of inertia in order to obtain the centroidal moment of inertia of the triangle. That is, this product is *added* when transferring *from* a centroidal axis to a parallel axis, but it is *subtracted* when transferring *to* a centroidal axis. In other words, the moment of inertia of an area is always smaller with respect to a centroidal axis than with respect to any parallel axis.

Returning to Fig. 9.11, we can obtain the moment of inertia of the triangle with respect to the line DD' (which is drawn through a vertex) by writing

$$I_{DD'} = \bar{I}_{BB'} + Ad'^2 = \frac{1}{36}bh^3 + \frac{1}{2}bh\left(\frac{2}{3}h\right)^2 = \frac{1}{4}bh^3$$

Note that we could not have obtained $I_{DD'}$ directly from $I_{AA'}$. We can apply the parallel-axis theorem only if one of the two parallel axes passes through the centroid of the area.

9.2B Moments of Inertia of Composite Areas

Consider a composite area A made of several component areas $A_1, A_2, A_3, \dots$. The integral representing the moment of inertia of A can be subdivided into integrals evaluated over $A_1, A_2, A_3, \dots$. Therefore, we can obtain the moment of inertia of A with respect to a given axis by adding the moments of inertia of the areas $A_1, A_2, A_3, \dots$ with respect to the same axis.



Photo 9.1 Figure 9.13 tabulates data for a small sample of the rolled-steel shapes that are readily available. Shown above are examples of wide-flange shapes that are commonly used in the construction of buildings.

Figure 9.12 shows several common geometric shapes along with formulas for the moments of inertia of each one. Before adding the moments of inertia of the component areas, however, you may have to use the parallel-axis theorem to transfer each moment of inertia to the desired axis. Sample Probs. 9.4 and 9.5 illustrate the technique.

Properties of the cross sections of various structural shapes are given in Fig. 9.13. As we noted in Sec. 9.1A, the moment of inertia of a beam

Rectangle		$\bar{I}_{x'} = \frac{1}{12}bh^3$ $\bar{I}_{y'} = \frac{1}{12}b^3h$ $I_x = \frac{1}{3}bh^3$ $I_y = \frac{1}{3}b^3h$ $J_C = \frac{1}{12}bh(b^2 + h^2)$
Triangle		$\bar{I}_{x'} = \frac{1}{36}bh^3$ $I_x = \frac{1}{12}bh^3$
Circle		$\bar{I}_x = \bar{I}_y = \frac{1}{4}\pi r^4$ $J_O = \frac{1}{2}\pi r^4$
Semicircle		$I_x = I_y = \frac{1}{8}\pi r^4$ $J_O = \frac{1}{4}\pi r^4$
Quarter circle		$I_x = I_y = \frac{1}{16}\pi r^4$ $J_O = \frac{1}{8}\pi r^4$
Ellipse		$\bar{I}_x = \frac{1}{4}\pi ab^3$ $\bar{I}_y = \frac{1}{4}\pi a^3b$ $J_O = \frac{1}{4}\pi ab(a^2 + b^2)$

Fig. 9.12 Moments of inertia of common geometric shapes.

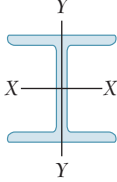
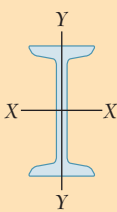
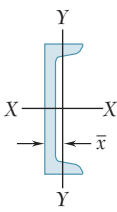
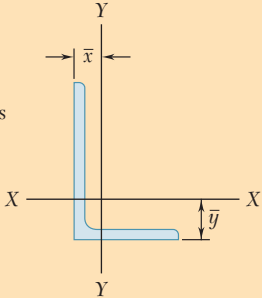
	Designation	Area in ²	Depth in.	Width in.	Axis X–X			Axis Y–Y		
					$\bar{I}_x$, in ⁴	$\bar{k}_x$, in.	$\bar{y}$, in.	$\bar{I}_y$, in ⁴	$\bar{k}_y$, in.	$\bar{x}$, in.
W Shapes (Wide-Flange Shapes) 	W18 × 76† W16 × 57 W14 × 38 W8 × 31	22.3 16.8 11.2 9.12	18.2 16.4 14.1 8.00	11.0 7.12 6.77 8.00	1330 758 385 110	7.73 6.72 5.87 3.47		152 43.1 26.7 37.1	2.61 1.60 1.55 2.02	
S Shapes (American Standard Shapes) 	S18 × 54.7† S12 × 31.8 S10 × 25.4 S6 × 12.5	16.0 9.31 7.45 3.66	18.0 12.0 10.0 6.00	6.00 5.00 4.66 3.33	801 217 123 22.0	7.07 4.83 4.07 2.45		20.7 9.33 6.73 1.80	1.14 1.00 0.950 0.702	
C Shapes (American Standard Channels) 	C12 × 20.7† C10 × 15.3 C8 × 11.5 C6 × 8.2	6.08 4.48 3.37 2.39	12.0 10.0 8.00 6.00	2.94 2.60 2.26 1.92	129 67.3 32.5 13.1	4.61 3.87 3.11 2.34		3.86 2.27 1.31 0.687	0.797 0.711 0.623 0.536	0.698 0.634 0.572 0.512
Angles 	L6 × 6 × 1‡ L4 × 4 × 1/2 L3 × 3 × 1/4 L6 × 4 × 1/2 L5 × 3 × 1/2 L3 × 2 × 1/4	11.0 3.75 1.44 4.75 3.75 1.19			35.4 5.52 1.23 17.3 9.43 1.09	1.79 1.21 0.926 1.91 1.58 0.953	1.86 1.18 0.836 1.98 1.74 0.980	35.4 5.52 1.23 6.22 2.55 0.390	1.79 1.21 0.926 1.14 0.824 0.569	1.86 1.18 0.836 0.981 0.746 0.487

Fig. 9.13A Properties of rolled-steel shapes (U.S. customary units).*

*Courtesy of the American Institute of Steel Construction, Chicago, Illinois

†Nominal depth in inches and weight in pounds per foot

‡Depth, width, and thickness in inches

section about its neutral axis is closely related to the computation of the bending moment in that section of the beam. Thus, determining moments of inertia is a prerequisite to the analysis and design of structural members.

Note that the radius of gyration of a composite area is *not* equal to the sum of the radii of gyration of the component areas. In order to determine the radius of gyration of a composite area, you must first compute the moment of inertia of the area.

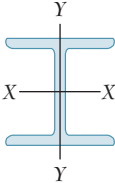
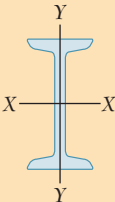
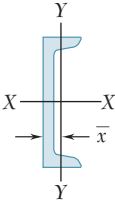
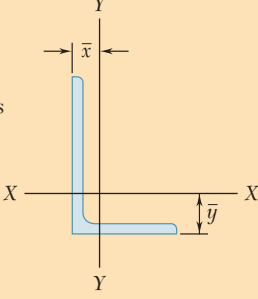
		Designation	Area mm ²	Depth mm	Width mm	Axis X-X			Axis Y-Y		
						$\bar{I}_x$ 10 ⁶ mm ⁴	$\bar{k}_x$ mm	$\bar{y}$ mm	$\bar{I}_y$ 10 ⁶ mm ⁴	$\bar{k}_y$ mm	$\bar{x}$ mm
W Shapes (Wide-Flange Shapes)		W460 × 113†	14 400	462	279	554	196		63.3	66.3	
		W410 × 85	10 800	417	181	316	171		17.9	40.6	
		W360 × 57.8	7230	358	172	160	149		11.1	39.4	
		W200 × 46.1	5880	203	203	45.8	88.1		15.4	51.3	
S Shapes (American Standard Shapes)		S460 × 81.4†	10 300	457	152	333	180		8.62	29.0	
		S310 × 47.3	6010	305	127	90.3	123		3.88	25.4	
		S250 × 37.8	4810	254	118	51.2	103		2.80	24.1	
		S150 × 18.6	2360	152	84.6	9.16	62.2		0.749	17.8	
C Shapes (American Standard Channels)		C310 × 30.8†	3920	305	74.7	53.7	117		1.61	20.2	17.7
		C250 × 22.8	2890	254	66.0	28.0	98.3		0.945	18.1	16.1
		C200 × 17.1	2170	203	57.4	13.5	79.0		0.545	15.8	14.5
		C150 × 12.2	1540	152	48.8	5.45	59.4		0.286	13.6	13.0
Angles		L152 × 152 × 25.4‡	7100			14.7	45.5	47.2	14.7	45.5	47.2
		L102 × 102 × 12.7	2420			2.30	30.7	30.0	2.30	30.7	30.0
		L76 × 76 × 6.4	929			0.512	23.5	21.2	0.512	23.5	21.2
		L152 × 102 × 12.7	3060			7.20	48.5	50.3	2.59	29.0	24.9
		L127 × 76 × 12.7	2420			3.93	40.1	44.2	1.06	20.9	18.9
		L76 × 51 × 6.4	768			0.454	24.2	24.9	0.162	14.5	12.4

Fig. 9.13B Properties of rolled-steel shapes (SI units).

†Nominal depth in millimeters and mass in kilograms per meter

‡Depth, width, and thickness in millimeters

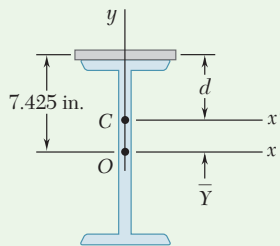
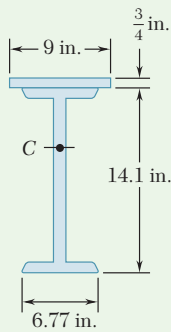


Fig. 1 Origin of coordinates placed at centroid of wide-flange shape.

Sample Problem 9.4

The strength of a $W14 \times 38$ rolled-steel beam is increased by attaching a $9 \times 3/4$ -in. plate to its upper flange as shown. Determine the moment of inertia and the radius of gyration of the composite section with respect to an axis that is parallel to the plate and passes through the centroid C of the section.

STRATEGY: This problem involves finding the moment of inertia of a composite area with respect to its centroid. You should first determine the location of this centroid. Then, using the parallel-axis theorem, you can determine the moment of inertia relative to this centroid for the overall section from the centroidal moment of inertia for each component part.

MODELING and ANALYSIS: Place the origin O of coordinates at the centroid of the wide-flange shape, and compute the distance $\bar{Y}$ to the centroid of the composite section by using the methods of Chap. 5 (Fig. 1). Refer to Fig. 9.13A for the area of the wide-flange shape. The area and the y coordinate of the centroid of the plate are

$$A = (9 \text{ in.})(0.75 \text{ in.}) = 6.75 \text{ in}^2$$

$$\bar{y} = \frac{1}{2}(14.1 \text{ in.}) + \frac{1}{2}(0.75 \text{ in.}) = 7.425 \text{ in.}$$

Section	Area, in^2	$\bar{y}$, in.	$\bar{y}A$, in^3
Plate	6.75	7.425	50.12
Wide-flange shape	11.2	0	0
	$\Sigma A = 17.95$		$\Sigma \bar{y}A = 50.12$

$$\bar{Y}\Sigma A = \Sigma \bar{y}A \quad \bar{Y}(17.95) = 50.12 \quad \bar{Y} = 2.792 \text{ in.}$$

Moment of Inertia. Use the parallel-axis theorem to determine the moments of inertia of the wide-flange shape and the plate with respect to the x' axis. This axis is a centroidal axis for the composite section but *not* for either of the elements considered separately. You can obtain the value of $\bar{I}_x$ for the wide-flange shape from Fig. 9.13A.

For the wide-flange shape,

$$I_{x'} = \bar{I}_x + A\bar{Y}^2 = 385 + (11.2)(2.792)^2 = 472.3 \text{ in}^4$$

For the plate,

$$I_{x'} = \bar{I}_x + Ad^2 = \left(\frac{1}{12}\right)(9)\left(\frac{3}{4}\right)^3 + (6.75)(7.425 - 2.792)^2 = 145.2 \text{ in}^4$$

For the composite area,

$$I_{x'} = 472.3 + 145.2 = 617.5 \text{ in}^4 \quad I_{x'} = 618 \text{ in}^4 \quad \blacktriangleleft$$

Radius of Gyration. From the moment of inertia and area just calculated, you obtain

$$k_{x'}^2 = \frac{I_{x'}}{A} = \frac{617.5 \text{ in}^4}{17.95 \text{ in}^2} \quad k_{x'} = 5.87 \text{ in.} \quad \blacktriangleleft$$

REFLECT and THINK: This is a common type of calculation for many different situations. It is often helpful to list data in a table to keep track of the numbers and identify which data you need.

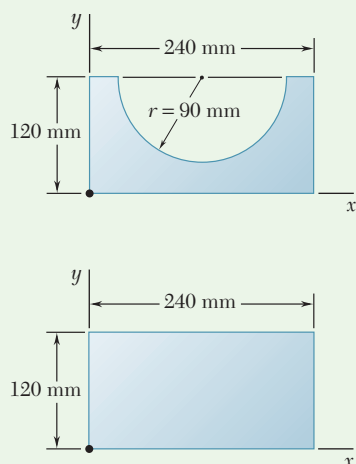
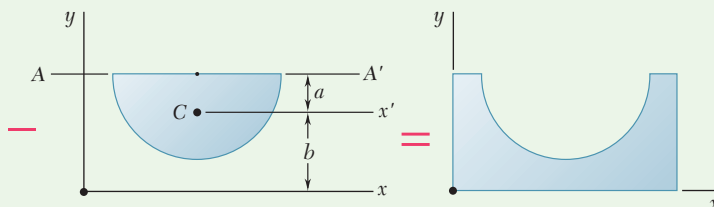


Fig. 1 Modeling given area by subtracting a half circle from a rectangle.

Sample Problem 9.5

Determine the moment of inertia of the shaded area with respect to the x axis.

STRATEGY: You can obtain the given area by subtracting a half circle from a rectangle (Fig. 1). Then compute the moments of inertia of the rectangle and the half circle separately.



MODELING and ANALYSIS:

Moment of Inertia of Rectangle. Referring to Fig. 9.12, you have

$$I_x = \frac{1}{3}bh^3 = \frac{1}{3}(240 \text{ mm})(120 \text{ mm})^3 = 138.2 \times 10^6 \text{ mm}^4$$

Moment of Inertia of Half Circle. Refer to Fig. 5.8 and determine the location of the centroid C of the half circle with respect to diameter AA' . As shown in Fig. 2, you have

$$a = \frac{4r}{3\pi} = \frac{(4)(90 \text{ mm})}{3\pi} = 38.2 \text{ mm}$$

The distance b from the centroid C to the x axis is

$$b = 120 \text{ mm} - a = 120 \text{ mm} - 38.2 \text{ mm} = 81.8 \text{ mm}$$

Referring now to Fig. 9.12, compute the moment of inertia of the half circle with respect to diameter AA' and then compute the area of the half circle.

$$I_{AA'} = \frac{1}{8}\pi r^4 = \frac{1}{8}\pi(90 \text{ mm})^4 = 25.76 \times 10^6 \text{ mm}^4$$

$$A = \frac{1}{2}\pi r^2 = \frac{1}{2}\pi(90 \text{ mm})^2 = 12.72 \times 10^3 \text{ mm}^2$$

Next, using the parallel-axis theorem, obtain the value of $\bar{I}_{x'}$ as

$$\begin{aligned} I_{AA'} &= \bar{I}_{x'} + Aa^2 \\ 25.76 \times 10^6 \text{ mm}^4 &= \bar{I}_{x'} + (12.72 \times 10^3 \text{ mm}^2)(38.2 \text{ mm})^2 \\ \bar{I}_{x'} &= 7.20 \times 10^6 \text{ mm}^4 \end{aligned}$$

Again using the parallel-axis theorem, obtain the value of I_x as

$$\begin{aligned} I_x &= \bar{I}_{x'} + Ab^2 = 7.20 \times 10^6 \text{ mm}^4 + (12.72 \times 10^3 \text{ mm}^2)(81.8 \text{ mm})^2 \\ &= 92.3 \times 10^6 \text{ mm}^4 \end{aligned}$$

Moment of Inertia of Given Area. Subtracting the moment of inertia of the half circle from that of the rectangle, you obtain

$$\begin{aligned} I_x &= 138.2 \times 10^6 \text{ mm}^4 - 92.3 \times 10^6 \text{ mm}^4 \\ &= 45.9 \times 10^6 \text{ mm}^4 \quad \blacktriangleleft \end{aligned}$$

REFLECT and THINK: Figures 5.8 and 9.12 are useful references for locating centroids and moments of inertia of common areas; don't forget to use them.

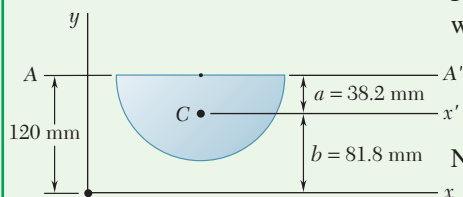


Fig. 2 Centroid location of the half circle.

SOLVING PROBLEMS ON YOUR OWN

In this section, we introduced the **parallel-axis theorem** and showed how to use it to simplify the computation of moments and polar moments of inertia of composite areas. The areas that you will consider in the following problems will consist of common shapes and rolled-steel shapes. You will also use the parallel-axis theorem to locate the point of application (the center of pressure) of the resultant of the hydrostatic forces acting on a submerged plane area.

1. Applying the parallel-axis theorem. In Sec. 9.2, we derived the parallel-axis theorem

$$I = \bar{I} + Ad^2 \quad (9.9)$$

which states that the moment of inertia I of an area A with respect to a given axis is equal to the sum of the moment of inertia $\bar{I}$ of that area with respect to a *parallel centroidal axis* and the product Ad^2 , where d is the distance between the two axes. It is important that you remember the following points as you use the parallel-axis theorem.

a. You can obtain the centroidal moment of inertia $\bar{I}$ of an area A by subtracting the product Ad^2 from the moment of inertia I of the area with respect to a parallel axis. It follows that the moment of inertia $\bar{I}$ is *smaller* than the moment of inertia I of the same area with respect to any parallel axis.

b. You can apply the parallel-axis theorem only if one of the two axes involved is a centroidal axis. Therefore, as we noted in Concept Application 9.3, to compute the moment of inertia of an area with respect to a *noncentroidal axis* when the moment of inertia of the area is known with respect to *another noncentroidal axis*, it is necessary to first compute the moment of inertia of the area with respect to a centroidal axis parallel to the two given axes.

2. Computing the moments and polar moments of inertia of composite areas. Sample Probs. 9.4 and 9.5 illustrate the steps you should follow to solve problems of this type. As with all composite-area problems, you should show on your sketch the common shapes or rolled-steel shapes that constitute the various elements of the given area, as well as the distances between the centroidal axes of the elements and the axes about which the moments of inertia are to be computed. In addition, it is important to note the following points.

a. The moment of inertia of an area is always positive, regardless of the location of the axis with respect to which it is computed. As pointed out in the comments for the preceding section, only when an area is *removed* (as in the case of a hole) should you enter its moment of inertia in your computations with a minus sign.

(continued)

b. The moments of inertia of a semiellipse and a quarter ellipse can be determined by dividing the moment of inertia of an ellipse by 2 and 4, respectively. Note, however, that the moments of inertia obtained in this manner are *with respect to the axes of symmetry of the ellipse*. To obtain the *centroidal* moments of inertia of these shapes, use the parallel-axis theorem. This remark also applies to a semicircle and to a quarter circle. Also note that the expressions given for these shapes in Fig. 9.12 are *not* centroidal moments of inertia.

c. To calculate the polar moment of inertia of a composite area, you can use either the expressions given in Fig. 9.12 for J_O or the relationship

$$J_O = I_x + I_y \quad (9.4)$$

depending on the shape of the given area.

d. Before computing the centroidal moments of inertia of a given area, you may find it necessary to first locate the centroid of the area using the methods of Chap. 5.

3. Locating the point of application of the resultant of a system of hydrostatic forces. In Sec. 9.1, we found that

$$R = \gamma \int y \, dA = \gamma \bar{y} A$$

$$M_x = \gamma \int y^2 \, dA = \gamma I_x$$

where $\bar{y}$ is the distance from the x axis to the centroid of the submerged plane area. Since $\mathbf{R}$ is equivalent to the system of elemental hydrostatic forces, it follows that

$$\Sigma M_x: \quad y_P R = M_x$$

where y_P is the depth of the point of application of $\mathbf{R}$. Then

$$y_P(\gamma \bar{y} A) = \gamma I_x \quad \text{or} \quad y_P = \frac{I_x}{\bar{y} A}$$

In closing, we encourage you to carefully study the notation used in Fig. 9.13 for the rolled-steel shapes, as you will likely encounter it again in subsequent engineering courses.

Problems

9.31 and 9.32 Determine the moment of inertia and the radius of gyration of the shaded area with respect to the x axis.

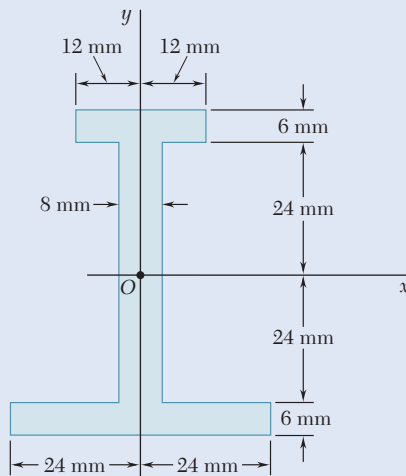


Fig. P9.31 and P9.33

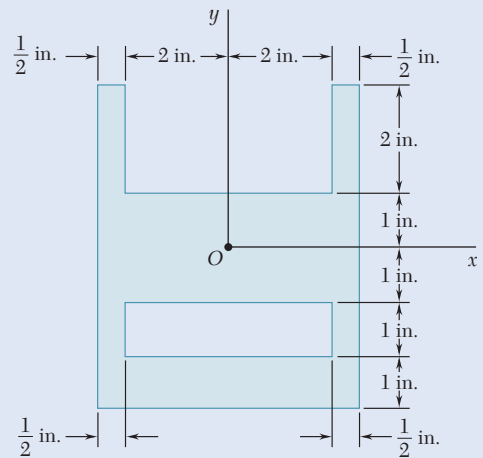


Fig. P9.32 and P9.34

9.33 and 9.34 Determine the moment of inertia and the radius of gyration of the shaded area with respect to the y axis.

9.35 and 9.36 Determine the moments of inertia of the shaded area shown with respect to the x and y axes.

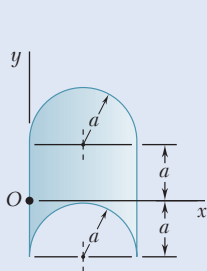


Fig. P9.35

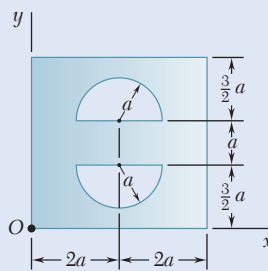


Fig. P9.36

9.37 The centroidal polar moment of inertia $\bar{J}_C$ of the 24-in² shaded area is 600 in⁴. Determine the polar moments of inertia J_B and J_D of the shaded area knowing that $J_D = 2J_B$ and $d = 5$ in.

9.38 Determine the centroidal polar moment of inertia $\bar{J}_C$ of the 25-in² shaded area knowing that the polar moments of inertia of the area with respect to points A , B , and D are, respectively, $J_A = 281$ in⁴, $J_B = 810$ in⁴, and $J_D = 1578$ in⁴.

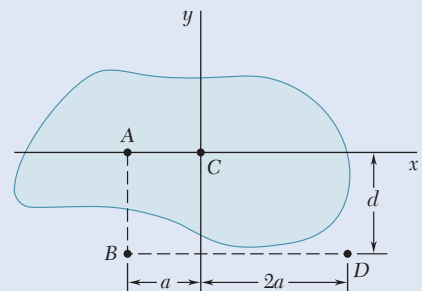


Fig. P9.37 and P9.38

- 9.39** Determine the shaded area and its moment of inertia with respect to the centroidal axis parallel to AA' knowing that $d_1 = 25$ mm and $d_2 = 10$ mm and that its moments of inertia with respect to AA' and BB' are 2.2×10^6 mm⁴ and 4×10^6 mm⁴, respectively.

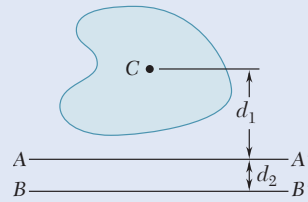


Fig. P9.39 and P9.40

- 9.40** Knowing that the shaded area is equal to 6000 mm² and that its moment of inertia with respect to AA' is 18×10^6 mm⁴, determine its moment of inertia with respect to BB' for $d_1 = 50$ mm and $d_2 = 10$ mm.

- 9.41 through 9.44** Determine the moments of inertia $\bar{I}_x$ and $\bar{I}_y$ of the area shown with respect to centroidal axes respectively parallel and perpendicular to side AB .

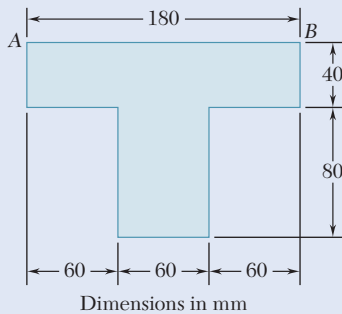


Fig. P9.41

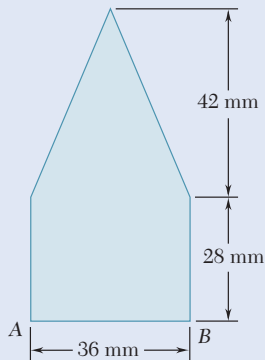


Fig. P9.42

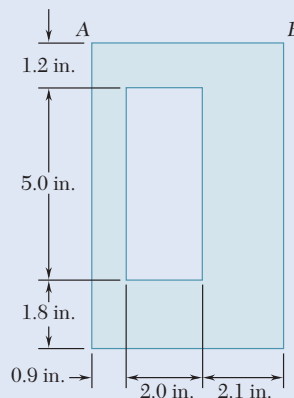


Fig. P9.43

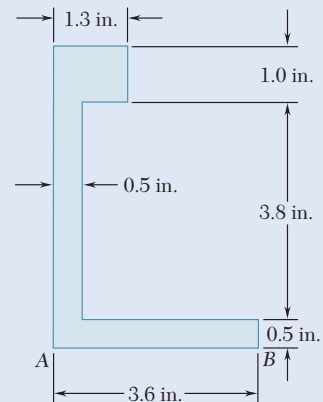


Fig. P9.44

- 9.45 and 9.46** Determine the polar moment of inertia of the area shown with respect to (a) point O , (b) the centroid of the area.

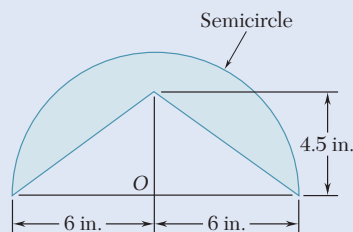


Fig. P9.45

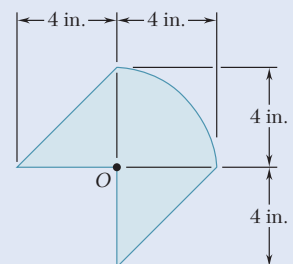


Fig. P9.46

9.47 and 9.48 Determine the polar moment of inertia of the area shown with respect to (a) point O , (b) the centroid of the area.

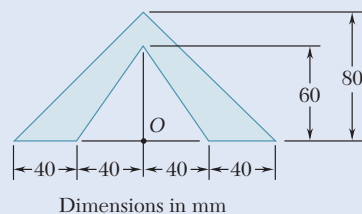


Fig. P9.47

9.49 Two channels and two plates are used to form the column section shown. For $b = 200$ mm, determine the moments of inertia and the radii of gyration of the combined section with respect to the centroidal x and y axes.

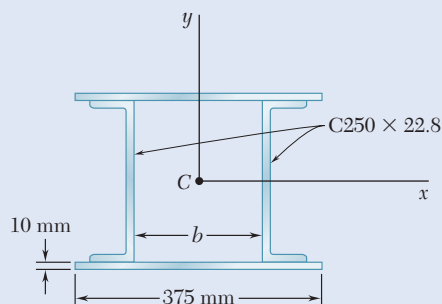


Fig. P9.49

9.50 Two $L6 \times 4 \times \frac{1}{2}$ -in. angles are welded together to form the section shown. Determine the moments of inertia and the radii of gyration of the combined section with respect to the centroidal x and y axes.

9.51 Four $L3 \times 3 \times \frac{1}{4}$ -in. angles are welded to a rolled W section as shown. Determine the moments of inertia and the radii of gyration of the combined section with respect to the centroidal x and y axes.

9.52 Two 20-mm steel plates are welded to a rolled S section as shown. Determine the moments of inertia and the radii of gyration of the combined section with respect to the centroidal x and y axes.

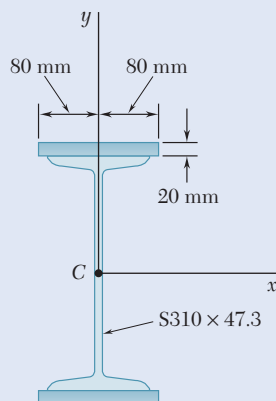


Fig. P9.52

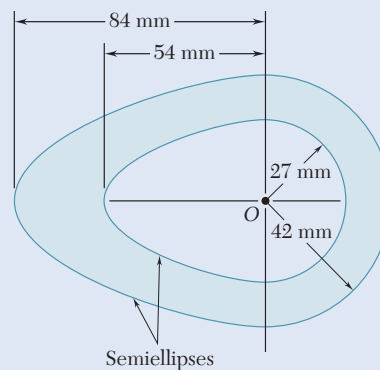


Fig. P9.48

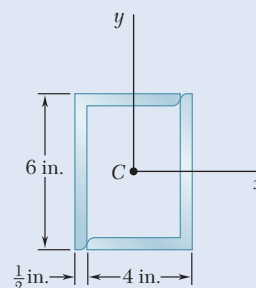


Fig. P9.50

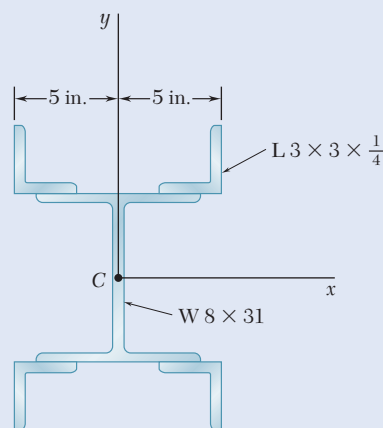


Fig. P9.51

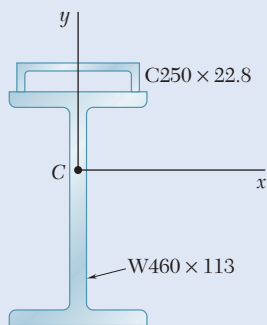


Fig. P9.54

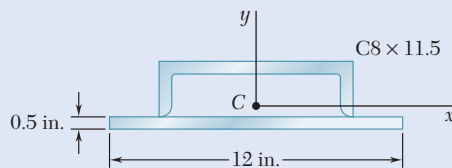


Fig. P9.53

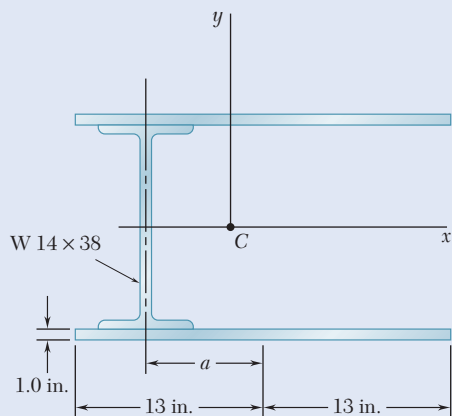


Fig. P9.56

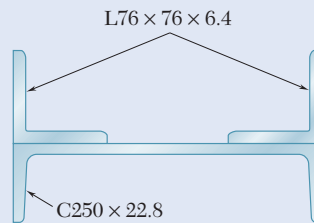


Fig. P9.55

9.56 Two steel plates are welded to a rolled W section as indicated. Knowing that the centroidal moments of inertia $\bar{I}_x$ and $\bar{I}_y$ of the combined section are equal, determine (a) the distance a , (b) the moments of inertia with respect to the centroidal x and y axes.

9.57 and 9.58 The panel shown forms the end of a trough that is filled with water to the line AA' . Referring to Sec. 9.1A, determine the depth of the point of application of the resultant of the hydrostatic forces acting on the panel (the center of pressure).

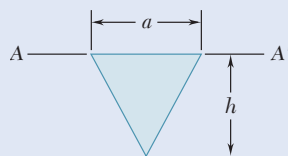


Fig. P9.57

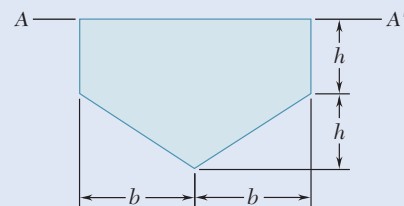


Fig. P9.58

9.59 and *9.60 The panel shown forms the end of a trough that is filled with water to the line AA' . Referring to Sec. 9.1A, determine the depth of the point of application of the resultant of the hydrostatic forces acting on the panel (the center of pressure).

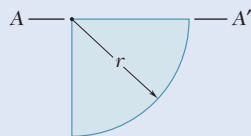


Fig. P9.59

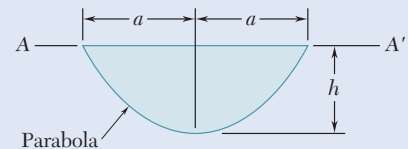


Fig. P9.60

9.61 A vertical trapezoidal gate that is used as an automatic valve is held shut by two springs attached to hinges located along edge AB . Knowing that each spring exerts a couple of magnitude $1470 \text{ N}\cdot\text{m}$, determine the depth d of water for which the gate will open.

9.62 The cover for a 0.5-m -diameter access hole in a water storage tank is attached to the tank with four equally spaced bolts as shown. Determine the additional force on each bolt due to the water pressure when the center of the cover is located 1.4 m below the water surface.

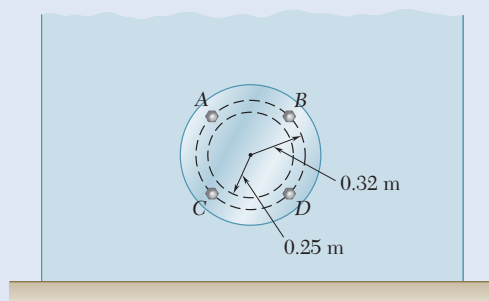


Fig. P9.62

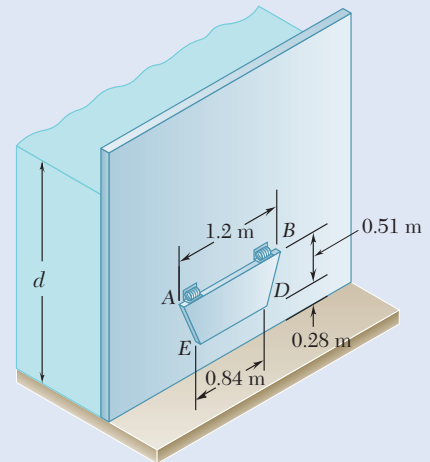


Fig. P9.61

***9.63** Determine the x coordinate of the centroid of the volume shown. (Hint: The height y of the volume is proportional to the x coordinate; consider an analogy between this height and the water pressure on a submerged surface.)

***9.64** Determine the x coordinate of the centroid of the volume shown; this volume was obtained by intersecting an elliptic cylinder with an oblique plane. (See hint of Prob. 9.63.)

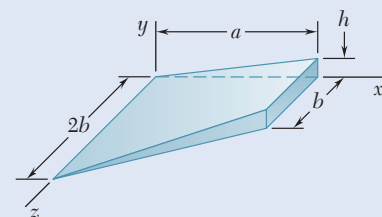


Fig. P9.63

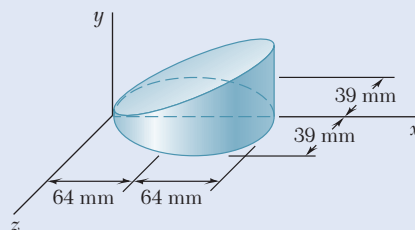


Fig. P9.64

- *9.65** Show that the system of hydrostatic forces acting on a submerged plane area A can be reduced to a force $\mathbf{P}$ at the centroid C of the area and two couples. The force $\mathbf{P}$ is perpendicular to the area and has a magnitude of $P = \gamma A \bar{y} \sin \theta$, where γ is the specific weight of the liquid. The couples are $\mathbf{M}_{x'} = (\gamma \bar{I}_{x'} \sin \theta) \mathbf{i}$ and $\mathbf{M}_{y'} = (\gamma \bar{I}_{y'} \sin \theta) \mathbf{j}$, where $\bar{I}_{x',y'} = \int x'^2 y'^2 dA$ (see Sec. 9.3). Note that the couples are independent of the depth at which the area is submerged.

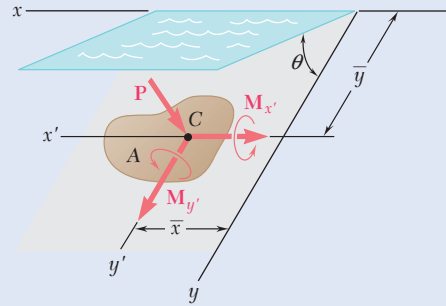


Fig. P9.65

- *9.66** Show that the resultant of the hydrostatic forces acting on a submerged plane area A is a force $\mathbf{P}$ perpendicular to the area and of magnitude $P = \gamma A \bar{y} \sin \theta = \bar{p} A$, where γ is the specific weight of the liquid and $\bar{p}$ is the pressure at the centroid C of the area. Show that $\mathbf{P}$ is applied at a point C_p , called the center of pressure, whose coordinates are $x_p = I_{xy}/A \bar{y}$ and $y_p = I_x/A \bar{y}$, where $I_{xy} = \int xy dA$ (see Sec. 9.3). Show also that the difference of ordinates $y_p - \bar{y}$ is equal to $\bar{k}_{x'}^2/\bar{y}$ and thus depends upon the depth at which the area is submerged.

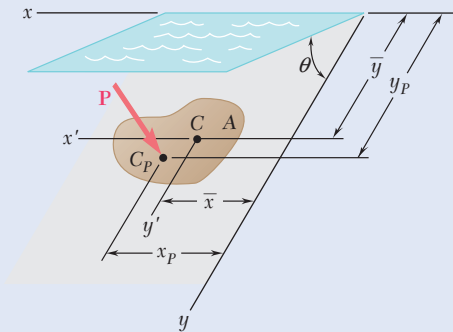


Fig. P9.66

*9.3 TRANSFORMATION OF MOMENTS OF INERTIA

The moments of inertia of an area can have different values depending on what axes we use to calculate them. It turns out that it is often important to determine the maximum and minimum values of the moments of inertia, which means finding the particular orientation of axes that produce these values. The first step in calculating moments of inertia with regard to rotated axes is to determine a new kind of second moment, called the product of inertia. In this section, we illustrate the procedures for this.

9.3A Product of Inertia

The **product of inertia** of an area A with respect to the x and y axes is defined by the integral

Product of inertia

$$I_{xy} = \int xy dA \quad (9.12)$$

We calculate it by multiplying each element dA of an area A by its coordinates x and y and integrating over the area (Fig. 9.14). Unlike the moments of inertia I_x and I_y , the product of inertia I_{xy} can be positive, negative, or zero. We will see shortly that the product of inertia is necessary for transforming moments of inertia with respect to a different set of axes; in a course on mechanics of materials, you will find other applications of this quantity.

When one or both of the x and y axes are axes of symmetry for the area A , the product of inertia I_{xy} is zero. Consider, for example, the channel section shown in Fig. 9.15. Since this section is symmetrical with respect to the x axis, we can associate with each element dA of coordinates x and y an element dA' of coordinates x and $-y$. Clearly, the contributions to I_{xy} of any pair of elements chosen in this way cancel out, and the integral of Eq. (9.12) reduces to zero.

We can derive a parallel-axis theorem for products of inertia similar to the one established in Sec. 9.2 for moments of inertia. Consider an area A and a system of rectangular coordinates x and y (Fig. 9.16). Through the centroid C of the area, with coordinates $\bar{x}$ and $\bar{y}$, we draw two centroidal axes x' and y' that are parallel, respectively, to the x and y axes. We denote the coordinates of an element of area dA with respect to the original axes by x and y , and the coordinates of the same element with respect to the centroidal axes by x' and y' . This gives us

$$x = x' + \bar{x} \quad \text{and} \quad y = y' + \bar{y}$$

Substituting into Eq. (9.12), we obtain the expression for the product of inertia I_{xy} as

$$\begin{aligned} I_{xy} &= \int xy dA = \int (x' + \bar{x})(y' + \bar{y}) dA \\ &= \int x'y' dA + \bar{y} \int x' dA + \bar{x} \int y' dA + \bar{x}\bar{y} \int dA \end{aligned}$$

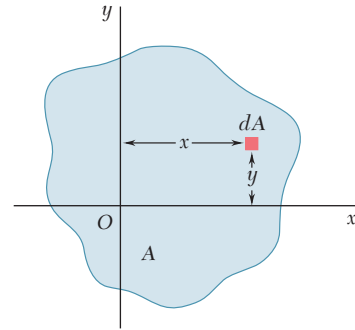


Fig. 9.14 An element of area dA with coordinates x and y .

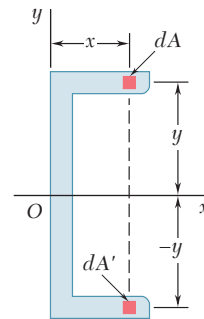


Fig. 9.15 If an area has an axis of symmetry, its product of inertia is zero.

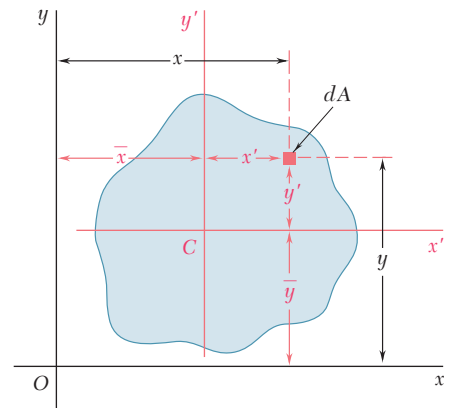


Fig. 9.16 An element of area dA with respect to x and y axes and the centroidal axes x' and y' for area A .

The first integral represents the product of inertia $\bar{I}_{xy}$ of the area A with respect to the centroidal axes x' and y' . The next two integrals represent first moments of the area with respect to the centroidal axes; they reduce to zero, since the centroid C is located on these axes. The last integral is equal to the total area A . Therefore, we have

Parallel-axis theorem for products of inertia

$$I_{xy} = \bar{I}_{x'y'} + \bar{x}\bar{y}A \quad (9.13)$$

9.3B Principal Axes and Principal Moments of Inertia

Consider an area A with coordinate axes x and y (Fig. 9.17) and assume that we know the moments and product of inertia of the area A . We have

$$I_x = \int y^2 dA \quad I_y = \int x^2 dA \quad I_{xy} = \int xy dA \quad (9.14)$$

We propose to determine the moments and product of inertia $I_{x'}$, $I_{y'}$, and $I_{x'y'}$ of A with respect to new axes x' and y' that we obtain by rotating the original axes about the origin through an angle θ .

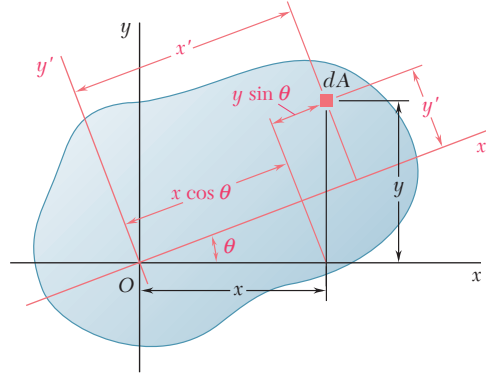


Fig. 9.17 An element of area dA with respect to x and y axes and a set of x' and y' axes rotated about the origin through an angle θ .

We first note that the relations between the coordinates x' , y' and x , y of an element of area dA are

$$x' = x \cos \theta + y \sin \theta \quad y' = y \cos \theta - x \sin \theta$$

Substituting for y' in the expression for $I_{x'}$, we obtain

$$\begin{aligned} I_{x'} &= \int (y')^2 dA = \int (y \cos \theta - x \sin \theta)^2 dA \\ &= \cos^2 \theta \int y^2 dA - 2 \sin \theta \cos \theta \int xy dA + \sin^2 \theta \int x^2 dA \end{aligned}$$

Using the relations in Eq. (9.14), we have

$$I_{x'} = I_x \cos^2 \theta - 2I_{xy} \sin \theta \cos \theta + I_y \sin^2 \theta \quad (9.15)$$

Similarly, we obtain for $I_{y'}$ and $I_{x'y'}$ the expressions

$$I_{y'} = I_x \sin^2 \theta + 2I_{xy} \sin \theta \cos \theta + I_y \cos^2 \theta \quad (9.16)$$

$$I_{x'y'} = (I_x - I_y) \sin \theta \cos \theta + I_{xy}(\cos^2 \theta - \sin^2 \theta) \quad (9.17)$$

Recalling the trigonometric relations

$$\sin 2\theta = 2 \sin \theta \cos \theta \quad \cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

and

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2} \quad \sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

we can write Eqs. (9.15), (9.16), and (9.17) as

$$I_{x'} = \frac{I_x + I_y}{2} + \frac{I_x - I_y}{2} \cos 2\theta - I_{xy} \sin 2\theta \quad (9.18)$$

$$I_{y'} = \frac{I_x + I_y}{2} - \frac{I_x - I_y}{2} \cos 2\theta + I_{xy} \sin 2\theta \quad (9.19)$$

$$I_{x'y'} = \frac{I_x - I_y}{2} \sin 2\theta + I_{xy} \cos 2\theta \quad (9.20)$$

Now, adding Eqs. (9.18) and (9.19), we observe that

$$I_{x'} + I_{y'} = I_x + I_y \quad (9.21)$$

We could have anticipated this result, since both members of Eq. (9.21) are equal to the polar moment of inertia J_O .

Equations (9.18) and (9.20) are the parametric equations of a circle. This means that, if we choose a set of rectangular axes and plot a point M of abscissa $I_{x'}$ and ordinate $I_{x'y'}$ for any given value of the parameter θ , all of the points will lie on a circle. To establish this property algebraically, we can eliminate θ from Eqs. (9.18) and (9.20) by transposing $(I_x + I_y)/2$ in Eq. (9.18), squaring both sides of Eqs. (9.18) and (9.20), and adding. The result is

$$\left(I_{x'} - \frac{I_x + I_y}{2}\right)^2 + I_{x'y'}^2 = \left(\frac{I_x - I_y}{2}\right)^2 + I_{xy}^2 \quad (9.22)$$

Setting

$$I_{\text{ave}} = \frac{I_x + I_y}{2} \quad \text{and} \quad R = \sqrt{\left(\frac{I_x - I_y}{2}\right)^2 + I_{xy}^2} \quad (9.23)$$

we can write the identity equation (9.22) in the form

$$(I_{x'} - I_{\text{ave}})^2 + I_{x'y'}^2 = R^2 \quad (9.24)$$

This is the equation of a circle of radius R centered at the point C whose x and y coordinates are I_{ave} and 0, respectively (Fig. 9.18a).

Note that Eqs. (9.19) and (9.20) are parametric equations of the same circle. Furthermore, because of the symmetry of the circle about the

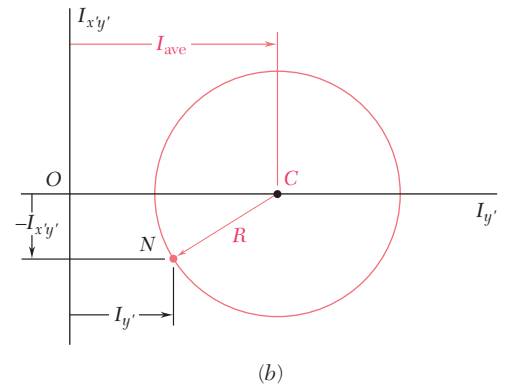
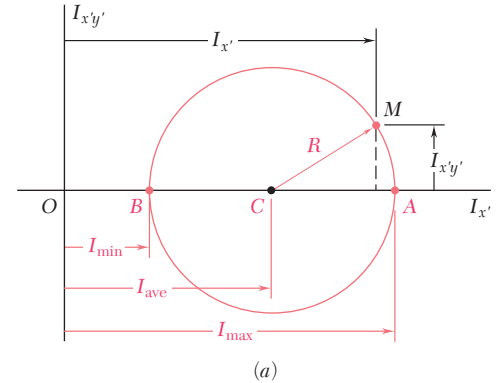


Fig. 9.18 Plots of $I_{x'y'}$ versus (a) $I_{x'}$ and (b) $I_{y'}$ for different values of the parameter θ are identical circles. The circle in part (a) indicates the average, maximum, and minimum values of the moment of inertia.

horizontal axis, we would obtain the same result if we plot a point N of coordinates $I_{y'}$ and $-I_{x'y'}$ (Fig. 9.18*b*) instead of plotting M . We will use this property in Sec. 9.4.

The two points A and B where this circle intersects the horizontal axis (Fig. 9.18*a*) are of special interest: Point A corresponds to the maximum value of the moment of inertia $I_{x'}$, whereas point B corresponds to its minimum value. In addition, both points correspond to a zero value of the product of inertia $I_{x'y'}$. Thus, we can obtain the values θ_m of the parameter θ corresponding to the points A and B by setting $I_{x'y'} = 0$ in Eq. (9.20). The result is[†]

$$\tan 2\theta_m = \frac{2I_{xy}}{I_x - I_y} \quad (9.25)$$

This equation defines two values ($2\theta_m$) that are 180° apart and thus two values (θ_m) that are 90° apart. One of these values corresponds to point A in Fig. 9.18*a* and to an axis through O in Fig. 9.17 with respect to which the moment of inertia of the given area is maximum. The other value corresponds to point B and to an axis through O with respect to which the moment of inertia of the area is minimum. These two perpendicular axes are called the **principal axes of the area about O** . The corresponding values $I_{\max}$ and $I_{\min}$ of the moment of inertia are called the **principal moments of inertia of the area about O** . Since we obtained the two values θ_m defined by Eq. (9.25) by setting $I_{x'y'} = 0$ in Eq. (9.20), it is clear that the product of inertia of the given area with respect to its principal axes is zero.

Note from Fig. 9.18*a* that

$$I_{\max} = I_{\text{ave}} + R \quad I_{\min} = I_{\text{ave}} - R \quad (9.26)$$

Using the values for I_{ave} and R from formulas (9.23), we obtain

$$I_{\max, \min} = \frac{I_x + I_y}{2} \pm \sqrt{\left(\frac{I_x - I_y}{2}\right)^2 + I_{xy}^2} \quad (9.27)$$

Unless you can tell by inspection which of the two principal axes corresponds to $I_{\max}$ and which corresponds to $I_{\min}$, you must substitute one of the values of θ_m into Eq. (9.18) in order to determine which of the two corresponds to the maximum value of the moment of inertia of the area about O .

Referring to Sec. 9.3A, note that, if an area possesses an axis of symmetry through a point O , this axis must be a principal axis of the area about O . On the other hand, a principal axis does not need to be an axis of symmetry; whether or not an area possesses any axes of symmetry, it will always have two principal axes of inertia about any point O .

The properties we have established hold for any point O located inside or outside the given area. If we choose the point O to coincide with the centroid of the area, any axis through O is a centroidal axis; the two principal axes of the area about its centroid are referred to as the **principal centroidal axes of the area**.

[†]We can also obtain this relation by differentiating $I_{x'}$ in Eq. (9.18) and setting $dI_{x'}/d\theta = 0$.

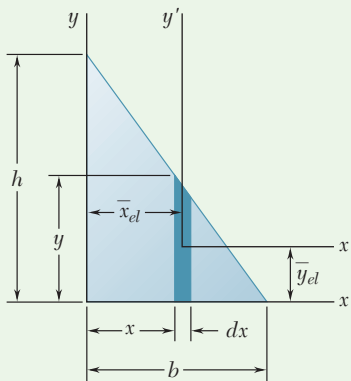
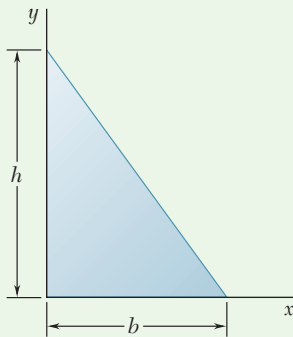


Fig. 1 Using a vertical rectangular strip as the differential element.

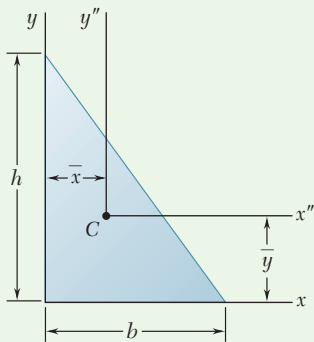


Fig. 2 Centroid of the triangular area.

Sample Problem 9.6

Determine the product of inertia of the right triangle shown (a) with respect to the x and y axes and (b) with respect to centroidal axes parallel to the x and y axes.

STRATEGY: You can approach this problem by using a vertical differential strip element. Because each point of the strip is at a different distance from the x axis, it is necessary to describe this strip mathematically using the parallel-axis theorem. Once you have completed the solution for the product of inertia with respect to the x and y axes, a second application of the parallel-axis theorem yields the product of inertia with respect to the centroidal axes.

MODELING and ANALYSIS:

a. Product of Inertia I_{xy} . Choose a vertical rectangular strip as the differential element of area (Fig. 1). Using a differential version of the parallel-axis theorem, you have

$$dI_{xy} = dI_{x'y'} + \bar{x}_{el}\bar{y}_{el} dA$$

The element is symmetrical with respect to the x' and y' axes, so $dI_{x'y'} = 0$. From the geometry of the triangle, you can express the variables in terms of x and y .

$$y = h\left(1 - \frac{x}{b}\right) \quad dA = y dx = h\left(1 - \frac{x}{b}\right) dx$$

$$\bar{x}_{el} = x \quad \bar{y}_{el} = \frac{1}{2}y = \frac{1}{2}h\left(1 - \frac{x}{b}\right)$$

Integrating dI_{xy} from $x = 0$ to $x = b$ gives you I_{xy} :

$$I_{xy} = \int dI_{xy} = \int \bar{x}_{el}\bar{y}_{el} dA = \int_0^b x\left(\frac{1}{2}\right)h^2\left(1 - \frac{x}{b}\right)^2 dx$$

$$= h^2 \int_0^b \left(\frac{x}{2} - \frac{x^2}{b} + \frac{x^3}{2b^2}\right) dx = h^2 \left[\frac{x^2}{4} - \frac{x^3}{3b} + \frac{x^4}{8b^2}\right]_0^b$$

$$I_{xy} = \frac{1}{24}b^2h^2 \quad \blacktriangleleft$$

b. Product of Inertia $\bar{I}_{x''y''}$. The coordinates of the centroid of the triangle relative to the x and y axes are (Fig. 2 and Fig. 5.8A)

$$\bar{x} = \frac{1}{3}b \quad \bar{y} = \frac{1}{3}h$$

Using the expression for I_{xy} obtained in part a, apply the parallel-axis theorem again:

$$I_{xy} = \bar{I}_{x''y''} + \bar{x}\bar{y}A$$

$$\frac{1}{24}b^2h^2 = \bar{I}_{x''y''} + \left(\frac{1}{3}b\right)\left(\frac{1}{3}h\right)\left(\frac{1}{2}bh\right)$$

$$\bar{I}_{x''y''} = \frac{1}{24}b^2h^2 - \frac{1}{18}b^2h^2$$

$$\bar{I}_{x''y''} = -\frac{1}{72}b^2h^2 \quad \blacktriangleleft$$

REFLECT and THINK: An equally effective alternative strategy would be to use a horizontal strip element. Again, you would need to use the parallel-axis theorem to describe this strip, since each point in the strip would be at a different distance from the y axis.

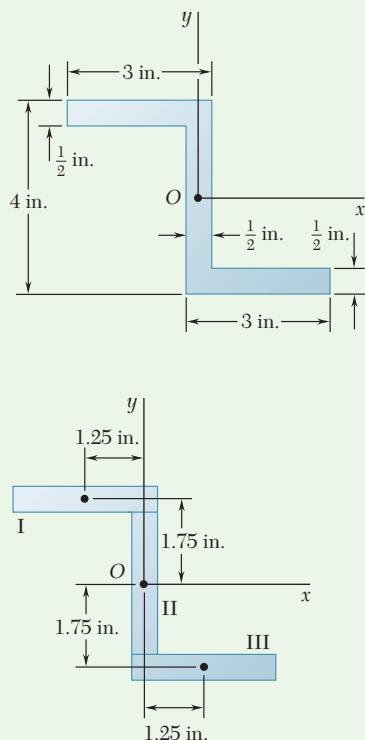


Fig. 1 Modeling the given area as three rectangles.

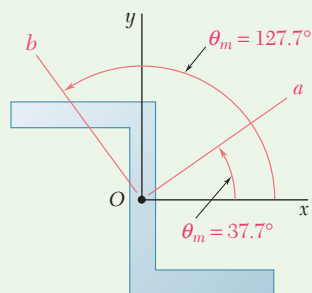


Fig. 2 Orientation of principal axes.

Sample Problem 9.7

For the section shown, the moments of inertia with respect to the x and y axes have been computed and are known to be

$$I_x = 10.38 \text{ in}^4 \quad I_y = 6.97 \text{ in}^4$$

Determine (a) the orientation of the principal axes of the section about O , (b) the values of the principal moments of inertia of the section about O .

STRATEGY: The first step is to compute the product of inertia with respect to the x and y axes, treating the section as a composite area of three rectangles. Then you can use Eq. (9.25) to find the principal axes and Eq. (9.27) to find the principal moments of inertia.

MODELING and ANALYSIS: Divide the area into three rectangles as shown (Fig. 1). Note that the product of inertia $I_{x'y'}$ with respect to centroidal axes parallel to the x and y axes is zero for each rectangle. Thus, using the parallel-axis theorem

$$I_{xy} = I_{x'y'} + \bar{x}\bar{y}A$$

you find that I_{xy} reduces to $\bar{x}\bar{y}A$ for each rectangle.

Rectangle	Area, in ²	$\bar{x}$, in.	$\bar{y}$, in.	$\bar{x}\bar{y}A$, in ⁴
I	1.5	-1.25	+1.75	-3.28
II	1.5	0	0	0
III	1.5	+1.25	-1.75	-3.28
				$\Sigma \bar{x}\bar{y}A = -6.56$

$$I_{xy} = \Sigma \bar{x}\bar{y}A = -6.56 \text{ in}^4$$

a. Principal Axes. Since you know the magnitudes of I_x , I_y , and I_{xy} , you can use Eq. (9.25) to determine the values of θ_m (Fig. 2):

$$\tan 2\theta_m = -\frac{2I_{xy}}{I_x - I_y} = -\frac{2(-6.56)}{10.38 - 6.97} = +3.85$$

$$2\theta_m = 75.4^\circ \text{ and } 255.4^\circ$$

$$\theta_m = 37.7^\circ \text{ and } \theta_m = 127.7^\circ \quad \blacktriangleleft$$

b. Principal Moments of Inertia. Using Eq. (9.27), you have

$$\begin{aligned}
 I_{\max, \min} &= \frac{I_x + I_y}{2} \pm \sqrt{\left(\frac{I_x - I_y}{2}\right)^2 + I_{xy}^2} \\
 &= \frac{10.38 + 6.97}{2} \pm \sqrt{\left(\frac{10.38 - 6.97}{2}\right)^2 + (-6.56)^2} \\
 I_{\max} &= 15.45 \text{ in}^4 \quad I_{\min} = 1.897 \text{ in}^4 \quad \blacktriangleleft
 \end{aligned}$$

REFLECT and THINK: Note that the elements of the area of the section are more closely distributed about the b axis than about the a axis. Therefore, you can conclude that $I_a = I_{\max} = 15.45 \text{ in}^4$ and $I_b = I_{\min} = 1.897 \text{ in}^4$. You can verify this conclusion by substituting $\theta = 37.7^\circ$ into Eqs. (9.18) and (9.19).

SOLVING PROBLEMS ON YOUR OWN

In the problems for this section, you will continue your work with **moments of inertia** and use various techniques for computing **products of inertia**. Although the problems are generally straightforward, several items are worth noting.

1. Calculating the product of inertia I_{xy} by integration. We defined this quantity as

$$I_{xy} = \int xy dA \quad (9.12)$$

and stated that its value can be positive, negative, or zero. You can compute the product of inertia directly from this equation using double integration, or you can find it by using single integration as shown in Sample Prob. 9.6. When applying single integration and using the parallel-axis theorem, it is important to remember that in the equation

$$dI_{xy} = dI_{x' y'} + \bar{x}_{el} \bar{y}_{el} dA$$

$\bar{x}_{el}$ and $\bar{y}_{el}$ are the coordinates of the centroid of the element of area dA . Thus, if dA is not in the first quadrant, one or both of these coordinates is negative.

2. Calculating the products of inertia of composite areas. You can easily compute these quantities from the products of inertia of their component parts by using the parallel-axis theorem, as

$$I_{xy} = \bar{I}_{x' y'} + \bar{x} \bar{y} A \quad (9.13)$$

The proper technique to use for problems of this type is illustrated in Sample Probs. 9.6 and 9.7. In addition to the usual rules for composite-area problems, it is essential that you remember the following points.

a. If either of the centroidal axes of a component area is an axis of symmetry for that area, the product of inertia $\bar{I}_{x' y'}$ for that area is zero. Thus, $\bar{I}_{x' y'}$ is zero for component areas such as circles, semicircles, rectangles, and isosceles triangles, which possess an axis of symmetry parallel to one of the coordinate axes.

b. Pay careful attention to the signs of the coordinates $\bar{x}$ and $\bar{y}$ of each component area when you use the parallel-axis theorem [Sample Prob. 9.7].

3. Determining the moments of inertia and the product of inertia for rotated coordinate axes. In Sec. 9.3B, we derived Eqs. (9.18), (9.19), and (9.20) from which you can compute the moments of inertia and the product of inertia for coordinate axes that have been rotated about the origin O . To apply these equations, you must know a set of values I_x , I_y , and I_{xy} for a given orientation of the axes, and you must remember that θ is positive for counterclockwise rotations of the axes and negative for clockwise rotations of the axes.

4. Computing the principal moments of inertia. We showed in Sec. 9.3B that a particular orientation of the coordinate axes exists for which the moments of inertia attain their maximum and minimum values, $I_{\max}$ and $I_{\min}$, and for which the product of inertia is zero. Equation (9.27) can be used to compute these values that are known as the **principal moments of inertia** of the area about O . The corresponding axes are referred to as the **principal axes** of the area about O , and their orientation is defined by Eq. (9.25). To determine which of the principal axes corresponds to $I_{\max}$ and which corresponds to $I_{\min}$, you can either follow the procedure outlined in the text after Eq. (9.27) or observe about which of the two principal axes the area is more closely distributed; that axis corresponds to $I_{\min}$ [Sample Prob. 9.7].

Problems

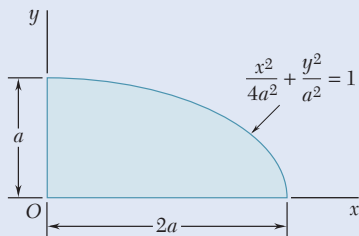


Fig. P9.67

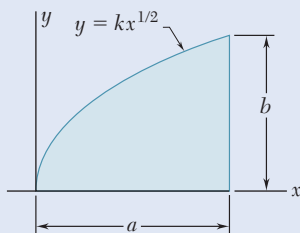


Fig. P9.69

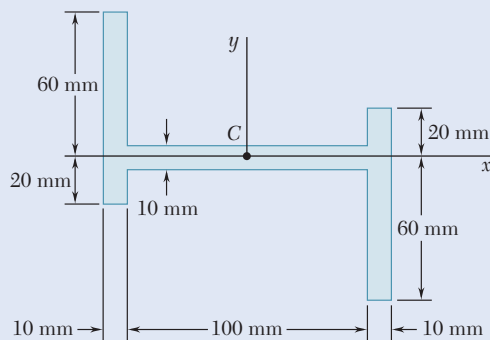


Fig. P9.71

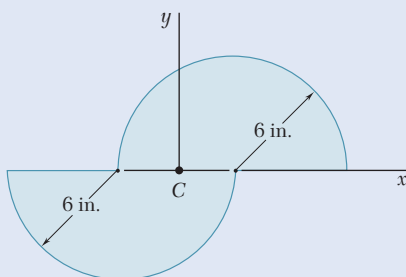


Fig. P9.73

9.67 through 9.70 Determine by direct integration the product of inertia of the given area with respect to the x and y axes.

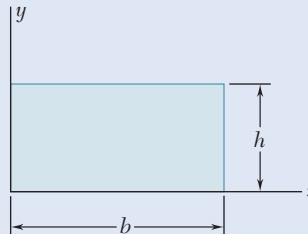


Fig. P9.68

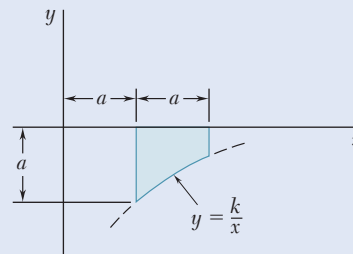


Fig. P9.70

9.71 through 9.74 Using the parallel-axis theorem, determine the product of inertia of the area shown with respect to the centroidal x and y axes.

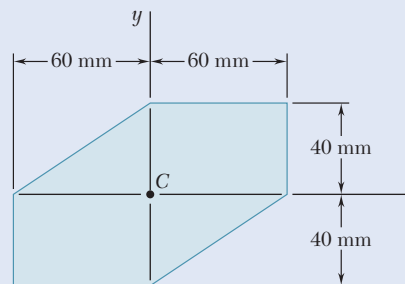


Fig. P9.72

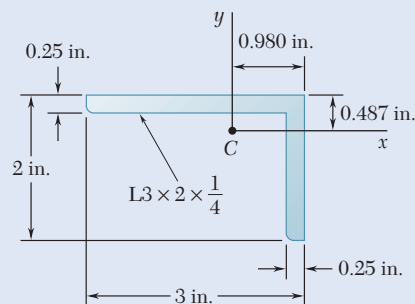


Fig. P9.74

9.75 through 9.78 Using the parallel-axis theorem, determine the product of inertia of the area shown with respect to the centroidal x and y axes.

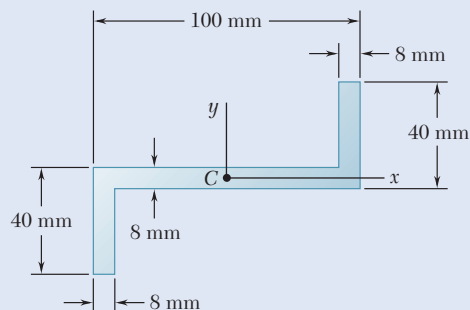


Fig. P9.75

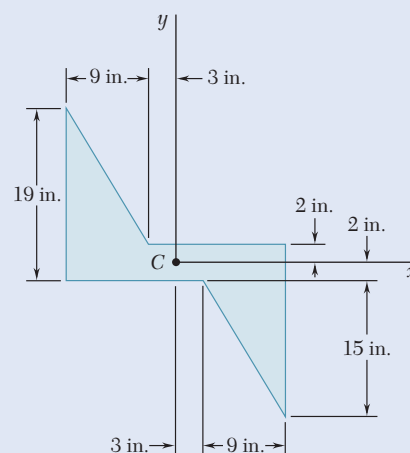


Fig. P9.76

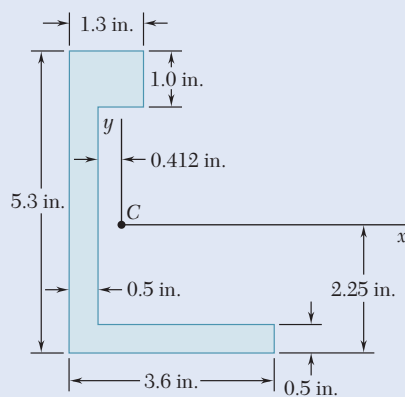


Fig. P9.77

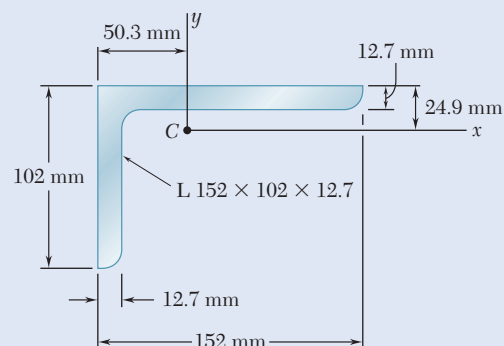


Fig. P9.78

- 9.79** Determine for the quarter ellipse of Prob. 9.67 the moments of inertia and the product of inertia with respect to new axes obtained by rotating the x and y axes about O (a) through 45° counterclockwise, (b) through 30° clockwise.
- 9.80** Determine the moments of inertia and the product of inertia of the area of Prob. 9.72 with respect to new centroidal axes obtained by rotating the x and y axes 30° counterclockwise.
- 9.81** Determine the moments of inertia and the product of inertia of the area of Prob. 9.73 with respect to new centroidal axes obtained by rotating the x and y axes 60° counterclockwise.
- 9.82** Determine the moments of inertia and the product of inertia of the area of Prob. 9.75 with respect to new centroidal axes obtained by rotating the x and y axes 45° clockwise.
- 9.83** Determine the moments of inertia and the product of inertia of the $L3 \times 2 \times \frac{1}{4}$ -in. angle cross section of Prob. 9.74 with respect to new centroidal axes obtained by rotating the x and y axes 30° clockwise.

- 9.84** Determine the moments of inertia and the product of inertia of the $L152 \times 102 \times 12.7$ -mm angle cross section of Prob. 9.78 with respect to new centroidal axes obtained by rotating the x and y axes 30° clockwise.
- 9.85** For the quarter ellipse of Prob. 9.67, determine the orientation of the principal axes at the origin and the corresponding values of the moments of inertia.
- 9.86 through 9.88** For the area indicated, determine the orientation of the principal axes at the origin and the corresponding values of the moments of inertia.
- 9.86** Area of Prob. 9.72
 - 9.87** Area of Prob. 9.73
 - 9.88** Area of Prob. 9.75
- 9.89 and 9.90** For the angle cross section indicated, determine the orientation of the principal axes at the origin and the corresponding values of the moments of inertia.
- 9.89** The $L3 \times 2 \times \frac{1}{4}$ -in. angle cross section of Prob. 9.74
 - 9.90** The $L152 \times 102 \times 12.7$ -mm angle cross section of Prob. 9.78

*9.4 MOHR'S CIRCLE FOR MOMENTS OF INERTIA

The circle introduced in the preceding section to illustrate the relations between the moments and products of inertia of a given area with respect to axes passing through a fixed point O was first introduced by the German engineer Otto Mohr (1835–1918) and is known as **Mohr's circle**. Here, we show that, if we know the moments and product of inertia of an area A with respect to two rectangular x and y axes that pass through a point O , we can use Mohr's circle to graphically determine (a) the principal axes and principal moments of inertia of the area about O and (b) the moments and product of inertia of the area with respect to any other pair of rectangular axes x' and y' through O .

Consider a given area A and two rectangular coordinate axes x and y (Fig. 9.19a). Assuming that we know the moments of inertia I_x and I_y and the product of inertia I_{xy} , we can represent them on a diagram by plotting a point X with coordinates I_x and I_{xy} and a point Y with coordinates I_y and $-I_{xy}$ (Fig. 9.19b). If I_{xy} is positive, as assumed in Fig. 9.19a, then point X is located above the horizontal axis and point Y is located below, as shown in Fig. 9.19b. If I_{xy} is negative, X is located below the horizontal axis and Y is located above. Joining X and Y with a straight line, we denote the point of intersection of line XY with the horizontal axis by C . Then we draw the circle of center C and diameter XY . Noting that the abscissa of C and the radius of the circle are respectively equal to the quantities I_{ave} and R defined by formula (9.23), we conclude that the circle obtained is Mohr's circle for the given area about point O . Thus, the abscissas of the points A and B where the circle intersects the horizontal axis represent, respectively, the principal moments of inertia I_{max} and I_{min} of the area.

Also note that, since $\tan(XCA) = 2I_{xy}/(I_x - I_y)$, the angle XCA is equal in magnitude to one of the angles $2\theta_m$ that satisfy Eq. (9.25). Thus, the angle θ_m , which in Fig. 9.19a defines the principal axis Oa corresponding

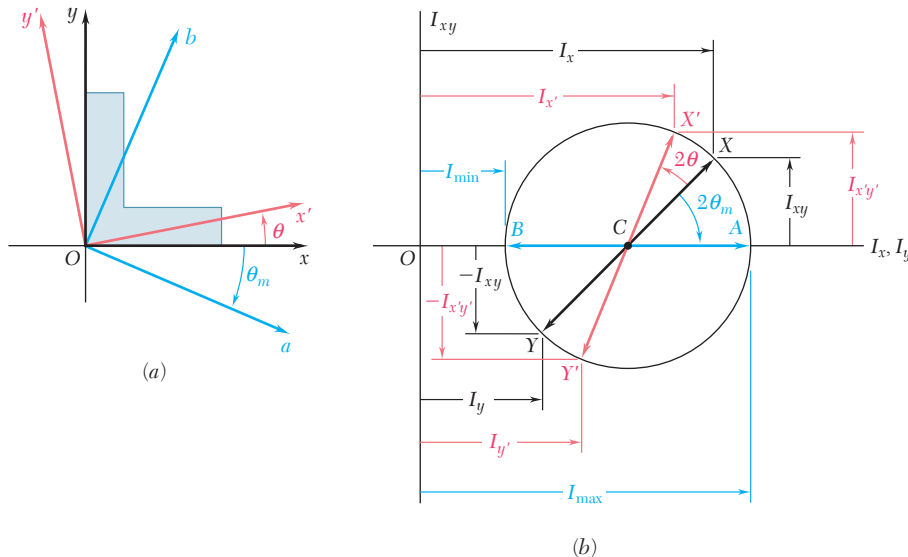
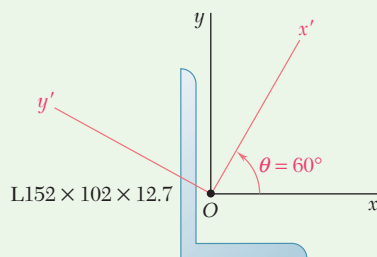


Fig. 9.19 (a) An area A with principal axes Oa and Ob and axes Ox' and Oy' obtained by rotation through an angle θ ; (b) Mohr's circle used to calculate angles and moments of inertia.

to point A in Fig. 9.19b, is equal to half of the angle XCA of Mohr's circle. In addition, if $I_x > I_y$ and $I_{xy} > 0$, as in the case considered here, the rotation that brings CX into CA is clockwise. Also, under these conditions, the angle θ_m obtained from Eq. (9.25) is negative; thus, the rotation that brings Ox into Oa is also clockwise. We conclude that the senses of rotation in both parts of Fig. 9.19 are the same. If a clockwise rotation through $2\theta_m$ is required to bring CX into CA on Mohr's circle, a clockwise rotation through θ_m will bring Ox into the corresponding principal axis Oa in Fig. 9.19a.

Since Mohr's circle is uniquely defined, we can obtain the same circle by considering the moments and product of inertia of the area A with respect to the rectangular axes x' and y' (Fig. 9.19a). The point X' with coordinates $I_{x'}$ and $I_{x'y'}$ and the point Y' with coordinates $I_{y'}$ and $-I_{x'y'}$ are thus located on Mohr's circle, and the angle $X'CA$ in Fig. 9.19b must be equal to twice the angle $x'Oa$ in Fig. 9.19a. Since, as noted previously, the angle XCA is twice the angle xOa , it follows that the angle XCX' in Fig. 9.19b is twice the angle xOx' in Fig. 9.19a. The diameter $X'Y'$, which defines the moments and product of inertia $I_{x'}$, $I_{y'}$, and $I_{x'y'}$ of the given area with respect to rectangular axes x' and y' forming an angle θ with the x and y axes, can be obtained by rotating through an angle 2θ the diameter XY , which corresponds to the moments and product of inertia I_x , I_y , and I_{xy} . Note that the rotation that brings the diameter XY into the diameter $X'Y'$ in Fig. 9.19b has the same sense as the rotation that brings the x and y axes into the x' and y' axes in Fig. 9.19a.

Finally, also note that the use of Mohr's circle is not limited to graphical solutions, i.e., to solutions based on the careful drawing and measuring of the various parameters involved. By merely sketching Mohr's circle and using trigonometry, you can easily derive the various relations required for a numerical solution of a given problem (see Sample Prob. 9.8).



Sample Problem 9.8

For the section shown, the moments and product of inertia with respect to the x and y axes are

$$I_x = 7.20 \times 10^6 \text{ mm}^4 \quad I_y = 2.59 \times 10^6 \text{ mm}^4 \quad I_{xy} = -2.54 \times 10^6 \text{ mm}^4$$

Using Mohr's circle, determine (a) the principal axes of the section about O , (b) the values of the principal moments of inertia of the section about O , and (c) the moments and product of inertia of the section with respect to the x' and y' axes that form an angle of 60° with the x and y axes.

STRATEGY: You should carefully draw Mohr's circle and use the geometry of the circle to determine the orientation of the principal axes. Then complete the analysis for the requested moments of inertia.

MODELING:

Drawing Mohr's Circle. First plot point X with coordinates $I_x = 7.20$, $I_{xy} = -2.54$, and plot point Y with coordinates $I_y = 2.59$, $-I_{xy} = +2.54$. Join X and Y with a straight line to define the center C

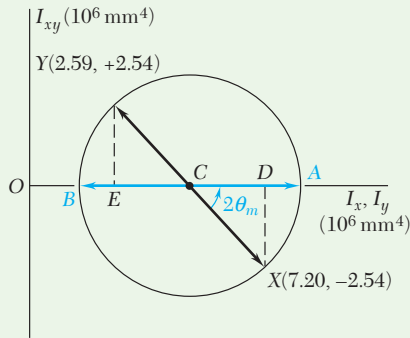


Fig. 1 Mohr's circle.

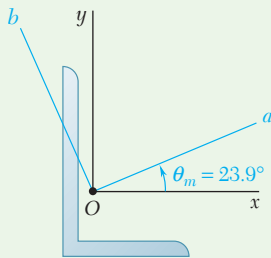


Fig. 2 Orientation of the principal axes.

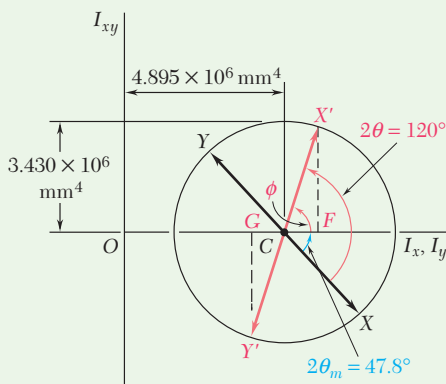


Fig. 3 Using Mohr's circle to determine the moments and product of inertia with respect to x' and y' axes.

of Mohr's circle (Fig. 1). You can measure the abscissa of C , which represents I_{ave} , and the radius R of the circle either directly or using

$$I_{ave} = OC = \frac{1}{2}(I_x + I_y) = \frac{1}{2}(7.20 \times 10^6 + 2.59 \times 10^6) = 4.895 \times 10^6 \text{ mm}^4$$

$$CD = \frac{1}{2}(I_x - I_y) = \frac{1}{2}(7.20 \times 10^6 - 2.59 \times 10^6) = 2.305 \times 10^6 \text{ mm}^4$$

$$R = \sqrt{(CD)^2 + (DX)^2} = \sqrt{(2.305 \times 10^6)^2 + (2.54 \times 10^6)^2} = 3.430 \times 10^6 \text{ mm}^4$$

ANALYSIS:

a. Principal Axes. The principal axes of the section correspond to points A and B on Mohr's circle, and the angle through which you should rotate CX to bring it into CA defines $2\theta_m$. You obtain

$$\tan 2\theta_m = \frac{DX}{CD} = \frac{2.54}{2.305} = 1.102 \quad 2\theta_m = 47.8^\circ \quad \theta_m = 23.9^\circ \quad \blacktriangleleft$$

Thus, the principal axis Oa corresponding to the maximum value of the moment of inertia is obtained by rotating the x axis through 23.9° counterclockwise; the principal axis Ob corresponding to the minimum value of the moment of inertia can be obtained by rotating the y axis through the same angle (Fig. 2).

b. Principal Moments of Inertia. The principal moments of inertia are represented by the abscissas of A and B . The results are

$$I_{max} = OA = OC + CA = I_{ave} + R = (4.895 + 3.430)10^6 \text{ mm}^4$$

$$I_{max} = 8.33 \times 10^6 \text{ mm}^4 \quad \blacktriangleleft$$

$$I_{min} = OB = OC - BC = I_{ave} - R = (4.895 - 3.430)10^6 \text{ mm}^4$$

$$I_{min} = 1.47 \times 10^6 \text{ mm}^4 \quad \blacktriangleleft$$

c. Moments and Product of Inertia with Respect to the x' and y' Axes. On Mohr's circle, you obtain the points X' and Y' , which correspond to the x' and y' axes, by rotating CX and CY through an angle $2\theta = 2(60^\circ) = 120^\circ$ counterclockwise (Fig. 3). The coordinates of X' and Y' yield the desired moments and product of inertia. Noting that the angle that CX' forms with the horizontal axis is $\phi = 120^\circ - 47.8^\circ = 72.2^\circ$, you have

$$I_{x'} = OF = OC + CF = 4.895 \times 10^6 \text{ mm}^4 + (3.430 \times 10^6 \text{ mm}^4) \cos 72.2^\circ$$

$$I_{x'} = 5.94 \times 10^6 \text{ mm}^4 \quad \blacktriangleleft$$

$$I_{y'} = OG = OC - GC = 4.895 \times 10^6 \text{ mm}^4 - (3.430 \times 10^6 \text{ mm}^4) \cos 72.2^\circ$$

$$I_{y'} = 3.85 \times 10^6 \text{ mm}^4 \quad \blacktriangleleft$$

$$I_{x'y'} = FX' = (3.430 \times 10^6 \text{ mm}^4) \sin 72.2^\circ$$

$$I_{x'y'} = 3.27 \times 10^6 \text{ mm}^4 \quad \blacktriangleleft$$

REFLECT and THINK: This problem illustrates typical calculations with Mohr's circle. The technique is a useful one to learn and remember.

SOLVING PROBLEMS ON YOUR OWN

In the problems for this section, you will use **Mohr's circle** to determine the moments and products of inertia of a given area for different orientations of the coordinate axes. Although in some cases using Mohr's circle may not be as direct as substituting into the appropriate equations [Eqs. (9.18) through (9.20)], this method of solution has the advantage of providing a visual representation of the relationships among the variables involved. Also, Mohr's circle shows all of the values of the moments and products of inertia that are possible for a given problem.

Using Mohr's circle. We presented the underlying theory in Sec. 9.3B, and we discussed the application of this method in Sec. 9.4 and in Sample Prob. 9.8. In the same problem, we presented the steps you should follow to determine the **principal axes**, the **principal moments of inertia**, and the **moments and product of inertia with respect to a specified orientation of the coordinates axes**. When you use Mohr's circle to solve problems, it is important that you remember the following points.

a. Mohr's circle is completely defined by the quantities R and I_{ave} , which represent, respectively, the radius of the circle and the distance from the origin O to the center C of the circle. You can obtain these quantities from Eqs. (9.23) if you know the moments and product of inertia for a given orientation of the axes. However, Mohr's circle can be defined by other combinations of known values [Probs. 9.103, 9.106, and 9.107]. For these cases, it may be necessary to first make one or more assumptions, such as choosing an arbitrary location for the center when I_{ave} is unknown, assigning relative magnitudes to the moments of inertia (for example, $I_x > I_y$), or selecting the sign of the product of inertia.

b. Point X of coordinates (I_x, I_{xy}) and point Y of coordinates $(I_y, -I_{xy})$ are both located on Mohr's circle and are diametrically opposite.

c. Since moments of inertia must be positive, all of Mohr's circle must lie to the right of the I_{xy} axis; it follows that $I_{ave} > R$ for all cases.

d. As the coordinate axes are rotated through an angle θ , the associated rotation of the diameter of Mohr's circle is equal to 2θ and is in the same sense (clockwise or counterclockwise). We strongly suggest that you label the known points on the circumference of the circle with the appropriate capital letter, as was done in Fig. 9.19b and for the Mohr circles of Sample Prob. 9.8. This will enable you to determine the sign of the corresponding product of inertia for each value of θ and which moment of inertia is associated with each of the coordinate axes [Sample Prob. 9.8, parts *a* and *c*].

Although we have introduced Mohr's circle within the specific context of the study of moments and products of inertia, the Mohr circle technique also applies to the solution of analogous but physically different problems in mechanics of materials. This multiple use of a specific technique is not unique, and as you pursue your engineering studies, you will encounter several methods of solution that can be applied to a variety of problems.

Problems

- 9.91** Using Mohr's circle, determine for the quarter ellipse of Prob. 9.67 the moments of inertia and the product of inertia with respect to new axes obtained by rotating the x and y axes about O (a) through 45° counterclockwise, (b) through 30° clockwise.
- 9.92** Using Mohr's circle, determine the moments of inertia and the product of inertia of the area of Prob. 9.72 with respect to new centroidal axes obtained by rotating the x and y axes 30° counterclockwise.
- 9.93** Using Mohr's circle, determine the moments of inertia and the product of inertia of the area of Prob. 9.73 with respect to new centroidal axes obtained by rotating the x and y axes 60° counterclockwise.
- 9.94** Using Mohr's circle, determine the moments of inertia and the product of inertia of the area of Prob. 9.75 with respect to new centroidal axes obtained by rotating the x and y axes 45° clockwise.
- 9.95** Using Mohr's circle, determine the moments of inertia and the product of inertia of the $L3 \times 2 \times \frac{1}{4}$ -in. angle cross section of Prob. 9.74 with respect to new centroidal axes obtained by rotating the x and y axes 30° clockwise.
- 9.96** Using Mohr's circle, determine the moments of inertia and the product of inertia of the $L152 \times 102 \times 12.7$ -mm angle cross section of Prob. 9.78 with respect to new centroidal axes obtained by rotating the x and y axes 30° clockwise.
- 9.97** For the quarter ellipse of Prob. 9.67, use Mohr's circle to determine the orientation of the principal axes at the origin and the corresponding values of the moments of inertia.
- 9.98 through 9.102** Using Mohr's circle, determine for the area indicated the orientation of the principal centroidal axes and the corresponding values of the moments of inertia.
- 9.98** Area of Prob. 9.72
 - 9.99** Area of Prob. 9.76
 - 9.100** Area of Prob. 9.73
 - 9.101** Area of Prob. 9.74
 - 9.102** Area of Prob. 9.77

(The moments of inertia $\bar{I}_x$ and $\bar{I}_y$ of the area of Prob. 9.102 were determined in Prob. 9.44)

- 9.103** The moments and product of inertia of an $L4 \times 3 \times \frac{1}{4}$ -in. angle cross section with respect to two rectangular axes x and y through C are, respectively, $\bar{I}_x = 1.33 \text{ in}^4$, $\bar{I}_y = 2.75 \text{ in}^4$, and $\bar{I}_{xy} < 0$, with the minimum value of the moment of inertia of the area with respect to any axis through C being $\bar{I}_{\min} = 0.692 \text{ in}^4$. Using Mohr's circle, determine (a) the product of inertia $\bar{I}_{xy}$ of the area, (b) the orientation of the principal axes, (c) the value of $\bar{I}_{\max}$.

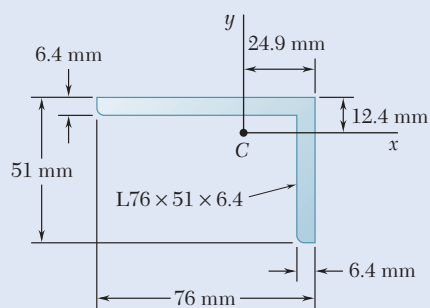


Fig. P9.104

9.104 and 9.105 Using Mohr's circle, determine the orientation of the principal centroidal axes and the corresponding values of the moments of inertia for the cross section of the rolled-steel angle shown. (Properties of the cross sections are given in Fig. 9.13.)

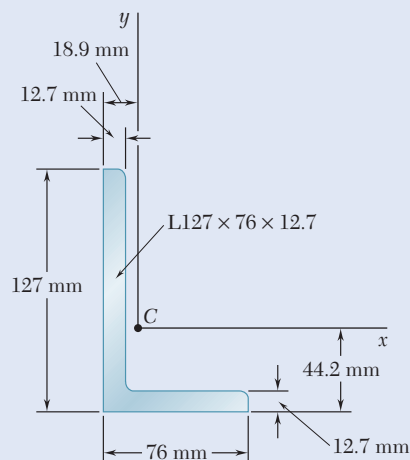


Fig. P9.105

- *9.106** For a given area, the moments of inertia with respect to two rectangular centroidal x and y axes are $\bar{I}_x = 1200 \text{ in}^4$ and $\bar{I}_y = 300 \text{ in}^4$, respectively. Knowing that, after rotating the x and y axes about the centroid 30° counterclockwise, the moment of inertia relative to the rotated x axis is 1450 in^4 , use Mohr's circle to determine (a) the orientation of the principal axes, (b) the principal centroidal moments of inertia.
- 9.107** It is known that for a given area $\bar{I}_y = 48 \times 10^6 \text{ mm}^4$ and $\bar{I}_{xy} = -20 \times 10^6 \text{ mm}^4$, where the x and y axes are rectangular centroidal axes. If the axis corresponding to the maximum product of inertia is obtained by rotating the x axis 67.5° counterclockwise about C , use Mohr's circle to determine (a) the moment of inertia $\bar{I}_x$ of the area, (b) the principal centroidal moments of inertia.
- 9.108** Using Mohr's circle, show that for any regular polygon (such as a pentagon) (a) the moment of inertia with respect to every axis through the centroid is the same, (b) the product of inertia with respect to every pair of rectangular axes through the centroid is zero.
- 9.109** Using Mohr's circle, prove that the expression $I_x I_{y'} - I_{x'y'}^2$ is independent of the orientation of the x' and y' axes, where I_x , $I_{y'}$, and $I_{x'y'}$ represent the moments and product of inertia, respectively, of a given area with respect to a pair of rectangular axes x' and y' through a given point O . Also show that the given expression is equal to the square of the length of the tangent drawn from the origin of the coordinate system to Mohr's circle.
- 9.110** Using the invariance property established in the preceding problem, express the product of inertia I_{xy} of an area A with respect to a pair of rectangular axes through O in terms of the moments of inertia I_x and I_y of A and the principal moments of inertia $I_{\min}$ and $I_{\max}$ of A about O . Use the formula obtained to calculate the product of inertia I_{xy} of the $L3 \times 2 \times \frac{1}{4}$ -in. angle cross section shown in Fig. 9.13A, knowing that its maximum moment of inertia is 1.257 in^4 .

9.5 MASS MOMENTS OF INERTIA

So far in this chapter, we have examined moments of inertia of areas. In the rest of this chapter, we consider moments of inertia associated with the masses of bodies. This will be an important concept in dynamics when studying the rotational motion of a rigid body about an axis.

9.5A Moment of Inertia of a Simple Mass

Consider a small mass Δm mounted on a rod of negligible mass that can rotate freely about an axis AA' (Fig. 9.20a). If we apply a couple to the system, the rod and mass (assumed to be initially at rest) start rotating about AA' . We will study the details of this motion later in dynamics. At present, we wish to indicate only that the time required for the system to reach a given speed of rotation is proportional to the mass Δm and to the square of the distance r . The product $r^2 \Delta m$ thus provides a measure of the **inertia** of the system; i.e., a measure of the resistance the system offers when we try to set it in motion. For this reason, the product $r^2 \Delta m$ is called the **moment of inertia** of the mass Δm with respect to axis AA' .

Now suppose a body of mass m is to be rotated about an axis AA' (Fig. 9.20b). Dividing the body into elements of mass Δm_1 , Δm_2 , etc., we find that the body's resistance to being rotated is measured by the sum $r_1^2 \Delta m_1 + r_2^2 \Delta m_2 + \dots$. This sum defines the moment of inertia of the body with respect to axis AA' . Increasing the number of elements, we find that the moment of inertia is equal, in the limit, to the integral

$$I = \int r^2 dm \quad (9.28)$$

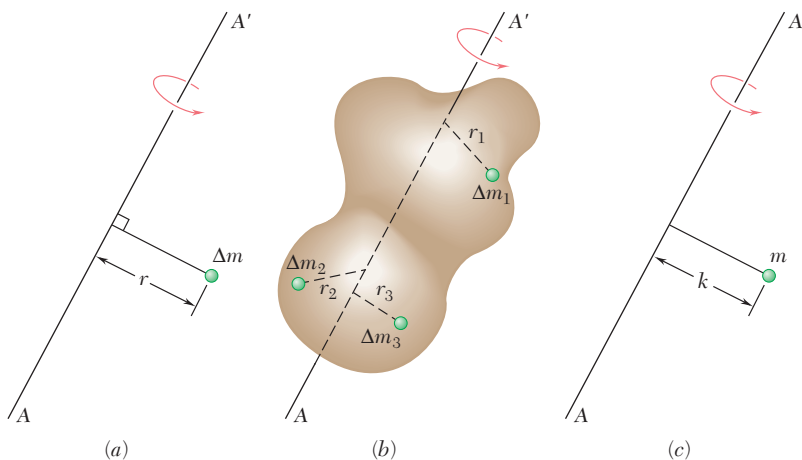


Fig. 9.20 (a) An element of mass Δm at a distance r from an axis AA' ; (b) the moment of inertia of a rigid body is the sum of the moments of inertia of many small masses; (c) the moment of inertia is unchanged if all the mass is concentrated at a point at a distance from the axis equal to the radius of gyration.

We define the **radius of gyration** k of the body with respect to axis AA' by the relation

**Radius of gyration
of a mass**

$$I = k^2 m \quad \text{or} \quad k = \sqrt{\frac{I}{m}} \quad (9.29)$$

The radius of gyration k represents the distance at which the entire mass of the body should be concentrated if its moment of inertia with respect to AA' is to remain unchanged (Fig. 9.20c). Whether it stays in its original shape (Fig. 9.20b) or is concentrated as shown in Fig. 9.20c, the mass m reacts in the same way to a rotation (or *gyration*) about AA' .

If SI units are used, the radius of gyration k is expressed in meters and the mass m in kilograms, so the unit for the moment of inertia of a mass is $\text{kg}\cdot\text{m}^2$. If U.S. customary units are used, the radius of gyration is expressed in feet and the mass in slugs (i.e., in $\text{lb}\cdot\text{s}^2/\text{ft}$), so the derived unit for the moment of inertia of a mass is $\text{lb}\cdot\text{ft}\cdot\text{s}^2$.[†]

We can express the moment of inertia of a body with respect to a coordinate axis in terms of the coordinates x, y, z of the element of mass dm (Fig. 9.21). Noting, for example, that the square of the distance r from the element dm to the y axis is $z^2 + x^2$, the moment of inertia of the body with respect to the y axis is

$$I_y = \int r^2 dm = \int (z^2 + x^2) dm$$

We obtain similar expressions for the moments of inertia with respect to the x and z axes.

**Moments of inertia with
respect to coordinate axes**

$$\begin{aligned} I_x &= \int (y^2 + z^2) dm \\ I_y &= \int (z^2 + x^2) dm \\ I_z &= \int (x^2 + y^2) dm \end{aligned} \quad (9.30)$$

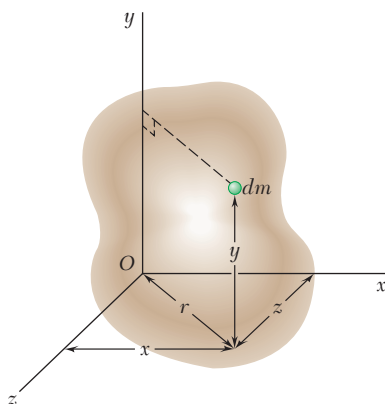


Fig. 9.21 An element of mass dm in an x, y, z coordinate system.



Photo 9.2 The rotational behavior of this crankshaft depends upon its mass moment of inertia with respect to its axis of rotation, as you will see in a dynamics course.

9.5B Parallel-Axis Theorem for Mass Moments of Inertia

Consider again a body of mass m and let $Oxyz$ be a system of rectangular coordinates whose origin is at the arbitrary point O . Let $Gx'y'z'$ be a system of parallel centroidal axes; i.e., a system whose origin is at the

[†]When converting the moment of inertia of a mass from U.S. customary units to SI units, keep in mind that the base unit (pound) used in the derived unit ($\text{lb}\cdot\text{ft}\cdot\text{s}^2$) is a unit of force (not of mass). Therefore, it should be converted into newtons. We have

$$1 \text{ lb}\cdot\text{ft}\cdot\text{s}^2 = (4.45 \text{ N})(0.3048 \text{ m})(1 \text{ s})^2 = 1.356 \text{ N}\cdot\text{m}\cdot\text{s}^2$$

or since $1 \text{ N} = 1 \text{ kg}\cdot\text{m}/\text{s}^2$

$$1 \text{ lb}\cdot\text{ft}\cdot\text{s}^2 = 1.356 \text{ kg}\cdot\text{m}^2$$

center of gravity G of the body and whose axes x' , y' , z' are parallel to the x , y , and z axes, respectively (Fig. 9.22). (Note that we use the term centroidal here to define axes passing through the center of gravity G of the body, regardless of whether or not G coincides with the centroid of the volume of the body.) We denote by $\bar{x}$, $\bar{y}$, $\bar{z}$ the coordinates of G with respect to $Oxyz$. Then we have the following relations between the coordinates x , y , z of the element dm with respect to $Oxyz$ and its coordinates x' , y' , z' with respect to the centroidal axes $Gx'y'z'$:

$$x = x' + \bar{x} \quad y = y' + \bar{y} \quad z = z' + \bar{z} \quad (9.31)$$

Referring to Eqs. (9.30), we can express the moment of inertia of the body with respect to the x axis as

$$\begin{aligned} I_x &= \int (y^2 + z^2) dm = \int [(y' + \bar{y})^2 + (z' + \bar{z})^2] dm \\ &= \int (y'^2 + z'^2) dm + 2\bar{y} \int y' dm + 2\bar{z} \int z' dm + (\bar{y}^2 + \bar{z}^2) \int dm \end{aligned}$$

The first integral in this expression represents the moment of inertia $\bar{I}_{x'}$ of the body with respect to the centroidal axis x' . The second and third integrals represent the first moment of the body with respect to the $z'x'$ and $x'y'$ planes, respectively, and since both planes contain G , these two integrals are zero. The last integral is equal to the total mass m of the body. Therefore, we have

$$I_x = \bar{I}_{x'} + m(\bar{y}^2 + \bar{z}^2) \quad (9.32)$$

Similarly,

$$I_y = \bar{I}_{y'} + m(\bar{z}^2 + \bar{x}^2) \quad I_z = \bar{I}_{z'} + m(\bar{x}^2 + \bar{y}^2) \quad (9.32')$$

We easily verify from Fig. 9.22 that the sum $\bar{z}^2 + \bar{x}^2$ represents the square of the distance OB between the y and y' axes. Similarly, $\bar{y}^2 + \bar{z}^2$ and $\bar{x}^2 + \bar{y}^2$ represent the squares of the distance between the x and x' axes and the z and z' axes, respectively. We denote the distance between an arbitrary axis AA' and a parallel centroidal axis BB' by d (Fig. 9.23). Then the general relation between the moment of inertia I of the body with respect to AA' and its moment of inertia $\bar{I}$ with respect to BB' , known as the parallel-axis theorem for mass moments of inertia, is

Parallel-axis theorem for mass moments of inertia

$$I = \bar{I} + md^2 \quad (9.33)$$

Expressing the moments of inertia in terms of the corresponding radii of gyration, we can also write

$$k^2 = \bar{k}^2 + d^2 \quad (9.34)$$

where k and $\bar{k}$ represent the radii of gyration of the body about AA' and BB' , respectively.

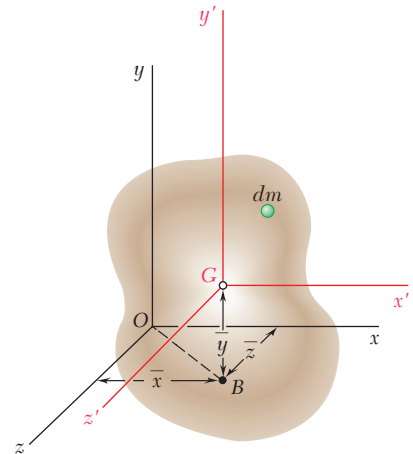


Fig. 9.22 A body of mass m with an arbitrary rectangular coordinate system at O and a parallel centroidal coordinate system at G . Also shown is an element of mass dm .

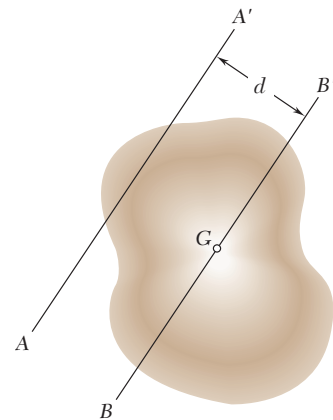


Fig. 9.23 We use d to denote the distance between an arbitrary axis AA' and a parallel centroidal axis BB' .

9.5C Moments of Inertia of Thin Plates

Now imagine a thin plate of uniform thickness t , made of a homogeneous material of density ρ (density = mass per unit volume). The mass moment of inertia of the plate with respect to an axis AA' contained in the plane of the plate (Fig. 9.24a) is

$$I_{AA', \text{ mass}} = \int r^2 dm$$

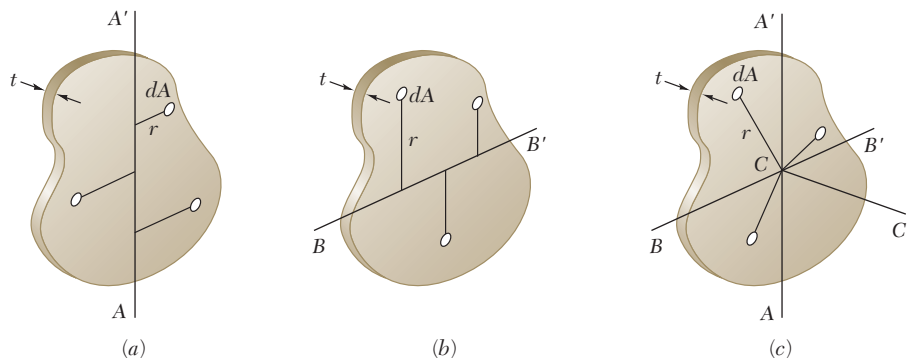


Fig. 9.24 (a) A thin plate with an axis AA' in the plane of the plate; (b) an axis BB' in the plane of the plate and perpendicular to AA' ; (c) an axis CC' perpendicular to the plate and passing through the intersection of AA' and BB' .

Since $dm = \rho t dA$, we have

$$I_{AA', \text{ mass}} = \rho t \int r^2 dA$$

However, r represents the distance of the element of area dA to the axis AA' . Therefore, the integral is equal to the moment of inertia of the area of the plate with respect to AA' .

$$I_{AA', \text{ mass}} = \rho t I_{AA', \text{ area}} \quad (9.35)$$

Similarly, for an axis BB' that is contained in the plane of the plate and is perpendicular to AA' (Fig. 9.24b), we have

$$I_{BB', \text{ mass}} = \rho t I_{BB', \text{ area}} \quad (9.36)$$

Consider now the axis CC' , which is *perpendicular* to the plate and passes through the point of intersection C of AA' and BB' (Fig. 9.24c). This time we have

$$I_{CC', \text{ mass}} = \rho t J_{C, \text{ area}} \quad (9.37)$$

where J_C is the polar moment of inertia of the area of the plate with respect to point C .

Recall the relation $J_C = I_{AA'} + I_{BB'}$ between the polar and rectangular moments of inertia of an area. We can use this to write the relation between the mass moments of inertia of a thin plate as

$$I_{CC'} = I_{AA'} + I_{BB'} \quad (9.38)$$

Rectangular Plate. In the case of a rectangular plate of sides a and b (Fig. 9.25), we obtain the mass moments of inertia with respect to axes through the center of gravity of the plate as

$$I_{AA', \text{ mass}} = \rho t I_{AA', \text{ area}} = \rho t \left(\frac{1}{12} a^3 b \right)$$

$$I_{BB', \text{ mass}} = \rho t I_{BB', \text{ area}} = \rho t \left(\frac{1}{12} a b^3 \right)$$

Since the product $\rho a b t$ is equal to the mass m of the plate, we can also write the mass moments of inertia of a thin rectangular plate as

$$I_{AA'} = \frac{1}{12} m a^2 \quad I_{BB'} = \frac{1}{12} m b^2 \quad (9.39)$$

$$I_{CC'} = I_{AA'} + I_{BB'} = \frac{1}{12} m (a^2 + b^2) \quad (9.40)$$

Circular Plate. In the case of a circular plate, or disk, of radius r (Fig. 9.26), Eq. (9.35) becomes

$$I_{AA', \text{ mass}} = \rho t I_{AA', \text{ area}} = \rho t \left(\frac{1}{4} \pi r^4 \right)$$

In this case, the product $\rho \pi r^2 t$ is equal to the mass m of the plate, and $I_{AA'} = I_{BB'}$. Therefore, we can write the mass moments of inertia of a circular plate as

$$I_{AA'} = I_{BB'} = \frac{1}{4} m r^2 \quad (9.41)$$

$$I_{CC'} = I_{AA'} + I_{BB'} = \frac{1}{2} m r^2 \quad (9.42)$$

9.5D Determining the Moment of Inertia of a Three-Dimensional Body by Integration

We obtain the moment of inertia of a three-dimensional body by evaluating the integral $I = \int r^2 dm$. If the body is made of a homogeneous material with a density ρ , the element of mass dm is equal to ρdV , and we have $I = \rho \int r^2 dV$. This integral depends only upon the shape of the body. Thus, in order to compute the moment of inertia of a three-dimensional body, it is generally necessary to perform a triple, or at least a double, integration.

However, if the body possesses two planes of symmetry, it is usually possible to determine the body's moment of inertia with a single integration. We do this by choosing as the element of mass dm a thin slab that is perpendicular to the planes of symmetry. In the case of bodies of revolution, for example, the element of mass is a thin disk (Fig. 9.27). Using formula (9.42), we can express the moment of inertia of the disk with respect to the axis of revolution as indicated in Fig. 9.27. Its moment of inertia with respect to each of the other two coordinate axes is obtained by using formula (9.41) and the parallel-axis theorem. Integration of these expressions yields the desired moment of inertia of the body.

9.5E Moments of Inertia of Composite Bodies

Figure 9.28 lists the moments of inertia of a few common shapes. For a body consisting of several of these simple shapes in combination, you can obtain the moment of inertia of the body with respect to a given axis by first computing the moments of inertia of its component parts about the desired axis and then adding them together. As was the case for areas, the radius of gyration of a composite body *cannot* be obtained by adding the radii of gyration of its component parts.

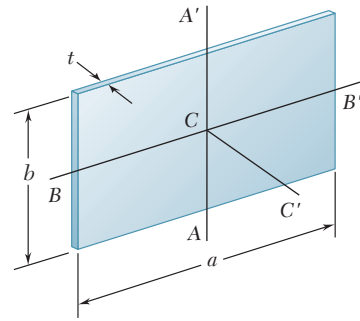


Fig. 9.25 A thin rectangular plate of sides a and b .

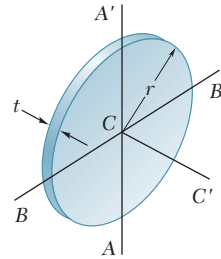
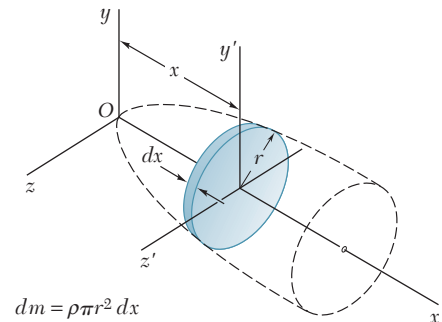


Fig. 9.26 A thin circular plate of radius r .



$$dm = \rho \pi r^2 dx$$

$$dI_x = \frac{1}{2} r^2 dm$$

$$dI_y = dI_{y'} + x^2 dm = \left(\frac{1}{4} r^2 + x^2 \right) dm$$

$$dI_z = dI_{z'} + x^2 dm = \left(\frac{1}{4} r^2 + x^2 \right) dm$$

Fig. 9.27 Using a thin disk to determine the moment of inertia of a body of revolution.

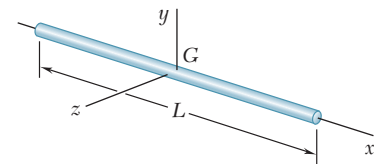
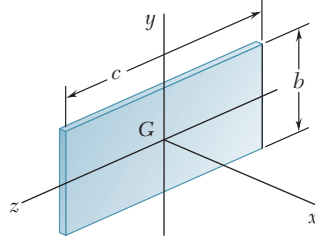
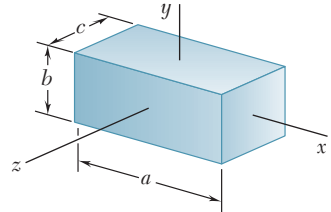
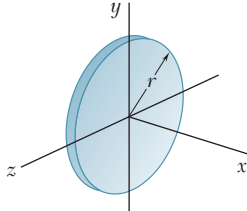
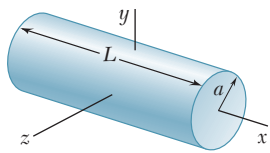
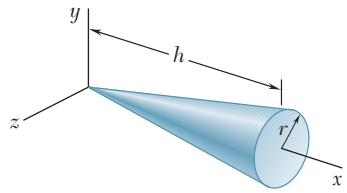
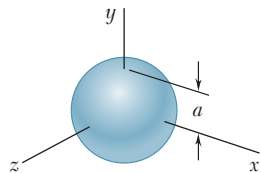
Slender rod		$I_y = I_z = \frac{1}{12} mL^2$
Thin rectangular plate		$I_x = \frac{1}{12} m(b^2 + c^2)$ $I_y = \frac{1}{12} mc^2$ $I_z = \frac{1}{12} mb^2$
Rectangular prism		$I_x = \frac{1}{12} m(b^2 + c^2)$ $I_y = \frac{1}{12} m(c^2 + a^2)$ $I_z = \frac{1}{12} m(a^2 + b^2)$
Thin disk		$I_x = \frac{1}{2} mr^2$ $I_y = I_z = \frac{1}{4} mr^2$
Circular cylinder		$I_x = \frac{1}{2} ma^2$ $I_y = I_z = \frac{1}{12} m(3a^2 + L^2)$
Circular cone		$I_x = \frac{3}{10} ma^2$ $I_y = I_z = \frac{3}{5} m(\frac{1}{4} a^2 + h^2)$
Sphere		$I_x = I_y = I_z = \frac{2}{5} ma^2$

Fig. 9.28 Mass moments of inertia of common geometric shapes.

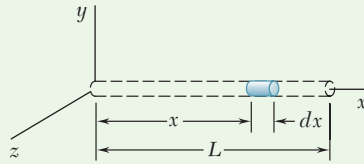
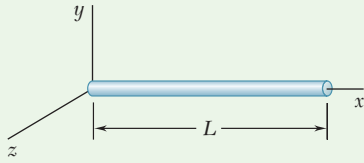


Fig. 1 Differential element of mass.

Sample Problem 9.9

Determine the moment of inertia of a slender rod of length L and mass m with respect to an axis that is perpendicular to the rod and passes through one end.

STRATEGY: Approximating the rod as a one-dimensional body enables you to solve the problem by a single integration.

MODELING and ANALYSIS: Choose the differential element of mass shown in Fig. 1 and express it as a mass per unit length.

$$dm = \frac{m}{L} dx$$

$$I_y = \int x^2 dm = \int_0^L x^2 \frac{m}{L} dx = \left[\frac{m}{L} \frac{x^3}{3} \right]_0^L \quad I_y = \frac{1}{3} mL^2 \quad \blacktriangleleft$$

REFLECT and THINK: This problem could also have been solved by starting with the moment of inertia for a slender rod with respect to its centroid, as given in Fig. 9.28, and using the parallel-axis theorem to obtain the moment of inertia with respect to an end of the rod.

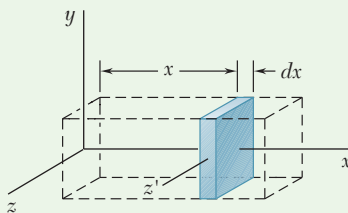
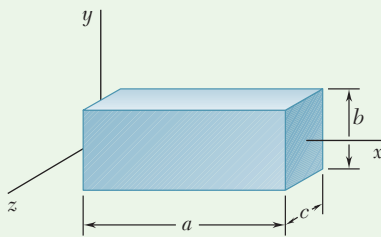


Fig. 1 Differential element of mass.

Sample Problem 9.10

For the homogeneous rectangular prism shown, determine the moment of inertia with respect to the z axis.

STRATEGY: You can approach this problem by choosing a differential element of mass perpendicular to the long axis of the prism; find its moment of inertia with respect to a centroidal axis parallel to the z axis; and then apply the parallel-axis theorem.

MODELING and ANALYSIS: Choose as the differential element of mass the thin slab shown in Fig. 1. Then

$$dm = \rho b c dx$$

Referring to Sec. 9.5C, the moment of inertia of the element with respect to the z' axis is

$$dI_{z'} = \frac{1}{12} b^2 dm$$

Applying the parallel-axis theorem, you can obtain the mass moment of inertia of the slab with respect to the z axis.

$$dI_z = dI_{z'} + x^2 dm = \frac{1}{12} b^2 dm + x^2 dm = \left(\frac{1}{12} b^2 + x^2 \right) \rho b c dx$$

Integrating from $x = 0$ to $x = a$ gives you

$$I_z = \int dI_z = \int_0^a \left(\frac{1}{12} b^2 + x^2 \right) \rho b c dx = \rho a b c \left(\frac{1}{12} b^2 + \frac{1}{3} a^2 \right)$$

Since the total mass of the prism is $m = \rho a b c$, you can write

$$I_z = m \left(\frac{1}{12} b^2 + \frac{1}{3} a^2 \right) \quad I_z = \frac{1}{12} m (4a^2 + b^2) \quad \blacktriangleleft$$

REFLECT and THINK: Note that if the prism is thin, b is small compared to a , and the expression for I_z reduces to $\frac{1}{3} m a^2$, which is the result obtained in Sample Prob. 9.9 when $L = a$.

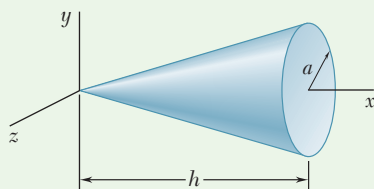


Fig. 1 Differential element of mass.

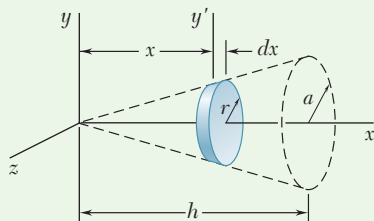


Fig. 2 Centroid of a right circular cone.

Sample Problem 9.11

Determine the moment of inertia of a right circular cone with respect to (a) its longitudinal axis, (b) an axis through the apex of the cone and perpendicular to its longitudinal axis, (c) an axis through the centroid of the cone and perpendicular to its longitudinal axis.

STRATEGY: For parts (a) and (b), choose a differential element of mass in the form of a thin circular disk perpendicular to the longitudinal axis of the cone. You can solve part (c) by an application of the parallel-axis theorem.

MODELING and ANALYSIS: Choose the differential element of mass shown in Fig. 1. Express the radius and mass of this disk as

$$r = a \frac{x}{h} \quad dm = \rho \pi r^2 dx = \rho \pi \frac{a^2}{h^2} x^2 dx$$

a. Moment of Inertia I_x . Using the expression derived in Sec. 9.5C for a thin disk, compute the mass moment of inertia of the differential element with respect to the x axis.

$$dI_x = \frac{1}{2} r^2 dm = \frac{1}{2} \left(a \frac{x}{h} \right)^2 \left(\rho \pi \frac{a^2}{h^2} x^2 dx \right) = \frac{1}{2} \rho \pi \frac{a^4}{h^4} x^4 dx$$

Integrating from $x = 0$ to $x = h$ gives you

$$I_x = \int dI_x = \int_0^h \frac{1}{2} \rho \pi \frac{a^4}{h^4} x^4 dx = \frac{1}{2} \rho \pi \frac{a^4}{h^4} \frac{h^5}{5} = \frac{1}{10} \rho \pi a^4 h$$

Since the total mass of the cone is $m = \frac{1}{3} \rho \pi a^2 h$, you can write this as

$$I_x = \frac{1}{10} \rho \pi a^4 h = \frac{3}{10} a^2 \left(\frac{1}{3} \rho \pi a^2 h \right) = \frac{3}{10} m a^2 \quad I_x = \frac{3}{10} m a^2 \quad \blacktriangleleft$$

b. Moment of Inertia I_y . Use the same differential element. Applying the parallel-axis theorem and using the expression derived in Sec. 9.5C for a thin disk, you have

$$dI_y = dI_{y'} + x^2 dm = \frac{1}{4} r^2 dm + x^2 dm = \left(\frac{1}{4} r^2 + x^2 \right) dm$$

Substituting the expressions for r and dm into this equation yields

$$dI_y = \left(\frac{1}{4} \frac{a^2}{h^2} x^2 + x^2 \right) \left(\rho \pi \frac{a^2}{h^2} x^2 dx \right) = \rho \pi \frac{a^2}{h^2} \left(\frac{a^2}{4h^2} + 1 \right) x^4 dx$$

$$I_y = \int dI_y = \int_0^h \rho \pi \frac{a^2}{h^2} \left(\frac{a^2}{4h^2} + 1 \right) x^4 dx = \rho \pi \frac{a^2}{h^2} \left(\frac{a^2}{4h^2} + 1 \right) \frac{h^5}{5}$$

Introducing the total mass of the cone m , you can rewrite I_y as

$$I_y = \frac{3}{5} \left(\frac{1}{4} a^2 + h^2 \right) \frac{1}{3} \rho \pi a^2 h \quad I_y = \frac{3}{5} m \left(\frac{1}{4} a^2 + h^2 \right) \quad \blacktriangleleft$$

c. Moment of Inertia $\bar{I}_{y''}$. Apply the parallel-axis theorem to obtain

$$I_y = \bar{I}_{y''} + m \bar{x}^2$$

Solve for $\bar{I}_{y''}$ and recall from Fig. 5.21 that $\bar{x} = \frac{3}{4} h$ (Fig. 2). The result is

$$\bar{I}_{y''} = I_y - m \bar{x}^2 = \frac{3}{5} m \left(\frac{1}{4} a^2 + h^2 \right) - m \left(\frac{3}{4} h \right)^2$$

$$\bar{I}_{y''} = \frac{3}{20} m \left(a^2 + \frac{1}{4} h^2 \right) \quad \blacktriangleleft$$

REFLECT and THINK: The parallel-axis theorem for masses can be just as useful as the version for areas. Don't forget to use the reference figures for centroids of volumes when needed.

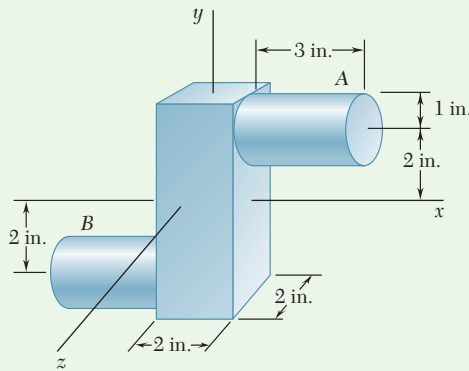


Fig. 1 Geometry of each component.

Sample Problem 9.12

A steel forging consists of a $6 \times 2 \times 2$ -in. rectangular prism and two cylinders with a diameter of 2 in. and length of 3 in. as shown. Determine the moments of inertia of the forging with respect to the coordinate axes. The specific weight of steel is 490 lb/ft^3 .

STRATEGY: Compute the moments of inertia of each component from Fig. 9.28 using the parallel-axis theorem when necessary. Note that all lengths should be expressed in feet to be consistent with the units for the given specific weight.

MODELING and ANALYSIS:

Computation of Masses.

Prism

$$\begin{aligned} V &= (2 \text{ in.})(2 \text{ in.})(6 \text{ in.}) = 24 \text{ in}^3 \\ W &= \frac{(24 \text{ in}^3)(490 \text{ lb/ft}^3)}{1728 \text{ in}^3/\text{ft}^3} = 6.81 \text{ lb} \\ m &= \frac{6.81 \text{ lb}}{32.2 \text{ ft/s}^2} = 0.211 \text{ lb}\cdot\text{s}^2/\text{ft} \end{aligned}$$

Each Cylinder

$$\begin{aligned} V &= \pi(1 \text{ in.})^2(3 \text{ in.}) = 9.42 \text{ in}^3 \\ W &= \frac{(9.42 \text{ in}^3)(490 \text{ lb/ft}^3)}{1728 \text{ in}^3/\text{ft}^3} = 2.67 \text{ lb} \\ m &= \frac{2.67 \text{ lb}}{32.2 \text{ ft/s}^2} = 0.0829 \text{ lb}\cdot\text{s}^2/\text{ft} \end{aligned}$$

Moments of Inertia (Fig. 1).

Prism

$$\begin{aligned} I_x = I_z &= \frac{1}{12}(0.211 \text{ lb}\cdot\text{s}^2/\text{ft})[(\frac{6}{12} \text{ ft})^2 + (\frac{2}{12} \text{ ft})^2] = 4.88 \times 10^{-3} \text{ lb}\cdot\text{ft}\cdot\text{s}^2 \\ I_y &= \frac{1}{12}(0.211 \text{ lb}\cdot\text{s}^2/\text{ft})[(\frac{2}{12} \text{ ft})^2 + (\frac{2}{12} \text{ ft})^2] = 0.977 \times 10^{-3} \text{ lb}\cdot\text{ft}\cdot\text{s}^2 \end{aligned}$$

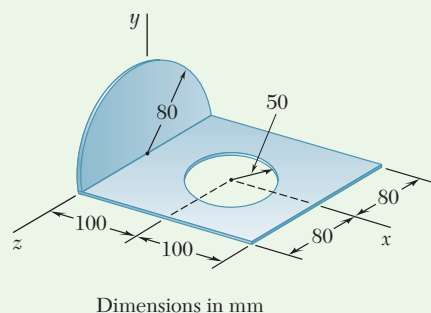
Each Cylinder

$$\begin{aligned} I_x &= \frac{1}{2}ma^2 + m\bar{y}^2 = \frac{1}{2}(0.0829 \text{ lb}\cdot\text{s}^2/\text{ft})(\frac{1}{12} \text{ ft})^2 \\ &\quad + (0.0829 \text{ lb}\cdot\text{s}^2/\text{ft})(\frac{2}{12} \text{ ft})^2 = 2.59 \times 10^{-3} \text{ lb}\cdot\text{ft}\cdot\text{s}^2 \\ I_y &= \frac{1}{12}m(3a^2 + L^2) = m\bar{x}^2 = \frac{1}{12}(0.0829 \text{ lb}\cdot\text{s}^2/\text{ft})[3(\frac{1}{12} \text{ ft})^2 + (\frac{3}{12} \text{ ft})^2] \\ &\quad + (0.0829 \text{ lb}\cdot\text{s}^2/\text{ft})(\frac{2.5}{12} \text{ ft})^2 = 4.17 \times 10^{-3} \text{ lb}\cdot\text{ft}\cdot\text{s}^2 \\ I_z &= \frac{1}{12}m(3a^2 + L^2) + m(\bar{x}^2 + \bar{y}^2) = \frac{1}{12}(0.0829 \text{ lb}\cdot\text{s}^2/\text{ft})[3(\frac{1}{12} \text{ ft})^2 + (\frac{3}{12} \text{ ft})^2] \\ &\quad + (0.0829 \text{ lb}\cdot\text{s}^2/\text{ft})[(\frac{2.5}{12} \text{ ft})^2 + (\frac{2}{12} \text{ ft})^2] = 6.48 \times 10^{-3} \text{ lb}\cdot\text{ft}\cdot\text{s}^2 \end{aligned}$$

Entire Body. Adding the values obtained for the prism and two cylinders, you have

$$\begin{aligned} I_x &= 4.88 \times 10^{-3} + 2(2.59 \times 10^{-3}) & I_x &= 10.06 \times 10^{-3} \text{ lb}\cdot\text{ft}\cdot\text{s}^2 \quad \blacktriangleleft \\ I_y &= 0.977 \times 10^{-3} + 2(4.17 \times 10^{-3}) & I_y &= 9.32 \times 10^{-3} \text{ lb}\cdot\text{ft}\cdot\text{s}^2 \quad \blacktriangleleft \\ I_z &= 4.88 \times 10^{-3} + 2(6.48 \times 10^{-3}) & I_z &= 17.84 \times 10^{-3} \text{ lb}\cdot\text{ft}\cdot\text{s}^2 \quad \blacktriangleleft \end{aligned}$$

REFLECT and THINK: The results indicate this forging has more resistance to rotation about the z axis (largest moment of inertia) than about the x or y axes. This makes intuitive sense, because more of the mass is farther from the z axis than from the x or y axes.



Sample Problem 9.13

A thin steel plate that is 4 mm thick is cut and bent to form the machine part shown. The density of the steel is 7850 kg/m^3 . Determine the moments of inertia of the machine part with respect to the coordinate axes.

STRATEGY: The machine part consists of a semicircular plate and a rectangular plate from which a circular plate has been removed (Fig. 1). After calculating the moments of inertia for each part, add those of the semicircular plate and the rectangular plate, then subtract those of the circular plate to determine the moments of inertia for the entire machine part.

MODELING and ANALYSIS:

Computation of Masses. Semicircular Plate

$$V_1 = \frac{1}{2}\pi r^2 t = \frac{1}{2}\pi (0.08 \text{ m})^2 (0.004 \text{ m}) = 40.21 \times 10^{-6} \text{ m}^3$$

$$m_1 = \rho V_1 = (7.85 \times 10^3 \text{ kg/m}^3)(40.21 \times 10^{-6} \text{ m}^3) = 0.3156 \text{ kg}$$

Rectangular Plate

$$V_2 = (0.200 \text{ m})(0.160 \text{ m})(0.004 \text{ m}) = 128 \times 10^{-6} \text{ m}^3$$

$$m_2 = \rho V_2 = (7.85 \times 10^3 \text{ kg/m}^3)(128 \times 10^{-6} \text{ m}^3) = 1.005 \text{ kg}$$

Circular Plate

$$V_3 = \pi a^2 t = \pi (0.050 \text{ m})^2 (0.004 \text{ m}) = 31.42 \times 10^{-6} \text{ m}^3$$

$$m_3 = \rho V_3 = (7.85 \times 10^3 \text{ kg/m}^3)(31.42 \times 10^{-6} \text{ m}^3) = 0.2466 \text{ kg}$$

Moments of Inertia. Compute the moments of inertia of each component, using the method presented in Sec. 9.5C.

Semicircular Plate. Observe from Fig. 9.28 that, for a circular plate of mass m and radius r ,

$$I_x = \frac{1}{2}mr^2 \quad I_y = I_z = \frac{1}{4}mr^2$$

Because of symmetry, halve these values for a semicircular plate. Thus,

$$I_x = \frac{1}{2}(\frac{1}{2}mr^2) \quad I_y = I_z = \frac{1}{2}(\frac{1}{4}mr^2)$$

Since the mass of the semicircular plate is $m_1 = \frac{1}{2}m$, you have

$$I_x = \frac{1}{2}m_1 r^2 = \frac{1}{2}(0.3156 \text{ kg})(0.08 \text{ m})^2 = 1.010 \times 10^{-3} \text{ kg}\cdot\text{m}^2$$

$$I_y = I_z = \frac{1}{4}(\frac{1}{2}mr^2) = \frac{1}{4}m_1 r^2 = \frac{1}{4}(0.3156 \text{ kg})(0.08 \text{ m})^2 = 0.505 \times 10^{-3} \text{ kg}\cdot\text{m}^2$$

Rectangular Plate

$$I_x = \frac{1}{12}m_2 c^2 = \frac{1}{12}(1.005 \text{ kg})(0.16 \text{ m})^2 = 2.144 \times 10^{-3} \text{ kg}\cdot\text{m}^2$$

$$I_z = \frac{1}{3}m_2 b^2 = \frac{1}{3}(1.005 \text{ kg})(0.2 \text{ m})^2 = 13.400 \times 10^{-3} \text{ kg}\cdot\text{m}^2$$

$$I_y = I_x + I_z = (2.144 + 13.400)(10^{-3}) = 15.544 \times 10^{-3} \text{ kg}\cdot\text{m}^2$$

Circular Plate

$$I_x = \frac{1}{4}m_3 a^2 = \frac{1}{4}(0.2466 \text{ kg})(0.05 \text{ m})^2 = 0.154 \times 10^{-3} \text{ kg}\cdot\text{m}^2$$

$$I_y = \frac{1}{2}m_3 a^2 + m_3 d^2$$

$$= \frac{1}{2}(0.2466 \text{ kg})(0.05 \text{ m})^2 + (0.2466 \text{ kg})(0.1 \text{ m})^2 = 2.774 \times 10^{-3} \text{ kg}\cdot\text{m}^2$$

$$I_z = \frac{1}{4}m_3 a^2 + m_3 d^2 = \frac{1}{4}(0.2466 \text{ kg})(0.05 \text{ m})^2 + (0.2466 \text{ kg})(0.1 \text{ m})^2$$

$$= 2.620 \times 10^{-3} \text{ kg}\cdot\text{m}^2$$

Entire Machine Part

$$I_x = (1.010 + 2.144 - 0.154)(10^{-3}) \text{ kg}\cdot\text{m}^2 \quad I_x = 3.00 \times 10^{-3} \text{ kg}\cdot\text{m}^2 \quad \blacktriangleleft$$

$$I_y = (0.505 + 15.544 - 2.774)(10^{-3}) \text{ kg}\cdot\text{m}^2 \quad I_y = 13.28 \times 10^{-3} \text{ kg}\cdot\text{m}^2 \quad \blacktriangleleft$$

$$I_z = (0.505 + 13.400 - 2.620)(10^{-3}) \text{ kg}\cdot\text{m}^2 \quad I_z = 11.29 \times 10^{-3} \text{ kg}\cdot\text{m}^2 \quad \blacktriangleleft$$

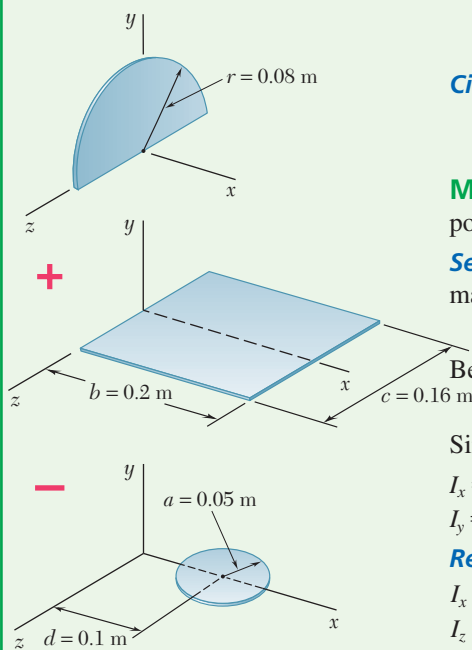


Fig. 1 Modeling the machine part as a combination of simple geometric shapes.

SOLVING PROBLEMS ON YOUR OWN

In this section, we introduced the **mass moment of inertia** and the **radius of gyration** of a three-dimensional body with respect to a given axis [Eqs. (9.28) and (9.29)]. We also derived a **parallel-axis theorem** for use with mass moments of inertia and discussed the computation of the mass moments of inertia of thin plates and three-dimensional bodies.

1. Computing mass moments of inertia. You can calculate the mass moment of inertia I of a body with respect to a given axis directly from the definition given in Eq. (9.28) for simple shapes [Sample Prob. 9.9]. In most cases, however, it is necessary to divide the body into thin slabs, compute the moment of inertia of a typical slab with respect to the given axis—using the parallel-axis theorem if necessary—and integrate the resulting expression.

2. Applying the parallel-axis theorem. In Sec. 9.5B, we derived the parallel-axis theorem for mass moments of inertia as

$$I = \bar{I} + md^2 \quad (9.33)$$

This theorem states that the moment of inertia I of a body of mass m with respect to a given axis is equal to the sum of the moment of inertia $\bar{I}$ of that body with respect to a parallel centroidal axis and the product md^2 , where d is the distance between the two axes. When you calculate the moment of inertia of a three-dimensional body with respect to one of the coordinate axes, you can replace d^2 by the sum of the squares of distances measured along the other two coordinate axes [Eqs. (9.32) and (9.32')].

3. Avoiding unit-related errors. To avoid errors, you must be consistent in your use of units. Thus, all lengths should be expressed in meters or feet, as appropriate, and for problems using U.S. customary units, masses should be given in $\text{lb}\cdot\text{s}^2/\text{ft}$. In addition, we strongly recommend that you include units as you perform your calculations [Sample Probs. 9.12 and 9.13].

4. Calculating the mass moment of inertia of thin plates. We showed in Sec. 9.5C that you can obtain the mass moment of inertia of a thin plate with respect to a given axis by multiplying the corresponding moment of inertia of the area of the plate by the density ρ and the thickness t of the plate [Eqs. (9.35) through (9.37)]. Note that, since the axis CC' in Fig. 9.24c is perpendicular to the plate, $I_{CC',\text{mass}}$ is associated with the *polar* moment of inertia $J_{C,\text{area}}$.

Instead of calculating the moment of inertia of a thin plate with respect to a specified axis directly, you may sometimes find it convenient to first compute its moment of inertia with respect to an axis parallel to the specified axis and to then apply the parallel-axis theorem. Furthermore, to determine the moment of inertia of a thin plate with respect to an axis perpendicular to the plate, you may wish to first determine its moments of inertia with respect to two perpendicular in-plane axes and to then use Eq. (9.38). Finally, remember that the mass of a plate consists of area A , thickness t , and density ρ , as $m = \rho tA$.

(continued)

5. Determining the moment of inertia of a body by direct single integration. We discussed in Sec. 9.5D and illustrated in Sample Probs. 9.10 and 9.11 how you can use single integration to compute the moment of inertia of a body that can be divided into a series of thin, parallel slabs. For such cases, you will often need to express the mass of the body in terms of the body's density and dimensions. Assuming that the body has been divided, as in the sample problems, into thin slabs perpendicular to the x axis, you will need to express the dimensions of each slab as functions of the variable x .

a. In the special case of a body of revolution, the elemental slab is a thin disk, and you can use the equations given in Fig. 9.27 to determine the moments of inertia of the body [Sample Prob. 9.11].

b. In the general case, when the body is not a solid of revolution, the differential element is not a disk but a thin slab of a different shape. You cannot use the equations of Fig. 9.27 in this case. See, for example, Sample Prob. 9.10, where the element was a thin, rectangular slab. For more complex configurations, you may want to use one or more of the following equations, which are based on Eqs. (9.32) and (9.32') of Sec. 9.5B.

$$\begin{aligned} dI_x &= dI_{x'} + (\bar{y}_{el}^2 + \bar{z}_{el}^2) dm \\ dI_y &= dI_{y'} + (\bar{z}_{el}^2 + \bar{x}_{el}^2) dm \\ dI_z &= dI_{z'} + (\bar{x}_{el}^2 + \bar{y}_{el}^2) dm \end{aligned}$$

Here, the primes denote the centroidal axes of each elemental slab and $\bar{x}_{el}$, $\bar{y}_{el}$, and $\bar{z}_{el}$ represent the coordinates of its centroid. Determine the centroidal moments of inertia of the slab in the manner described earlier for a thin plate: Refer to Fig. 9.12, calculate the corresponding moments of inertia of the area of the slab, and multiply the result by the density ρ and the thickness t of the slab. Also, assuming that the body has been divided into thin slabs perpendicular to the x axis, remember that you can obtain $dI_{x'}$ by adding $dI_{y'}$ and $dI_{z'}$ instead of computing it directly. Finally, using the geometry of the body, express the result obtained in terms of the single variable x , and integrate in x .

6. Computing the moment of inertia of a composite body. As stated in Sec. 9.5E, the moment of inertia of a composite body with respect to a specified axis is equal to the sum of the moments of its components with respect to that axis. Sample Probs. 9.12 and 9.13 illustrate the appropriate method of solution. Also remember that the moment of inertia of a component is negative only if the component is *removed* (as in the case of a hole).

Although the composite-body problems in this section are relatively straightforward, you will have to work carefully to avoid computational errors. In addition, if some of the moments of inertia that you need are not given in Fig. 9.28, you will have to derive your own formulas, using the techniques described in this section.

Problems

- 9.111** A thin plate with a mass m is cut in the shape of an equilateral triangle of side a . Determine the mass moment of inertia of the plate with respect to (a) the centroidal axes AA' and BB' , (b) the centroidal axis CC' that is perpendicular to the plate.

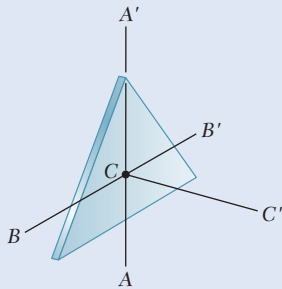


Fig. P9.111

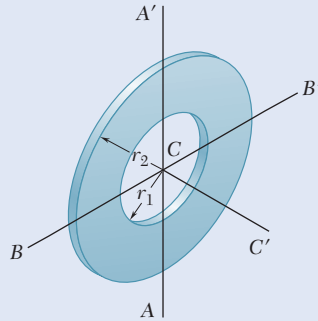


Fig. P9.112

- 9.112** A ring with a mass m is cut from a thin uniform plate. Determine the mass moment of inertia of the ring with respect to (a) the axis AA' , (b) the centroidal axis CC' that is perpendicular to the plane of the ring.

- 9.113** A thin, semielliptical plate has a mass m . Determine the mass moment of inertia of the plate with respect to (a) the centroidal axis BB' , (b) the centroidal axis CC' that is perpendicular to the plate.

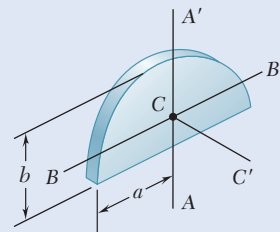


Fig. P9.113

- 9.114** The parabolic spandrel shown was cut from a thin, uniform plate. Denoting the mass of the spandrel by m , determine its mass moment of inertia with respect to (a) the axis BB' , (b) the axis DD' that is perpendicular to the spandrel. (Hint: See Sample Prob. 9.3.)

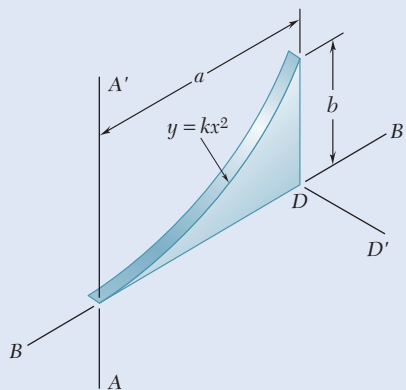


Fig. P9.114

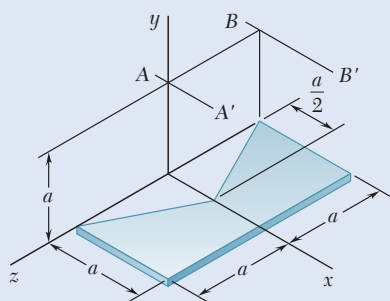


Fig. P9.115 and P9.116

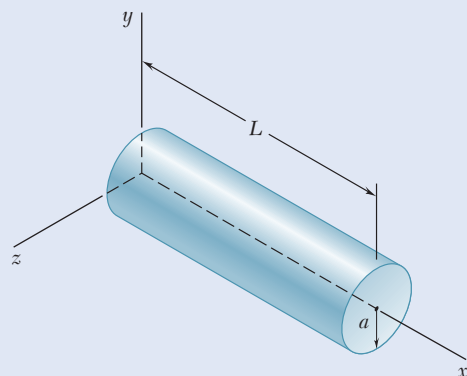


Fig. P9.119

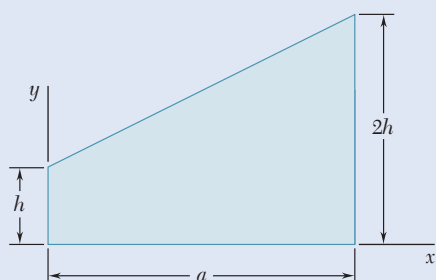


Fig. P9.120

9.115 A piece of thin, uniform sheet metal is cut to form the machine component shown. Denoting the mass of the component by m , determine its mass moment of inertia with respect to (a) the x axis, (b) the y axis.

9.116 A piece of thin, uniform sheet metal is cut to form the machine component shown. Denoting the mass of the component by m , determine its mass moment of inertia with respect to (a) the axis AA' , (b) the axis BB' , where the AA' and BB' axes are parallel to the x axis and lie in a plane parallel to and at a distance a above the xz plane.

9.117 A thin plate with a mass m has the trapezoidal shape shown. Determine the mass moment of inertia of the plate with respect to (a) the x axis, (b) the y axis.

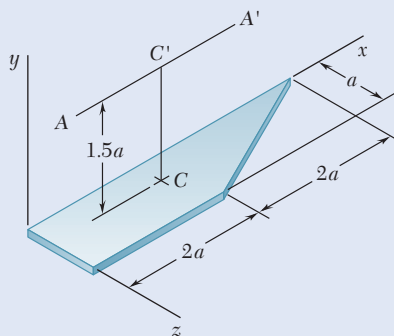


Fig. P9.117 and P9.118

9.118 A thin plate with a mass m has the trapezoidal shape shown. Determine the mass moment of inertia of the plate with respect to (a) the centroidal axis CC' that is perpendicular to the plate, (b) the axis AA' that is parallel to the x axis and is located at a distance $1.5a$ from the plate.

9.119 Determine by direct integration the mass moment of inertia with respect to the z axis of the right circular cylinder shown, assuming that it has a uniform density and a mass m .

9.120 The area shown is revolved about the x axis to form a homogeneous solid of revolution of mass m . Using direct integration, express the mass moment of inertia of the solid with respect to the x axis in terms of m and h .

9.121 The area shown is revolved about the x axis to form a homogeneous solid of revolution of mass m . Determine by direct integration the mass moment of inertia of the solid with respect to (a) the x axis, (b) the y axis. Express your answers in terms of m and the dimensions of the solid.

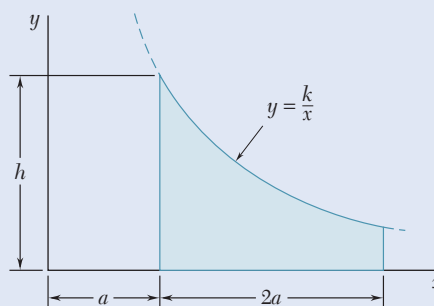


Fig. P9.121

- 9.122** Determine by direct integration the mass moment of inertia with respect to the x axis of the tetrahedron shown, assuming that it has a uniform density and a mass m .

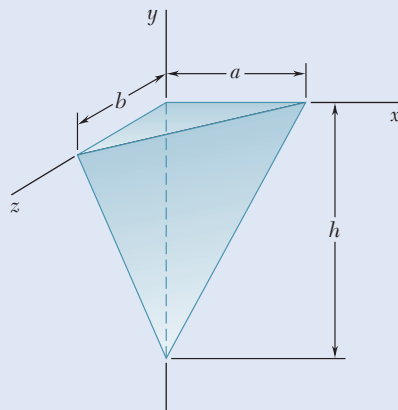


Fig. P9.122 and P9.123

- 9.123** Determine by direct integration the mass moment of inertia with respect to the y axis of the tetrahedron shown, assuming that it has a uniform density and a mass m .

- 9.124** Determine by direct integration the mass moment of inertia and the radius of gyration with respect to the x axis of the paraboloid shown, assuming that it has a uniform density and a mass m .

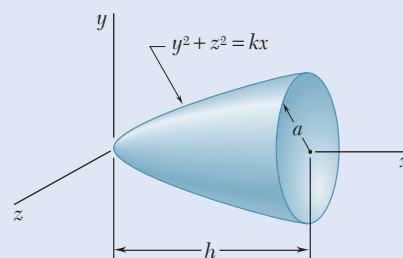


Fig. P9.124

- 9.125** A thin, rectangular plate with a mass m is welded to a vertical shaft AB as shown. Knowing that the plate forms an angle θ with the y axis, determine by direct integration the mass moment of inertia of the plate with respect to (a) the y axis, (b) the z axis.

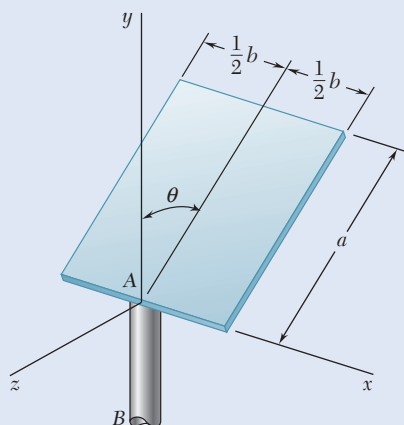


Fig. P9.125

- *9.126** A thin steel wire is bent into the shape shown. Denoting the mass per unit length of the wire by m' , determine by direct integration the mass moment of inertia of the wire with respect to each of the coordinate axes.

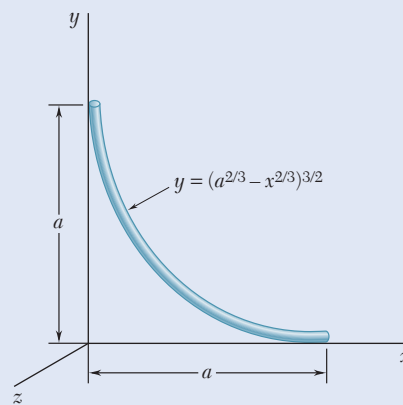


Fig. P9.126

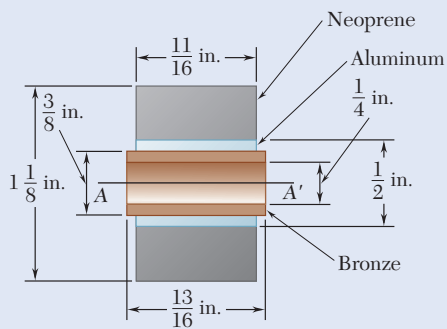


Fig. P9.127

9.127 Shown is the cross section of an idler roller. Determine its mass moment of inertia and its radius of gyration with respect to the axis AA' . (The specific weight of bronze is 0.310 lb/in^3 ; of aluminum, 0.100 lb/in^3 ; and of neoprene, 0.0452 lb/in^3 .)

9.128 Shown is the cross section of a molded flat-belt pulley. Determine its mass moment of inertia and its radius of gyration with respect to the axis AA' . (The density of brass is 8650 kg/m^3 , and the density of the fiber-reinforced polycarbonate used is 1250 kg/m^3 .)

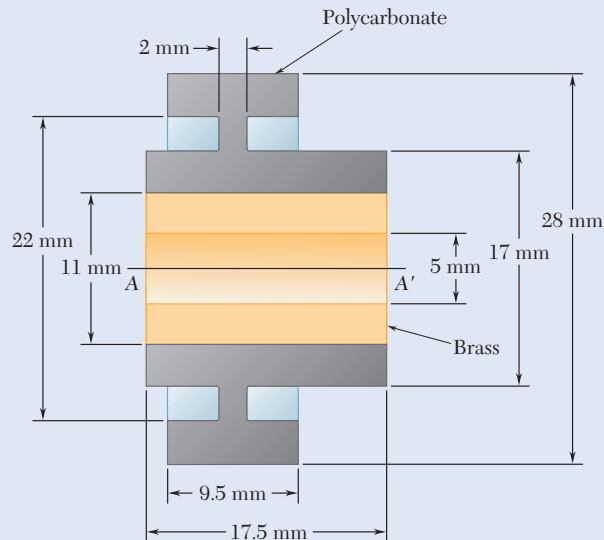


Fig. P9.128

9.129 The machine part shown is formed by machining a conical surface into a circular cylinder. For $b = \frac{1}{2}h$, determine the mass moment of inertia and the radius of gyration of the machine part with respect to the y axis.

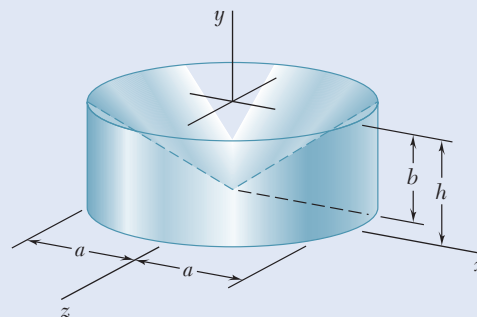


Fig. P9.129

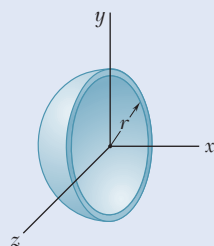


Fig. P9.130

9.130 Knowing that the thin hemispherical shell shown has a mass m and thickness t , determine the mass moment of inertia and the radius of gyration of the shell with respect to the x axis. (Hint: Consider the shell as formed by removing a hemisphere of radius r from a hemisphere of radius $r + t$; then neglect the terms containing t^2 and t^3 and keep those terms containing t .)

- 9.131** A square hole is centered in and extends through the aluminum machine component shown. Determine (a) the value of a for which the mass moment of inertia of the component is maximum with respect to the axis AA' that bisects the top surface of the hole, (b) the corresponding values of the mass moment of inertia and the radius of gyration with respect to the axis AA' . (The specific weight of aluminum is 0.100 lb/in^3 .)

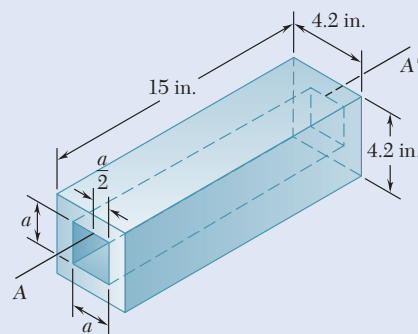


Fig. P9.131

- 9.132** The cups and the arms of an anemometer are fabricated from a material of density ρ . Knowing that the mass moment of inertia of a thin, hemispherical shell with a mass m and thickness t with respect to its centroidal axis GG' is $5ma^2/12$, determine (a) the mass moment of inertia of the anemometer with respect to the axis AA' , (b) the ratio of a to l for which the centroidal moment of inertia of the cups is equal to 1 percent of the moment of inertia of the cups with respect to the axis AA' .

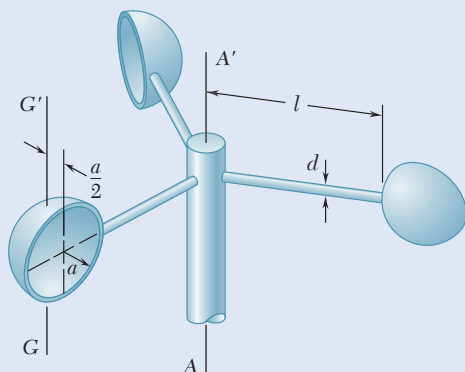


Fig. P9.132

- 9.133** After a period of use, one of the blades of a shredder has been worn to the shape shown and is of mass 0.18 kg . Knowing that the mass moments of inertia of the blade with respect to the AA' and BB' axes are $0.320 \text{ g}\cdot\text{m}^2$ and $0.680 \text{ g}\cdot\text{m}^2$, respectively, determine (a) the location of the centroidal axis GG' , (b) the radius of gyration with respect to axis GG' .

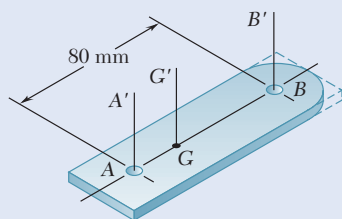


Fig. P9.133

- 9.134** Determine the mass moment of inertia of the 0.9-lb machine component shown with respect to the axis AA' .

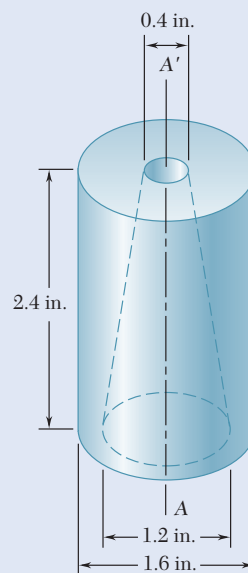


Fig. P9.134

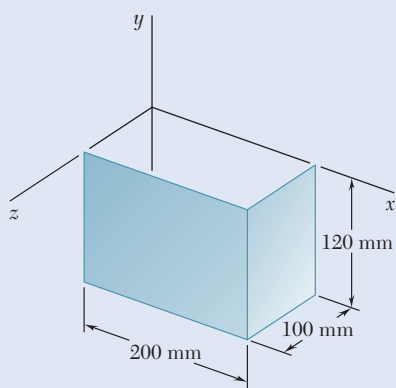


Fig. P9.135

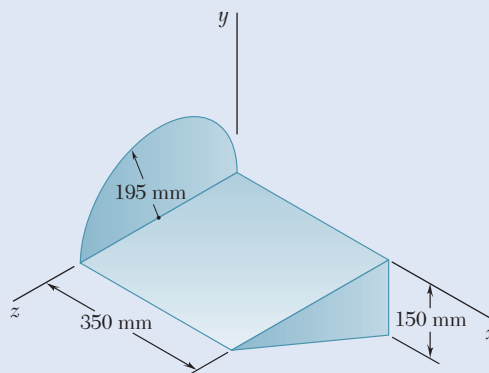


Fig. P9.136

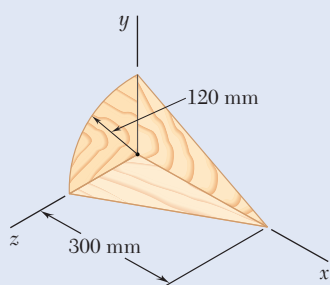


Fig. P9.137

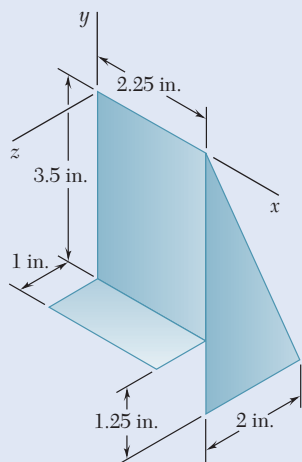


Fig. P9.139

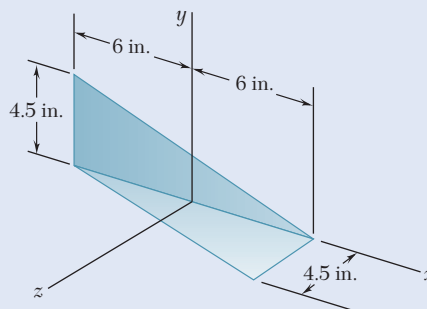


Fig. P9.138

9.135 and 9.136 A 2-mm thick piece of sheet steel is cut and bent into the machine component shown. Knowing that the density of steel is 7850 kg/m^3 , determine the mass moment of inertia of the component with respect to each of the coordinate axes.

9.137 A subassembly for a model airplane is fabricated from three pieces of 1.5-mm plywood. Neglecting the mass of the adhesive used to assemble the three pieces, determine the mass moment of inertia of the subassembly with respect to each of the coordinate axes. (The density of the plywood is 780 kg/m^3 .)

9.138 A section of sheet steel 0.03 in. thick is cut and bent into the sheet metal machine component shown. Determine the mass moment of inertia of the component with respect to each of the coordinate axes. (The specific weight of steel is 490 lb/ft^3 .)

9.139 A framing anchor is formed of 0.05-in.-thick galvanized steel. Determine the mass moment of inertia of the anchor with respect to each of the coordinate axes. (The specific weight of galvanized steel is 470 lb/ft^3 .)

- *9.140** A farmer constructs a trough by welding a rectangular piece of 2-mm-thick sheet steel to half of a steel drum. Knowing that the density of steel is 7850 kg/m^3 and that the thickness of the walls of the drum is 1.8 mm, determine the mass moment of inertia of the trough with respect to each of the coordinate axes. Neglect the mass of the welds.

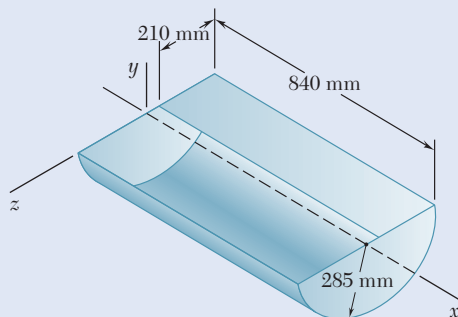


Fig. P9.140

- 9.141** The machine element shown is fabricated from steel. Determine the mass moment of inertia of the assembly with respect to (a) the x axis, (b) the y axis, (c) the z axis. (The density of steel is 7850 kg/m^3 .)

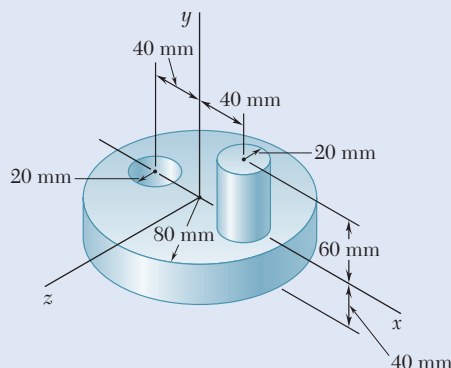


Fig. P9.141

- 9.142** Determine the mass moments of inertia and the radii of gyration of the steel machine element shown with respect to the x and y axes. (The density of steel is 7850 kg/m^3 .)

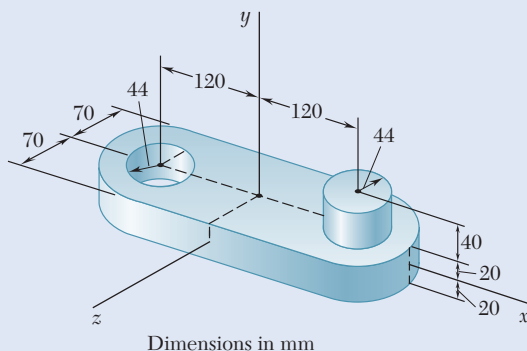


Fig. P9.142

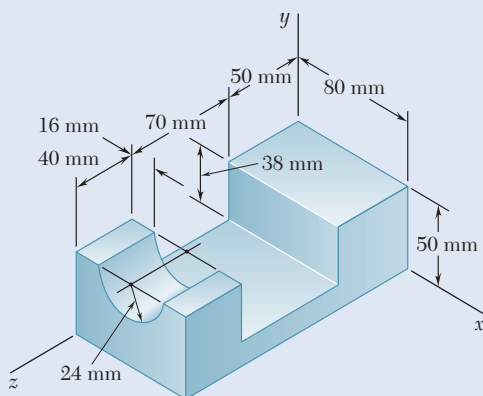


Fig. P9.145

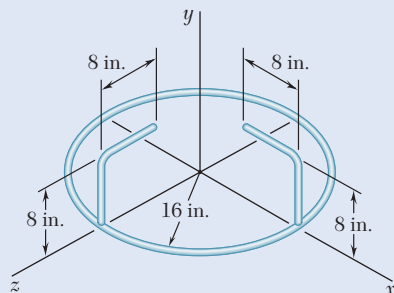


Fig. P9.146

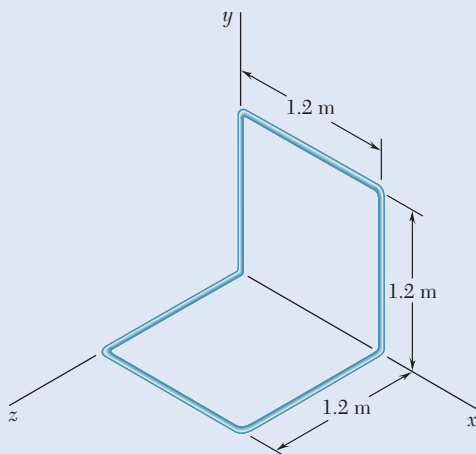


Fig. P9.148

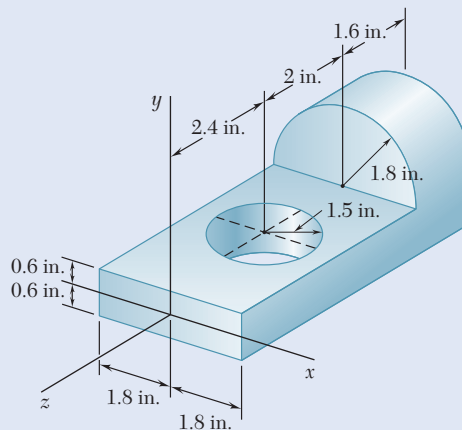


Fig. P9.143 and P.9.144

9.143 Determine the mass moment of inertia of the steel machine element shown with respect to the x axis. (The specific weight of steel is 490 lb/ft^3 .)

9.144 Determine the mass moment of inertia of the steel machine element shown with respect to the y axis. (The specific weight of steel is 490 lb/ft^3 .)

9.145 Determine the mass moment of inertia of the steel fixture shown with respect to (a) the x axis, (b) the y axis, (c) the z axis. (The density of steel is 7850 kg/m^3 .)

9.146 Aluminum wire with a weight per unit length of 0.033 lb/ft is used to form the circle and the straight members of the figure shown. Determine the mass moment of inertia of the assembly with respect to each of the coordinate axes.

9.147 The figure shown is formed of $\frac{1}{8}$ -in.-diameter steel wire. Knowing that the specific weight of the steel is 490 lb/ft^3 , determine the mass moment of inertia of the wire with respect to each of the coordinate axes.

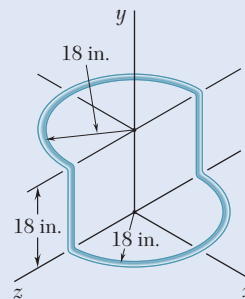


Fig. P9.147

9.148 A homogeneous wire with a mass per unit length of 0.056 kg/m is used to form the figure shown. Determine the mass moment of inertia of the wire with respect to each of the coordinate axes.

*9.6 ADDITIONAL CONCEPTS OF MASS MOMENTS OF INERTIA

In this final section of the chapter, we present several concepts involving mass moments of inertia that are analogous to material presented in Sec. 9.4 involving moments of inertia of areas. These ideas include mass products of inertia, principal axes of inertia, and principal moments of inertia for masses, which are necessary for the study of the dynamics of rigid bodies in three dimensions.

9.6A Mass Products of Inertia

In this section, you will see how to determine the moment of inertia of a body with respect to an arbitrary axis OL through the origin (Fig. 9.29) if its moments of inertia with respect to the three coordinate axes, as well as certain other quantities defined here, have already been determined.

The moment of inertia I_{OL} of the body with respect to OL is equal to $\int p^2 dm$, where p denotes the perpendicular distance from the element of mass dm to the axis OL . If we denote the unit vector along OL by λ and the position vector of the element dm by $\mathbf{r}$, the perpendicular distance p is equal to $r \sin \theta$, which is the magnitude of the vector product $\lambda \times \mathbf{r}$. We therefore have

$$I_{OL} = \int p^2 dm = \int |\lambda \times \mathbf{r}|^2 dm \quad (9.43)$$

Expressing $|\lambda \times \mathbf{r}|^2$ in terms of the rectangular components of the vector product, we have

$$I_{OL} = \int [(\lambda_x y - \lambda_y x)^2 + (\lambda_y z - \lambda_z y)^2 + (\lambda_z x - \lambda_x z)^2] dm$$

Here, the components $\lambda_x, \lambda_y, \lambda_z$ of the unit vector λ represent the direction cosines of the axis OL , and the components x, y, z of $\mathbf{r}$ represent the coordinates of the element of mass dm . Expanding the squares and rearranging the terms, we obtain

$$\begin{aligned} I_{OL} = & \lambda_x^2 \int (y^2 + z^2) dm + \lambda_y^2 \int (z^2 + x^2) dm + \lambda_z^2 \int (x^2 + y^2) dm \\ & - 2\lambda_x \lambda_y \int xy dm - 2\lambda_y \lambda_z \int yz dm - 2\lambda_z \lambda_x \int zx dm \end{aligned} \quad (9.44)$$

Referring to Eqs. (9.30), note that the first three integrals in Eq. (9.44) represent, respectively, the moments of inertia I_x, I_y , and I_z of the body with respect to the coordinate axes. The last three integrals in Eq. (9.44), which involve products of coordinates, are called the **products of inertia** of the body with respect to the x and y axes, the y and z axes, and the z and x axes, respectively.

Mass products of inertia

$$I_{xy} = \int xy dm \quad I_{yz} = \int yz dm \quad I_{zx} = \int zx dm \quad (9.45)$$

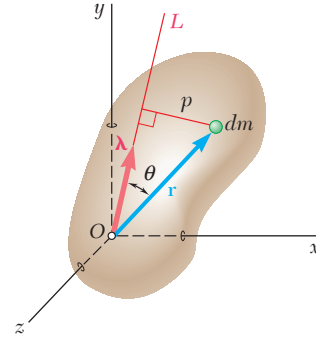


Fig. 9.29 An element of mass dm of a body and its perpendicular distance to an arbitrary axis OL through the origin.

Rewriting Eq. (9.44) in terms of the integrals defined in Eqs. (9.30) and (9.45), we have

$$I_{OL} = I_x \lambda_x^2 + I_y \lambda_y^2 + I_z \lambda_z^2 - 2I_{xy} \lambda_x \lambda_y - 2I_{yz} \lambda_y \lambda_z - 2I_{zx} \lambda_z \lambda_x \quad (9.46)$$

The definition of the products of inertia of a mass given in Eqs. (9.45) is an extension of the definition of the product of inertia of an area (Sec. 9.3). Mass products of inertia reduce to zero under the same conditions of symmetry as do products of inertia of areas, and the parallel-axis theorem for mass products of inertia is expressed by relations similar to the formula derived for the product of inertia of an area. Substituting the expressions for x , y , and z given in Eqs. (9.31) into Eqs. (9.45), we find that

Parallel-axis theorem for mass products of inertia

$$\begin{aligned} I_{xy} &= \bar{I}_{x'y'} + m\bar{x}\bar{y} \\ I_{yz} &= \bar{I}_{y'z'} + m\bar{y}\bar{z} \\ I_{zx} &= \bar{I}_{z'x'} + m\bar{z}\bar{x} \end{aligned} \quad (9.47)$$

Here $\bar{x}$, $\bar{y}$, $\bar{z}$ are the coordinates of the center of gravity G of the body and $\bar{I}_{x'y'}$, $\bar{I}_{y'z'}$, $\bar{I}_{z'x'}$ denote the products of inertia of the body with respect to the centroidal axes x' , y' , and z' (see Fig. 9.22).

9.6B Principal Axes and Principal Moments of Inertia

Let us assume that we have determined the moment of inertia of the body considered in the preceding section with respect to a large number of axes OL through the fixed point O . Suppose that we plot a point Q on each axis OL at a distance $OQ = 1/\sqrt{I_{OL}}$ from O . The locus of the points Q forms a surface (Fig. 9.30). We can obtain the equation of that surface by substituting $1/(OQ)^2$ for I_{OL} in Eq. (9.46) and then multiplying both sides of the equation by $(OQ)^2$. Observing that

$$(OQ)\lambda_x = x \quad (OQ)\lambda_y = y \quad (OQ)\lambda_z = z$$

where x , y , z denote the rectangular coordinates of Q , we have

$$I_x x^2 + I_y y^2 + I_z z^2 - 2I_{xy} xy - 2I_{yz} yz - 2I_{zx} zx = 1 \quad (9.48)$$

This is the equation of a *quadric surface*. Since the moment of inertia I_{OL} is different from zero for every axis OL , no point Q can be at an infinite distance from O . Thus, the quadric surface obtained is an *ellipsoid*. This ellipsoid, which defines the moment of inertia of the body with respect to any axis through O , is known as the **ellipsoid of inertia** of the body at O .

Observe that, if we rotate the axes in Fig. 9.30, the coefficients of the equation defining the ellipsoid change, since they are equal to the moments and products of inertia of the body with respect to the rotated coordinate axes. However, the *ellipsoid itself remains unaffected*, since its

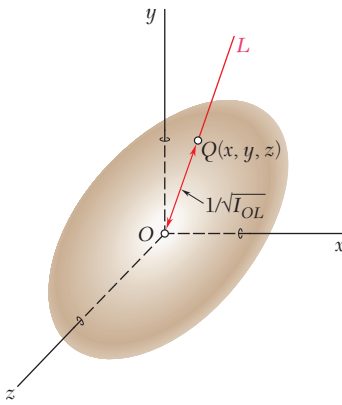


Fig. 9.30 The ellipsoid of inertia defines the moment of inertia of a body with respect to any axis through O .

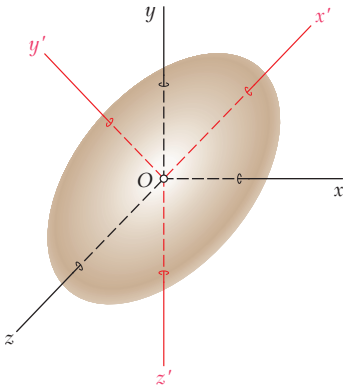


Fig. 9.31 Principal axes of inertia x' , y' , z' of the body at O .

shape depends only upon the distribution of mass in the given body. Suppose that we choose as coordinate axes the principal axes x' , y' , and z' of the ellipsoid of inertia (Fig. 9.31). The equation of the ellipsoid with respect to these coordinate axes is known to be of the form

$$I_{x'}x'^2 + I_{y'}y'^2 + I_{z'}z'^2 = 1 \quad (9.49)$$

which does not contain any products of the coordinates. Comparing Eqs. (9.48) and (9.49), we observe that the products of inertia of the body with respect to the x' , y' , and z' axes must be zero. The x' , y' , and z' axes are known as the **principal axes of inertia** of the body at O , and the coefficients $I_{x'}$, $I_{y'}$, and $I_{z'}$ are referred to as the **principal moments of inertia** of the body at O . Note that, given a body of arbitrary shape and a point O , it is always possible to find principal axes of inertia of the body at O ; that is, axes with respect to which the products of inertia of the body are zero. Indeed, whatever the shape of the body, the moments and products of inertia of the body with respect to the x , y , and z axes through O define an ellipsoid, and this ellipsoid has principal axes that, by definition, are the principal axes of inertia of the body at O .

If the principal axes of inertia x' , y' , and z' are used as coordinate axes, the expression in Eq. (9.46) for the moment of inertia of a body with respect to an arbitrary axis through O reduces to

$$I_{OL} = I_{x'}\lambda_{x'}^2 + I_{y'}\lambda_{y'}^2 + I_{z'}\lambda_{z'}^2 \quad (9.50)$$

The determination of the principal axes of inertia of a body of arbitrary shape is somewhat involved and is discussed in the next section. In many cases, however, these axes can be spotted immediately. Consider, for instance, the homogeneous cone of elliptical base shown in Fig. 9.32; this cone possesses two mutually perpendicular planes of symmetry OAA' and OBB' . From the definition of Eq. (9.45), we observe that if we choose the $x'y'$ and $y'z'$ planes to coincide with the two planes of symmetry, all of the products of inertia are zero. The x' , y' , and z' axes selected in this way are therefore the principal axes of inertia of the cone at O . In the

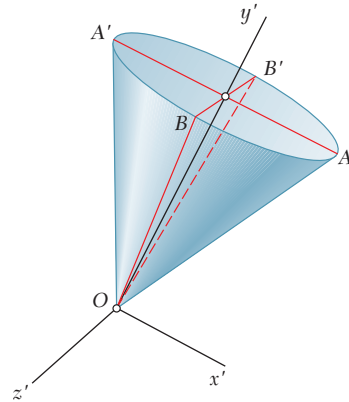


Fig. 9.32 A homogeneous cone with elliptical base has two mutually perpendicular planes of symmetry.

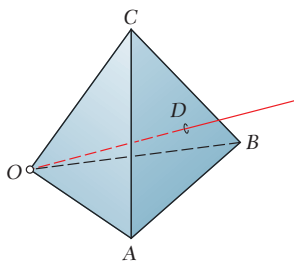


Fig. 9.33 A line drawn from a corner to the center of the opposite face of a homogeneous regular tetrahedron is a principal axis, since each 120° rotation of the body about this axis leaves its shape and mass distribution unchanged.

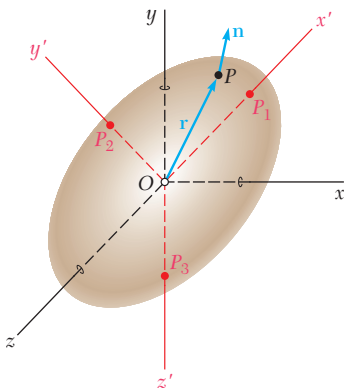


Fig. 9.34 The principal axes intersect an ellipsoid of inertia at points where the radius vectors are collinear with the unit normal vectors to the surface.

case of the homogeneous regular tetrahedron $OABC$ shown in Fig. 9.33, the line joining the corner O to the center D of the opposite face is a principal axis of inertia at O , and any line through O perpendicular to OD is also a principal axis of inertia at O . This property is apparent if we observe that rotating the tetrahedron through 120° about OD leaves its shape and mass distribution unchanged. It follows that the ellipsoid of inertia at O also remains unchanged under this rotation. The ellipsoid, therefore, is a body of revolution whose axis of revolution is OD , and the line OD , as well as any perpendicular line through O , must be a principal axis of the ellipsoid.

9.6C Principal Axes and Moments of Inertia for a Body of Arbitrary Shape

The method of analysis described in this section extends the analysis in the preceding section. However, generally speaking, you should use it only when the body under consideration has no obvious property of symmetry.

Consider the ellipsoid of inertia of a body at a given point O (Fig. 9.34). Let $\mathbf{r}$ be the radius vector of a point P on the surface of the ellipsoid, and let $\mathbf{n}$ be the unit vector along the normal to that surface at P . We observe that the only points where $\mathbf{r}$ and $\mathbf{n}$ are collinear are points P_1 , P_2 , and P_3 , where the principal axes intersect the visible portion of the surface of the ellipsoid (along with the corresponding points on the other side of the ellipsoid).

Recall from calculus that the direction of the normal to a surface of equation $f(x, y, z) = 0$ at a point $P(x, y, z)$ is defined by the gradient ∇f of the function f at that point. To obtain the points where the principal axes intersect the surface of the ellipsoid of inertia, we must therefore express that $\mathbf{r}$ and ∇f are collinear,

$$\nabla f = (2K)\mathbf{r} \quad (9.51)$$

where K is a constant, $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, and

$$\nabla f = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j} + \frac{\partial f}{\partial z}\mathbf{k}$$

Recalling Eq. (9.48), we note that the function $f(x, y, z)$ corresponding to the ellipsoid of inertia is

$$f(x, y, z) = I_x x^2 + I_y y^2 + I_z z^2 - 2I_{xy}xy - 2I_{yz}yz - 2I_{zx}zx - 1$$

Substituting for $\mathbf{r}$ and ∇f into Eq. (9.51) and equating the coefficients of the unit vectors, we obtain

$$\begin{aligned} I_x x - I_{xy}y - I_{zx}z &= Kx \\ -I_{xy}x + I_y y - I_{yz}z &= Ky \\ -I_{zx}x - I_{yz}y + I_z z &= Kz \end{aligned} \quad (9.52)$$

Dividing each term by the distance r from O to P , we obtain similar equations involving the direction cosines λ_x , λ_y , and λ_z :

$$\begin{aligned} I_x \lambda_x - I_{xy} \lambda_y - I_{zx} \lambda_z &= K \lambda_x \\ -I_{xy} \lambda_x + I_y \lambda_y - I_{yz} \lambda_z &= K \lambda_y \\ -I_{zx} \lambda_x - I_{yz} \lambda_y + I_z \lambda_z &= K \lambda_z \end{aligned} \quad (9.53)$$

Transposing the right-hand members leads to the homogeneous linear equations, as

$$\begin{aligned} (I_x - K) \lambda_x - I_{xy} \lambda_y - I_{zx} \lambda_z &= 0 \\ -I_{xy} \lambda_x + (I_y - K) \lambda_y - I_{yz} \lambda_z &= 0 \\ -I_{zx} \lambda_x - I_{yz} \lambda_y + (I_z - K) \lambda_z &= 0 \end{aligned} \quad (9.54)$$

For this system of equations to have a solution different from $\lambda_x = \lambda_y = \lambda_z = 0$, its discriminant must be zero. Thus,

$$\begin{vmatrix} I_x - K & -I_{xy} & -I_{zx} \\ -I_{xy} & I_y - K & -I_{yz} \\ -I_{zx} & -I_{yz} & I_z - K \end{vmatrix} = 0 \quad (9.55)$$

Expanding this determinant and changing signs, we have

$$\begin{aligned} K^3 - (I_x + I_y + I_z)K^2 + (I_x I_y + I_y I_z + I_z I_x - I_{xy}^2 - I_{yz}^2 - I_{zx}^2)K \\ - (I_x I_y I_z - I_x I_{yz}^2 - I_y I_{zx}^2 - I_z I_{xy}^2 - 2I_{xy} I_{yz} I_{zx}) = 0 \end{aligned} \quad (9.56)$$

This is a cubic equation in K , which yields three real, positive roots: K_1 , K_2 , and K_3 .

To obtain the direction cosines of the principal axis corresponding to the root K_1 , we substitute K_1 for K in Eqs. (9.54). Since these equations are now linearly dependent, only two of them may be used to determine λ_x , λ_y , and λ_z . We can obtain an additional equation, however, by recalling from Sec. 2.4A that the direction cosines must satisfy the relation

$$\lambda_x^2 + \lambda_y^2 + \lambda_z^2 = 1 \quad (9.57)$$

Repeating this procedure with K_2 and K_3 , we obtain the direction cosines of the other two principal axes.

We now show that *the roots K_1 , K_2 , and K_3 of Eq. (9.56) are the principal moments of inertia of the given body*. Let us substitute for K in Eqs. (9.53) the root K_1 , and for λ_x , λ_y , and λ_z the corresponding values $(\lambda_x)_1$, $(\lambda_y)_1$, and $(\lambda_z)_1$ of the direction cosines; the three equations are satisfied. We now multiply by $(\lambda_x)_1$, $(\lambda_y)_1$, and $(\lambda_z)_1$, respectively, each term in the first, second, and third equation and add the equations obtained in this way. The result is

$$\begin{aligned} I_x^2 (\lambda_x)_1^2 + I_y^2 (\lambda_y)_1^2 + I_z^2 (\lambda_z)_1^2 - 2I_{xy} (\lambda_x)_1 (\lambda_y)_1 \\ - 2I_{yz} (\lambda_y)_1 (\lambda_z)_1 - 2I_{zx} (\lambda_z)_1 (\lambda_x)_1 = K_1 [(\lambda_x)_1^2 + (\lambda_y)_1^2 + (\lambda_z)_1^2] \end{aligned}$$

Recalling Eq. (9.46), we observe that the left-hand side of this equation represents the moment of inertia of the body with respect to the principal axis corresponding to K_1 ; it is thus the principal moment of inertia corresponding to that root. On the other hand, recalling Eq. (9.57), we note that the right-hand member reduces to K_1 . Thus, K_1 itself is the principal moment of inertia. In the same fashion, we can show that K_2 and K_3 are the other two principal moments of inertia of the body.

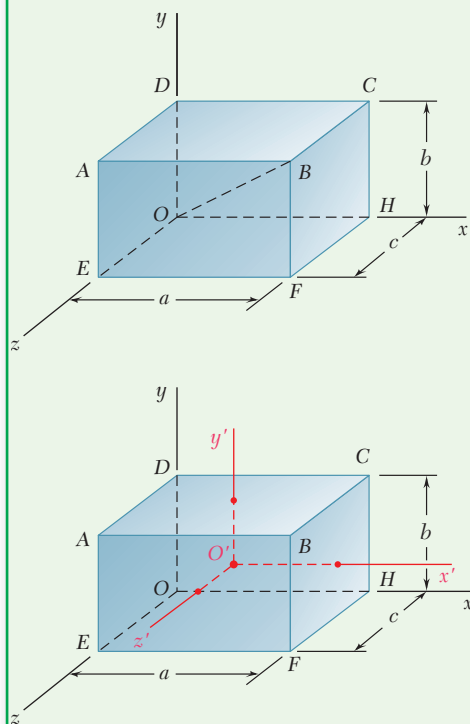


Fig. 1 Centroidal axes for the rectangular prism.

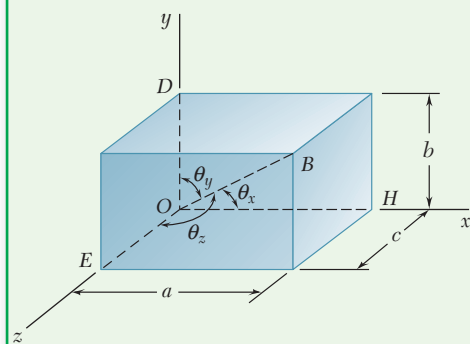


Fig. 2 Direction angles for OB .

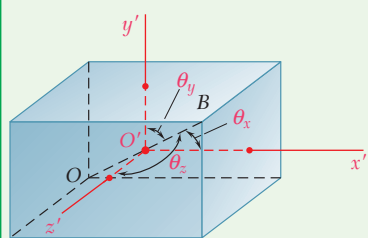


Fig. 3 Line OB passes through the centroid O' .

Sample Problem 9.14

Consider a rectangular prism with a mass of m and sides a , b , and c . Determine (a) the moments and products of inertia of the prism with respect to the coordinate axes shown, (b) its moment of inertia with respect to the diagonal OB .

STRATEGY: For part (a), you can introduce centroidal axes and apply the parallel-axis theorem. For part (b), determine the direction cosines of line OB from the given geometry and use either Eq. (9.46) or (9.50).

MODELING and ANALYSIS: a. **Moments and Products of Inertia with Respect to the Coordinate Axes.**

Moments of Inertia. Introduce the centroidal axes x' , y' , and z' with respect to which the moments of inertia are given in Fig. 9.28, and then apply the parallel-axis theorem (Fig. 1). Thus,

$$I_x = \bar{I}_{x'} + m(\bar{y}^2 + \bar{z}^2) = \frac{1}{12}m(b^2 + c^2) + m(\frac{1}{4}b^2 + \frac{1}{4}c^2)$$

$$I_x = \frac{1}{3}m(b^2 + c^2) \quad \blacktriangleleft$$

Similarly,

$$I_y = \frac{1}{3}m(c^2 + a^2) \quad I_z = \frac{1}{3}m(a^2 + b^2) \quad \blacktriangleleft$$

Products of Inertia. Because of symmetry, the products of inertia with respect to the centroidal axes x' , y' , and z' are zero, and these axes are principal axes of inertia. Using the parallel-axis theorem, you have

$$I_{xy} = \bar{I}_{x'y'} + m\bar{x}\bar{y} = 0 + m(\frac{1}{2}a)(\frac{1}{2}b) \quad I_{xy} = \frac{1}{4}mab \quad \blacktriangleleft$$

Similarly,

$$I_{yz} = \frac{1}{4}mbc \quad I_{zx} = \frac{1}{4}mca \quad \blacktriangleleft$$

b. **Moment of Inertia with Respect to OB .** Recall Eq. (9.46):

$$I_{OB} = I_x\lambda_x^2 + I_y\lambda_y^2 + I_z\lambda_z^2 - 2I_{xy}\lambda_x\lambda_y - 2I_{yz}\lambda_y\lambda_z - 2I_{zx}\lambda_z\lambda_x$$

where the direction cosines of OB are (Fig. 2)

$$\lambda_x = \cos \theta_x = \frac{OH}{OB} = \frac{a}{(a^2 + b^2 + c^2)^{1/2}}$$

$$\lambda_y = \frac{b}{(a^2 + b^2 + c^2)^{1/2}} \quad \lambda_z = \frac{c}{(a^2 + b^2 + c^2)^{1/2}}$$

Substituting the values obtained in part (a) for the moments and products of inertia and for the direction cosines into the equation for I_{OB} , you obtain

$$I_{OB} = \frac{1}{a^2 + b^2 + c^2} \left[\frac{1}{3}m(b^2 + c^2)a^2 + \frac{1}{3}m(c^2 + a^2)b^2 + \frac{1}{3}m(a^2 + b^2)c^2 \right. \\ \left. - \frac{1}{2}ma^2b^2 - \frac{1}{2}mb^2c^2 - \frac{1}{2}mc^2a^2 \right]$$

$$I_{OB} = \frac{m}{6} \frac{a^2b^2 + b^2c^2 + c^2a^2}{a^2 + b^2 + c^2} \quad \blacktriangleleft$$

REFLECT and THINK: You can also obtain the moment of inertia I_{OB} directly from the principal moments of inertia $\bar{I}_{x'}$, $\bar{I}_{y'}$, and $\bar{I}_{z'}$, since the line OB passes through the centroid O' . Since the x' , y' , and z' axes are principal axes of inertia (Fig. 3), use Eq. (9.50) to write

$$I_{OB} = \bar{I}_{x'}\lambda_x^2 + \bar{I}_{y'}\lambda_y^2 + \bar{I}_{z'}\lambda_z^2$$

$$= \frac{1}{a^2 + b^2 + c^2} \left[\frac{m}{12}(b^2 + c^2)a^2 + \frac{m}{12}(c^2 + a^2)b^2 + \frac{m}{12}(a^2 + b^2)c^2 \right]$$

$$I_{OB} = \frac{m}{6} \frac{a^2b^2 + b^2c^2 + c^2a^2}{a^2 + b^2 + c^2} \quad \blacktriangleleft$$

Sample Problem 9.15

If $a = 3c$ and $b = 2c$ for the rectangular prism of Sample Prob. 9.14, determine (a) the principal moments of inertia at the origin O , (b) the principal axes of inertia at O .

STRATEGY: Substituting the data into the results from Sample Prob. 9.14 gives you values you can use with Eq. (9.56) to determine the principal moments of inertia. You can then use these values to set up a system of equations for finding the direction cosines of the principal axes.

MODELING and ANALYSIS:

a. Principal Moments of Inertia at the Origin O . Substituting $a = 3c$ and $b = 2c$ into the solution to Sample Prob. 9.14 gives you

$$\begin{aligned} I_x &= \frac{5}{3}mc^2 & I_y &= \frac{10}{3}mc^2 & I_z &= \frac{13}{3}mc^2 \\ I_{xy} &= \frac{3}{2}mc^2 & I_{yz} &= \frac{1}{2}mc^2 & I_{zx} &= \frac{3}{4}mc^2 \end{aligned}$$

Substituting the values of the moments and products of inertia into Eq. (9.56) and collecting terms yields

$$K^3 - \left(\frac{28}{3}mc^2\right)K^2 + \left(\frac{3479}{144}m^2c^4\right)K - \frac{589}{54}m^3c^6 = 0$$

Now solve for the roots of this equation; from the discussion in Sec. 9.6C, it follows that these roots are the principal moments of inertia of the body at the origin.

$$\begin{aligned} K_1 &= 0.568867mc^2 & K_2 &= 4.20885mc^2 & K_3 &= 4.55562mc^2 \\ K_1 &= 0.569mc^2 & K_2 &= 4.21mc^2 & K_3 &= 4.56mc^2 \end{aligned} \quad \blacktriangleleft$$

b. Principal Axes of Inertia at O . To determine the direction of a principal axis of inertia, first substitute the corresponding value of K into two of the equations (9.54). The resulting equations, together with Eq. (9.57), constitute a system of three equations from which you can determine the direction cosines of the corresponding principal axis. Thus, for the first principal moment of inertia K_1 , you have

$$\begin{aligned} \left(\frac{5}{3} - 0.568867\right)mc^2(\lambda_x)_1 - \frac{3}{2}mc^2(\lambda_y)_1 - \frac{3}{4}mc^2(\lambda_z)_1 &= 0 \\ -\frac{3}{2}mc^2(\lambda_x)_1 + \left(\frac{10}{3} - 0.568867\right)mc^2(\lambda_y)_1 - \frac{1}{2}mc^2(\lambda_z)_1 &= 0 \\ (\lambda_x)_1^2 + (\lambda_y)_1^2 + (\lambda_z)_1^2 &= 1 \end{aligned}$$

Solving yields

$$(\lambda_x)_1 = 0.836600 \quad (\lambda_y)_1 = 0.496001 \quad (\lambda_z)_1 = 0.232557$$

The angles that the first principal axis of inertia forms with the coordinate axes are then

$$(\theta_x)_1 = 33.2^\circ \quad (\theta_y)_1 = 60.3^\circ \quad (\theta_z)_1 = 76.6^\circ \quad \blacktriangleleft$$

Using the same set of equations successively with K_2 and K_3 , you can find that the angles associated with the second and third principal moments of inertia at the origin are, respectively,

$$(\theta_x)_2 = 57.8^\circ \quad (\theta_y)_2 = 146.6^\circ \quad (\theta_z)_2 = 98.0^\circ \quad \blacktriangleleft$$

and

$$(\theta_x)_3 = 82.8^\circ \quad (\theta_y)_3 = 76.1^\circ \quad (\theta_z)_3 = 164.3^\circ \quad \blacktriangleleft$$

SOLVING PROBLEMS ON YOUR OWN

In this section, we defined the **mass products of inertia** I_{xy} , I_{yz} , and I_{zx} of a body and showed you how to determine the moments of inertia of that body with respect to an arbitrary axis passing through the origin O . You also saw how to determine at the origin O the **principal axes of inertia** of a body and the corresponding **principal moments of inertia**.

1. Determining the mass products of inertia of a composite body. You can express the mass products of inertia of a composite body with respect to the coordinate axes as the sums of the products of inertia of its component parts with respect to those axes. For each component part, use the parallel-axis theorem to write Eqs. (9.47)

$$I_{xy} = \bar{I}_{x'y'} + m\bar{x}\bar{y} \quad I_{yz} = \bar{I}_{y'z'} + m\bar{y}\bar{z} \quad I_{zx} = \bar{I}_{z'x'} + m\bar{z}\bar{x}$$

Here the primes denote the centroidal axes of each component part, and $\bar{x}$, $\bar{y}$, and $\bar{z}$ represent the coordinates of its center of gravity. Keep in mind that the mass products of inertia can be positive, negative, or zero, and be sure to take into account the signs of $\bar{x}$, $\bar{y}$, and $\bar{z}$.

a. From the properties of symmetry of a component part, you can deduce that two or all three of its centroidal mass products of inertia are zero. For instance, you can verify for a thin plate parallel to the xy plane, a wire lying in a plane parallel to the xy plane, a body with a plane of symmetry parallel to the xy plane, and a body with an axis of symmetry parallel to the z axis that the products of inertia $\bar{I}_{y'z'}$ and $\bar{I}_{z'x'}$ are zero.

For rectangular, circular, or semicircular plates with axes of symmetry parallel to the coordinate axes, straight wires parallel to a coordinate axis, circular and semicircular wires with axes of symmetry parallel to the coordinate axes, and rectangular prisms with axes of symmetry parallel to the coordinate axes, the products of inertia $\bar{I}_{x'y'}$, $\bar{I}_{y'z'}$, and $\bar{I}_{z'x'}$ are all zero.

b. Mass products of inertia that are different from zero can be computed from Eqs. (9.45). Although, in general, you need a triple integration to determine a mass product of inertia, you can use a single integration if you can divide the given body into a series of thin, parallel slabs. The computations are then similar to those discussed in the preceding section for moments of inertia.

2. Computing the moment of inertia of a body with respect to an arbitrary axis OL . In Sec. 9.6A, we derived an expression for the moment of inertia I_{OL} that was given in Eq. (9.46). Before computing I_{OL} , you must first determine the mass moments and products of inertia of the body with respect to the given coordinate axes, as well as the direction cosines of the unit vector λ along OL .

3. Calculating the principal moments of inertia of a body and determining its principal axes of inertia. You saw in Sec. 9.6B that it is always possible to find an orientation of the coordinate axes for which the mass products of inertia are zero. These axes are referred to as the **principal axes of inertia**, and the corresponding moments of inertia are known as the **principal moments of inertia** of the body. In many cases, you can determine the principal axes of inertia of a body from its properties of symmetry. The procedure required to determine the principal moments and principal axes of inertia of a body with no obvious property of symmetry was discussed in Sec. 9.6C and was illustrated in Sample Prob. 9.15. It consists of the following steps.

a. Expand the determinant in Eq. (9.55) and solve the resulting cubic equation. You can obtain the solution by trial and error or (preferably) with an advanced scientific calculator or appropriate computer software. The roots K_1 , K_2 , and K_3 of this equation are the principal moments of inertia of the body.

b. To determine the direction of the principal axis corresponding to K_1 , substitute this value for K in two of the equations (9.54) and solve these equations, together with Eq. (9.57), for the direction cosines of the principal axis corresponding to K_1 .

c. Repeat this procedure with K_2 and K_3 to determine the directions of the other two principal axes. As a check of your computations, you may wish to verify that the scalar product of any two of the unit vectors along the three axes you have obtained is zero and, thus, that these axes are perpendicular to each other.

Problems

- 9.149** Determine the mass products of inertia I_{xy} , I_{yz} , and I_{zx} of the steel fixture shown. (The density of steel is 7850 kg/m^3 .)

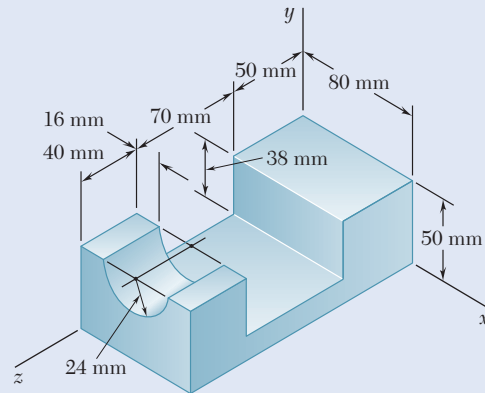


Fig. P9.149

- 9.150** Determine the mass products of inertia I_{xy} , I_{yz} , and I_{zx} of the steel machine element shown. (The density of steel is 7850 kg/m^3 .)

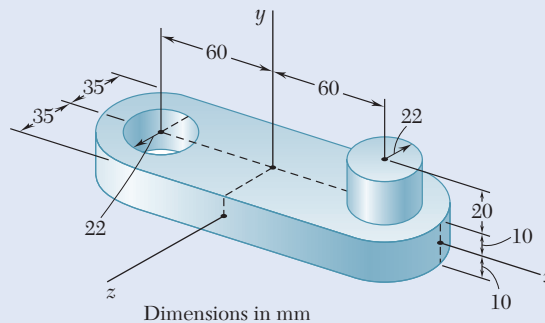


Fig. P9.150

- 9.151 and 9.152** Determine the mass products of inertia I_{xy} , I_{yz} , and I_{zx} of the cast aluminum machine component shown. (The specific weight of aluminum is 0.100 lb/in^3 .)

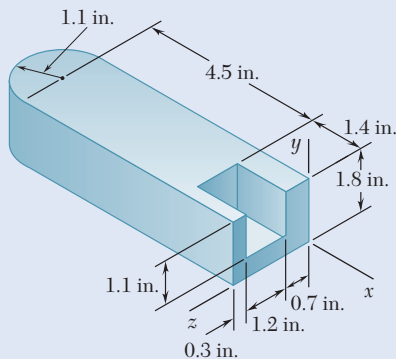


Fig. P9.151

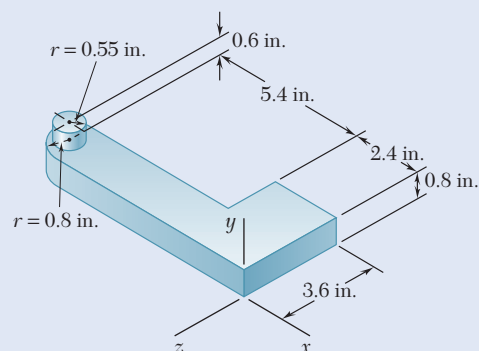


Fig. P9.152

9.153 through 9.156 A section of sheet steel 2 mm thick is cut and bent into the machine component shown. Knowing that the density of steel is 7850 kg/m^3 , determine the mass products of inertia I_{xy} , I_{yz} , and I_{zx} of the component.

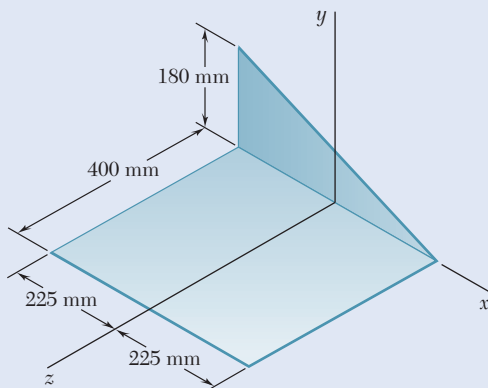


Fig. P9.153

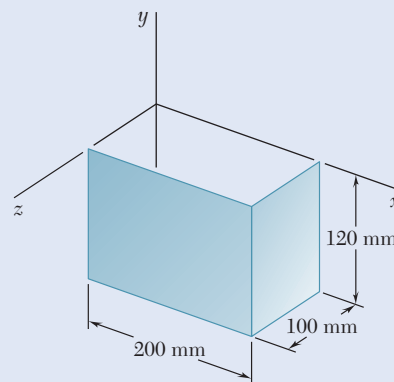


Fig. P9.154

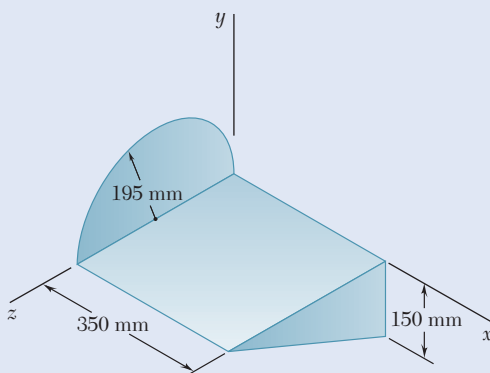


Fig. P9.155

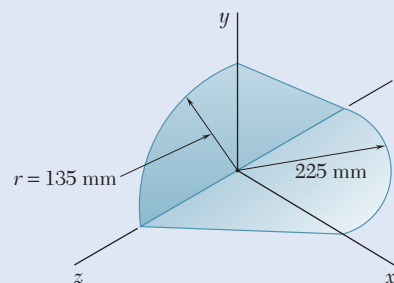


Fig. P9.156

9.157 The figure shown is formed of 1.5-mm-diameter aluminum wire. Knowing that the density of aluminum is 2800 kg/m^3 , determine the mass products of inertia I_{xy} , I_{yz} , and I_{zx} of the wire figure.

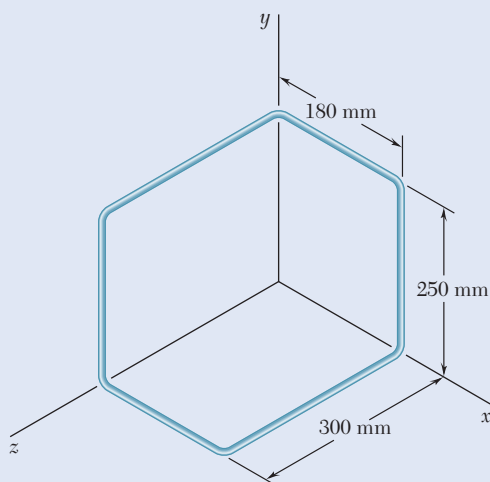


Fig. P9.157

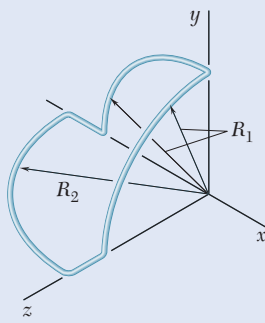


Fig. P9.158

9.158 Thin aluminum wire of uniform diameter is used to form the figure shown. Denoting the mass per unit length of the wire by m' , determine the mass products of inertia I_{xy} , I_{yz} , and I_{zx} of the wire figure.

9.159 and 9.160 Brass wire with a weight per unit length w is used to form the figure shown. Determine the mass products of inertia I_{xy} , I_{yz} , and I_{zx} of the wire figure.

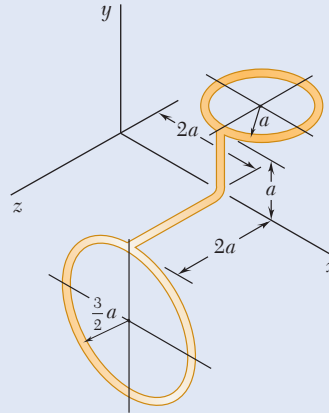


Fig. P9.159

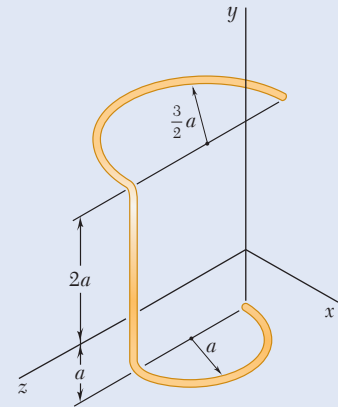


Fig. P9.160

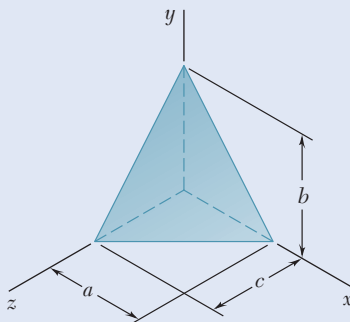


Fig. P9.162

9.161 Complete the derivation of Eqs. (9.47) that expresses the parallel-axis theorem for mass products of inertia.

9.162 For the homogeneous tetrahedron of mass m shown, (a) determine by direct integration the mass product of inertia I_{zx} , (b) deduce I_{yz} and I_{xy} from the result obtained in part a.

9.163 The homogeneous circular cone shown has a mass m . Determine the mass moment of inertia of the cone with respect to the line joining the origin O and point A .

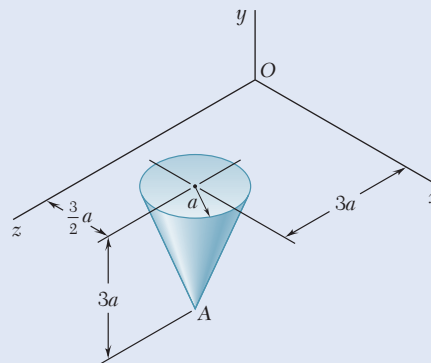


Fig. P9.163

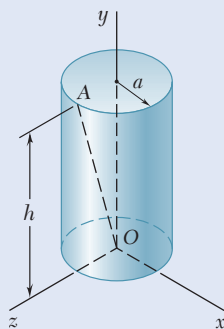


Fig. P9.164

9.164 The homogeneous circular cylinder shown has a mass m . Determine the mass moment of inertia of the cylinder with respect to the line joining the origin O and point A that is located on the perimeter of the top surface of the cylinder.

- 9.165** Shown is the machine element of Prob. 9.141. Determine its mass moment of inertia with respect to the line joining the origin O and point A .

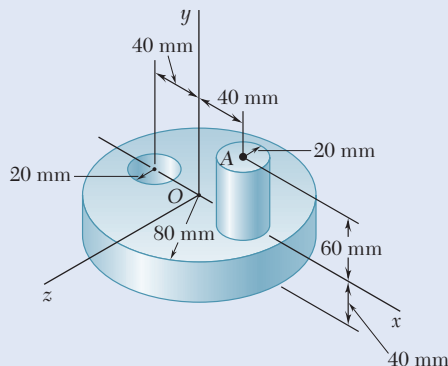


Fig. P9.165

- 9.166** Determine the mass moment of inertia of the steel fixture of Probs. 9.145 and 9.149 with respect to the axis through the origin that forms equal angles with the x , y , and z axes.
- 9.167** The thin, bent plate shown is of uniform density and weight W . Determine its mass moment of inertia with respect to the line joining the origin O and point A .

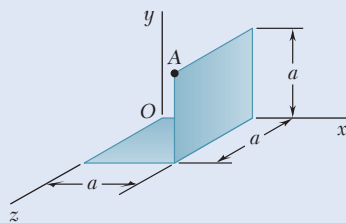


Fig. P9.167

- 9.168** A piece of sheet steel with thickness t and specific weight γ is cut and bent into the machine component shown. Determine the mass moment of inertia of the component with respect to the line joining the origin O and point A .
- 9.169** Determine the mass moment of inertia of the machine component of Probs. 9.136 and 9.155 with respect to the axis through the origin characterized by the unit vector $\lambda = (-4\mathbf{i} + 8\mathbf{j} + \mathbf{k})/9$.
- 9.170 through 9.172** For the wire figure of the problem indicated, determine the mass moment of inertia of the figure with respect to the axis through the origin characterized by the unit vector $\lambda = (-3\mathbf{i} - 6\mathbf{j} + 2\mathbf{k})/7$.

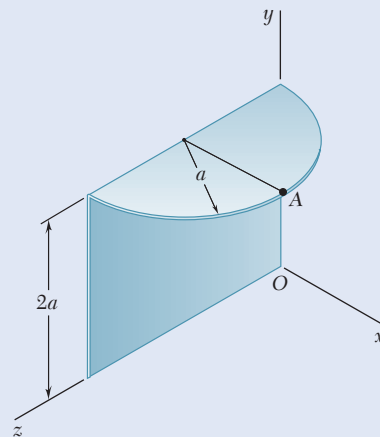


Fig. P9.168

9.170 Prob. 9.148

9.171 Prob. 9.147

9.172 Prob. 9.146

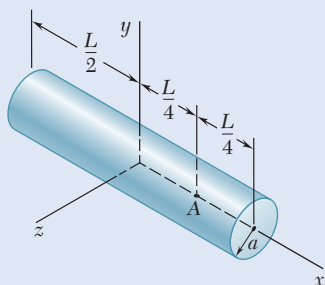


Fig. P9.173

9.173 For the homogeneous circular cylinder shown with radius a and length L , determine the value of the ratio a/L for which the ellipsoid of inertia of the cylinder is a sphere when computed (a) at the centroid of the cylinder, (b) at point A.

9.174 For the rectangular prism shown, determine the values of the ratios b/a and c/a so that the ellipsoid of inertia of the prism is a sphere when computed (a) at point A, (b) at point B.

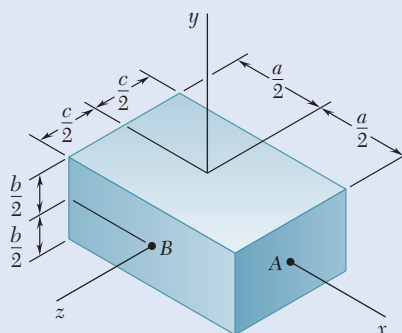


Fig. P9.174

9.175 For the right circular cone of Sample Prob. 9.11, determine the value of the ratio a/h for which the ellipsoid of inertia of the cone is a sphere when computed (a) at the apex of the cone, (b) at the center of the base of the cone.

9.176 Given an arbitrary body and three rectangular axes x , y , and z , prove that the mass moment of inertia of the body with respect to any one of the three axes cannot be larger than the sum of the mass moments of inertia of the body with respect to the other two axes. That is, prove that the inequality $I_x \leq I_y + I_z$ and the two similar inequalities are satisfied. Furthermore, prove that $I_y \geq \frac{1}{2}I_x$ if the body is a homogeneous solid of revolution, where x is the axis of revolution and y is a transverse axis.

9.177 Consider a cube with mass m and side a . (a) Show that the ellipsoid of inertia at the center of the cube is a sphere, and use this property to determine the moment of inertia of the cube with respect to one of its diagonals. (b) Show that the ellipsoid of inertia at one of the corners of the cube is an ellipsoid of revolution, and determine the principal moments of inertia of the cube at that point.

9.178 Given a homogeneous body of mass m and of arbitrary shape and three rectangular axes x , y , and z with origin at O , prove that the sum $I_x + I_y + I_z$ of the mass moments of inertia of the body cannot be smaller than the similar sum computed for a sphere of the same mass and the same material centered at O . Furthermore, using the result of Prob. 9.176, prove that, if the body is a solid of revolution where x is the axis of revolution, its mass moment of inertia I_y about a transverse axis y cannot be smaller than $3ma^2/10$, where a is the radius of the sphere of the same mass and the same material.

- *9.179** The homogeneous circular cylinder shown has a mass m , and the diameter OB of its top surface forms 45° angles with the x and z axes. (a) Determine the principal mass moments of inertia of the cylinder at the origin O . (b) Compute the angles that the principal axes of inertia at O form with the coordinate axes. (c) Sketch the cylinder, and show the orientation of the principal axes of inertia relative to the x , y , and z axes.

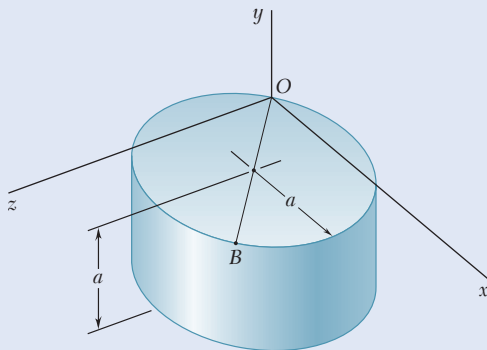


Fig. P9.179

- 9.180 through 9.184** For the component described in the problem indicated, determine (a) the principal mass moments of inertia at the origin, (b) the principal axes of inertia at the origin. Sketch the body and show the orientation of the principal axes of inertia relative to the x , y , and z axes.

***9.180** Prob. 9.165

***9.181** Probs. 9.145 and 9.149

***9.182** Prob. 9.167

***9.183** Prob. 9.168

***9.184** Probs. 9.148 and 9.170

Review and Summary

In the first half of this chapter, we discussed how to determine the resultant $\mathbf{R}$ of forces $\Delta\mathbf{F}$ distributed over a plane area A when the magnitudes of these forces are proportional to both the areas ΔA of the elements on which they act and the distances y from these elements to a given x axis; we thus had $\Delta F = ky \Delta A$. We found that the magnitude of the resultant $\mathbf{R}$ is proportional to the first moment $Q_x = \int y dA$ of area A , whereas the moment of $\mathbf{R}$ about the x axis is proportional to the **second moment**, or **moment of inertia**, $I_x = \int y^2 dA$ of A with respect to the same axis [Sec. 9.1A].

Rectangular Moments of Inertia

The **rectangular moments of inertia** I_x and I_y of an area [Sec. 9.1B] are obtained by evaluating the integrals

$$I_x = \int y^2 dA \quad I_y = \int x^2 dA \quad (9.1)$$

We can reduce these computations to single integrations by choosing dA to be a thin strip parallel to one of the coordinate axes. We also recall that it is possible to compute I_x and I_y from the same elemental strip (Fig. 9.35) using the formula for the moment of inertia of a rectangular area [Sample Prob. 9.3].

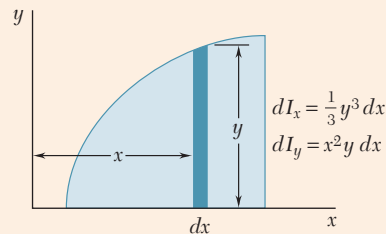


Fig. 9.35

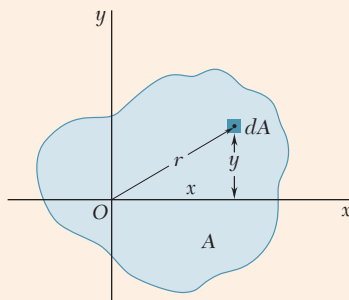


Fig. 9.36

Polar Moment of Inertia

We defined the **polar moment of inertia of an area** A with respect to the pole O [Sec. 9.1C] as

$$J_O = \int r^2 dA \quad (9.3)$$

where r is the distance from O to the element of area dA (Fig. 9.36). Observing that $r^2 = x^2 + y^2$, we established the relation

$$J_O = I_x + I_y \quad (9.4)$$

Radius of Gyration

We defined the **radius of gyration of an area** A with respect to the x axis [Sec. 9.1D] as the distance k_x , where $I_x = k_x^2 A$. With similar definitions for the radii of gyration of A with respect to the y axis and with respect to O , we have

$$k_x = \sqrt{\frac{I_x}{A}} \quad k_y = \sqrt{\frac{I_y}{A}} \quad k_O = \sqrt{\frac{J_O}{A}} \quad (9.5-9.7)$$

Parallel-Axis Theorem

The **parallel-axis theorem**, presented in Sec. 9.2A, states that the moment of inertia I of an area with respect to any given axis AA' (Fig. 9.37) is equal to the moment of inertia $\bar{I}$ of the area with respect to the centroidal axis BB' that is parallel to AA' *plus* the product of the area A and the square of the distance d between the two axes:

$$I = \bar{I} + Ad^2 \quad (9.9)$$

You can use this formula to determine the moment of inertia $\bar{I}$ of an area with respect to a centroidal axis BB' if you know its moment of inertia I with respect to a parallel axis AA' . In this case, however, the product Ad^2 should be *subtracted* from the known moment of inertia I .

A similar relation holds between the polar moment of inertia J_O of an area about a point O and the polar moment of inertia $\bar{J}_C$ of the same area about its centroid C . Letting d be the distance between O and C , we have

$$J_C = \bar{J}_C + Ad^2 \quad (9.11)$$

Composite Areas

The parallel-axis theorem can be used very effectively to compute the **moment of inertia of a composite area** with respect to a given axis [Sec. 9.2B]. Considering each component area separately, we first compute the moment of inertia of each area with respect to its centroidal axis, using the data provided in Figs. 9.12 and 9.13 whenever possible. Then apply the parallel-axis theorem to determine the moment of inertia of each component area with respect to the desired axis, and add the values [Sample Probs. 9.4 and 9.5].

Product of Inertia

Section 9.3 was devoted to the transformation of the moments of inertia of an area under a rotation of the coordinate axes. First, we defined the **product of inertia of an area** A as

$$I_{xy} = \int xy dA \quad (9.12)$$

and showed that $I_{xy} = 0$ if the area A is symmetrical with respect to either or both of the coordinate axes. We also derived the **parallel-axis theorem for products of inertia** as

$$I_{xy} = \bar{I}_{x'y'} + \bar{x}\bar{y}A \quad (9.13)$$

where $\bar{I}_{x'y'}$ is the product of inertia of the area with respect to the centroidal axes x' and y' that are parallel to the x and y axes and $\bar{x}$ and $\bar{y}$ are the coordinates of the centroid of the area [Sec. 9.3A].

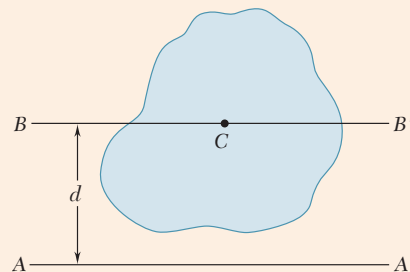


Fig. 9.37

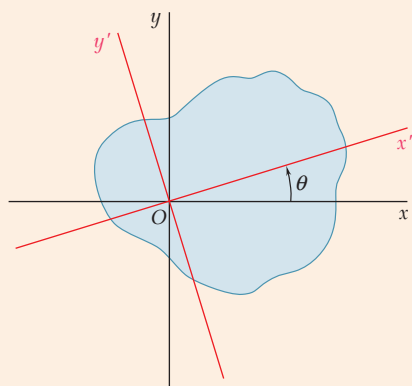


Fig. 9.38

Rotation of Axes

In Sec. 9.3B, we determined the moments and product of inertia $I_{x'}$, $I_{y'}$, and $I_{x'y'}$ of an area with respect to x' and y' axes obtained by rotating the original x and y coordinate axes counterclockwise through an angle θ (Fig. 9.38). We expressed $I_{x'}$, $I_{y'}$, and $I_{x'y'}$ in terms of the moments and product of inertia I_x , I_y , and I_{xy} computed with respect to the original x and y axes.

$$I_{x'} = \frac{I_x + I_y}{2} + \frac{I_x - I_y}{2} \cos 2\theta - I_{xy} \sin 2\theta \quad (9.18)$$

$$I_{y'} = \frac{I_x + I_y}{2} - \frac{I_x - I_y}{2} \cos 2\theta + I_{xy} \sin 2\theta \quad (9.19)$$

$$I_{x'y'} = \frac{I_x - I_y}{2} \sin 2\theta + I_{xy} \cos 2\theta \quad (9.20)$$

Principal Axes

We defined the **principal axes of the area about O** as the two axes perpendicular to each other with respect to which the moments of inertia of the area are maximum and minimum. The corresponding values of θ , denoted by θ_m , were obtained from

$$\tan 2\theta_m = -\frac{2I_{xy}}{I_x - I_y} \quad (9.25)$$

Principal Moments of Inertia

The corresponding maximum and minimum values of I are called the **principal moments of inertia** of the area about O :

$$I_{\max, \min} = \frac{I_x + I_y}{2} \pm \sqrt{\left(\frac{I_x - I_y}{2}\right)^2 + I_{xy}^2} \quad (9.27)$$

We also noted that the corresponding value of the product of inertia is zero.

Mohr's Circle

The transformation of the moments and product of inertia of an area under a rotation of axes can be represented graphically by drawing **Mohr's circle** [Sec. 9.4]. Given the moments and product of inertia I_x , I_y , and I_{xy} of the area with respect to the x and y coordinate axes, we plot points $X (I_x, I_{xy})$ and $Y (I_y, -I_{xy})$ and draw the line joining these two points (Fig. 9.39). This line is a diameter of Mohr's circle and thus defines this circle. As the coordinate axes are rotated through θ , the diameter rotates through *twice that angle*, and the coordinates of X' and Y' yield the new values $I_{x'}$, $I_{y'}$, and $I_{x'y'}$ of the moments and product of inertia of the area. Also, the angle θ_m and the coordinates of points A and B define the principal axes a and b and the principal moments of inertia of the area [Sample Prob. 9.8].

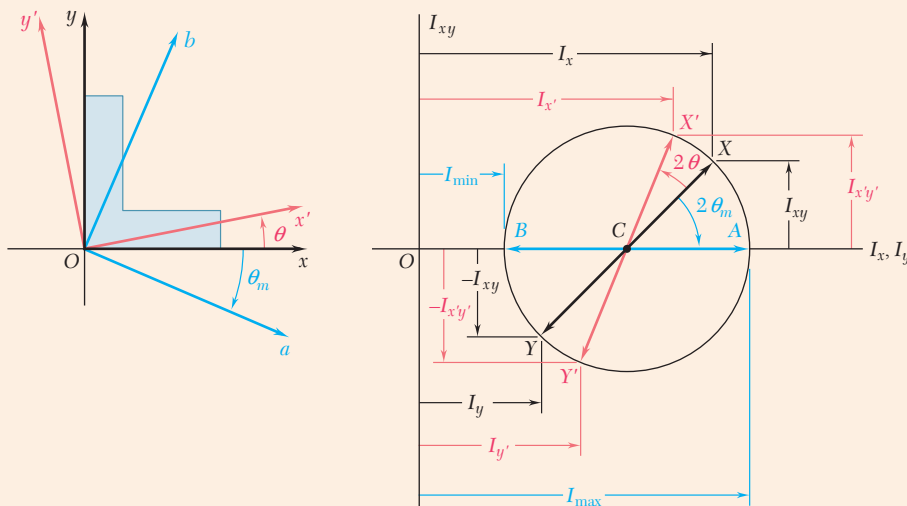


Fig. 9.39

Moments of Inertia of Masses

The second half of the chapter was devoted to determining **moments of inertia of masses**, which are encountered in dynamics problems involving the rotation of a rigid body about an axis. We defined the mass moment of inertia of a body with respect to an axis AA' (Fig. 9.40) as

$$I = \int r^2 dm \quad (9.28)$$

where r is the distance from AA' to the element of mass [Sec. 9.5A]. We defined the **radius of gyration** of the body as

$$k = \sqrt{\frac{I}{m}} \quad (9.29)$$

The moments of inertia of a body with respect to the coordinate axes were expressed as

$$\begin{aligned} I_x &= \int (y^2 + z^2) dm \\ I_y &= \int (z^2 + x^2) dm \\ I_z &= \int (x^2 + y^2) dm \end{aligned} \quad (9.30)$$

Parallel-Axis Theorem

We saw that the **parallel-axis theorem** also applies to mass moments of inertia [Sec. 9.5B]. Thus, the moment of inertia I of a body with respect to an arbitrary axis AA' (Fig. 9.41) can be expressed as

$$I = \bar{I} + md^2 \quad (9.33)$$

where $\bar{I}$ is the moment of inertia of the body with respect to the centroidal axis BB' that is parallel to the axis AA' , m is the mass of the body, and d is the distance between the two axes.

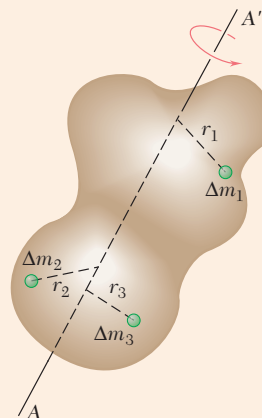


Fig. 9.40

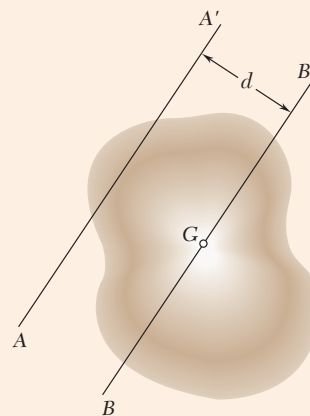


Fig. 9.41

Moments of Inertia of Thin Plates

We can readily obtain the moments of inertia of thin plates from the moments of inertia of their areas [Sec. 9.5C]. We found that for a rectangular plate the moments of inertia with respect to the axes shown (Fig. 9.42) are

$$I_{AA'} = \frac{1}{12}ma^2 \quad I_{BB'} = \frac{1}{12}mb^2 \quad (9.39)$$

$$I_{CC'} = I_{AA'} + I_{BB'} = \frac{1}{12}m(a^2 + b^2) \quad (9.40)$$

whereas for a circular plate (Fig. 9.43), they are

$$I_{AA'} = I_{BB'} = \frac{1}{4}mr^2 \quad (9.41)$$

$$I_{CC'} = I_{AA'} + I_{BB'} = \frac{1}{2}mr^2 \quad (9.42)$$

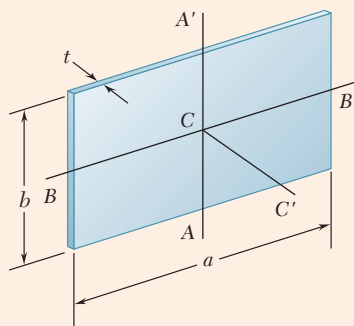


Fig. 9.42

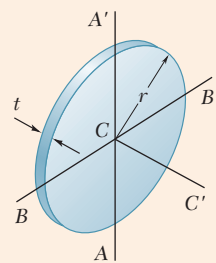


Fig. 9.43

Composite Bodies

When a body possesses two planes of symmetry, it is usually possible to use a single integration to determine its moment of inertia with respect to a given axis by selecting the element of mass dm to be a thin plate [Sample Probs. 9.10 and 9.11]. On the other hand, when a body consists of several common geometric shapes, we can obtain its moment of inertia with respect to a given axis by using the formulas given in Fig. 9.28 together with the parallel-axis theorem [Sample Probs. 9.12 and 9.13].

Moment of Inertia with Respect to an Arbitrary Axis

In the last section of the chapter, we described how to determine the moment of inertia of a body with respect to an arbitrary axis OL that is drawn through the origin O [Sec. 9.6A]. We denoted the components of the unit vector λ along OL by λ_x , λ_y , and λ_z (Fig. 9.44) and introduced the **products of inertia** as

$$I_{xy} = \int xy \, dm \quad I_{yz} = \int yz \, dm \quad I_{zx} = \int zx \, dm \quad (9.45)$$

We found that the moment of inertia of the body with respect to OL could be expressed as

$$I_{OL} = I_x\lambda_x^2 + I_y\lambda_y^2 + I_z\lambda_z^2 - 2I_{xy}\lambda_x\lambda_y - 2I_{yz}\lambda_y\lambda_z - 2I_{zx}\lambda_z\lambda_x \quad (9.46)$$

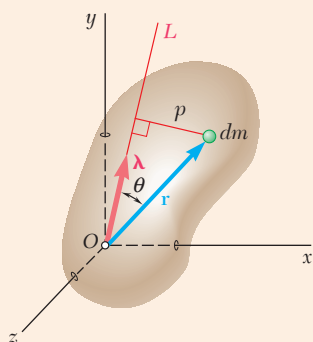


Fig. 9.44

Ellipsoid of Inertia

By plotting a point Q along each axis OL at a distance $OQ = 1/\sqrt{I_{OL}}$ from O [Sec. 9.6B], we obtained the surface of an ellipsoid, known as the **ellipsoid of inertia** of the body at point O .

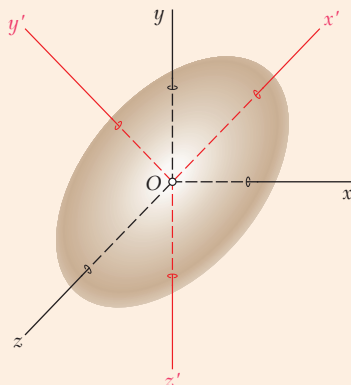


Fig. 9.45

Principal Axes and Principal Moments of Inertia

The principal axes x' , y' , and z' of this ellipsoid (Fig. 9.45) are the **principal axes of inertia** of the body; that is, the products of inertia $I_{x'y'}$, $I_{y'z'}$, and $I_{z'x'}$ of the body with respect to these axes are all zero. In many situations, you can deduce the principal axes of inertia of a body from its properties of symmetry. Choosing these axes to be the coordinate axes, we can then express I_{OL} as

$$I_{OL} = I_{x'}\lambda_{x'}^2 + I_{y'}\lambda_{y'}^2 + I_{z'}\lambda_{z'}^2 \quad (9.50)$$

where $I_{x'}$, $I_{y'}$, and $I_{z'}$ are the **principal moments of inertia** of the body at O .

When the principal axes of inertia cannot be obtained by observation [Sec. 9.6B], it is necessary to solve the cubic equation

$$K^3 - (I_x + I_y + I_z)K^2 + (I_x I_y + I_y I_z + I_z I_x - I_{xy}^2 - I_{yz}^2 - I_{zx}^2)K - (I_x I_y I_z - I_x I_{yz}^2 - I_y I_{zx}^2 - I_z I_{xy}^2 - 2I_{xy} I_{yz} I_{zx}) = 0 \quad (9.56)$$

We found [Sec. 9.6C] that the roots K_1 , K_2 , and K_3 of this equation are the principal moments of inertia of the given body. The direction cosines $(\lambda_x)_1$, $(\lambda_y)_1$, and $(\lambda_z)_1$ of the principal axis corresponding to the principal moment of inertia K_1 are then determined by substituting K_1 into Eqs. (9.54) and by solving two of these equations and Eq. (9.57) simultaneously. The same procedure is then repeated using K_2 and K_3 to determine the direction cosines of the other two principal axes [Sample Prob. 9.15].

Review Problems

- 9.185** Determine by direct integration the moments of inertia of the shaded area with respect to the x and y axes.

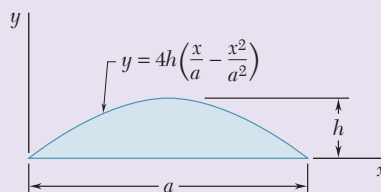


Fig. P9.185

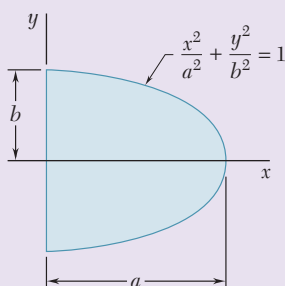


Fig. P9.186

- 9.186** Determine the moment of inertia and the radius of gyration of the shaded area shown with respect to the y axis.

- 9.187** Determine the moment of inertia and the radius of gyration of the shaded area shown with respect to the x axis.

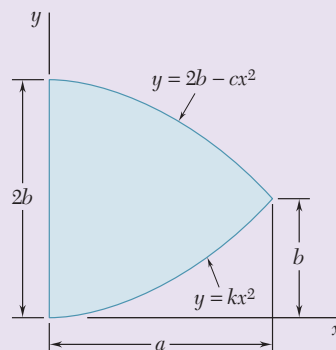


Fig. P9.187

- 9.188** Determine the moments of inertia $\bar{I}_x$ and $\bar{I}_y$ of the area shown with respect to centroidal axes respectively parallel and perpendicular to side AB .

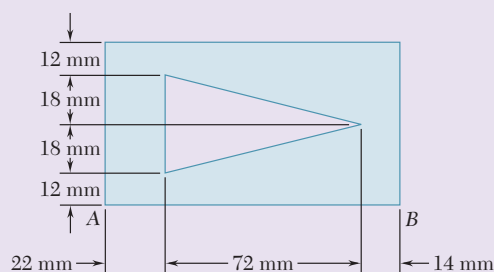


Fig. P9.188

9.189 Determine the polar moment of inertia of the area shown with respect to (a) point O , (b) the centroid of the area.

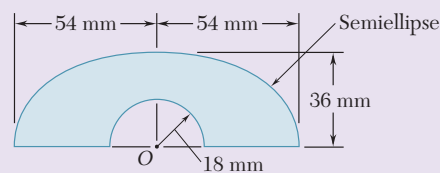


Fig. P9.189

9.190 Two $L4 \times 4 \times \frac{1}{2}$ -in. angles are welded to a steel plate as shown. Determine the moments of inertia of the combined section with respect to centroidal axes respectively parallel and perpendicular to the plate.

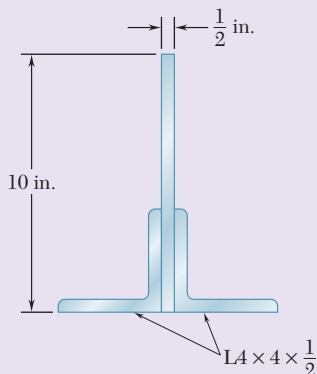


Fig. P9.190

9.191 Using the parallel-axis theorem, determine the product of inertia of the $L5 \times 3 \times \frac{1}{2}$ -in. angle cross section shown with respect to the centroidal x and y axes.

9.192 For the $L5 \times 3 \times \frac{1}{2}$ -in. angle cross section shown, use Mohr's circle to determine (a) the moments of inertia and the product of inertia with respect to new centroidal axes obtained by rotating the x and y axes 30° clockwise, (b) the orientation of the principal axes through the centroid and the corresponding values of the moments of inertia.

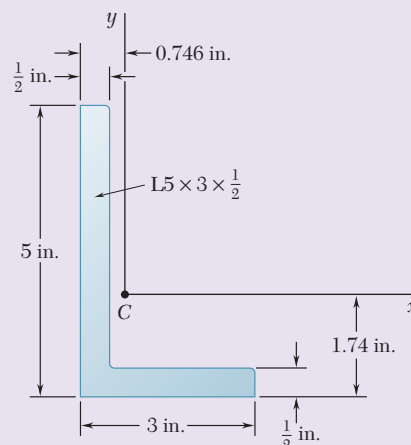


Fig. P9.191 and P9.192

9.193 A thin plate with a mass m was cut in the shape of a parallelogram as shown. Determine the mass moment of inertia of the plate with respect to (a) the x axis, (b) the axis BB' that is perpendicular to the plate.

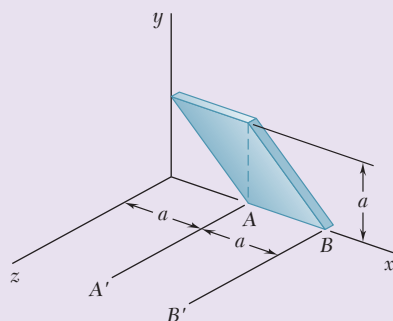


Fig. P9.193 and P9.194

9.194 A thin plate with mass m was cut in the shape of a parallelogram as shown. Determine the mass moment of inertia of the plate with respect to (a) the y axis, (b) the axis AA' that is perpendicular to the plate.

- 9.195** A 2-mm-thick piece of sheet steel is cut and bent into the machine component shown. Knowing that the density of steel is 7850 kg/m^3 , determine the mass moment of inertia of the component with respect to each of the coordinate axes.

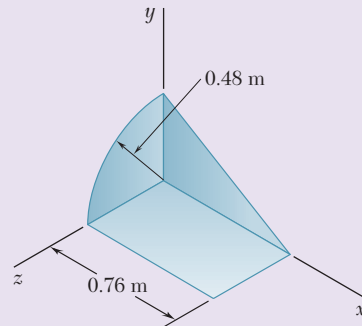


Fig. P9.195

- 9.196** Determine the mass moment of inertia of the steel machine element shown with respect to the z axis. (The specific weight of steel is 490 lb/ft^3 .)

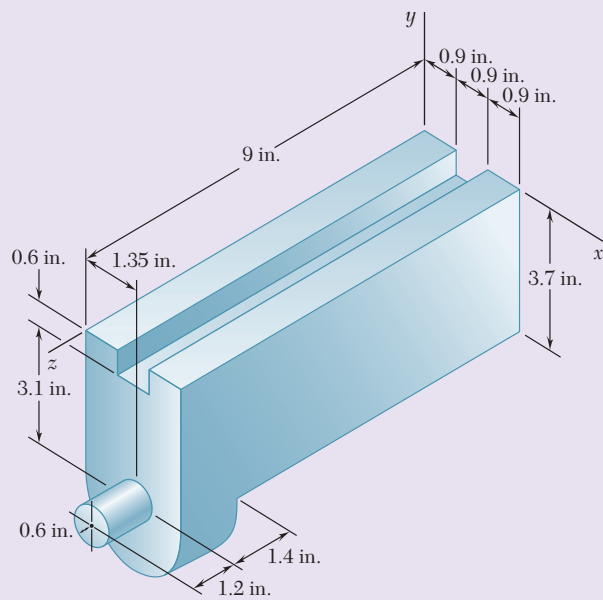


Fig. P9.196



10

Method of Virtual Work

The method of virtual work is particularly effective when a simple relation can be found among the displacements of the points of application of the various forces involved. This is the case for the scissor lift platform being used by workers to gain access to a highway bridge under construction.

Introduction**10.1 THE BASIC METHOD****10.1A** Work of a Force**10.1B** Principle of Virtual Work**10.1C** Applying the Principle of Virtual Work**10.1D** Mechanical Efficiency of Real Machines***10.2 WORK, POTENTIAL ENERGY, AND STABILITY****10.2A** Work of a Force During a Finite Displacement**10.2B** Potential Energy**10.2C** Potential Energy and Equilibrium**10.2D** Stability of Equilibrium**Objectives**

- **Define** the work of a force, and consider the circumstances when a force does no work.
- **Examine** the principle of virtual work, and apply it to analyze the equilibrium of machines and mechanisms.
- **Apply** the concept of potential energy to determine the equilibrium position of a rigid body or a system of rigid bodies.
- **Evaluate** the mechanical efficiency of machines, and consider the stability of equilibrium.

***Introduction**

In the preceding chapters, we solved problems involving the equilibrium of rigid bodies by expressing the balance of external forces acting on the bodies. We wrote the equations of equilibrium $\Sigma F_x = 0$, $\Sigma F_y = 0$, and $\Sigma M_A = 0$ and solved them for the desired unknowns. We now consider a different method, which turns out to be more effective for solving certain types of equilibrium problems. This method, based on the **principle of virtual work**, was first formally used by the Swiss mathematician Jean Bernoulli in the eighteenth century.

As you will see in Sec. 10.1B, the principle of virtual work considers a particle or rigid body (or more generally, a system of connected rigid bodies) that is in equilibrium under various external forces. The principle states that, if the body is given an arbitrary displacement from that position of equilibrium, the total work done by the external forces during the displacement is zero. This principle is particularly effective when applied to the solution of problems involving the equilibrium of machines or mechanisms consisting of several connected members.

In the second part of this chapter, we apply the method of virtual work in an alternative form based on the concept of **potential energy**. We will show in Sec. 10.2 that, if a particle, rigid body, or system of rigid bodies is in equilibrium, the derivative of its potential energy with respect to a variable defining its position must be zero.

You will also learn in this chapter to evaluate the mechanical efficiency of a machine (Sec. 10.1D) and to determine whether a given position of equilibrium is stable, unstable, or neutral (Sec. 10.2D).

***10.1 THE BASIC METHOD**

The first step in explaining the method of virtual work is to define the terms displacement and work as they are used in mechanics. Then we can state the principle of virtual work and show how to apply it in practical situations. We also take the opportunity to define mechanical efficiency, which is a useful and important parameter for the design of real machines.

10.1A Work of a Force

Consider a particle that moves from a point A to a neighboring point A' (Fig. 10.1). If $\mathbf{r}$ denotes the position vector corresponding to point A , we denote the small vector joining A and A' by the differential $d\mathbf{r}$; we call the vector $d\mathbf{r}$ the **displacement** of the particle.

Now let us assume that a force $\mathbf{F}$ is acting on the particle. The **work** dU of force $\mathbf{F}$ corresponding to the displacement $d\mathbf{r}$ is defined as the quantity

Definition of work

$$dU = \mathbf{F} \cdot d\mathbf{r} \quad (10.1)$$

That is, dU is the scalar product of the force $\mathbf{F}$ and the displacement $d\mathbf{r}$. Suppose we denote the magnitudes of the force by F , the displacement by ds , and the angle formed by $\mathbf{F}$ and $d\mathbf{r}$ by α . Then, recalling the definition of the scalar product of two vectors (Sec. 3.2A), we have

$$dU = F ds \cos \alpha \quad (10.1')$$

Work is a scalar quantity, so it has a magnitude and a sign, but no direction. Note that work should be expressed in units obtained by multiplying units of length by units of force. Thus, if we use U.S. customary units, we should express work in ft·lb or in·lb. If we use SI units, we express work in N·m. This unit of work is called a **joule** (J).[†]

It follows from (10.1') that work dU is positive if the angle α is acute and negative if α is obtuse. Three particular cases are of special interest.

- If the force $\mathbf{F}$ has the same direction as $d\mathbf{r}$, the work dU reduces to $F ds$.
- If $\mathbf{F}$ has a direction opposite to that of $d\mathbf{r}$, the work is $dU = -F ds$.
- Finally, if $\mathbf{F}$ is perpendicular to $d\mathbf{r}$, the work dU is zero.

We can also consider the work dU of a force $\mathbf{F}$ during a displacement $d\mathbf{r}$ to be the product of F and the component $ds \cos \alpha$ of the displacement $d\mathbf{r}$ along $\mathbf{F}$ (Fig. 10.2a). This view is particularly useful in computing the work done by the weight $\mathbf{W}$ of a body (Fig. 10.2b). The work of $\mathbf{W}$ is equal to the product of W and the vertical displacement dy of the center of gravity G of the body. If the displacement is downward, the work is positive; if the displacement is upward, the work is negative.

Some forces frequently encountered in statics do no work, such as forces applied to fixed points ($ds = 0$) or acting in a direction perpendicular to the displacement ($\cos \alpha = 0$). Among these forces are the reaction at a frictionless pin when the body supported rotates about the pin; the reaction at a frictionless surface when the body in contact moves along the surface; the reaction at a roller moving along its track; the weight of a body when its center of gravity moves horizontally; and the friction force

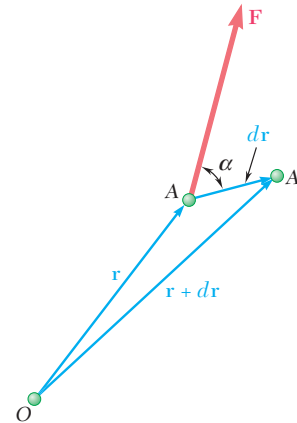


Fig. 10.1 The work of a force acting on a particle is the scalar product of the force and the particle's displacement.

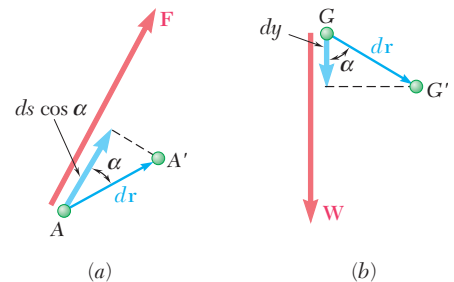


Fig. 10.2 (a) You can think of work as the product of a force and the component of displacement in the direction of the force. (b) This is useful for computing the work done by an object's weight.

[†]The joule is the SI unit of *energy*, whether in mechanical form (work, potential energy, kinetic energy) or in chemical, electrical, or thermal form. Note that even though $1 \text{ N} \cdot \text{m} = 1 \text{ J}$, we must express the moment of a force in $\text{N} \cdot \text{m}$, and not in joules, since the moment of a force is not a form of energy.

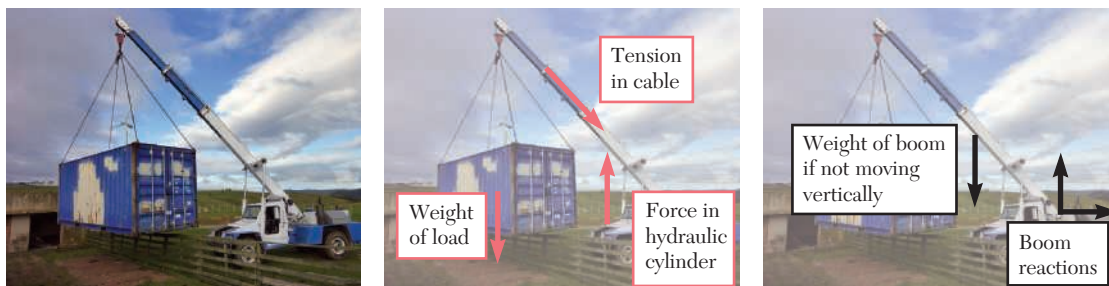


Photo 10.1 (a) In analyzing a crane, we might consider displacements associated with vertical movement of a container. (b) A force does work if it has a component in the direction of a displacement. (c) A force does no work if there is no displacement or if the force is perpendicular to a displacement.

acting on a wheel rolling without slipping (since at any instant the point of contact does not move). Examples of forces that do work are the weight of a body (except in the case considered previously), the friction force acting on a body sliding on a rough surface, and most forces applied on a moving body.

In certain cases, the sum of the work done by several forces is zero. Consider, for example, two rigid bodies AC and BC that are connected at C by a *frictionless pin* (Fig. 10.3a). Among the forces acting on AC is the force $\mathbf{F}$ exerted at C by BC . In general, the work of this force is not zero, but it is equal in magnitude and opposite in sign to the work of the force $-\mathbf{F}$ exerted by AC on BC , since these forces are equal and opposite and are applied to the same particle. Thus, when the total work done by all the forces acting on AC and BC is considered, the work of the two internal forces at C cancels out. We obtain a similar result if we consider a system consisting of two blocks connected by a *cord* AB that is not extensible (Fig. 10.3b). The work of the tension force $\mathbf{T}$ at A is equal in magnitude to the work of the tension force $\mathbf{T}'$ at B , since these forces have the same magnitude and the points A and B move through the same distance; but in one case, the work is positive, and in the other, it is negative. Thus, the work of the internal forces again cancels out.

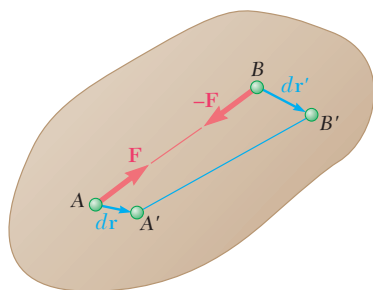


Fig. 10.4 As demonstrated here for an arbitrary pair of particles, the total work of the internal forces holding a rigid body together is zero.

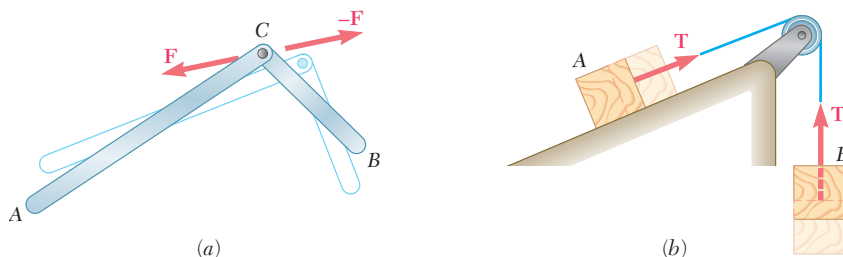


Fig. 10.3 (a) For a frictionless pin or (b) a cord that is not extensible, the total work done by the pairs of internal forces is zero.

We can show that the total work of the internal forces holding together the particles of a rigid body is zero. Consider two particles A and B of a rigid body and the two equal and opposite forces $\mathbf{F}$ and $-\mathbf{F}$ they exert on each other (Fig. 10.4). Although, in general, small displacements $d\mathbf{r}$ and $d\mathbf{r}'$ of the two particles are different, the components of these displacements along AB must be equal; otherwise, the particles would not

remain at the same distance from each other, so the body would not be rigid. Therefore, the work of $\mathbf{F}$ is equal in magnitude and opposite in sign to the work of $-\mathbf{F}$, and their sum is zero.

In computing the work of the external forces acting on a rigid body, it is often convenient to determine the work of a couple without considering separately the work of each of the two forces forming the couple. Consider the two forces $\mathbf{F}$ and $-\mathbf{F}$ forming a couple of moment $\mathbf{M}$ and acting on a rigid body (Fig. 10.5). Any small displacement of the rigid body bringing A and B , respectively, into A' and B'' can be divided into two parts: one in which points A and B undergo equal displacements $d\mathbf{r}_1$, the other in which A' remains fixed while B' moves into B'' through a displacement $d\mathbf{r}_2$ with a magnitude of $ds_2 = r d\theta$. In the first part of the motion, the work of $\mathbf{F}$ is equal in magnitude and opposite in sign to the work of $-\mathbf{F}$, and their sum is zero. In the second part of the motion, only force $\mathbf{F}$ works, and its work is $dU = F ds_2 = Fr d\theta$. But the product Fr is equal to the magnitude M of the moment of the couple. Thus, the work of a couple of moment $\mathbf{M}$ acting on a rigid body is

Work of a couple

$$dU = M d\theta \quad (10.2)$$

where $d\theta$ is the small angle (expressed in radians) through which the body rotates. We again note that work should be expressed in units obtained by multiplying units of force by units of length.

10.1B Principle of Virtual Work

Consider a particle acted upon by several forces $\mathbf{F}_1, \mathbf{F}_2, \dots, \mathbf{F}_n$ (Fig. 10.6). We can imagine that the particle undergoes a small displacement from A to A' . This displacement is possible, but it does not necessarily take place. The forces may be balanced and the particle remains at rest, or the particle may move under the action of the given forces in a direction different from that of AA' . Since the considered displacement does not actually occur, it is called a **virtual displacement**, which is denoted by $\delta\mathbf{r}$. The symbol $\delta\mathbf{r}$ represents a differential of the first order; it is used to distinguish the virtual displacement from the displacement $d\mathbf{r}$ that would take place under actual motion. As you will see, we can use virtual displacements to determine whether the conditions of equilibrium of a particle are satisfied.

The work of each of the forces $\mathbf{F}_1, \mathbf{F}_2, \dots, \mathbf{F}_n$ during the virtual displacement $\delta\mathbf{r}$ is called **virtual work**. The virtual work of all the forces acting on the particle of Fig. 10.6 is

$$\begin{aligned} \delta U &= \mathbf{F}_1 \cdot \delta\mathbf{r} + \mathbf{F}_2 \cdot \delta\mathbf{r} + \dots + \mathbf{F}_n \cdot \delta\mathbf{r} \\ &= (\mathbf{F}_1 + \mathbf{F}_2 + \dots + \mathbf{F}_n) \cdot \delta\mathbf{r} \end{aligned}$$

or

$$\delta U = \mathbf{R} \cdot \delta\mathbf{r} \quad (10.3)$$

where $\mathbf{R}$ is the resultant of the given forces. Thus, the total virtual work of the forces $\mathbf{F}_1, \mathbf{F}_2, \dots, \mathbf{F}_n$ is equal to the virtual work of their resultant $\mathbf{R}$.

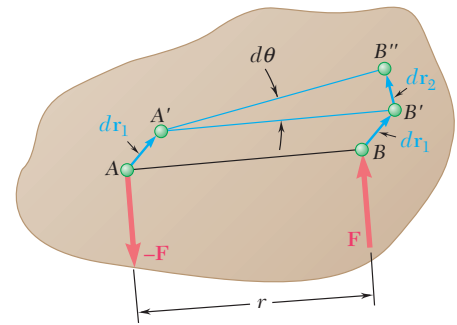


Fig. 10.5 The work of a couple acting on a rigid body is the moment of the couple times the angular rotation.

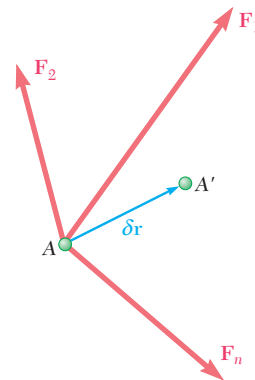


Fig. 10.6 Forces acting on a particle that goes through a virtual displacement.

The principle of virtual work for a particle states:

If a particle is in equilibrium, the total virtual work of the forces acting on the particle is zero for any virtual displacement of the particle.

This condition is necessary: if the particle is in equilibrium, the resultant $\mathbf{R}$ of the forces is zero, and it follows from Eq. (10.3) that the total virtual work δU is zero. The condition is also sufficient: if the total virtual work δU is zero for any virtual displacement, the scalar product $\mathbf{R} \cdot \delta \mathbf{r}$ is zero for any $\delta \mathbf{r}$, and the resultant $\mathbf{R}$ must be zero.

In the case of a rigid body, the principle of virtual work states:

If a rigid body is in equilibrium, the total virtual work of the external forces acting on the rigid body is zero for any virtual displacement of the body.

The condition is necessary: if the body is in equilibrium, all the particles forming the body are in equilibrium and the total virtual work of the forces acting on all the particles must be zero. However, we have seen in the preceding section that the total work of the internal forces is zero; therefore, the total work of the external forces also must be zero. The condition can also be proven to be sufficient.

The principle of virtual work can be extended to the case of a **system of connected rigid bodies**. If the system remains connected during the virtual displacement, **only the work of the forces external to the system need be considered**, since the total work of the internal forces at the various connections is zero.

10.1C Applying the Principle of Virtual Work

The principle of virtual work is particularly effective when applied to the solution of problems involving machines or mechanisms consisting of several connected rigid bodies. Consider, for instance, the toggle vise ACB of Fig. 10.7a used to compress a wooden block. Suppose we wish to determine the force exerted by the vise on the block when a given force $\mathbf{P}$ is applied at C , assuming there is no friction. Denoting the reaction of the block on the vise by $\mathbf{Q}$, we draw the free-body diagram of the vise and

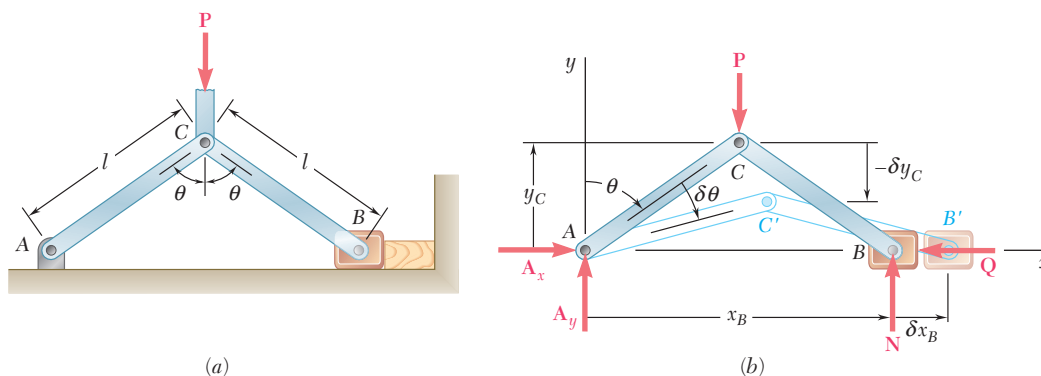


Fig. 10.7 (a) A toggle vise used to compress a wooden block, assuming no friction; (b) a virtual displacement of the vise.

consider the virtual displacement obtained by giving a positive increment $\delta\theta$ to angle θ (Fig. 10.7*b*). Choosing a system of coordinate axes with origin at A , we note that x_B increases as y_C decreases. This is indicated in the figure, where we indicate a positive increment δx_B and a negative increment $-\delta y_C$. The reactions $\mathbf{A}_x$, $\mathbf{A}_y$, and $\mathbf{N}$ do no work during the virtual displacement considered, so we need only compute the work done by $\mathbf{P}$ and $\mathbf{Q}$. Since $\mathbf{Q}$ and δx_B have opposite senses, the virtual work of $\mathbf{Q}$ is $\delta U_Q = -Q \delta x_B$. Since $\mathbf{P}$ and the increment shown ($-\delta y_C$) have the same sense, the virtual work of $\mathbf{P}$ is $\delta U_P = +P(-\delta y_C) = -P \delta y_C$. (We could have predicted the minus signs by simply noting that the forces $\mathbf{Q}$ and $\mathbf{P}$ are directed opposite to the positive x and y axes, respectively.) Expressing the coordinates x_B and y_C in terms of the angle θ and differentiating, we obtain

$$\begin{aligned} x_B &= 2l \sin \theta & y_C &= l \cos \theta \\ \delta x_B &= 2l \cos \theta \delta\theta & \delta y_C &= -l \sin \theta \delta\theta \end{aligned} \quad (10.4)$$

The total virtual work of the forces $\mathbf{Q}$ and $\mathbf{P}$ is thus

$$\begin{aligned} \delta U &= \delta U_Q + \delta U_P = -Q \delta x_B - P \delta y_C \\ &= -2Ql \cos \theta \delta\theta + Pl \sin \theta \delta\theta \end{aligned}$$

Setting $\delta U = 0$, we obtain

$$2Ql \cos \theta \delta\theta = Pl \sin \theta \delta\theta \quad (10.5)$$

and

$$Q = \frac{1}{2}P \tan \theta \quad (10.6)$$

The superiority of the method of virtual work over the conventional equilibrium equations in the problem considered here is clear: by using the method of virtual work, we were able to eliminate all unknown reactions, whereas the equation $\Sigma M_A = 0$ would have eliminated only two of the unknown reactions. This property of the method of virtual work can be used in solving many problems involving machines and mechanisms.

If the virtual displacement considered is consistent with the constraints imposed by the supports and connections, all reactions and internal forces are eliminated and only the work of the loads, applied forces, and friction forces need be considered.

We can also use the method of virtual work to solve problems involving completely constrained structures, although the virtual displacements considered never actually take place. Consider, for example, the frame ACB shown in Fig. 10.8*a*. If point A is kept fixed while point B is given a horizontal virtual displacement (Fig. 10.8*b*), we need consider only the work of $\mathbf{P}$ and $\mathbf{B}_x$. We can thus determine the reaction component $\mathbf{B}_x$ in the same way as the force $\mathbf{Q}$ of the preceding example (Fig. 10.7*b*); we have

$$B_x = \frac{1}{2}P \tan \theta$$

By keeping B fixed and giving a horizontal virtual displacement to A , we can similarly determine the reaction component $\mathbf{A}_x$. Then we can determine the components $\mathbf{A}_y$ and $\mathbf{B}_y$ by rotating the frame ACB as a rigid body about B and A , respectively.



Photo 10.2 The method of virtual work is useful for determining the forces exerted by the hydraulic cylinders positioning the bucket lift. The reason is that a simple relation exists among the displacements of the points of application of the forces acting on the members of the lift.

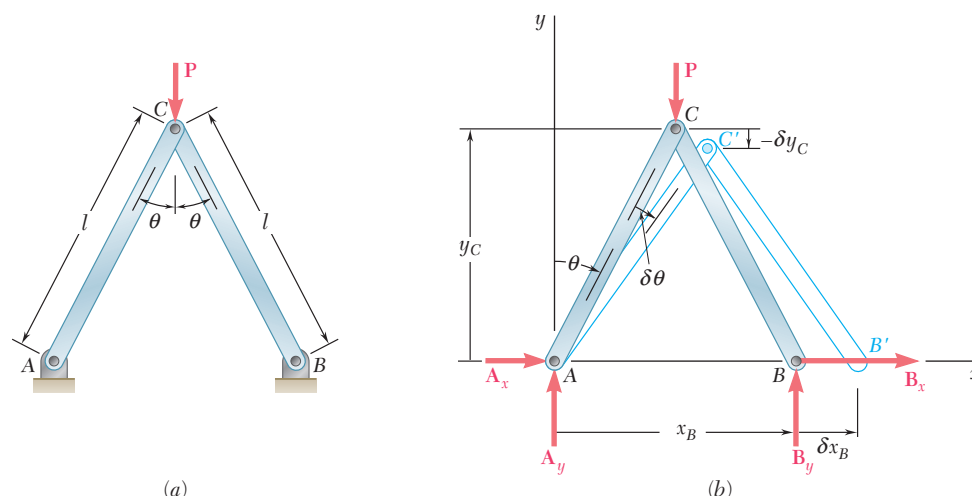


Fig. 10.8 (a) A completely constrained frame ACB; (b) a virtual displacement of the frame in order to determine B_x , keeping A fixed.

We can also use the method of virtual work to determine the configuration of a system in equilibrium under given forces. For example, we can obtain the value of the angle θ for which the linkage of Fig. 10.7 is in equilibrium under two given forces $\mathbf{P}$ and $\mathbf{Q}$ by solving Eq. (10.6) for $\tan \theta$.

Note, however, that the attractiveness of the method of virtual work depends to a large extent upon the existence of simple geometric relations between the various virtual displacements involved in the solution of a given problem. When no such simple relations exist, it is usually advisable to revert to the conventional method of Chap. 6.

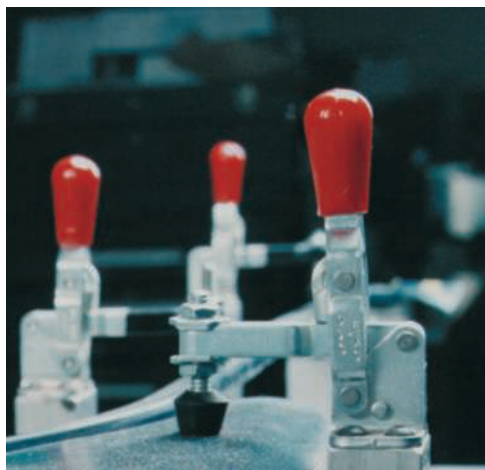


Photo 10.3 The clamping force of the toggle clamp shown can be expressed as a function of the force applied to the handle by first establishing the geometric relations among the members of the clamp and then applying the method of virtual work.

10.1D Mechanical Efficiency of Real Machines

In analyzing the toggle vise of Fig. 10.7, we assumed that no friction forces were involved. Thus, the virtual work consisted only of the work of the applied force $\mathbf{P}$ and of the reaction $\mathbf{Q}$. However, the work of reaction $\mathbf{Q}$ is equal in magnitude and opposite in sign to the work of the force exerted by the vise on the block. Therefore, Equation (10.5) states that the **output work** $2Ql \cos \theta \delta\theta$ is equal to the **input work** $Pl \sin \theta \delta\theta$. A machine in which input and output work are equal is said to be an “ideal” machine. In a “real” machine, friction forces always do some work, and the output work is smaller than the input work.

Consider again the toggle vise of Fig. 10.7a. and now assume that a friction force $\mathbf{F}$ develops between the sliding block B and the horizontal plane (Fig. 10.9). Using the conventional methods of statics and summing moments about A, we find $N = P/2$. Denoting the coefficient of friction between block B and the horizontal plane by μ , we have $F = \mu N = \mu P/2$.

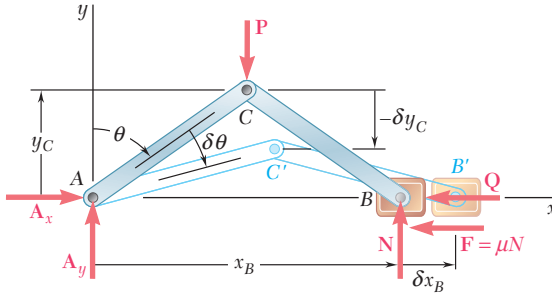


Fig. 10.9 A virtual displacement of the toggle vise with friction.

Recalling formulas (10.4), we find that the total virtual work of the forces **Q**, **P**, and **F** during the virtual displacement shown in Fig. 10.9 is

$$\begin{aligned}\delta U &= -Q \delta x_B - P \delta y_C - F \delta x_B \\ &= -2Ql \cos \theta \delta \theta + Pl \sin \theta \delta \theta - \mu Pl \cos \theta \delta \theta\end{aligned}$$

Setting $\delta U = 0$, we obtain

$$2Ql \cos \theta \delta \theta = Pl \sin \theta \delta \theta - \mu Pl \cos \theta \delta \theta \quad (10.7)$$

This equation states that the output work is equal to the input work minus the work of the friction force. Solving for Q , we have

$$Q = \frac{1}{2}P (\tan \theta - \mu) \quad (10.8)$$

Note that $Q = 0$ when $\tan \theta = \mu$, that is, when θ is equal to the angle of friction ϕ , and that $Q < 0$ when $\theta < \phi$. Thus, we can use the toggle vise only for values of θ larger than the angle of friction.

We define the **mechanical efficiency** η of a machine as the ratio

Mechanical efficiency

$$\eta = \frac{\text{output work}}{\text{input work}} \quad (10.9)$$

Clearly, the mechanical efficiency of an ideal machine is $\eta = 1$ when input and output work are equal, whereas the mechanical efficiency of a real machine is always less than 1.

In the case of the toggle vise we have just analyzed, we have

$$\eta = \frac{\text{output work}}{\text{input work}} = \frac{2Ql \cos \theta \delta \theta}{Pl \sin \theta \delta \theta} \quad (10.10)$$

We can check that, in the absence of friction forces, we would have $\mu = 0$ and $\eta = 1$. In the general case when μ is different from zero, the efficiency η becomes zero for $\mu \cot \theta = 1$, that is, for $\tan \theta = \mu$ or $\theta = \tan^{-1} \mu = \phi$. We note again that the toggle vise can be used only for values of θ larger than the angle of friction ϕ .

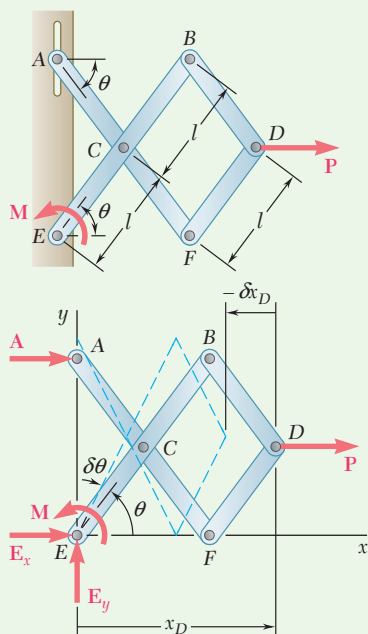


Fig. 1 Free-body diagram of mechanism showing a virtual displacement.

Sample Problem 10.1

Using the method of virtual work, determine the magnitude of the couple $\mathbf{M}$ required to maintain the equilibrium of the mechanism shown.

STRATEGY: For a virtual displacement consistent with the constraints, the reactions do no work, so you can focus solely on the force $\mathbf{P}$ and the moment $\mathbf{M}$. You can solve for $\mathbf{M}$ in terms of $\mathbf{P}$ and the geometric parameters.

MODELING: Choose a coordinate system with origin at E (Fig. 1). Then

$$x_D = 3l \cos \theta \quad \delta x_D = -3l \sin \theta \delta \theta$$

ANALYSIS: Principle of Virtual Work. Since the reactions $\mathbf{A}$, $\mathbf{E}_x$, and $\mathbf{E}_y$ do no work during the virtual displacement, the total virtual work done by $\mathbf{M}$ and $\mathbf{P}$ must be zero. Notice that $\mathbf{P}$ acts in the positive x direction and $\mathbf{M}$ acts in the positive θ direction. You obtain

$$\begin{aligned} \delta U = 0: \quad & +M \delta \theta + P \delta x_D = 0 \\ & +M \delta \theta + P(-3l \sin \theta \delta \theta) = 0 \end{aligned}$$

$$M = 3Pl \sin \theta \quad \blacktriangleleft$$

REFLECT and THINK: This problem illustrates that the principle of virtual work can help determine a moment as well as a force in a straightforward computation.

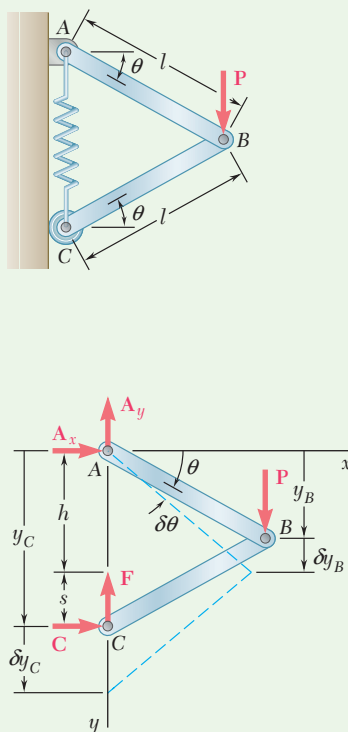


Fig. 1 Free-body diagram of mechanism showing a virtual displacement.

Sample Problem 10.2

Determine the expressions for θ and for the tension in the spring that correspond to the equilibrium position of the mechanism. The unstretched length of the spring is h , and the spring constant is k . Neglect the weight of the mechanism.

STRATEGY: The tension in the spring is a force $\mathbf{F}$ exerted at C . Applying the principle of virtual work, you can obtain a relationship between $\mathbf{F}$ and the applied force $\mathbf{P}$.

MODELING: With the coordinate system shown in Fig. 1,

$$\begin{aligned} y_B &= l \sin \theta & y_C &= 2l \sin \theta \\ \delta y_B &= l \cos \theta \delta \theta & \delta y_C &= 2l \cos \theta \delta \theta \end{aligned}$$

The elongation of the spring is $s = y_C - h = 2l \sin \theta - h$. The magnitude of the force exerted at C by the spring is

$$F = ks = k(2l \sin \theta - h) \quad (1)$$

ANALYSIS: Principle of Virtual Work. Since the reactions $\mathbf{A}_x$, $\mathbf{A}_y$, and $\mathbf{C}$ do no work, the total virtual work done by $\mathbf{P}$ and $\mathbf{F}$ must be zero.

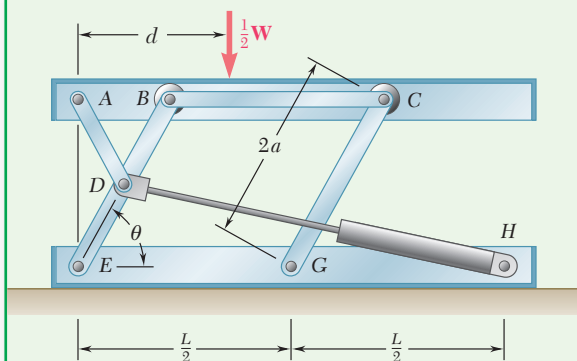
$$\begin{aligned} \delta U = 0: \quad & P \delta y_B - F \delta y_C = 0 \\ & P(l \cos \theta \delta \theta) - k(2l \sin \theta - h)(2l \cos \theta \delta \theta) = 0 \end{aligned}$$

$$\sin \theta = \frac{P + 2kh}{4kl} \quad \blacktriangleleft$$

$$F = \frac{1}{2}P \quad \blacktriangleleft$$

Substituting this expression into Eq. (1), you obtain

REFLECT and THINK: You can verify these results by applying the appropriate equations of equilibrium.



Sample Problem 10.3

A hydraulic-lift table is used to raise a 1000-kg crate. The table consists of a platform and two identical linkages on which hydraulic cylinders exert equal forces. (Only one linkage and one cylinder are shown.) Members EDB and CG are each of length $2a$, and member AD is pinned to the midpoint of EDB . If the crate is placed on the table so that half of its weight is supported by the system shown, determine the force exerted by each cylinder in raising the crate for $\theta = 60^\circ$, $a = 0.70$ m, and $L = 3.20$ m. (This mechanism was previously considered in Sample Prob. 6.7.)

STRATEGY: The principle of virtual work allows you to find a relationship between the force applied by the cylinder and the weight without involving the reactions. However, you need a relationship between the virtual displacement and the change in angle θ , which is found from the law of cosines applied to the given geometry.

MODELING: The free body consists of the platform and the linkage (Fig. 1), with an input force $\mathbf{F}_{DH}$ exerted by the cylinder and an output force equal and opposite to $\frac{1}{2}\mathbf{W}$.

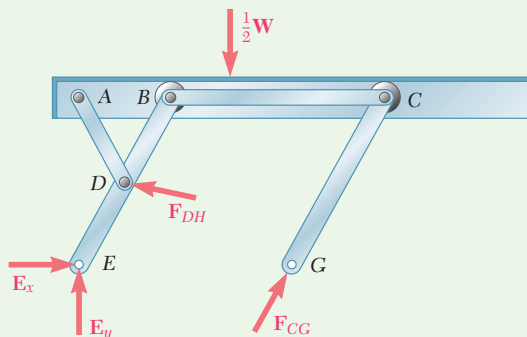


Fig. 1 Free-body diagram of the platform and linkage.

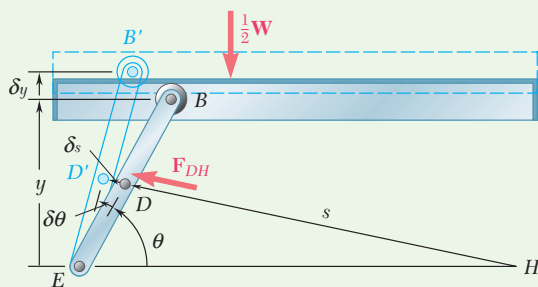


Fig. 2 Virtual displacement of the machine.

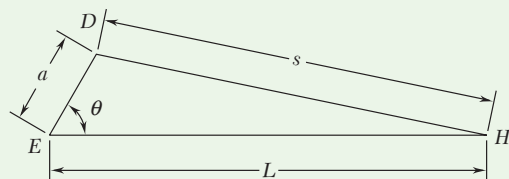


Fig. 3 Geometry associated with the cylinder-and-piston assembly.

ANALYSIS: Principle of Virtual Work. First observe that the reactions at E and G do no work. Denoting the elevation of the platform above the base by y and the length DH of the cylinder-and-piston assembly by s (Fig. 2), you have

$$\delta U = 0: \quad -\frac{1}{2}W \delta y + F_{DH} \delta s = 0 \quad (1)$$

You can express the vertical displacement δy of the platform in terms of the angular displacement $\delta\theta$ of EDB as

$$y = (EB) \sin \theta = 2a \sin \theta$$

$$\delta y = 2a \cos \theta \delta\theta$$

To express δs similarly in terms of $\delta\theta$, first note that by the law of cosines (Fig. 3),

$$s^2 = a^2 + L^2 - 2aL \cos \theta$$

Differentiating,

$$2s \delta s = -2aL(-\sin \theta) \delta \theta$$

$$\delta s = \frac{aL \sin \theta}{s} \delta \theta$$

Substituting for δy and δs into Eq. (1), you have

$$(-\tfrac{1}{2}W)2a \cos \theta \delta \theta + F_{DH} \frac{aL \sin \theta}{s} \delta \theta = 0$$

$$F_{DH} = W \frac{s}{L} \cot \theta$$

With the given numerical data, you obtain

$$W = mg = (1000 \text{ kg})(9.81 \text{ m/s}^2) = 9810 \text{ N} = 9.81 \text{ kN}$$

$$s^2 = a^2 + L^2 - 2aL \cos \theta$$

$$= (0.70)^2 + (3.20)^2 - 2(0.70)(3.20) \cos 60^\circ = 8.49$$

$$s = 2.91 \text{ m}$$

$$F_{DH} = W \frac{s}{L} \cot \theta = (9.81 \text{ kN}) \frac{2.91 \text{ m}}{3.20 \text{ m}} \cot 60^\circ$$

$$F_{DH} = 5.15 \text{ kN} \quad \blacktriangleleft$$

REFLECT and THINK: The principle of virtual work gives you a relationship between forces, but sometimes you need to review the geometry carefully to find a relationship between the displacements.

SOLVING PROBLEMS ON YOUR OWN

In this section, we described how to use the **method of virtual work**, which is a different way of solving problems involving the equilibrium of rigid bodies.

The work done by a force during a displacement of its point of application or by a couple during a rotation is found, respectively, by using:

$$dU = F ds \cos \alpha \quad (10.1)$$

$$dU = M d\theta \quad (10.2)$$

Principle of virtual work. In its more general and more useful form, this principle can be stated as:

If a system of connected rigid bodies is in equilibrium, the total virtual work of the external forces applied to the system is zero for any virtual displacement of the system.

As you apply the principle of virtual work, keep in mind the following points.

1. Virtual displacement. A machine or mechanism in equilibrium has no tendency to move. However, we can cause—or imagine—a small displacement. Since it does not actually occur, such a displacement is called a **virtual displacement**.

2. Virtual work. The work done by a force or couple during a virtual displacement is called **virtual work**.

3. You need consider only the forces that do work during the virtual displacement.

4. Forces that do no work during a virtual displacement that are consistent with the constraints imposed on the system are

- a. Reactions at supports
- b. Internal forces at connections
- c. Forces exerted by inextensible cords and cables

None of these forces need be considered when you use the method of virtual work.

5. Be sure to express the various virtual displacements involved in your computations in terms of a single virtual displacement. This is done in each of the three preceding sample problems, where the virtual displacements are all expressed in terms of $\delta\theta$.

6. Remember that the method of virtual work is effective only in those cases where the geometry of the system makes it relatively easy to relate the displacements involved.

Problems

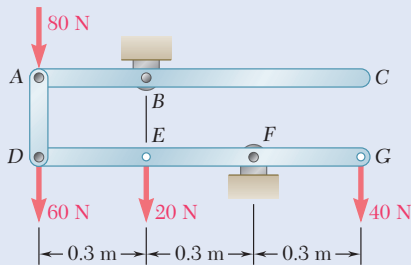


Fig. P10.1 and P10.3

10.1 Determine the vertical force P that must be applied at C to maintain the equilibrium of the linkage.

10.2 Determine the horizontal force P that must be applied at A to maintain the equilibrium of the linkage.

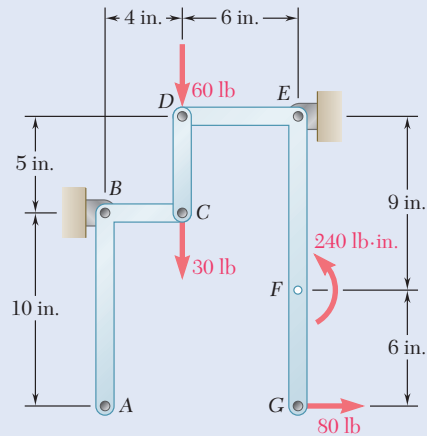


Fig. P10.2 and P10.4

10.3 and 10.4 Determine the couple M that must be applied to member ABC to maintain the equilibrium of the linkage.

10.5 A spring of constant 15 kN/m connects points C and F of the linkage shown. Neglecting the weight of the spring and linkage, determine the force in the spring and the vertical motion of point G when a vertical downward 120-N force is applied (a) at point C , (b) at points C and H .

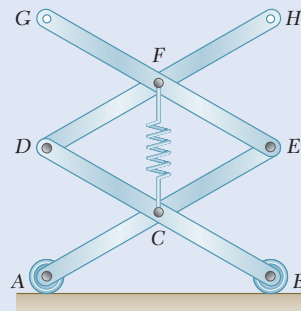


Fig. P10.5 and P10.6

10.6 A spring of constant 15 kN/m connects points C and F of the linkage shown. Neglecting the weight of the spring and linkage, determine the force in the spring and the vertical motion of point G when a vertical downward 120-N force is applied (a) at point E , (b) at points E and F .

- 10.7** The two-bar linkage shown is supported by a pin and bracket at B and a collar at D that slides freely on a vertical rod. Determine the force $\mathbf{P}$ required to maintain the equilibrium of the linkage.

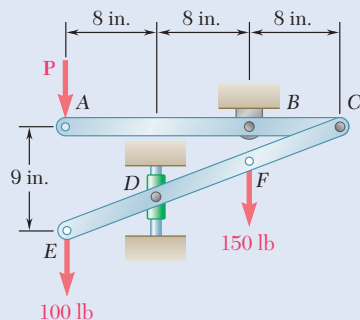


Fig. P10.7

- 10.8** Knowing that the maximum friction force exerted by the bottle on the cork is 60 lb, determine (a) the force $\mathbf{P}$ that must be applied to the corkscrew to open the bottle, (b) the maximum force exerted by the base of the corkscrew on the top of the bottle.

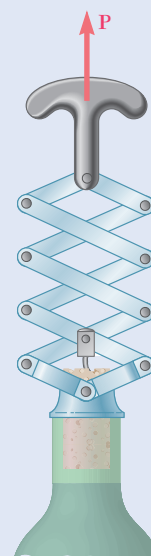


Fig. P10.8

- 10.9** Rod AD is acted upon by a vertical force $\mathbf{P}$ at end A and by two equal and opposite horizontal forces of magnitude Q at points B and C . Derive an expression for the magnitude Q of the horizontal forces required for equilibrium.

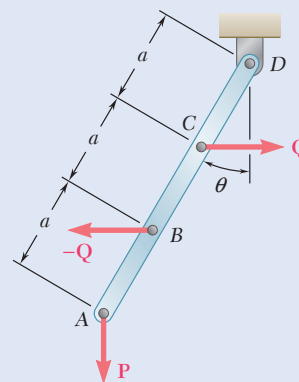


Fig. P10.9

- 10.10 and 10.11** The slender rod AB is attached to a collar A and rests on a small wheel at C . Neglecting the radius of the wheel and the effect of friction, derive an expression for the magnitude of the force Q required to maintain the equilibrium of the rod.

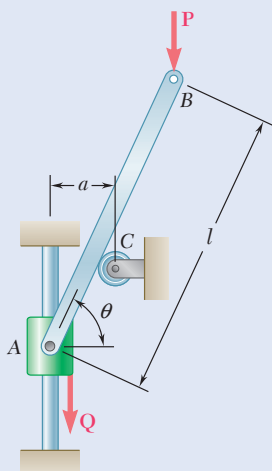


Fig. P10.10

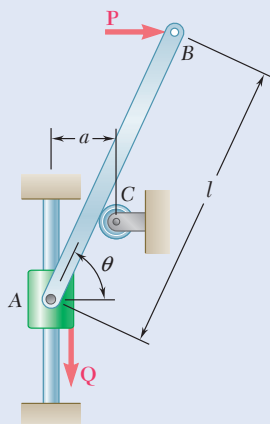


Fig. P10.11

- 10.12** Knowing that the line of action of the force Q passes through point C , derive an expression for the magnitude of Q required to maintain equilibrium.

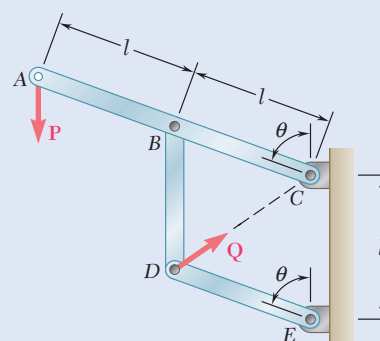


Fig. P10.12

- 10.13** Solve Prob. 10.12 assuming that the force $\mathbf{P}$ applied at point A acts horizontally to the left.

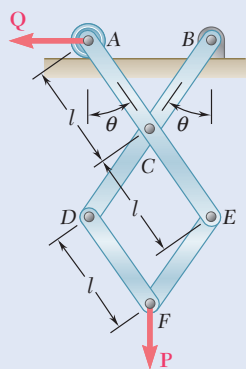


Fig. P10.14

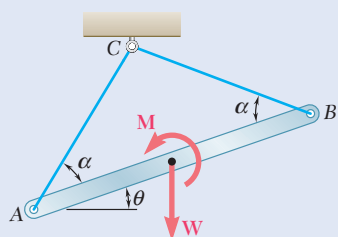


Fig. P10.17

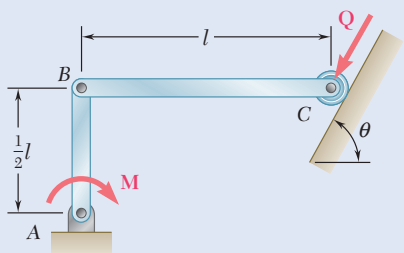


Fig. P10.19 and P10.20

10.14 The mechanism shown is acted upon by the force $\mathbf{P}$; derive an expression for the magnitude of the force $\mathbf{Q}$ required to maintain equilibrium.

10.15 and 10.16 Derive an expression for the magnitude of the couple $\mathbf{M}$ required to maintain the equilibrium of the linkage shown.

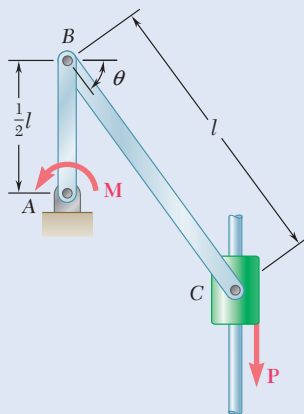


Fig. P10.15

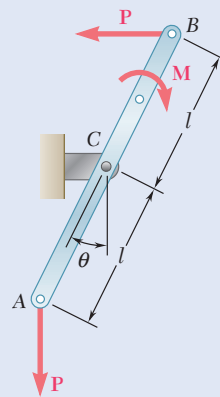


Fig. P10.16

10.17 A uniform rod AB with length l and weight W is suspended from two cords AC and BC of equal length. Derive an expression for the magnitude of the couple $\mathbf{M}$ required to maintain equilibrium of the rod in the position shown.

10.18 The pin at C is attached to member BCD and can slide along a slot cut in the fixed plate shown. Neglecting the effect of friction, derive an expression for the magnitude of the couple $\mathbf{M}$ required to maintain equilibrium when the force $\mathbf{P}$ that acts at D is directed (a) as shown, (b) vertically downward, (c) horizontally to the right.

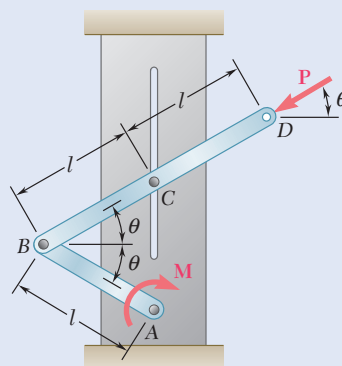


Fig. P10.18

10.19 For the linkage shown, determine the couple $\mathbf{M}$ required for equilibrium when $l = 1.8$ ft, $Q = 40$ lb, and $\theta = 65^\circ$.

10.20 For the linkage shown, determine the force $\mathbf{Q}$ required for equilibrium when $l = 18$ in., $M = 600$ lb·in., and $\theta = 70^\circ$.

10.21 A 4-kN force $\mathbf{P}$ is applied as shown to the piston of the engine system. Knowing that $AB = 50$ mm and $BC = 200$ mm, determine the couple $\mathbf{M}$ required to maintain the equilibrium of the system when (a) $\theta = 30^\circ$, (b) $\theta = 150^\circ$.

10.22 A couple $\mathbf{M}$ with a magnitude of 100 N·m is applied as shown to the crank of the engine system. Knowing that $AB = 50$ mm and $BC = 200$ mm, determine the force $\mathbf{P}$ required to maintain the equilibrium of the system when (a) $\theta = 60^\circ$, (b) $\theta = 120^\circ$.

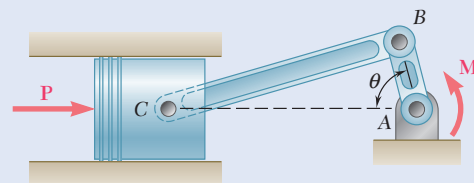


Fig. P10.21 and P10.22

10.23 Rod AB is attached to a block at A that can slide freely in the vertical slot shown. Neglecting the effect of friction and the weights of the rods, determine the value of θ corresponding to equilibrium.

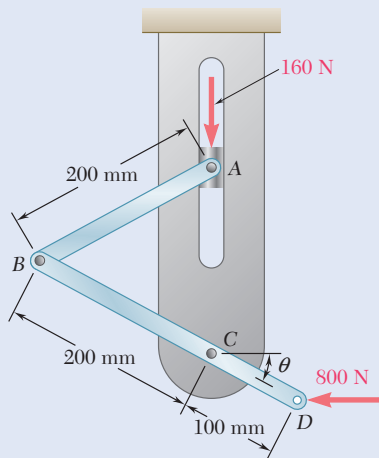


Fig. P10.23

10.24 Solve Prob. 10.23 assuming that the 800-N force is replaced by a 24 N·m clockwise couple applied at D .

10.25 Determine the value of θ corresponding to the equilibrium position of the rod of Prob. 10.10 when $l = 30$ in., $a = 5$ in., $P = 25$ lb, and $Q = 40$ lb.

10.26 Determine the values of θ corresponding to the equilibrium position of the rod of Prob. 10.11 when $l = 24$ in., $a = 4$ in., $P = 10$ lb, and $Q = 18$ lb.

10.27 Determine the value of θ corresponding to the equilibrium position of the mechanism of Prob. 10.12 when $P = 80$ N and $Q = 100$ N.

10.28 Determine the value of θ corresponding to the equilibrium position of the mechanism of Prob. 10.14 when $P = 270$ N and $Q = 960$ N.

10.29 Two rods AC and CE are connected by a pin at C and by a spring AE . The constant of the spring is k , and the spring is unstretched when $\theta = 30^\circ$. For the loading shown, derive an equation in P , θ , l , and k that must be satisfied when the system is in equilibrium.

10.30 Two rods AC and CE are connected by a pin at C and by a spring AE . The constant of the spring is 1.5 lb/in., and the spring is unstretched when $\theta = 30^\circ$. Knowing that $l = 10$ in. and neglecting the weight of the rods, determine the value of θ corresponding to equilibrium when $P = 40$ lb.

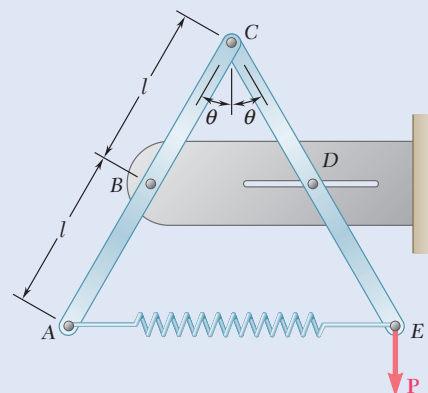


Fig. P10.29 and P10.30

10.31 Solve Prob. 10.30 assuming that force $\mathbf{P}$ is moved to C and acts vertically downward.

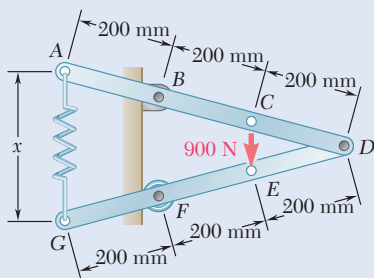


Fig. P10.32

10.32 Two bars AD and DG are connected by a pin at D and by a spring AG . Knowing that the spring is 300 mm long when unstretched and that the constant of the spring is 5 kN/m, determine the value of x corresponding to equilibrium when a 900-N load is applied at E as shown.

10.33 Solve Prob. 10.32 assuming that the 900-N vertical force is applied at C instead of E .

10.34 Two 5-kg bars AB and BC are connected by a pin at B and by a spring DE . Knowing that the spring is 150 mm long when unstretched and that the constant of the spring is 1 kN/m, determine the value of x corresponding to equilibrium.

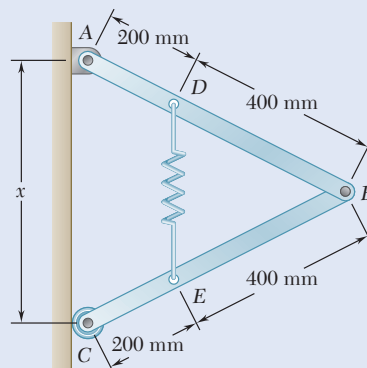


Fig. P10.34

10.35 A vertical force P with a magnitude of 150 N is applied to end E of cable CDE that passes over a small pulley D and is attached to the mechanism at C . The constant of the spring is $k = 4$ kN/m, and the spring is unstretched when $\theta = 0$. Neglecting the weight of the mechanism and the radius of the pulley, determine the value of θ corresponding to equilibrium.

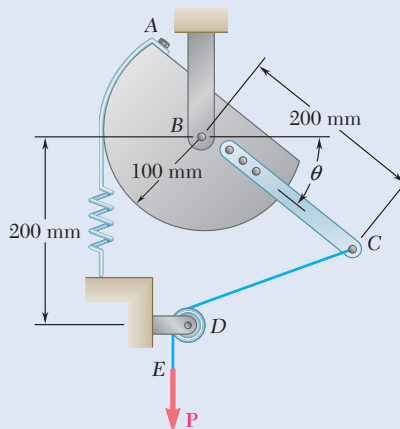


Fig. P10.35

10.36 A load W with a magnitude of 72 lb is applied to the mechanism at C . Neglecting the weight of the mechanism, determine the value of θ corresponding to equilibrium. The constant of the spring is $k = 20$ lb/in., and the spring is unstretched when $\theta = 0$.

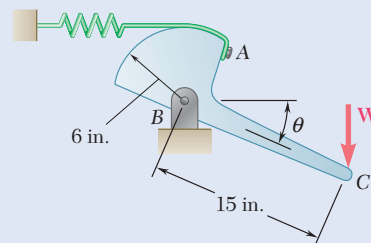


Fig. P10.36

10.37 and P10.38 Knowing that the constant of spring CD is k and that the spring is unstretched when rod ABC is horizontal, determine the value of θ corresponding to equilibrium for the data indicated.

10.37 $P = 300$ N, $l = 400$ mm, and $k = 5$ kN/m

10.38 $P = 75$ lb, $l = 15$ in., and $k = 20$ lb/in.

10.39 The lever AB is attached to the horizontal shaft BC that passes through a bearing and is welded to a fixed support at C . The torsional spring constant of the shaft BC is K ; that is, a couple of magnitude K is required to rotate end B through 1 rad. Knowing that the shaft is untwisted when AB is horizontal, determine the value of θ corresponding to the position of equilibrium when $P = 100$ N, $l = 250$ mm, and $K = 12.5$ N·m/rad.

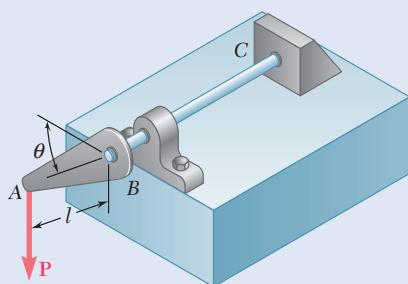


Fig. P10.39

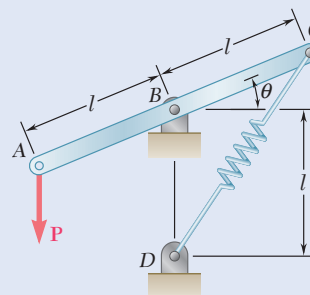


Fig. P10.37 and P10.38

10.40 Solve Prob. 10.39 assuming that $P = 350$ N, $l = 250$ mm, and $K = 12.5$ N·m/rad. Obtain answers in each of the following quadrants: $0 < \theta < 90^\circ$, $270^\circ < \theta < 360^\circ$, and $360^\circ < \theta < 450^\circ$.

10.41 The position of boom ABC is controlled by the hydraulic cylinder BD . For the loading shown, determine the force exerted by the hydraulic cylinder on pin B when $\theta = 70^\circ$.

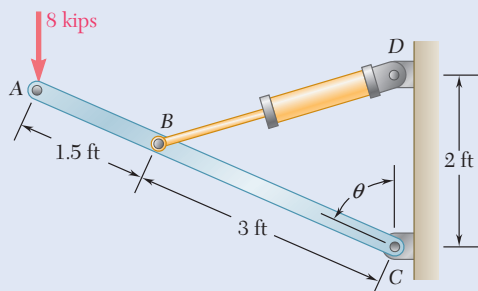


Fig. P10.41 and P10.42

10.42 The position of boom ABC is controlled by the hydraulic cylinder BD . For the loading shown, determine the largest allowable value of the angle θ if the maximum force that the cylinder can exert on pin B is 25 kips.

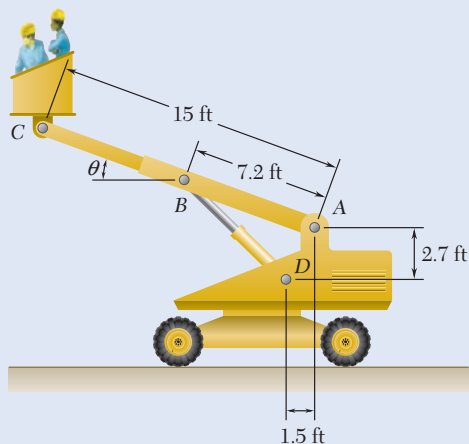


Fig. P10.45

- 10.43** The position of member ABC is controlled by the hydraulic cylinder CD . For the loading shown, determine the force exerted by the hydraulic cylinder on pin C when $\theta = 55^\circ$.

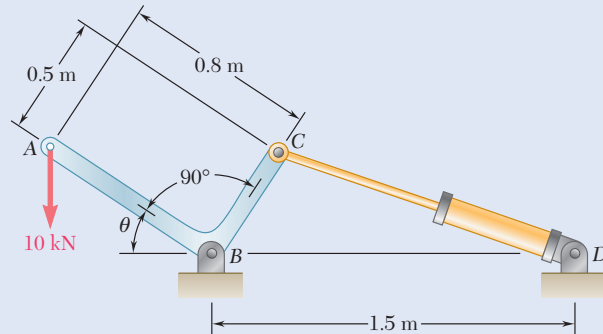


Fig. P10.43 and P10.44

- 10.44** The position of member ABC is controlled by the hydraulic cylinder CD . Determine angle θ , knowing that the hydraulic cylinder exerts a 15-kN force on pin C .

- 10.45** The telescoping arm ABC is used to provide an elevated platform for construction workers. The workers and the platform together weigh 500 lb, and their combined center of gravity is located directly above C . For the position when $\theta = 20^\circ$, determine the force exerted on pin B by the single hydraulic cylinder BD .

- 10.46** Solve Prob. 10.45 assuming that the workers are lowered to a point near the ground so that $\theta = -20^\circ$.

- 10.47** Denoting the coefficient of static friction between collar C and the vertical rod by μ_s , derive an expression for the magnitude of the largest couple M for which equilibrium is maintained in the position shown. Explain what happens if $\mu_s \geq \tan \theta$.

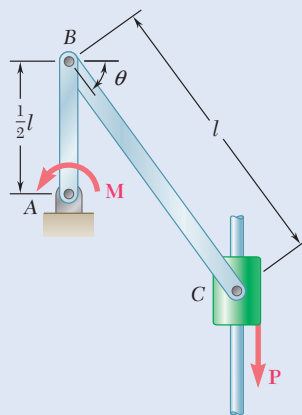


Fig. P10.47 and P10.48

- 10.48** Knowing that the coefficient of static friction between collar C and the vertical rod is 0.40, determine the magnitude of the largest and smallest couple M for which equilibrium is maintained in the position shown, when $\theta = 35^\circ$, $l = 600$ mm, and $P = 300$ N.

- 10.49** A block with weight W is pulled up a plane forming an angle α with the horizontal by a force $\mathbf{P}$ directed along the plane. If μ is the coefficient of friction between the block and the plane, derive an expression for the mechanical efficiency of the system. Show that the mechanical efficiency cannot exceed $\frac{1}{2}$ if the block is to remain in place when the force $\mathbf{P}$ is removed.
- 10.50** Derive an expression for the mechanical efficiency of the jack discussed in Sec. 8.2B. Show that if the jack is to be self-locking, the mechanical efficiency cannot exceed $\frac{1}{2}$.
- 10.51** Denoting the coefficient of static friction between the block attached to rod ACE and the horizontal surface by μ_s , derive expressions in terms of P , μ_s , and θ for the largest and smallest magnitude of the force Q for which equilibrium is maintained.

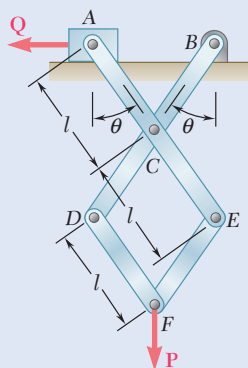


Fig. P10.51 and P10.52

- 10.52** Knowing that the coefficient of static friction between the block attached to rod ACE and the horizontal surface is 0.15, determine the magnitude of the largest and smallest force Q for which equilibrium is maintained when $\theta = 30^\circ$, $l = 0.2$ m, and $P = 40$ N.
- 10.53** Using the method of virtual work, determine separately the force and couple representing the reaction at A .

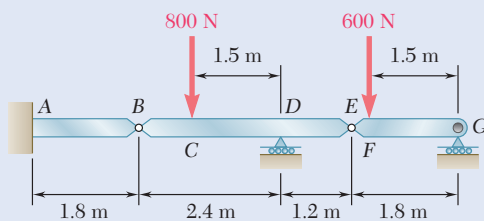


Fig. P10.53 and P10.54

- 10.54** Using the method of virtual work, determine the reaction at D .

10.55 Referring to Prob. 10.43 and using the value found for the force exerted by the hydraulic cylinder CD , determine the change in the length of CD required to raise the 10-kN load by 15 mm.

10.56 Referring to Prob. 10.45 and using the value found for the force exerted by the hydraulic cylinder BD , determine the change in the length of BD required to raise the platform attached at C by 2.5 in.

10.57 Determine the vertical movement of joint D if the length of member BF is increased by 1.5 in. (*Hint:* Apply a vertical load at joint D , and using the methods of Chap. 6, compute the force exerted by member BF on joints B and F . Then apply the method of virtual work for a virtual displacement resulting in the specified increase in length of member BF . This method should be used only for small changes in the lengths of members.)

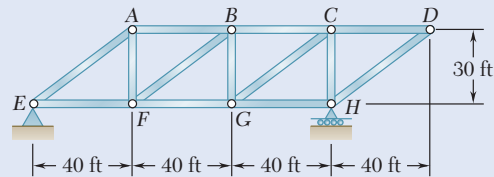


Fig. P10.57 and P10.58

10.58 Determine the horizontal movement of joint D if the length of member BF is increased by 1.5 in. (See the hint for Prob. 10.57.)

*10.2 WORK, POTENTIAL ENERGY, AND STABILITY

The concept of virtual work has another important connection with equilibrium, leading to criteria for conditions of stable, unstable, and neutral equilibrium. However, to explain this connection, we first need to introduce expressions for the work of a force during a finite displacement and then to define the concept of potential energy.

10.2A Work of a Force During a Finite Displacement

Consider a force $\mathbf{F}$ acting on a particle. In Sec. 10.1A, we defined the work of $\mathbf{F}$ corresponding to an infinitesimal displacement $d\mathbf{r}$ of the particle as

$$dU = \mathbf{F} \cdot d\mathbf{r} \quad (10.1)$$

We obtain the work of $\mathbf{F}$ corresponding to a finite displacement of the particle from A_1 to A_2 (Fig. 10.10a) that is denoted by $U_{1 \rightarrow 2}$ by integrating Eq. (10.1) along the curve described by the particle. Thus,

Work during a finite displacement

$$U_{1 \rightarrow 2} = \int_{A_1}^{A_2} \mathbf{F} \cdot d\mathbf{r} \quad (10.11)$$

Using the alternative expression

$$dU = F ds \cos \alpha \quad (10.1')$$

given in Sec. 10.1 for the elementary work dU , we can also express the work $U_{1 \rightarrow 2}$ as

$$U_{1 \rightarrow 2} = \int_{s_1}^{s_2} (F \cos \alpha) ds \quad (10.11')$$

Here, the variable of integration s measures the distance along the path traveled by the particle. We can represent the work $U_{1 \rightarrow 2}$ by the area under the curve obtained by plotting $F \cos \alpha$ against s (Fig. 10.10b). In the case of a force $\mathbf{F}$ of constant magnitude acting in the direction of motion, formula (10.11') yields $U_{1 \rightarrow 2} = F(s_2 - s_1)$.

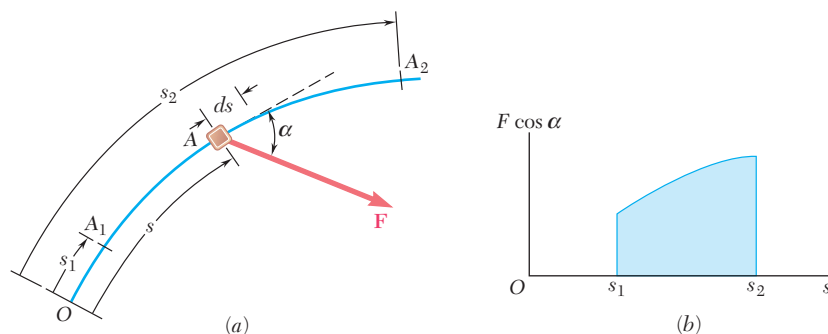


Fig. 10.10 (a) A force acting on a particle moving along a path from A_1 to A_2 ; (b) the work done by the force in (a) equals the area under the graph of $F \cos \alpha$ versus s .

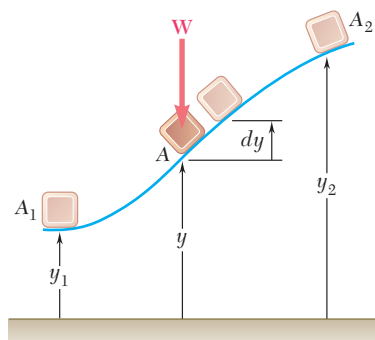


Fig. 10.11 The work done by the weight of a body equals the magnitude of the weight times the vertical displacement of its center of gravity.

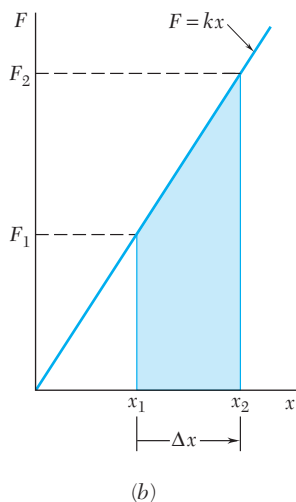
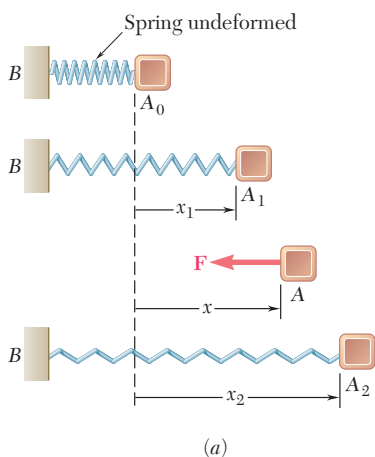


Fig. 10.12 (a) When a body is attached to a fixed point by a spring, the force on it is the product of the spring constant and the displacement from the undeformed position; (b) the work of the force equals the area under the graph of F versus x between x_1 and x_2 .

Recall from Sec. 10.1 that the work of a couple of moment M during an infinitesimal rotation $d\theta$ of a rigid body is

$$dU = M d\theta \quad (10.2)$$

Therefore, we can express the work of the couple during a finite rotation of the body as

Work during a finite rotation

$$U_{1 \rightarrow 2} = \int_{\theta_1}^{\theta_2} M d\theta \quad (10.12)$$

In the case of a constant couple, formula (10.12) yields

$$U_{1 \rightarrow 2} = M(\theta_2 - \theta_1)$$

Work of a Weight. We stated in Sec. 10.1 that the work of a body's weight W during an infinitesimal displacement of the body is equal to the product of W and the vertical displacement of the body's center of gravity. With the y axis pointing upward, we obtain the work of W during a finite displacement of the body (Fig. 10.11) from

$$dU = -W dy$$

Integrating from A_1 to A_2 , we have

$$U_{1 \rightarrow 2} = - \int_{y_1}^{y_2} W dy = Wy_1 - Wy_2 \quad (10.13)$$

or

$$U_{1 \rightarrow 2} = -W(y_2 - y_1) = -W \Delta y \quad (10.13')$$

where Δy is the vertical displacement from A_1 to A_2 . The work of the weight W is thus equal to **the product of W and the vertical displacement of the center of gravity of the body**. The work is *positive* when $\Delta y < 0$, that is, when the body moves down.

Work of the Force Exerted by a Spring. Consider a body A attached to a fixed point B by a spring. We assume that the spring is undeformed when the body is at A_0 (Fig. 10.12a). Experimental evidence shows that the magnitude of the force F exerted by the spring on a body A is proportional to the deflection x of the spring measured from position A_0 . We have

$$F = kx \quad (10.14)$$

where k is the **spring constant** expressed in SI units of N/m or U.S. customary units of lb/ft or lb/in. The work of force F exerted by the spring during a finite displacement of the body from A_1 ($x = x_1$) to A_2 ($x = x_2$) is obtained from

$$dU = -F dx = -kx dx$$

$$U_{1 \rightarrow 2} = - \int_{x_1}^{x_2} kx dx = \frac{1}{2} kx_1^2 - \frac{1}{2} kx_2^2 \quad (10.15)$$

You should take care to express k and x in consistent units. For example, if you use U.S. customary units, k should be expressed in lb/ft and x expressed in feet, or k is given in lb/in. and x in inches. In the first case, the work is obtained in ft·lb; in the second case, it is in in·lb. We note that the work of

the force $\mathbf{F}$ exerted by the spring on the body is *positive* when $x_2 < x_1$, that is, *when the spring is returning to its undeformed position*.

Since Eq. (10.14) is the equation of a straight line of slope k passing through the origin, we can obtain the work $U_{1 \rightarrow 2}$ of $\mathbf{F}$ during the displacement from A_1 to A_2 by evaluating the area of the trapezoid shown in Fig. 10.12b. This is done by computing the values F_1 and F_2 and multiplying the base Δx of the trapezoid by its mean height as $\frac{1}{2}(F_1 + F_2)$. Since the work of the force $\mathbf{F}$ exerted by the spring is positive for a negative value of Δx , we have

$$U_{1 \rightarrow 2} = -\frac{1}{2}(F_1 + F_2) \Delta x \quad (10.16)$$

Equation (10.16) is usually more convenient to use than Eq. (10.15) and affords fewer chances of confusing the units involved.

10.2B Potential Energy

Let's consider again the body of Fig. 10.11. Using Eq. (10.13), we obtain the work of weight $\mathbf{W}$ during a finite displacement by subtracting the value of the function W_y corresponding to the second position of the body from its value corresponding to the first position. Thus, the work of $\mathbf{W}$ is independent of the actual path followed; it depends only upon the initial and final values of the function W_y . This function is called the **potential energy** of the body with respect to the force due to gravity $\mathbf{W}$ and is denoted by V_g . Thus,

$$U_{1 \rightarrow 2} = (V_g)_1 - (V_g)_2 \quad \text{with } V_g = W_y \quad (10.17)$$

Note that if $(V_g)_2 > (V_g)_1$, that is, *if the potential energy increases* during the displacement (as in the case considered here), *the work $U_{1 \rightarrow 2}$ is negative*. If, on the other hand, the work of $\mathbf{W}$ is positive, the potential energy decreases. Therefore, the potential energy V_g of the body provides a measure of *the work that can be done* by its weight $\mathbf{W}$. Since only the *change* in potential energy—not the actual value of V_g —is involved in formula (10.17), we can add an arbitrary constant to the expression obtained for V_g . In other words, the level from which the elevation y is measured can be chosen arbitrarily. Note that potential energy is expressed in the same units as work, i.e., in joules (J) if SI units are used[†] and in ft·lb or in·lb if U.S. customary units are used.

Now consider the body of Fig. 10.12a. Using Eq. (10.15), we obtain the work of the elastic force $\mathbf{F}$ by subtracting the value of the function $\frac{1}{2}kx^2$ corresponding to the second position of the body from its value corresponding to the first position. This function, denoted by V_e , is called the **potential energy** of the body with respect to the **elastic force $\mathbf{F}$** . We have

$$U_{1 \rightarrow 2} = (V_e)_1 - (V_e)_2 \quad \text{with } V_e = \frac{1}{2}kx^2 \quad (10.18)$$

Note that during the displacement considered, the work of force $\mathbf{F}$ exerted by the spring on the body is negative and the potential energy V_e increases. Also note that the expression obtained for V_e is valid only if the deflection of the spring is measured from its undeformed position.

We can use the concept of potential energy when forces other than gravity forces and elastic forces are involved. It remains valid as long as

[†]See footnote, p. 575.

the elementary work dU of the force considered is an *exact differential*. It is then possible to find a function V , called potential energy, such that

$$dU = -dV \quad (10.19)$$

Integrating Eq. (10.19) over a finite displacement, we obtain

**Potential energy,
general formulation**

$$U_{1 \rightarrow 2} = V_1 - V_2 \quad (10.20)$$

This equation says that **the work of the force is independent of the path followed and is equal to minus the change in potential energy**. A force that satisfies Eq. (10.20) is called a **conservative force**.[†]

10.2C Potential Energy and Equilibrium

Applying the principle of virtual work is considerably simplified if we know the potential energy of a system. In the case of a virtual displacement, formula (10.19) becomes $\delta U = -\delta V$. Moreover, if the position of the system is defined by a single independent variable θ , we can write $\delta V = (dV/d\theta) \delta\theta$. Since $\delta\theta$ must be different from zero, the condition $\delta U = 0$ for the equilibrium of the system becomes

Equilibrium condition

$$\frac{dV}{d\theta} = 0 \quad (10.21)$$

In terms of potential energy, therefore, the principle of virtual work states:

If a system is in equilibrium, the derivative of its total potential energy is zero.

If the position of the system depends upon several independent variables (the system is then said to possess *several degrees of freedom*), the partial derivatives of V with respect to each of the independent variables must be zero.

Consider, for example, a structure made of two members AC and CB and carrying a load W at C . The structure is supported by a pin at A and a roller at B , and a spring BD connects B to a fixed point D (Fig. 10.13a). The constant of the spring is k , and we assume that the natural length of the spring is equal to AD , so that the spring is undeformed when B coincides with A . Neglecting friction forces and the weights of the members, we find that the only forces that do work during a virtual displacement of the structure are the weight W and the force F exerted by the spring at point B (Fig. 10.13b). Therefore, we can obtain the total potential energy of the system by adding the potential energy V_g corresponding to the gravity force W and the potential energy V_e corresponding to the elastic force F .

Choosing a coordinate system with the origin at A and noting that the deflection of the spring measured from its undeformed position is $AB = x_B$, we have

$$V_e = \frac{1}{2} k x_B^2 \quad \text{and} \quad V_g = W y_C$$

[†]A detailed discussion of conservative forces is given in Sec. 13.2B of *Dynamics*.

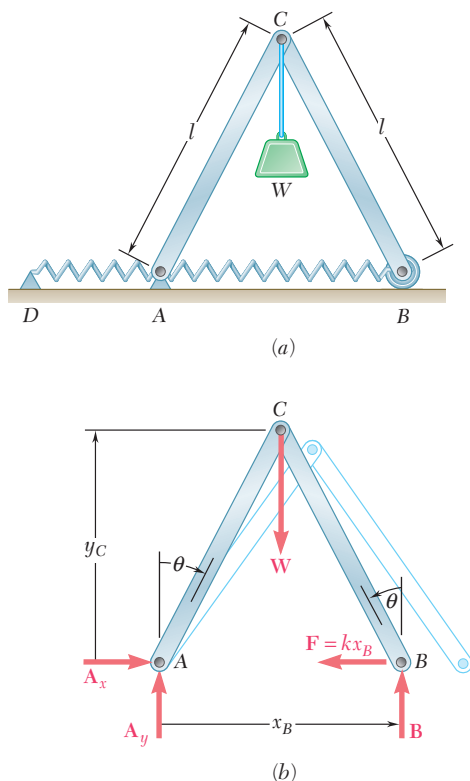


Fig. 10.13 (a) Structure carrying a load at C with a spring from B to D ; (b) free-body diagram of the structure, and a virtual displacement.

Expressing the coordinates x_B and y_C in terms of the angle θ , we have

$$\begin{aligned}x_B &= 2l \sin \theta & y_C &= l \cos \theta \\V_e &= \frac{1}{2}k(2l \sin \theta)^2 & V_g &= W(l \cos \theta) \\V &= V_e + V_g = 2kl^2 \sin^2 \theta + Wl \cos \theta\end{aligned}\quad (10.22)$$

We obtain the positions of equilibrium of the system by setting the derivative of the potential energy V to zero, as

$$\frac{dV}{d\theta} = 4kl^2 \sin \theta \cos \theta - Wl \sin \theta = 0$$

or, factoring out $l \sin \theta$, as

$$\frac{dV}{d\theta} = l \sin \theta (4kl \cos \theta - W) = 0$$

There are therefore two positions of equilibrium corresponding to the values $\theta = 0$ and $\theta = \cos^{-1}(W/4kl)$, respectively.[†]

10.2D Stability of Equilibrium

Consider the three uniform rods with a length of $2a$ and weight $\mathbf{W}$ shown in Fig. 10.14. Although each rod is in equilibrium, there is an important difference between the three cases considered. Suppose that each rod is slightly disturbed from its position of equilibrium and then released. Rod a moves back toward its original position; rod b keeps moving away from its original position; and rod c remains in its new position. In case a , the equilibrium of the rod is said to be **stable**; in case b , it is **unstable**; and in case c , it is **neutral**.

Recall from Sec. 10.2B that the potential energy V_g with respect to gravity is equal to Wy , where y is the elevation of the point of application of $\mathbf{W}$ measured from an arbitrary level. We observe that the potential energy of rod a is minimum in the position of equilibrium considered, that the potential energy of rod b is maximum, and that the potential energy of rod c is constant. Equilibrium is thus *stable*, *unstable*, or *neutral* according to whether the potential energy is *minimum*, *maximum*, or *constant* (Fig. 10.15).

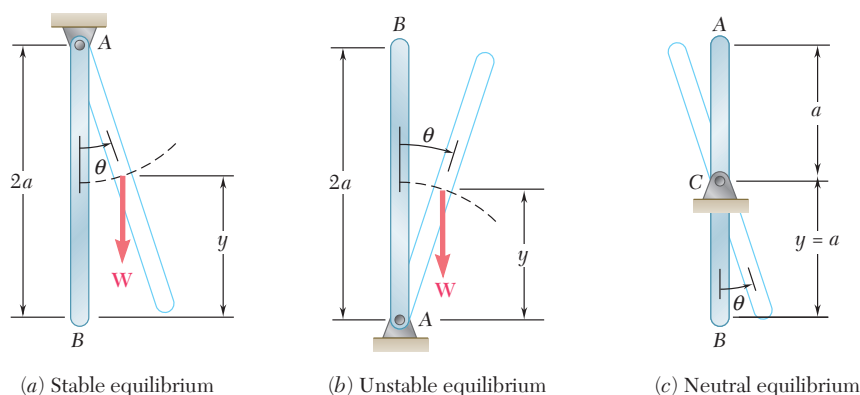


Fig. 10.14 (a) Rod supported from above, stable equilibrium; (b) rod supported from below, unstable equilibrium; (c) rod supported at its midpoint, neutral equilibrium.

[†]The second position does not exist if $W > 4kl$.

This result is quite general, as we now show. We first observe that a force always tends to do positive work and thus to decrease the potential energy of the system on which it is applied. Therefore, when a system is disturbed from its position of equilibrium, the forces acting on the system tend to bring it back to its original position if V is minimum (Fig. 10.15a) and to move it farther away if V is maximum (Fig. 10.15b). If V is constant (Fig. 10.15c), the forces do not tend to move the system either way.

Recall from calculus that a function is minimum or maximum according to whether its second derivative is positive or negative. Therefore, we can summarize the conditions for the equilibrium of a system with one degree of freedom (i.e., a system for which the position is defined by a single independent variable θ) as

$$\begin{aligned} \frac{dV}{d\theta} = 0 \quad \frac{d^2V}{d\theta^2} > 0 &: \text{stable equilibrium} \\ \frac{dV}{d\theta} = 0 \quad \frac{d^2V}{d\theta^2} < 0 &: \text{unstable equilibrium} \end{aligned} \quad (10.23)$$

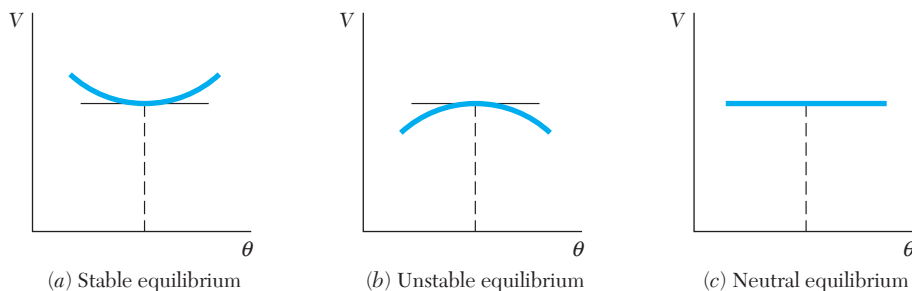


Fig. 10.15 Stable, unstable, and neutral equilibria correspond to potential energy values that are minimum, maximum, or constant, respectively.

If both the first and the second derivatives of V are zero, it is necessary to examine derivatives of a higher order to determine whether the equilibrium is stable, unstable, or neutral. The equilibrium is neutral if all derivatives are zero, since the potential energy V is then a constant. The equilibrium is stable if the first derivative found to be different from zero is of even order and positive. In all other cases, the equilibrium is unstable.

If the system of interest possesses *several degrees of freedom*, the potential energy V depends upon several variables. Thus, it becomes necessary to apply the theory of functions of several variables to determine whether V is minimum. It can be verified that a system with two degrees of freedom is stable, and the corresponding potential energy $V(\theta_1, \theta_2)$ is minimum, if the following relations are satisfied simultaneously:

$$\begin{aligned} \frac{\partial V}{\partial \theta_1} = \frac{\partial V}{\partial \theta_2} &= 0 \\ \left(\frac{\partial^2 V}{\partial \theta_1 \partial \theta_2} \right)^2 - \frac{\partial^2 V}{\partial \theta_1^2} \frac{\partial^2 V}{\partial \theta_2^2} &< 0 \\ \frac{\partial^2 V}{\partial \theta_1^2} &> 0 \quad \text{or} \quad \frac{\partial^2 V}{\partial \theta_2^2} > 0 \end{aligned} \quad (10.24)$$

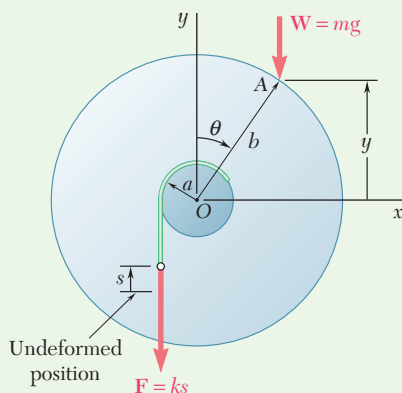
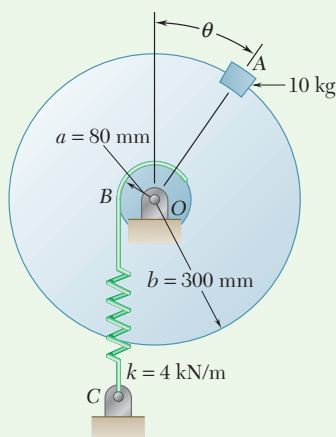


Fig. 1 Free-body diagram of rotated disk, showing only those forces that do work.

Sample Problem 10.4

A 10-kg block is attached to the rim of a 300-mm-radius disk as shown. Knowing that spring BC is unstretched when $\theta = 0$, determine the position or positions of equilibrium, and state in each case whether the equilibrium is stable, unstable, or neutral.

STRATEGY: The first step is to determine a potential energy function V for the system. You can find the positions of equilibrium by determining where the derivative of V is zero. You can find the types of stability by finding where V is maximum or minimum.

MODELING and ANALYSIS:

Potential Energy. Denote the deflection of the spring from its undeformed position by s , and place the origin of coordinates at O (Fig. 1). You obtain

$$V_e = \frac{1}{2}ks^2 \quad V_g = Wy = mgy$$

Measuring θ in radians, you have

$$s = a\theta \quad y = b \cos \theta$$

Substituting for s and y in the expressions for V_e and V_g gives you

$$V_e = \frac{1}{2}ka^2\theta^2 \quad V_g = mgb \cos \theta$$

$$V = V_e + V_g = \frac{1}{2}ka^2\theta^2 + mgb \cos \theta$$

Positions of Equilibrium. Setting $dV/d\theta = 0$, you obtain

$$\begin{aligned} \frac{dV}{d\theta} &= ka^2\theta - mgb \sin \theta = 0 \\ \sin \theta &= \frac{ka^2}{mgb} \theta \end{aligned}$$

Now substitute $a = 0.08$ m, $b = 0.3$ m, $k = 4$ kN/m, and $m = 10$ kg. The result is

$$\sin \theta = \frac{(4 \text{ kN/m})(0.08 \text{ m})^2}{(10 \text{ kg})(9.81 \text{ m/s}^2)(0.3 \text{ m})} \theta$$

$$\sin \theta = 0.8699 \theta$$

where θ is expressed in radians. Solving by trial and error for θ , you find

$$\begin{aligned} \theta &= 0 & \text{and} & \quad \theta = 0.902 \text{ rad} \\ \theta &= 0 & \text{and} & \quad \theta = 51.7^\circ \end{aligned}$$

Stability of Equilibrium. The second derivative of the potential energy V with respect to θ is

$$\begin{aligned} \frac{d^2V}{d\theta^2} &= ka^2 - mgb \cos \theta \\ &= (4 \text{ kN/m})(0.08 \text{ m})^2 - (10 \text{ kg})(9.81 \text{ m/s}^2)(0.3 \text{ m}) \cos \theta \\ &= 25.6 - 29.43 \cos \theta \end{aligned}$$

$$\text{For } \theta = 0, \quad \frac{d^2V}{d\theta^2} = 25.6 - 29.43 \cos 0^\circ = -3.83 < 0$$

The equilibrium is unstable for $\theta = 0$ ◀

$$\text{For } \theta = 51.7^\circ, \quad \frac{d^2V}{d\theta^2} = 25.6 - 29.43 \cos 51.7^\circ = +7.36 > 0$$

The equilibrium is stable for $\theta = 51.7^\circ$ ◀

REFLECT and THINK: If you just let the block-and-disk system fall on its own, it will come to rest at $\theta = 51.7^\circ$. If you balance the system at $\theta = 0$, the slightest touch will put it in motion.

SOLVING PROBLEMS ON YOUR OWN

In this section, we defined the **work of a force during a finite displacement** and the **potential energy** of a rigid body or a system of rigid bodies. You saw how to use the concept of potential energy to determine the **equilibrium position** of a rigid body or a system of rigid bodies.

1. The potential energy V of a system is the sum of the potential energies associated with the various forces acting on the system that do work as the system moves. In the problems of this section, you will determine the following energies.

a. Potential energy of a weight. This is the potential energy due to *gravity*, $V_g = Wy$, where y is the elevation of the weight W measured from some arbitrary reference level. You can use the potential energy V_g with any vertical force $\mathbf{P}$ of constant magnitude directed downward; we write $V_g = Py$.

b. Potential energy of a spring. This is the potential energy due to the *elastic* force exerted by a spring, $V_e = \frac{1}{2}kx^2$, where k is the constant of the spring and x is the deformation of the spring measured from its unstretched position.

Reactions at fixed supports, internal forces at connections, forces exerted by inextensible cords and cables, and other forces that do no work do not contribute to the potential energy of the system.

2. Express all distances and angles in terms of a single variable, such as an angle θ , when computing the potential energy V of a system. This is necessary because determining the equilibrium position of the system requires computing the derivative $dV/d\theta$.

3. When a system is in equilibrium, the first derivative of its potential energy is zero. Therefore,

a. To determine a position of equilibrium of a system, first express its potential energy V in terms of the single variable θ , and then compute its derivative and solve the equation $dV/d\theta = 0$ for θ .

b. To determine the force or couple required to maintain a system in a given position of equilibrium, substitute the known value of θ in the equation $dV/d\theta = 0$, and solve this equation for the desired force or couple.

4. Stability of equilibrium. The following rules generally apply:

a. Stable equilibrium occurs when the potential energy of the system is *minimum*, that is, when $dV/d\theta = 0$ and $d^2V/d\theta^2 > 0$ (Figs. 10.14a and 10.15a).

b. Unstable equilibrium occurs when the potential energy of the system is *maximum*, that is, when $dV/d\theta = 0$ and $d^2V/d\theta^2 < 0$ (Figs. 10.14b and 10.15b).

c. Neutral equilibrium occurs when the potential energy of the system is *constant*; $dV/d\theta$, $d^2V/d\theta^2$, and all the successive derivatives of V are then equal to zero (Figs. 10.14c and 10.15c).

See page 600 for a discussion of the case when $dV/d\theta$, $d^2V/d\theta^2$, but *not all* of the successive derivatives of V are equal to zero.

Problems

- 10.59** Using the method of Sec. 10.2C, solve Prob. 10.29.
- 10.60** Using the method of Sec. 10.2C, solve Prob. 10.30.
- 10.61** Using the method of Sec. 10.2C, solve Prob. 10.31.
- 10.62** Using the method of Sec. 10.2C, solve Prob. 10.32.
- 10.63** Using the method of Sec. a 10.2C, solve Prob. 10.34.
- 10.64** Using the method of Sec. 10.2C, solve Prob. 10.35.
- 10.65** Using the method of Sec. 10.2C, solve Prob. 10.37.
- 10.66** Using the method of Sec. 10.2C, solve Prob. 10.38.
- 10.67** Show that equilibrium is neutral in Prob. 10.1.
- 10.68** Show that equilibrium is neutral in Prob. 10.7.
- 10.69** Two uniform rods, each with a mass m , are attached to gears of equal radii as shown. Determine the positions of equilibrium of the system and state in each case whether the equilibrium is stable, unstable, or neutral.
- 10.70** Two uniform rods, AB and CD , are attached to gears of equal radii as shown. Knowing that $W_{AB} = 8$ lb and $W_{CD} = 4$ lb, determine the positions of equilibrium of the system and state in each case whether the equilibrium is stable, unstable, or neutral.
- 10.71** Two uniform rods AB and CD , of the same length l , are attached to gears as shown. Knowing that rod AB weighs 3 lb and that rod CD weighs 2 lb, determine the positions of equilibrium of the system and state in each case whether the equilibrium is stable, unstable, or neutral.

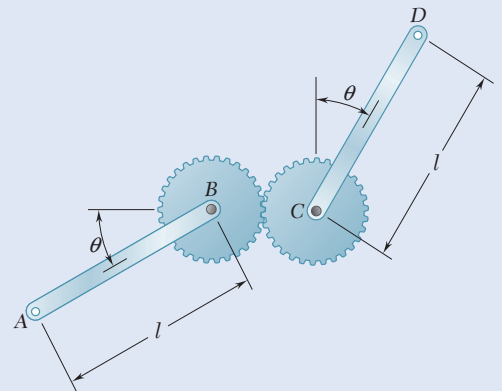


Fig. P10.69 and P10.70

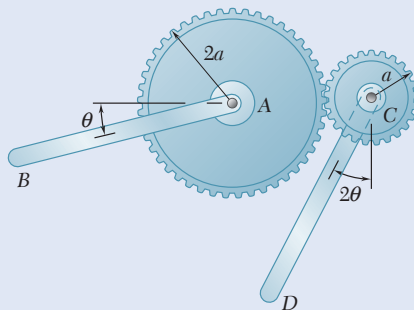


Fig. P10.71

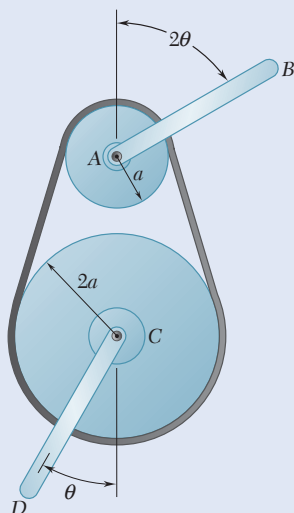


Fig. P10.72

10.72 Two uniform rods, each of mass m and length l , are attached to drums that are connected by a belt as shown. Assuming that no slipping occurs between the belt and the drums, determine the positions of equilibrium of the system and state in each case whether the equilibrium is stable, unstable, or neutral.

10.73 Using the method of Sec. 10.2C, solve Prob. 10.39. Determine whether the equilibrium is stable, unstable, or neutral. (*Hint:* The potential energy corresponding to the couple exerted by a torsion spring is $\frac{1}{2}K\theta^2$, where K is the torsional spring constant and θ is the angle of twist.)

10.74 In Prob. 10.40, determine whether each of the positions of equilibrium is stable, unstable, or neutral. (See hint for Prob. 10.73.)

10.75 A load W with a magnitude of 100 lb is applied to the mechanism at C . Knowing that the spring is unstretched when $\theta = 15^\circ$, determine that value of θ corresponding to equilibrium and check that the equilibrium is stable.

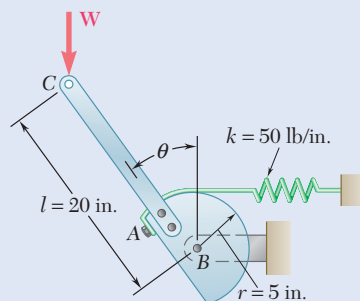


Fig. P10.75 and P10.76

10.76 A load W with a magnitude of 100 lb is applied to the mechanism at C . Knowing that the spring is unstretched when $\theta = 30^\circ$, determine that value of θ corresponding to equilibrium and check that the equilibrium is stable.

10.77 A slender rod AB with a weight W is attached to two blocks A and B that can move freely in the guides shown. Knowing that the spring is unstretched when $y = 0$, determine the value of y corresponding to equilibrium when $W = 80$ N, $l = 500$ mm, and $k = 600$ N/m.

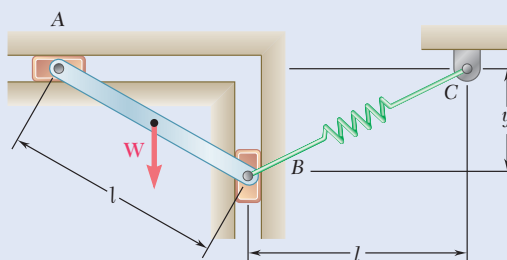


Fig. P10.77

- 10.78** A slender rod AB with a weight W is attached to two blocks A and B that can move freely in the guides shown. Knowing that both springs are unstretched when $y = 0$, determine the value of y corresponding to equilibrium when $W = 80 \text{ N}$, $l = 500 \text{ mm}$, and $k = 600 \text{ N/m}$.

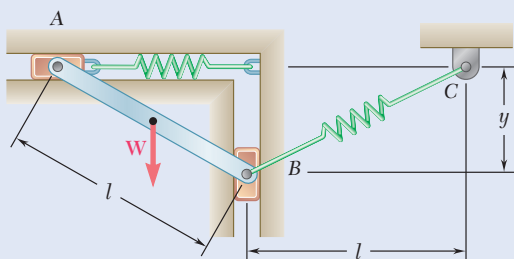


Fig. P10.78

- 10.79** A slender rod AB with a weight W is attached to two blocks A and B that can move freely in the guides shown. The constant of the spring is k , and the spring is unstretched when AB is horizontal. Neglecting the weight of the blocks, derive an equation in θ , W , l , and k that must be satisfied when the rod is in equilibrium.

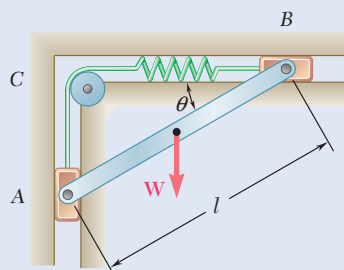
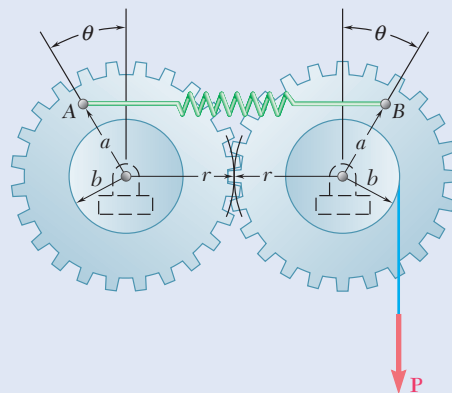


Fig. P10.79 and P10.80

- 10.80** A slender rod AB with a weight W is attached to two blocks A and B that can move freely in the guides shown. Knowing that the spring is unstretched when AB is horizontal, determine three values of θ corresponding to equilibrium when $W = 300 \text{ lb}$, $l = 16 \text{ in.}$, and $k = 75 \text{ lb/in.}$ State in each case whether the equilibrium is stable, unstable, or neutral.
- 10.81** A spring AB of constant k is attached to two identical gears as shown. Knowing that the spring is undeformed when $\theta = 0$, determine two values of the angle θ corresponding to equilibrium when $P = 30 \text{ lb}$, $a = 4 \text{ in.}$, $b = 3 \text{ in.}$, $r = 6 \text{ in.}$, and $k = 5 \text{ lb/in.}$ State in each case whether the equilibrium is stable, unstable, or neutral.



- 10.82** A spring AB of constant k is attached to two identical gears as shown. Knowing that the spring is undeformed when $\theta = 0$, and given that $a = 60 \text{ mm}$, $b = 45 \text{ mm}$, $r = 90 \text{ mm}$, and $k = 6 \text{ kN/m}$, determine (a) the range of values of P for which a position of equilibrium exists, (b) two values of θ corresponding to equilibrium if the value of P is equal to half the upper limit of the range found in part a.

Fig. P10.81 and P10.82

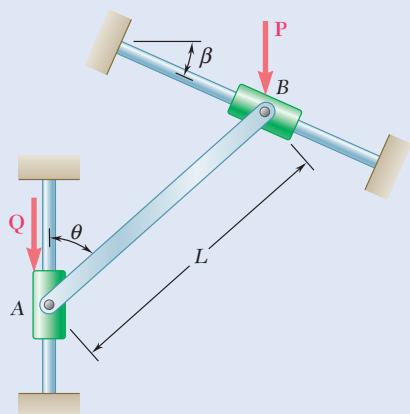


Fig. P10.83 and P10.84

10.83 A slender rod AB is attached to two collars A and B that can move freely along the guide rods shown. Knowing that $\beta = 30^\circ$ and $P = Q = 400$ N, determine the value of the angle θ corresponding to equilibrium.

10.84 A slender rod AB is attached to two collars A and B that can move freely along the guide rods shown. Knowing that $\beta = 30^\circ$, $P = 100$ N, and $Q = 25$ N, determine the value of the angle θ corresponding to equilibrium.

10.85 and 10.86 Cart B , which weighs 75 kN, rolls along a sloping track that forms an angle β with the horizontal. The spring constant is 5 kN/m, and the spring is unstretched when $x = 0$. Determine the distance x corresponding to equilibrium for the angle β indicated.

10.85 Angle $\beta = 30^\circ$

10.86 Angle $\beta = 60^\circ$

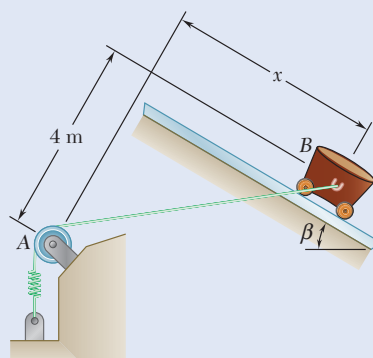


Fig. P10.85 and P10.86

10.87 and 10.88 Collar A can slide freely on the semicircular rod shown. Knowing that the constant of the spring is k and that the unstretched length of the spring is equal to the radius r , determine the value of θ corresponding to equilibrium when $W = 50$ lb, $r = 9$ in., and $k = 15$ lb/in.

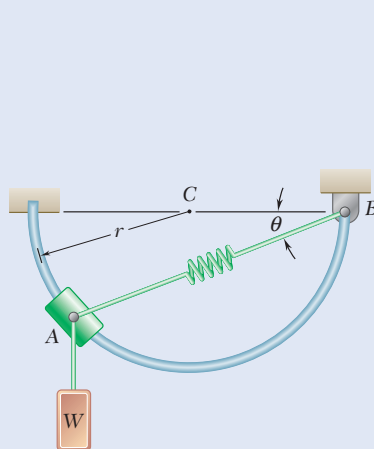


Fig. P10.87

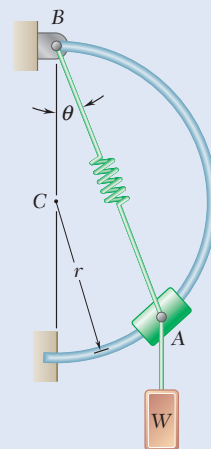


Fig. P10.88

- 10.89** Two bars AB and BC of negligible weight are attached to a single spring of constant k that is unstretched when the bars are horizontal. Determine the range of values of the magnitude P of two equal and opposite forces $\mathbf{P}$ and $-\mathbf{P}$ for which the equilibrium of the system is stable in the position shown.

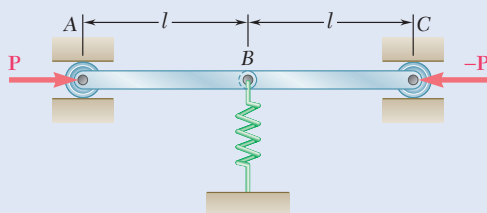


Fig. P10.89

- 10.90** A vertical bar AD is attached to two springs of constant k and is in equilibrium in the position shown. Determine the range of values of the magnitude P of two equal and opposite vertical forces $\mathbf{P}$ and $-\mathbf{P}$ for which the equilibrium position is stable if (a) $AB = CD$, (b) $AB = 2CD$.

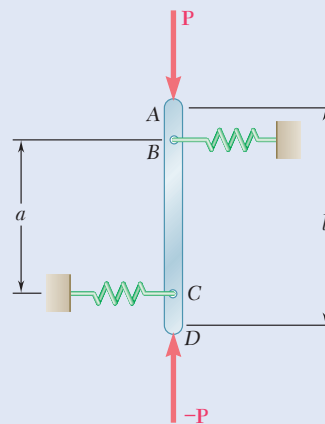


Fig. P10.90

- 10.91** Rod AB is attached to a hinge at A and to two springs, each of constant k . If $h = 25$ in., $d = 12$ in., and $W = 80$ lb, determine the range of values of k for which the equilibrium of the rod is stable in the position shown. Each spring can act in either tension or compression.

- 10.92** Rod AB is attached to a hinge at A and to two springs, each of constant k . If $h = 45$ in., $k = 6$ lb/in., and $W = 60$ lb, determine the smallest distance d for which the equilibrium of the rod is stable in the position shown. Each spring can act in either tension or compression.

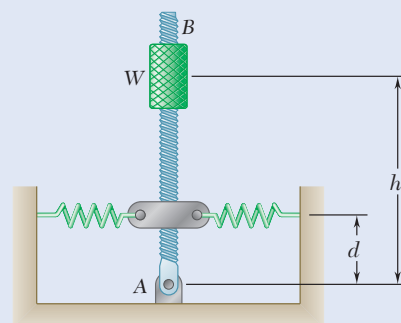


Fig. P10.91 and P10.92

- 10.93 and 10.94** Two bars are attached to a single spring of constant k that is unstretched when the bars are vertical. Determine the range of values of P for which the equilibrium of the system is stable in the position shown.

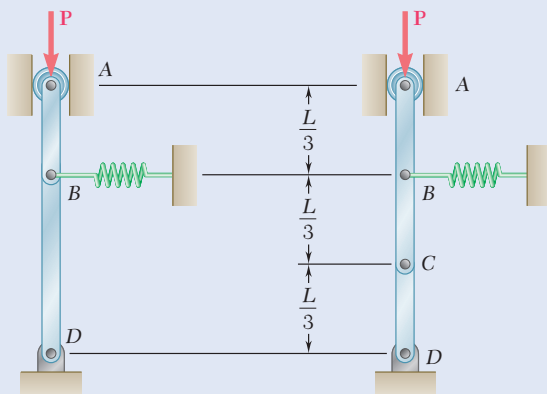


Fig. P10.93

Fig. P10.94

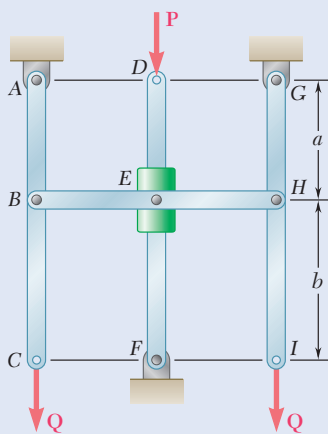


Fig. P10.95 and P10.96

10.95 The horizontal bar BEH is connected to three vertical bars. The collar at E can slide freely on bar DF . Determine the range of values of Q for which the equilibrium of the system is stable in the position shown when $a = 24$ in., $b = 20$ in., and $P = 150$ lb.

10.96 The horizontal bar BEH is connected to three vertical bars. The collar at E can slide freely on bar DF . Determine the range of values of P for which the equilibrium of the system is stable in the position shown when $a = 150$ mm, $b = 200$ mm, and $Q = 45$ N.

***10.97** Bars AB and BC , each with a length l and of negligible weight, are attached to two springs, each of constant k . The springs are undeformed, and the system is in equilibrium when $\theta_1 = \theta_2 = 0$. Determine the range of values of P for which the equilibrium position is stable.

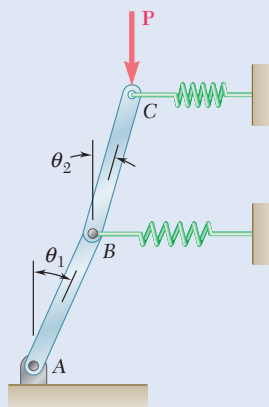


Fig. P10.97

***10.98** Solve Prob. 10.97 knowing that $l = 800$ mm and $k = 2.5$ kN/m.

***10.99** Two rods of negligible weight are attached to drums of radius r that are connected by a belt and spring of constant k . Knowing that the spring is undeformed when the rods are vertical, determine the range of values of P for which the equilibrium position $\theta_1 = \theta_2 = 0$ is stable.

***10.100** Solve Prob. 10.99 knowing that $k = 20$ lb/in., $r = 3$ in., $l = 6$ in., and (a) $W = 15$ lb, (b) $W = 60$ lb.

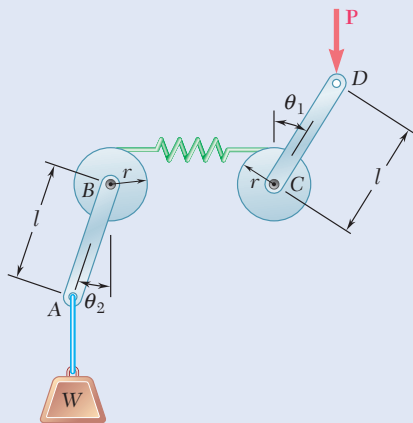


Fig. P10.99

Review and Summary

Work of a Force

The first section of this chapter was devoted to the **principle of virtual work** and to its direct application to the solution of equilibrium problems. We first defined the **work of a force $\mathbf{F}$ corresponding to the small displacement $d\mathbf{r}$** [Sec. 10.1A] as the quantity

$$dU = \mathbf{F} \cdot d\mathbf{r} \quad (10.1)$$

obtained by forming the scalar product of the force $\mathbf{F}$ and the displacement $d\mathbf{r}$ (Fig. 10.16). Denoting the magnitudes of the force and of the displacement by F and ds , respectively, and the angle formed by $\mathbf{F}$ and $d\mathbf{r}$ by α , we have

$$dU = F ds \cos \alpha \quad (10.19)$$

The work dU is positive if $\alpha < 90^\circ$, zero if $\alpha = 90^\circ$, and negative if $\alpha > 90^\circ$. We also found that the **work of a couple of moment $\mathbf{M}$** acting on a rigid body is

$$dU = M d\theta \quad (10.2)$$

where $d\theta$ is the small angle expressed in radians through which the body rotates.

Virtual Displacement

Considering a particle located at A and acted upon by several forces $\mathbf{F}_1, \mathbf{F}_2, \dots, \mathbf{F}_n$ [Sec. 10.1B], we imagined that the particle moved to a new position A' (Fig. 10.17). Since this displacement does not actually take place, we refer to it to as a **virtual displacement** denoted by $\delta\mathbf{r}$. The corresponding work of the forces is called **virtual work** and is denoted by δU . We have

$$\delta U = \mathbf{F}_1 \cdot \delta\mathbf{r} + \mathbf{F}_2 \cdot \delta\mathbf{r} + \dots + \mathbf{F}_n \cdot \delta\mathbf{r}$$

Principle of Virtual Work

The **principle of virtual work** states that **if a particle is in equilibrium, the total virtual work δU of the forces acting on the particle is zero for any virtual displacement of the particle.**

The principle of virtual work can be extended to the case of rigid bodies and systems of rigid bodies. Since it involves only forces that do work, its application provides a useful alternative to the use of the equilibrium equations in the solution of many engineering problems. It is particularly effective in the case of machines and mechanisms consisting of connected rigid bodies, since the work of the reactions at the supports is zero and the work of the internal forces at the pin connections cancels out [Sec. 10.1C; Sample Probs. 10.1, 10.2, and 10.3].



Fig. 10.16

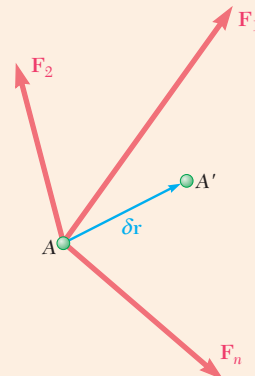


Fig. 10.17

Mechanical Efficiency

In the case of real machines, however [Sec. 10.1D], the work of the friction forces should be taken into account with the result that the **output work is less than the input work**. We defined the **mechanical efficiency** of a machine as the ratio

$$\eta = \frac{\text{output work}}{\text{input work}} \quad (10.9)$$

We noted that, for an ideal machine (no friction), $\eta = 1$, whereas for a real machine, $\eta < 1$.

Work of a Force over a Finite Displacement

In the second section of this chapter, we considered the work of forces corresponding to finite displacements of their points of application. We obtained the work $U_{1 \rightarrow 2}$ of the force $\mathbf{F}$ corresponding to a displacement of the particle A from A_1 to A_2 (Fig. 10.18) by integrating the right-hand side of Eqs. (10.1) or (10.1') along the curve described by the particle [Sec. 10.2A]. Thus,

$$U_{1 \rightarrow 2} = \int_{A_1}^{A_2} \mathbf{F} \cdot d\mathbf{r} \quad (10.11)$$

or

$$U_{1 \rightarrow 2} = \int_{s_1}^{s_2} (F \cos \alpha) ds \quad (10.11')$$

Similarly, we expressed the work of a couple of moment $\mathbf{M}$ corresponding to a finite rotation from θ_1 to θ_2 of a rigid body as

$$U_{1 \rightarrow 2} = \int_{\theta_1}^{\theta_2} M d\theta \quad (10.12)$$

Work of a Weight

We obtained the **work of the weight $\mathbf{W}$ of a body** as its center of gravity moves from the elevation y_1 to y_2 (Fig. 10.19) by setting $F = W$ and $\alpha = 180^\circ$ in Eq. (10.11') as

$$U_{1 \rightarrow 2} = - \int_{y_1}^{y_2} W dy = Wy_1 - Wy_2 \quad (10.13)$$

The work of $\mathbf{W}$ is therefore positive when the elevation y decreases.

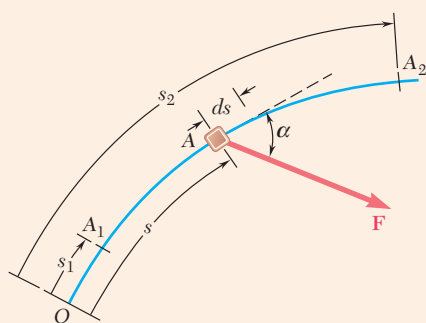


Fig. 10.18

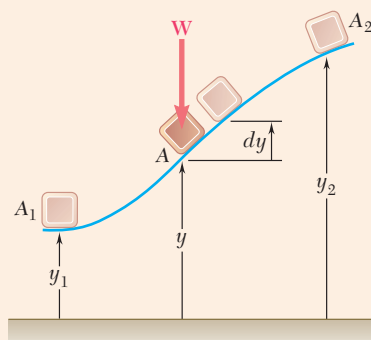


Fig. 10.19

Work of the Force Exerted by a Spring

The **work of the force $\mathbf{F}$ exerted by a spring** on a body A as the spring is stretched from x_1 to x_2 (Fig. 10.20) can be obtained by setting $F = kx$, where k is the constant of the spring, and $\alpha = 180^\circ$ in Eq. (10.11'). Hence,

$$U_{1 \rightarrow 2} = - \int_{x_1}^{x_2} kx \, dx = \frac{1}{2}kx_1^2 - \frac{1}{2}kx_2^2 \quad (10.15)$$

The work of $\mathbf{F}$ is therefore positive when the spring is returning to its undeformed position.

Potential Energy

When the work of a force $\mathbf{F}$ is independent of the path actually followed between A_1 and A_2 , the force is said to be a **conservative force**, and we can express its work as

$$U_{1 \rightarrow 2} = V_1 - V_2 \quad (10.20)$$

Here V is the **potential energy** associated with $\mathbf{F}$, and V_1 and V_2 represent the values of V at A_1 and A_2 , respectively [Sec. 10.2B]. We found the potential energies associated, respectively, with the force of gravity $\mathbf{W}$ and the elastic force $\mathbf{F}$ exerted by a spring to be

$$V_g = Wy \quad \text{and} \quad V_e = \frac{1}{2}kx^2 \quad (10.17, 10.18)$$

Alternative Expression for the Principle of Virtual Work

When the position of a mechanical system depends upon a single independent variable θ , the potential energy of the system is a function $V(\theta)$ of that variable, and it follows from Eq. (10.20) that $\delta U = -\delta V = -(dV/d\theta) \delta\theta$. The condition $\delta U = 0$ required by the principle of virtual work for the equilibrium of the system thus can be replaced by the condition

$$\frac{dV}{d\theta} = 0 \quad (10.21)$$

When all the forces involved are conservative, it may be preferable to use Eq. (10.21) rather than apply the principle of virtual work directly [Sec. 10.2C; Sample Prob. 10.4].

Stability of Equilibrium

This alternative approach presents another advantage, since it is possible to determine from the sign of the second derivative of V whether the equilibrium of the system is *stable*, *unstable*, or *neutral* [Sec. 10.2D]. If $d^2V/d\theta^2 > 0$, V is *minimum* and the equilibrium is *stable*; if $d^2V/d\theta^2 < 0$, V is *maximum* and the equilibrium is *unstable*; if $d^2V/d\theta^2 = 0$, it is necessary to examine derivatives of a higher order.

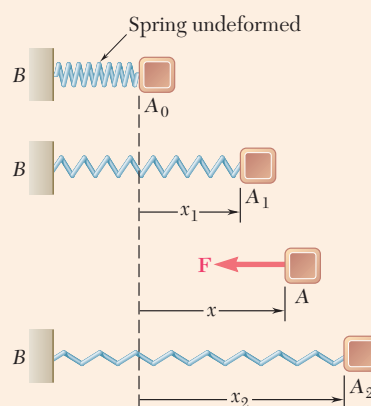


Fig. 10.20

Review Problems

- 10.101** Determine the vertical force P that must be applied at G to maintain the equilibrium of the linkage.

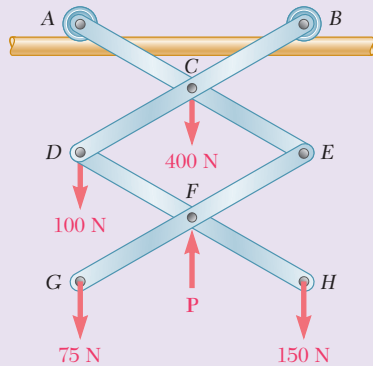


Fig. P10.103

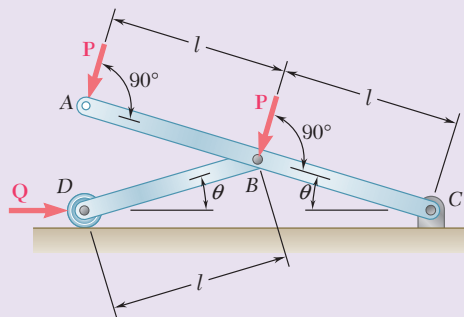


Fig. P10.104

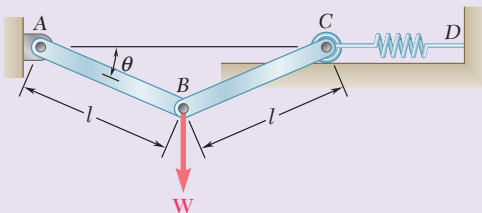


Fig. P10.106

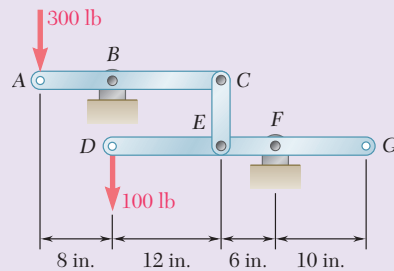


Fig. P10.101 and P10.102

- 10.102** Determine the couple M that must be applied to member $DEFG$ to maintain the equilibrium of the linkage.

- 10.103** Determine the force P required to maintain the equilibrium of the linkage shown. All members are of the same length, and the wheels at A and B roll freely on the horizontal rod.

- 10.104** Derive an expression for the magnitude of the force Q required to maintain the equilibrium of the mechanism shown.

- 10.105** Derive an expression for the magnitude of the couple M required to maintain the equilibrium of the linkage shown.

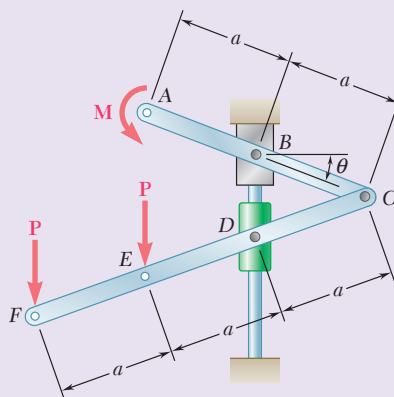


Fig. P10.105

- 10.106** A vertical load W is applied to the linkage at B . The constant of the spring is k , and the spring is unstretched when AB and BC are horizontal. Neglecting the weight of the linkage, derive an equation in θ , W , l , and k that must be satisfied when the linkage is in equilibrium.

- 10.107** A force $\mathbf{P}$ with a magnitude of 240 N is applied to end E of cable CDE , which passes under pulley D and is attached to the mechanism at C . Neglecting the weight of the mechanism and the radius of the pulley, determine the value of θ corresponding to equilibrium. The constant of the spring is $k = 4 \text{ kN/m}$, and the spring is unstretched when $\theta = 90^\circ$.

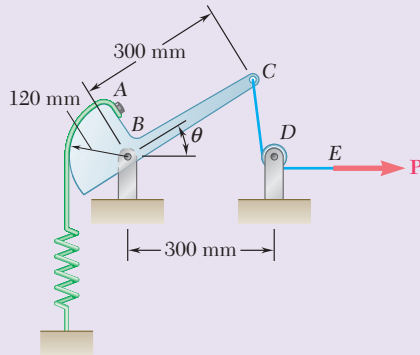


Fig. P10.107

- 10.108** Two identical rods ABC and DBE are connected by a pin at B and by a spring CE . Knowing that the spring is 4 in. long when unstretched and that the constant of the spring is 8 lb/in., determine the distance x corresponding to equilibrium when a 24-lb load is applied at E as shown.
- 10.109** Solve Prob. 10.108 assuming that the 24-lb load is applied at C instead of E .
- 10.110** Two uniform rods each with a mass m and length l are attached to gears as shown. For the range $0 \leq \theta \leq 180^\circ$, determine the positions of equilibrium of the system, and state in each case whether the equilibrium is stable, unstable, or neutral.
- 10.111** A homogeneous hemisphere with a radius r is placed on an incline as shown. Assuming that friction is sufficient to prevent slipping between the hemisphere and the incline, determine the angle θ corresponding to equilibrium when $\beta = 10^\circ$.

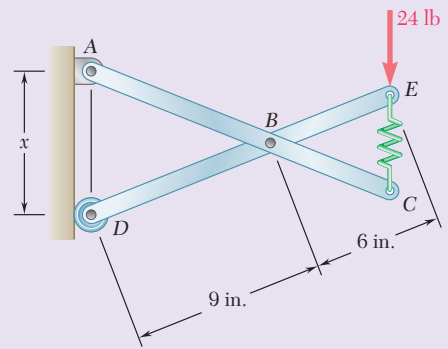


Fig. P10.108

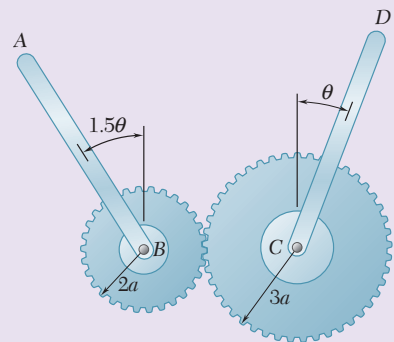


Fig. P10.110

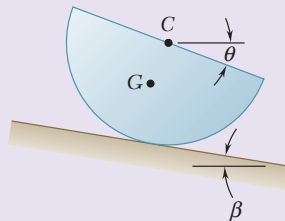


Fig. P10.111 and P10.112

- 10.112** A homogeneous hemisphere with a radius r is placed on an incline as shown. Assuming that friction is sufficient to prevent slipping between the hemisphere and the incline, determine (a) the largest angle β for which a position of equilibrium exists, (b) the angle θ corresponding to equilibrium when the angle β is equal to half the value found in part a.

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11

Kinematics of Particles

The motion of the paraglider can be described in terms of its *position*, *velocity*, and *acceleration*. When landing, the pilot of the paraglider needs to consider the wind velocity and the *relative motion* of the glider with respect to the wind. The study of motion is known as *kinematics* and is the subject of this chapter.

Introduction

11.1 RECTILINEAR MOTION OF PARTICLES

11.1A Position, Velocity, and Acceleration

11.1B Determining the Motion of a Particle

11.2 SPECIAL CASES AND RELATIVE MOTION

11.2A Uniform Rectilinear Motion

11.2B Uniformly Accelerated Rectilinear Motion

11.2C Motion of Several Particles

*11.3 GRAPHICAL SOLUTIONS

11.4 CURVILINEAR MOTION OF PARTICLES

11.4A Position, Velocity, and Acceleration Vectors

11.4B Derivatives of Vector Functions

11.4C Rectangular Components of Velocity and Acceleration

11.4D Motion Relative to a Frame in Translation

11.5 NON-RECTANGULAR COMPONENTS

11.5A Tangential and Normal Components

11.5B Radial and Transverse Components

Objectives

- **Describe** the basic kinematic relationships between position, velocity, acceleration, and time.
- **Solve** problems using these basic kinematic relationships and calculus or graphical methods.
- **Define** position, velocity, and acceleration in terms of Cartesian, tangential and normal, and radial and transverse coordinates.
- **Analyze** the relative motion of multiple particles by using a translating coordinate system.
- **Determine** the motion of a particle that depends on the motion of another particle.
- **Determine** which coordinate system is most appropriate for solving a curvilinear kinematics problem.
- **Calculate** the position, velocity, and acceleration of a particle undergoing curvilinear motion using Cartesian, tangential and normal, and radial and transverse coordinates.

Introduction

Chapters 1 to 10 were devoted to **statics**, i.e., to the analysis of bodies at rest. We now begin the study of **dynamics**, which is the part of mechanics that deals with the analysis of bodies in motion.

Although the study of statics goes back to the time of the Greek philosophers, the first significant contribution to dynamics was made by Galileo (1564–1642). Galileo's experiments on uniformly accelerated bodies led Newton (1642–1727) to formulate his fundamental laws of motion.

Dynamics includes two broad areas of study:

1. **Kinematics**, which is the study of the geometry of motion. The principles of kinematics relate the displacement, velocity, acceleration, and time of a body's motion, without reference to the cause of the motion.
2. **Kinetics**, which is the study of the relation between the forces acting on a body, the mass of the body, and the motion of the body. We use kinetics to predict the motion caused by given forces or to determine the forces required to produce a given motion.

Chapters 11 through 14 describe the **dynamics of particles**; in Chap. 11, we consider the **kinematics of particles**. The use of the word *particles* does not mean that our study is restricted to small objects; rather, it indicates that in these first chapters we study the motion of bodies—possibly as large as cars, rockets, or airplanes—without regard to their size or shape. By saying that we analyze the bodies as particles, we mean that we consider only their motion as an entire unit; we neglect any rotation about their own centers of mass. In some cases, however, such a rotation is not negligible, and we cannot treat the bodies as particles. Such motions are analyzed in later chapters dealing with the **dynamics of rigid bodies**.

In the first part of Chap. 11, we describe the rectilinear motion of a particle; that is, we determine the position, velocity, and acceleration of a particle at every instant as it moves along a straight line. We first use general methods of analysis to study the motion of a particle; we then consider two important particular cases, namely, the uniform motion and the uniformly accelerated motion of a particle (Sec. 11.2). We then discuss the simultaneous motion of several particles and introduce the concept of the relative motion of one particle with respect to another. The first part of this chapter concludes with a study of graphical methods of analysis and their application to the solution of problems involving the rectilinear motion of particles.

In the second part of this chapter, we analyze the motion of a particle as it moves along a curved path. We define the position, velocity, and acceleration of a particle as vector quantities and introduce the derivative of a vector function to add to our mathematical tools. We consider applications in which we define the motion of a particle by the rectangular components of its velocity and acceleration; at this point, we analyze the motion of a projectile (Sec. 11.4C). Then we examine the motion of a particle relative to a reference frame in translation. Finally, we analyze the curvilinear motion of a particle in terms of components other than rectangular. In Sec. 11.5, we introduce the tangential and normal components of an object's velocity and acceleration and then examine the radial and transverse components.

11.1 RECTILINEAR MOTION OF PARTICLES

A particle moving along a straight line is said to be in **rectilinear motion**. The only variables we need to describe this motion are the time, t , and the distance along the line, x , as a function of time. With these variables, we can define the particle's position, velocity, and acceleration, which completely describe the particle's motion. When we study the motion of a particle moving in a plane (two dimensions) or in space (three dimensions), we will use a more general position vector rather than simply the distance along a line.

11.1A Position, Velocity, and Acceleration

At any given instant t , a particle in rectilinear motion occupies some position on the straight line. To define the particle's position P , we choose a fixed origin O on the straight line and a positive direction along the line. We measure the distance x from O to P and record it with a plus or minus sign, according to whether we reach P from O by moving along the line in the positive or negative direction. The distance x , with the appropriate sign, completely defines the position of the particle; it is called the **position coordinate** of the particle. For example, the position coordinate corresponding to P in Fig. 11.1a is $x = +5$ m; the coordinate corresponding to P' in Fig. 11.1b is $x' = -2$ m.

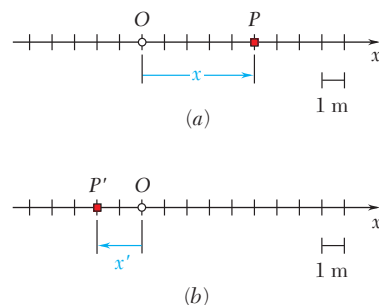


Fig. 11.1 Position is measured from a fixed origin. (a) A positive position coordinate; (b) a negative position coordinate.

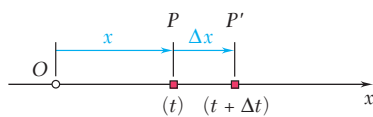


Fig. 11.2 A small displacement Δx from time t to time $t + \Delta t$.



Photo 11.1 The motion of this solar car can be described by its position, velocity, and acceleration.

When we know the position coordinate x of a particle for every value of time t , we say that the motion of the particle is known. We can provide a “timetable” of the motion in the form of an equation in x and t , such as $x = 6t^2 - t^3$, or in the form of a graph of x versus t , as shown in Fig. 11.6. The units most often used to measure the position coordinate x are the meter (m) in the SI system of units[†] and the foot (ft) in the U.S. customary system of units. Time t is usually measured in seconds (s).

Now consider the position P occupied by the particle at time t and the corresponding coordinate x (Fig. 11.2). Consider also the position P' occupied by the particle at a later time $t + \Delta t$. We can obtain the position coordinate of P' by adding the small displacement Δx to the coordinate x of P . This displacement is positive or negative according to whether P' is to the right or to the left of P . We define the **average velocity** of the particle over the time interval Δt as the quotient of the displacement Δx and the time interval Δt as

$$\text{Average velocity} = \frac{\Delta x}{\Delta t}$$

If we use SI units, Δx is expressed in meters and Δt in seconds; the average velocity is then expressed in meters per second (m/s). If we use U.S. customary units, Δx is expressed in feet and Δt in seconds; the average velocity is then expressed in feet per second (ft/s).

We can determine the **instantaneous velocity** v of a particle at the instant t by allowing the time interval Δt to become infinitesimally small. Thus,

$$\text{Instantaneous velocity} = v = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t}$$

The instantaneous velocity is also expressed in m/s or ft/s. Observing that the limit of the quotient is equal, by definition, to the derivative of x with respect to t , we have

**Velocity of a particle
along a line**

$$v = \frac{dx}{dt} \quad (11.1)$$

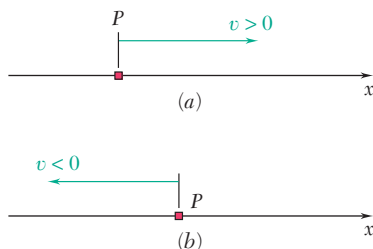


Fig. 11.3 In rectilinear motion, velocity can be only (a) positive or (b) negative along the line.

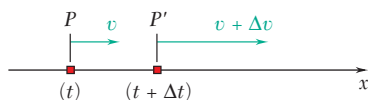


Fig. 11.4 A change in velocity from v to $v + \Delta v$ corresponding to a change in time from t to $t + \Delta t$.

We represent the velocity v by an algebraic number that can be positive or negative.[‡] A positive value of v indicates that x increases, i.e., that the particle moves in the positive direction (Fig. 11.3a). A negative value of v indicates that x decreases, i.e., that the particle moves in the negative direction (Fig. 11.3b). The magnitude of v is known as the **speed** of the particle.

Consider the velocity v of the particle at time t and also its velocity $v + \Delta v$ at a later time $t + \Delta t$ (Fig. 11.4). We define the **average acceleration** of the particle over the time interval Δt as the quotient of Δv and Δt as

$$\text{Average acceleration} = \frac{\Delta v}{\Delta t}$$

[†]See Sec. 1.3.

[‡]As you will see in Sec. 11.4A, velocity is actually a vector quantity. However, since we are considering here the rectilinear motion of a particle where the velocity has a known and fixed direction, we need only specify its sense and magnitude. We can do this conveniently by using a scalar quantity with a plus or minus sign. This is also true of the acceleration of a particle in rectilinear motion.

If we use SI units, Δv is expressed in m/s and Δt in seconds; the average acceleration is then expressed in m/s^2 . If we use U.S. customary units, Δv is expressed in ft/s and Δt in seconds; the average acceleration is then expressed in ft/s^2 .

We obtain the **instantaneous acceleration** a of the particle at the instant t by again allowing the time interval Δt to approach zero. Thus,

$$\text{Instantaneous acceleration} = a = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t}$$

The instantaneous acceleration is also expressed in m/s^2 or ft/s^2 . The limit of the quotient, which is by definition the derivative of v with respect to t , measures the rate of change of the velocity. We have

Acceleration of a particle along a line

$$a = \frac{dv}{dt} \quad (11.2)$$

or substituting for v from Eq. (11.1),

$$a = \frac{d^2x}{dt^2} \quad (11.3)$$

We represent the acceleration a by an algebraic number that can be positive or negative (see the footnote on the preceding page). A positive value of a indicates that the velocity (i.e., the algebraic number v) increases. This may mean that the particle is moving faster in the positive direction (Fig. 11.5a) or that it is moving more slowly in the negative direction (Fig. 11.5b); in both cases, Δv is positive. A negative value of a indicates that the velocity decreases; either the particle is moving more slowly in the positive direction (Fig. 11.5c), or it is moving faster in the negative direction (Fig. 11.5d).

Sometimes we use the term *deceleration* to refer to a when the speed of the particle (i.e., the magnitude of v) decreases; the particle is then moving more slowly. For example, the particle of Fig. 11.5 is decelerating in parts b and c; it is truly accelerating (i.e., moving faster) in parts a and d.

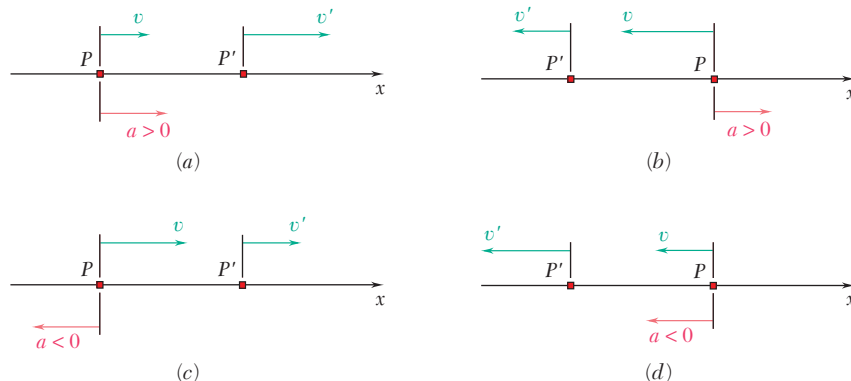


Fig. 11.5 Velocity and acceleration can be in the same or different directions. (a, d) When a and v are in the same direction, the particle speeds up; (b, c) when a and v are in opposite directions, the particle slows down.

We can obtain another expression for the acceleration by eliminating the differential dt in Eqs. (11.1) and (11.2). Solving Eq. (11.1) for dt , we have $dt = dx/v$; substituting into Eq. (11.2) gives us

$$a = v \frac{dv}{dx} \quad (11.4)$$

Concept Application 11.1

Consider a particle moving in a straight line, and assume that its position is defined by

$$x = 6t^2 - t^3$$

where t is in seconds and x in meters. We can obtain the velocity v at any time t by differentiating x with respect to t as

$$v = \frac{dx}{dt} = 12t - 3t^2$$

We can obtain the acceleration a by differentiating again with respect to t . Hence,

$$a = \frac{dv}{dt} = 12 - 6t$$

In Fig. 11.6, we have plotted the position coordinate, the velocity, and the acceleration. These curves are known as *motion curves*. Keep in mind, however, that the particle does not move along any of these curves; the particle moves in a straight line.

Since the derivative of a function measures the slope of the corresponding curve, the slope of the x - t curve at any given time is equal to the value of v at that time. Similarly, the slope of the v - t curve is equal to the value of a . Since $a = 0$ at $t = 2$ s, the slope of the v - t curve must be zero at $t = 2$ s; the velocity reaches a maximum at this instant. Also, since $v = 0$ at $t = 0$ and at $t = 4$ s, the tangent to the x - t curve must be horizontal for both of these values of t .

A study of the three motion curves of Fig. 11.6 shows that the motion of the particle from $t = 0$ to $t = \infty$ can be divided into four phases:

1. The particle starts from the origin, $x = 0$, with no velocity but with a positive acceleration. Under this acceleration, the particle gains a positive velocity and moves in the positive direction. From $t = 0$ to $t = 2$ s, x , v , and a are all positive.
2. At $t = 2$ s, the acceleration is zero; the velocity has reached its maximum value. From $t = 2$ s to $t = 4$ s, v is positive, but a is negative. The particle still moves in the positive direction but more slowly; the particle is decelerating.
3. At $t = 4$ s, the velocity is zero; the position coordinate x has reached its maximum value (32 m). From then on, both v and a are negative; the particle is accelerating and moves in the negative direction with increasing speed.
4. At $t = 6$ s, the particle passes through the origin; its coordinate x is then zero, while the total distance traveled since the beginning of the motion is 64 m (i.e., twice its maximum value). For values of t larger than 6 s, x , v , and a are all negative. The particle keeps moving in the negative direction—away from O —faster and faster. ■

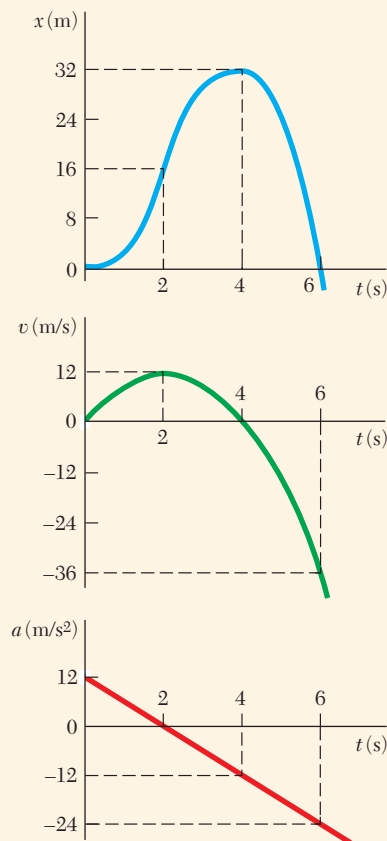


Fig. 11.6 Graphs of position, velocity, and acceleration as functions of time for Concept Application 11.1.

11.1B Determining the Motion of a Particle

We have just seen that the motion of a particle is said to be known if we know its position for every value of the time t . In practice, however, a motion is seldom defined by a relation between x and t . More often, the conditions of the motion are specified by the type of acceleration that the particle possesses. For example, a freely falling body has a constant acceleration that is directed downward and equal to 9.81 m/s^2 or 32.2 ft/s^2 , a mass attached to a stretched spring has an acceleration proportional to the instantaneous elongation of the spring measured from its equilibrium position, etc. In general, we can express the acceleration of the particle as a function of one or more of the variables x , v , and t . Thus, in order to determine the position coordinate x in terms of t , we need to perform two successive integrations.

Let us consider three common classes of motion.

1. $a = f(t)$. **The Acceleration Is a Given Function of t .** Solving Eq. (11.2) for dv and substituting $f(t)$ for a , we have

$$\begin{aligned} dv &= a dt \\ dv &= f(t) dt \end{aligned}$$

Integrating both sides of the equation, we obtain

$$\int dv = \int f(t) dt$$

This equation defines v in terms of t . Note, however, that an arbitrary constant is introduced after the integration is performed. This is due to the fact that many motions correspond to the given acceleration $a = f(t)$. In order to define the motion of the particle uniquely, it is necessary to specify the **initial conditions** of the motion, i.e., the value v_0 of the velocity and the value x_0 of the position coordinate at $t = 0$. Rather than use an arbitrary constant that is determined by the initial conditions, it is often more convenient to replace the indefinite integrals with **definite integrals**. Definite integrals have lower limits corresponding to the initial conditions $t = 0$ and $v = v_0$ and upper limits corresponding to $t = t$ and $v = v$. This gives us

$$\begin{aligned} \int_{v_0}^v dv &= \int_0^t f(t) dt \\ v - v_0 &= \int_0^t f(t) dt \end{aligned}$$

which yields v in terms of t .

We can now solve Eq. (11.1) for dx as

$$dx = v dt$$

and substitute for v the expression obtained from the first integration. Then we integrate both sides of this equation via the left-hand side with respect to x from $x = x_0$ to $x = x$ and the right-hand side with respect to t from $t = 0$ to $t = t$. In this way, we obtain the position coordinate x in terms of t ; the motion is completely determined.

We will study two important cases in greater detail in Sec. 11.2: the case when $a = 0$, corresponding to a *uniform motion*, and the case when $a = \text{constant}$, corresponding to a *uniformly accelerated motion*.

2. $a = f(x)$. **The Acceleration Is a Given Function of x .** Rearranging Eq. (11.4) and substituting $f(x)$ for a , we have

$$\begin{aligned} v \, dv &= a \, dx \\ v \, dv &= f(x) \, dx \end{aligned}$$

Since each side contains only one variable, we can integrate the equation. Denoting again the initial values of the velocity and of the position coordinate by v_0 and x_0 , respectively, we obtain

$$\begin{aligned} \int_{v_0}^v v \, dv &= \int_{x_0}^x f(x) \, dx \\ \frac{1}{2}v^2 - \frac{1}{2}v_0^2 &= \int_{x_0}^x f(x) \, dx \end{aligned}$$

which yields v in terms of x . We now solve Eq. (11.1) for dt , giving

$$dt = \frac{dx}{v}$$

and substitute for v the expression just obtained. We can then integrate both sides to obtain the desired relation between x and t . However, in most cases, this last integration cannot be performed analytically, and we must resort to a numerical method of integration.

3. $a = f(v)$. **The Acceleration Is a Given Function of v .** We can now substitute $f(v)$ for a in either Eqs. (11.2) or (11.4) to obtain either

$$\begin{aligned} f(v) &= \frac{dv}{dt} & f(v) &= v \frac{dv}{dx} \\ dt &= \frac{dv}{f(v)} & dx &= \frac{v \, dv}{f(v)} \end{aligned}$$

Integration of the first equation yields a relation between v and t ; integration of the second equation yields a relation between v and x . Either of these relations can be used in conjunction with Eq. (11.1) to obtain the relation between x and t that characterizes the motion of the particle.